

Excursions from Hyperplanes for the α -stable Lévy Process

submitted by

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Summary

This thesis focuses on advancing the theoretical understanding of isotropic α -stable Lévy processes, Brownian motion, and self-similar Markov processes (ssMps) through an excursion theory framework and skew-product representation. Each chapter addresses a specific set of problems, culminating in a collection of results with theoretical and practical significance.

In Chapter 1, we extend results on radial excursions of isotropic α -stable Lévy processes to Brownian motion ($\alpha = 2$), deriving a Deep Wiener-Hopf factorisation and explicit expressions for ladder potential measures. Additionally, we establish the Hunt-Nagasawa duality for isotropic α -stable processes and transient Brownian motion ($d \geq 3$), using it to time-reverse laws at first entry and last exit of a ball.

Chapter 2 develops an excursion theory for isotropic α -stable Lévy processes ($d \geq 2$, $\alpha \in (0, 2)$) relative to hyperplanes, characterising key distributions such as an n -tuple law at the moment of crossing a hyperplane. These results are extended to Brownian motion in $d \geq 2$. Additionally, conditional laws of processes attracted, repelled, or constrained by hyperplanes are explored.

In Chapter 3, we introduce the cylindrical Lamperti-Kiu transform to represent ssMps in $\mathbb{R}^d \setminus \mathbb{H}_0$ as Markov Additive Processes (MAPs). This transform enables the study of fluctuations and excursions relative to hyperplanes and slabs. The framework is applied to isotropic α -stable processes and extends to cylindrically positive self-similar Markov processes (cpssMps), generalising positive ssMp theory to higher dimensions.

Chapter 4 investigates first-entry and first-exit problems for isotropic α -stable processes in higher dimensions, specifically within infinite strips. A probabilistic approach is used to characterise the distribution of first hit or first exit to a slab. Numerical methods, including Monte Carlo sampling algorithms, are developed to overcome the challenges of solving integral equations in high dimensions.

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Preliminaries

In this chapter, we introduce the primary tools used to develop what we refer to as the *excursion from hyperplanes* framework for the isotropic α -stable Lévy process in dimension $d \geq 2$. This process is a Lévy process with the distinctive property that each of its d coordinates behaves as a one-dimensional symmetric α -stable process. To build a foundation, we begin by reviewing the relevant definitions and properties of Lévy processes in $\mathbb{R}$.

We then explore key concepts such as path variation, regularity, hitting probabilities, overshoot and undershoot distributions, and both point-recurrence and recurrence, among others. These ideas are subsequently extended to the d -dimensional setting. The chapter concludes by presenting the results from [37], which study excursions from radial maxima and minima for the isotropic α -stable Lévy process. These results served as the inspiration for the excursions that give name to this thesis.

The results in all but the last section of this chapter can be found in the books of Bertoin [4] and Kyprianou [32].

0.1 Lévy processes

A one-dimensional Lévy process is a stochastic process issued from the origin with stationary and independent increments and càdlàg paths, whose state space is $\mathbb{R} \cup \{\partial\}$. The state ∂ is an absorbing state, also called killing state or cemetery. If $X = (X_t : t \geq 0)$ is a one-dimensional Lévy process with law $\mathbb{P}$, there is a known expression for its characteristic exponent Ψ , called the Lévy-Khintchine formula, for $\theta \in \mathbb{R}$

$$\Psi(\theta) := -\frac{1}{t} \log \mathbb{E} \left[e^{i\theta X_t} \right] = ia\theta + \frac{1}{2}\sigma^2\theta^2 + \int_{\mathbb{R}} [1 - e^{i\theta x} + i\theta x \mathbf{1}_{\{x \in [-1,1]\}}] \Pi(dx) + q,$$

where $a \in \mathbb{R}$ is called the drift coefficient, $\sigma \geq 0$ is the Brownian coefficient, Π is a measure with no atom at 0 such that $\int_{\mathbb{R}} (1 \wedge x^2) \Pi(dx) < \infty$ called the Lévy measure and

the value $q \geq 0$ is the *killing rate* of the process, i.e. the parameter of an independent exponential clock which sends X to ∂ . The triplet (a, σ, Π) is called the characteristic triplet of X .

There is an interpretation to each of the terms in the Lévy-Khintchine formula, which roughly speaking says that any Lévy process can be written as the sum of three independent processes: the first one is a Brownian motion with drift, whose characteristic exponent corresponds to $ia\theta + \frac{1}{2}\sigma^2\theta^2$. The second process is a Compound Poisson process with jumps larger than one. And the third process a limit of compensated Compound Poisson process with jumps smaller than one. The characteristic exponent of the sum of these two processes correspond to the term $\int_{\mathbb{R}} [1 - e^{i\theta x} + i\theta x \mathbf{1}_{\{x \in [-1,1]\}}] \Pi(dx)$. This decomposition is useful for determining some path-related behaviours, and its precise formulation and proof can be found in Chapter 2 of [32].

Using the Lévy-Khintchine formula, it becomes straightforward to observe that, when the derivative $\Psi'(0) := \lim_{z \rightarrow 0} \Psi'(z)$ is well-defined, differentiating gives, for $t \geq 0$, the expression

$$\mathbb{E}[X_t] = i\Psi'(0)t = \mathbb{E}[X_1]t. \quad (0.1)$$

We refer to $\mathbb{E}[X_1]$ as the mean of the Lévy process, as it determines $\mathbb{E}[X_t]$ for any $t \geq 0$. When the mean is both well-defined and finite, together with the properties of stationary and independent increments, (0.1) strongly suggests the occurrence of a law of large numbers for X . This, in turn, provides insight into the long-term behaviour of X . The following theorem presents a fairly complete picture, showing a strict trichotomy: either X drifts to ∞ ($\lim_{t \rightarrow \infty} X_t = \infty$), drifts to $-\infty$ ($\lim_{t \rightarrow \infty} X_t = -\infty$), or oscillates ($\limsup_{t \rightarrow \infty} X_t = -\liminf_{t \rightarrow \infty} X_t = \infty$).

Long-term behaviour

Theorem 0.1. *Suppose that X is a one-dimensional Lévy process with characteristic measure Π .*

1. *If $\mathbb{E}[X_1]$ is defined and valued in $[-\infty, 0)$ or if $\mathbb{E}[X_1]$ is undefined and*

$$\int_{(1,\infty)} \frac{x\Pi(dx)}{\int_0^x \Pi(-\infty, -y)dy} < \infty,$$

then $\lim_{t \rightarrow \infty} X_t/t = \gamma_-$, where $\gamma_- = \mathbb{E}[X_1]$ in the first case and $\gamma_- = -\infty$ in the second case. In particular, in both cases,

$$\lim_{t \rightarrow \infty} X_t = -\infty.$$

2. If $\mathbb{E}[X_1]$ is defined and valued in $(0, \infty]$, or if $\mathbb{E}[X_1]$ is undefined and

$$\int_{(-\infty, -1)} \frac{|x| \Pi(dx)}{\int_0^{|x|} \Pi(y, \infty) dy} < \infty,$$

then $\lim_{t \rightarrow \infty} X_t/t = \gamma_+$, where $\gamma_+ = \mathbb{E}[X_1]$ in the first case and $\gamma_+ = \infty$ in the second case. In particular, in both cases,

$$\lim_{t \rightarrow \infty} X_t = \infty.$$

3. If $\mathbb{E}[X_1]$ is defined and equal to zero, or if $\mathbb{E}[X_1]$ is undefined and both of the integral tests in parts 1 and 2 fail, then $\lim_{t \rightarrow \infty} X_t/t = 0$ in the first case and $\limsup_{t \rightarrow \infty} X_t/t = -\liminf_{t \rightarrow \infty} X_t/t = \infty$ in the second case. Moreover, in both cases $\limsup_{t \rightarrow \infty} X_t = -\liminf_{t \rightarrow \infty} X_t = \infty$.

Paths of bounded variation

A càdlàg function $f : [0, t] \rightarrow \mathbb{R}$ is said to be of bounded (or finite) variation if

$$\sup_{s_1, \dots, s_n} \sum_{j=1}^n |f(s_j) - f(s_{j-1})| < \infty,$$

where the supremum is taken over all finite partitions $\{s_1 = 0 < s_2 < \dots < s_n = t\}$ of $[0, t]$. Then, it is known (cf. Section 2.6 in [32]) that a Lévy process X has paths which are almost surely of bounded variation on every set $[0, t]$ if and only if $\sigma = 0$ and either $\Pi(\mathbb{R}) < \infty$ or $\int_{|x| \leq 1} |x| \Pi(dx) < \infty$. In this case, the characteristic exponent can be written as

$$\Psi(\theta) = i\theta + \int_{\mathbb{R}} (1 - e^{i\theta x}) \Pi(dx) + q, \quad \theta \in \mathbb{R}.$$

A particular case of bounded variation occurs when X has almost surely increasing paths. In this case, X is said to be a subordinator, and instead of the characteristic exponent, it is more common to work with the Laplace exponent ψ , given by

$$\psi(\lambda) = -\log \mathbb{E} \left[e^{-\lambda X_1} \right].$$

For a subordinator we have

$$\psi(\lambda) = d\lambda + \int_{(0, \infty)} (1 - e^{-\lambda x}) \Pi(dx) + q.$$

Subordinators play a fundamental role in the theory of Lévy processes, as many broader results can be reduced to the analysis of appropriately selected subordinators. Their importance is particularly evident in the Wiener-Hopf factorisation (cf. Chapter 6 of [32]), where a Lévy process is decomposed into the ascending and descending ladder processes, both of which are subordinators. This factorisation is a key tool for studying the extremal behaviour of Lévy processes, such as the distribution of the supremum and infimum over a time interval.

Hitting points

To explore the behaviour of the isotropic α -stable Lévy process in relation to hyperplanes, it is useful to first consider how a Lévy process behaves with respect to the probability of hitting specific points. A Lévy process is said to hit a point $x \in \mathbb{R}$ if

$$\mathbb{P}(X_t = x \text{ for at least one } t > 0) > 0.$$

Let $C = \{x \in \mathbb{R} : \mathbb{P}(X_t = x \text{ for at least one } t > 0) > 0\}$ be the set of points that a Lévy process can hit. We say that a Lévy process can hit points if $C \neq \emptyset$. We have the following classification.

Theorem 0.2. *Suppose that X is not a compound Poisson process. Then X can hit points if and only if*

$$\int_{\mathbb{R}} \operatorname{Re} \left(\frac{1}{1 + \Psi(z)} \right) dz < \infty.$$

Moreover, we have that

- (i) *If $\sigma > 0$, then X can hit points and $C = \mathbb{R}$*
- (ii) *If $\sigma = 0$ but X is of unbounded variation and X can hit points, then $C = \mathbb{R}$*
- (iii) *If X is of bounded variation, then X can hit points if and only if $a + \int_{|x| < 1} x \Pi(dx) \neq 0$. In that case $C = \mathbb{R}$ unless X or $-X$ is a subordinator and then $C = (0, \infty)$ or $C = (-\infty, 0)$ respectively.*

Regularity of the half-line

For a one-dimensional Lévy process X (which starts at zero) we say that 0 is regular for $(0, \infty)$ (equiv. the upper half-line) if X enters $(0, \infty)$ immediately. That is to say, if $\mathbb{P}(\tau_0^+ = 0) = 1$, where $\tau_0^+ = \inf\{t > 0 : X_t > 0\}$. Because of Blumenthal's 0-1 law, the probability $\mathbb{P}(\tau_0^+ = 0)$ is necessarily zero or one. When this probability is zero, we say that 0 is irregular for $(0, \infty)$. We also say that 0 is regular for $(-\infty, 0)$ (equiv. the

lower half-line) if $-X$ is regular for the upper half-line.

Theorem 0.3. *For a Lévy process X , the point 0 is regular for $(0, \infty)$ if and only if one of the following three situations occurs:*

- (i) X is a process of unbounded variation,
- (ii) X is a process of bounded variation and $a + \int_{|x|<1} x\Pi(dx) < 0$,
- (iii) X is a process of bounded variation $a + \int_{|x|<1} x\Pi(dx) = 0$ and

$$\int_0^1 \frac{x\Pi(dx)}{\int_0^x \Pi(-\infty, -y)dy} = \infty.$$

Let us assume henceforth that neither X nor $-X$ is an unkilled subordinator. Define the running maximum and minimum processes by

$$\overline{X}_t = \sup_{s \leq t} X_s, \quad \text{and} \quad \underline{X}_t = \inf_{s \leq t} X_s, \quad t \geq 0,$$

respectively. If we take, for example, the range of the process $\overline{X} = (\overline{X}_t : t \geq 0)$, then it corresponds precisely to the range of the first passage points on $(0, \infty)$ given by $\{X_{\tau_x^+} : x > 0\}$. Therefore, we can expect an understanding of how Lévy processes undergo the process of first passage over/under a level to be closely related to the distributional properties of $\overline{X}$ and $\underline{X}$.

It can be shown that there exists a random measure L on $[0, \infty)$ with the property that its support agrees with the closure of the set $\{t \geq 0 : X_t = \overline{X}_t\}$ and with the property that, if T is any $\mathbb{F}$ -stopping time¹ such that $X_T = \overline{X}_T$ on $\{T < \infty\}$, then $((X_{T+t} - X_T, L_{T+t} - L_T) : t \geq 0)$ has the same law as $((X_t, L_t) : t \geq 0)$ under $\mathbb{P}$. Moreover, the complement of its support consists of a countable union of open intervals, each one corresponding to an excursion of X from its running maximum.

It is a straightforward exercise to show that the reflected process $\overline{X} - X$ is a strong Markov process. The process $L = (L_t : t \geq 0)$, referred to as the local time at zero of the reflected process (or equivalently, the local time of X at its maximum), has the notable property that its inverse, $(L_t^{-1} : t \geq 0)$, given by $L_t^{-1} = \inf\{s > 0 : L_s > t\}$, $t \geq 0$, is a subordinator. This subordinator is killed if $L_\infty < \infty$, which happens precisely when $\lim_{t \rightarrow \infty} X_t = -\infty$. Moreover, due to the regenerative nature of the point zero for the process $\overline{X} - X$, the random variable L_∞ follows an exponential distribution (with a

¹assume that $\mathbb{F}$ is the completion of the filtration under which X is defined.

rate parameter that may be zero, in which case L_∞ is infinite almost surely).

We define the process $H = (H_t : t \geq 0)$ by the relation

$$H_t = X_{L_t^{-1}}, \quad t < L_\infty,$$

and otherwise $H_t = \infty$, where ∞ can be considered as a cemetery state. The need for a cemetery state appears if and only if the Lévy process drifts to $-\infty$. The range of the process H in $[0, \infty)$ agrees with the range of $\bar{X}$. Moreover, the regenerative property of the state 0 for the reflected process $\bar{X} - X$ ensures that H is a subordinator, killed at a constant rate which may be zero (when $L_\infty = \infty$). The process H may also possess a drift component, corresponding to there being Lebesgue mass in the range of H .

Naturally, by considering the process $-X$ (which is still a Lévy process), it is immediately obvious that everything we have described above can also be constructed for the process reflected in its infimum, $X - \underline{X}$. If $\underline{L}$ is the local time at zero for $X - \underline{X}$, then we define $\hat{H} = -X_{\underline{L}_t^{-1}}$. We call H the ascending ladder height process and $\hat{H}$ the descending ladder height process.

Suppose that we denote the Laplace exponent of H by κ , and by $\hat{\kappa}$ the Laplace exponent of $\hat{H}$.

Theorem 0.4 (Wiener-Hopf factorisation). *For $z \in \mathbb{R}$, we have*

$$\Psi(z) = \kappa(-iz)\hat{\kappa}(iz).$$

Other identities also go by the name *Wiener-Hopf factorisation*, some of which involve the Laplace exponents of local times. For the proof of Theorem 0.4 and a more comprehensive discussion on this topic, we refer the reader to Chapter 6 of [32].

0.2 One-dimensional α -stable processes

An α -stable Lévy process, $\alpha \in (0, 2)$, is a Lévy process $X = (X_t : t \geq 0)$ with laws $(\mathbb{P}_x : x \in \mathbb{R})$ (this means $X_0 = x$ $\mathbb{P}_x$ -a.s.), for which the *scaling property* holds, i.e. for $c > 0$, $(cX_{c^{-\alpha}t} : t \geq 0)$ under $\mathbb{P}_x$ is equal in law to X under $\mathbb{P}_{cx}$.

As a Lévy process it can be defined via its characteristic triplet which takes the form $(0, 0, \Pi)$, and Π , the Lévy measure is given by

$$\Pi(dx) = |x|^{-1-\alpha} (c_1 \mathbf{1}_{\{x>0\}} + c_2 \mathbf{1}_{\{x<0\}}) dx \quad x \in \mathbb{R}, \quad (0.2)$$

such that $c_1 \geq 0$, $c_2 \geq 0$ and $\alpha \in (0, 2)$. However, our preferred way of referring to one-dimensional stable processes will be through a normalised version of these processes, which reduces the number of parameters from three, (α, c_1, c_2) to two (α, ρ) . For the aforesaid normalised version of X we have

$$\Psi(z) = |z|^\alpha \left(e^{\pi i \alpha (\frac{1}{2} - \rho)} \mathbf{1}_{\{z > 0\}} + e^{-\pi i \alpha (\frac{1}{2} - \rho)} \mathbf{1}_{\{z < 0\}} \right) \quad z \in \mathbb{R},$$

where $\rho = \mathbb{P}(X_1 > 0)$ is called the positivity parameter. If X is a normalised stable process, the coefficients c_1 and c_2 defining the Lévy measure, must be given by

$$c_1 = \Gamma(1 + \alpha) \frac{\sin(\pi \alpha \rho)}{\pi}, \quad c_2 = \Gamma(1 + \alpha) \frac{\sin(\pi \alpha \hat{\rho})}{\pi},$$

where $\hat{\rho} = 1 - \rho$. With this parametrisation, the two parameters defining the stable process belong to the admissible set

$$\begin{aligned} \mathcal{A} := & \{ \alpha \in (0, 1), \rho \in [0, 1] \} \\ & \cup \{ \alpha = 1, \rho = 1/2 \} \\ & \cup \{ \alpha \in (1, 2), \rho \in [1 - 1/\alpha, 1/\alpha] \}. \end{aligned} \quad (0.3)$$

In the one-dimensional setting, we will often find ourselves in future analysis dealing with three different regimes of α hinted in (0.3) : $\alpha \in (0, 1)$, $\alpha = 1$ and $\alpha \in (1, 2)$. This is due to the different distributional and path properties for the stable process that we detail below. For a more detailed treatment of such properties we refer the reader to [35].

Path variation. An α -stable Lévy process has paths of bounded variation if and only if $\alpha \in (0, 1)$. When $\alpha \in [1, 2)$, as the associated stable process has paths of unbounded variation, the case of monotone paths is ruled out.

Asymmetry and creeping. If the process has no negative jumps ($c_2 = 0$) and $\alpha \in (0, 1)$, then necessarily $\rho = 1$, or in other words X must be a subordinator. Similarly, if $c_1 = 0$ and $\alpha \in (0, 1)$, then $-X$ is a subordinator. A stable subordinator has no drift. This is easy to see from the law of large numbers for Lévy processes and the fact that $\Psi(z)/z \rightarrow 0$ as $|z| \rightarrow \infty$. Due to the lack of drift, we can say that a stable subordinator does not creep upwards. In other words $\mathbb{P}(X_{\tau_x^+} = x) = 0$ for $\tau_x^+ = \inf\{t > 0 : X_t > x\}$.

The Laplace exponent of an α -stable subordinator is given by

$$\psi(\lambda) = \lambda^\alpha \quad \lambda \geq 0.$$

From the set of admissible parameters (0.3) we observe that if $\alpha \in (1, 2)$, then ρ is bounded to be at most $1/\alpha$. This corresponds to the case $c_2 = 0$, making the associated process spectrally positive.

Drifting and oscillating. It is known (cf. Chapter 3 in [35]) that when $\alpha \in (0, 1]$, stable processes have no moments. But a first moment exists when $\alpha \in (1, 2)$. Moreover $\mathbb{E}[X_t] = i\Psi'(0)t = 0$ for all $t \geq 0$.

When $\alpha \in (0, 1]$ and the process experiments two-sided jumps ($c_1, c_2 > 0$), both positive and negative parts of the Lévy measure have a density which is proportional to $1/|x|^{1+\alpha}$. A straightforward application of theorem 0.1 for the limit behaviour of the process forces us to conclude that

$$\limsup_{t \rightarrow \infty} X_t = -\liminf_{t \rightarrow \infty} X_t = \infty.$$

This is also the case for $\alpha \in (1, 2)$, regardless of the value of ρ .

Transience and recurrence. It can be proved that X is transient, i.e. for all $\epsilon > 0$,

$$\mathbb{P} \left(\int_0^\infty \mathbf{1}_{\{|X_t| < \epsilon\}} dt < \infty \right) = 1,$$

whenever $\alpha \in (0, 1)$ and recurrent when $\alpha \in [1, 2)$. It is known that a recurrent process is necessarily oscillating, confirming what Theorem 0.1 claimed previously. However, the case of $\alpha \in (0, 1)$ is an example that an oscillating behaviour does not necessarily imply recurrence.

Hitting points. Using Theorem 0.2 we can see that X hits points if and only if $\alpha \in (1, 2)$.

Regularity of the half lines. If X is not a subordinator, then Theorem 0.3 establishes that 0 is regular for $(0, \infty)$ and for $(-\infty, 0)$ for $\alpha \in (0, 2)$.

Wiener-Hopf factorisation and the first passage problem

If we consider X a symmetric α -stable process, for which $\rho = \mathbb{P}(X_t > 0) = 1/2$, we easily observe from (0.2) that $\Psi(z) = |z|^\alpha$. Then we can verify that the Wiener-Hopf factorisation of this process is given by

$$|z|^\alpha = |-iz|^{\alpha/2} |iz|^{\alpha/2}, \tag{0.4}$$

indicating that the ascending and descending ladder height processes H and $\widehat{H}$ given by

$$H_t = X_{L_t^{-1}}, \quad \widehat{H}_t := X_{\underline{L}_t^{-1}} \quad t \geq 0, \quad (0.5)$$

respectively, are stable subordinators of index $\alpha/2$. It is also possible to show (see Theorem 6.12 in [32]) that $L^{-1} = (L_t^{-1} : t \geq 0)$ is a $1/2$ -stable subordinator.

For $z > 0$ the associated renewal measure $U(dz) := \mathbb{E}(\int_0^\infty \mathbf{1}_{\{H_t \in dz\}} dt)$ and Lévy measure of the ladder height processes, called the ascending and descending ladder potentials, are given respectively by

$$U(dz) = \frac{z^{\alpha/2-1}}{\Gamma(\alpha/2)} dz \quad \text{and} \quad \Pi(dz) = \frac{z^{-1-\alpha/2}}{-\Gamma(-\alpha/2)} dz. \quad (0.6)$$

A well known result for the first passage of a subordinator over a level (cf. Chapter 3 of [4]) claims that, if H is a subordinator with renewal measure $U(dz)$, Lévy measure Π_H and $T_a = \inf\{t > 0 : H_t > a\}$, then

$$\mathbb{P}(H_{T_a-} \in dy, H_{T_a} \in dz) = U(dy)\Pi(dz - y), \quad 0 \leq y \leq a < z.$$

For H , the $\frac{\alpha}{2}$ -stable subordinator mentioned before, this reads as

$$\mathbb{P}(H_{T_a-} \in dy, H_{T_a} \in dz) = \frac{\alpha/2 \sin(\pi\alpha/2)}{\pi} y^{\alpha/2-1} (z-y)^{-\alpha/2-1} dy dz, \quad 0 < y < a < z.$$

A more general result can be stated appealing to the Wiener-Hopf factorisation in the case of the first passage of symmetric α -stable process. Let us define $\tau_a^+ = \inf\{t > 0 : X_t > a\}$ and $\overline{X} := (\overline{X}_t : t \geq 0)$ is the process given by $\overline{X}_t = \sup_{s \leq t} X_s$, $t \geq 0$.

Theorem 0.5. *Suppose that X is a symmetric α -stable process. Then for all $y \in [0, a]$, $v \geq y$ and $u > 0$,*

$$\begin{aligned} & \mathbb{P}(X_{\tau_a^+} - a \in du, a - X_{\tau_a^+} \in dv, a - \overline{X}_{\tau_a^+} \in dy) \\ &= \frac{\sin(\pi\alpha/2)}{\pi} \frac{\Gamma(\alpha/2 + 1)}{\Gamma(\alpha/2)^2} \frac{(a-y)^{\alpha/2-1} (v-y)^{\alpha/2-1}}{(v+u)^{1+\alpha}} dy dv du. \end{aligned} \quad (0.7)$$

The random variable $\overline{X}_{\tau_a^+}$ will be of particular interest for us due to its connection to excursions from the maximum of X , and the two equivalent interpretations we will make of it are: the point of closest reach to the level a before crossing it, and the maximum at an instant before crossing a given level from below. In the next chapter

we will generalise this notion to higher dimensions and it will turn out that the density of this random variable is related to a renewal measure which we will call the ascending potential measure.

We will now consider a stable process in $\mathbb{R}^d$, $d \geq 2$ with a symmetry property with respect to all axes. The notion of symmetry is then generalised by that of isotropy. All the results considered in Chapter 2 will marginalise correctly if $d = 1$ is considered, however these are well known results and for our purposes we shall always consider $d \geq 2$.

0.3 The isotropic α -stable process

We say that $X = (X_t : t \geq 0)$, a Lévy process equipped with probabilities $(\mathbb{P}_x : x \in \mathbb{R}^d)$ ($\mathbb{P}_x$ indicates that $X_0 = x$ $\mathbb{P}_x$ -a.s.), is a d -dimensional isotropic α -stable process if its characteristic exponent is given by

$$\Psi(\theta) = -\frac{1}{t} \log \mathbb{E} \left[e^{i\theta, X_t} \right] = |\theta|^\alpha \quad \theta \in \mathbb{R}^d, \quad (0.8)$$

where we write $\mathbb{P}$ in place of $\mathbb{P}_0$. Across this work we will always consider $\alpha \in (0, 2)$ unless it is clearly indicated. The isotropic α -stable process X with $\alpha \in (0, 2)$ is a pure-jump Lévy process, characterised then by its jump measure Π , which satisfies

$$\begin{aligned} \Pi(B) &= \frac{2^\alpha \Gamma((d+\alpha)/2)}{\pi^{d/2} |\Gamma(-\alpha/2)|} \int_B \frac{\ell_d(dy)}{|y|^{\alpha+d}}, \\ &= 2^{\alpha-1} \pi^{-d} \frac{\Gamma((d+\alpha)/2) \Gamma(d/2)}{|\Gamma(-\alpha/2)|} \int_{\mathbb{S}^{d-1}} r^{d-1} \sigma_1(d\theta) \int_0^\infty \mathbf{1}_B(r\theta) \frac{1}{r^{\alpha+d}} dr, \end{aligned} \quad (0.9)$$

where $B \in \mathcal{B}(\mathbb{R}^d)$, ℓ_d is the d -dimensional Lebesgue measure and $\sigma_1(d\theta)$ is the surface measure on $\mathbb{S}^{d-1} := \{x \in \mathbb{R}^d : |x| = 1\}$ normalised to have unit mass. The changes in the constant come from the Jacobian in the change of coordinate system.

Additionally to the properties of a Lévy process, the α -stable process possesses the following self-similarity property. For any $c > 0$ and $x \in \mathbb{R}^d$, the law of $(cX_{c^{-\alpha}t} : t \geq 0)$ under $\mathbb{P}_x$, is that of X under $\mathbb{P}_{cx}$. The isotropy of X means that for every orthogonal transformation $U : \mathbb{R}^d \rightarrow \mathbb{R}^d$ and $x \in \mathbb{R}^d$, the law of

$$(UX_t : t \geq 0) \text{ under } \mathbb{P}_x \text{ is equal to the law of } (X_t : t \geq 0) \text{ under } \mathbb{P}_{Ux}.$$

Observe that using isotropy it is possible to translate an event originally stated for

a given hyperplane $\mathbb{H}_v = \{x \in \mathbb{R}^d : \langle x, v \rangle = 0\}$, into an event now regarding the hyperplane $\mathbb{H}_0 = \{x \in \mathbb{R}^d : x^{(1)} = 0\}$, where $x^{(1)}$ represents the first coordinate of x .

An example of a Lévy process with the self-similarity and isotropy properties is the d -dimensional Brownian motion. The characteristic exponent of Brownian motion is given by the function (0.8) taking $\alpha = 2$. We will separate the study of Brownian motion and present it separately due to the continuity of its paths. This is the reason why many of the techniques used for stable processes do not extend in a simple way to the Brownian case.

To describe how the paths of X behave in relation to hyperplanes and other related domains in $\mathbb{R}^d$ we need to point out that a subset of $k \in \{1, 2, \dots, d\}$ coordinates of X , under $\mathbb{P}_x$, is in fact a k -dimensional isotropic α -stable Lévy process. This follows by evaluating (0.8) on the vector $\theta_k := (\theta^{(1)}, \theta^{(2)}, \dots, \theta^{(k)}, 0, \dots, 0)$ and noting that

$$\Psi(\theta_k) = |\theta_k|^\alpha.$$

In particular $X^{(1)}$ is a symmetric real valued α -stable process. By using isotropy, the previous property about the coordinates of X and recalling some path properties of symmetric real-valued stable processes from the previous section we deduce the following.

- The isotropic α -stable process touches $\mathbb{H}_0$ (and, by isotropy, any other hyperplane) in finite time with positive probability when $\alpha \in (1, 2)$ and with probability zero when $\alpha \in (0, 1]$. A similar story holds if we replace $\mathbb{H}_0$ by the sphere $\mathbb{S}^{d-1}$. In that case if $\alpha \in (0, 1)$ the process will not hit the unit sphere in finite time but if $1 < \alpha < d$ there is a positive probability of hitting the sphere in finite time. The latter assertion was proved in [47] by studying the renewal measure of the radial process $|X| = (|X_t| : t \geq 0)$.
- Although X is transient in the sense that $\lim_{t \rightarrow \infty} |X_t| = \infty$, due to the oscillation of $X^{(1)}$, i.e. $\limsup_{t \rightarrow \infty} X_t = -\liminf_{t \rightarrow \infty} X_t = +\infty$, for any $x \in \mathbb{R}^d$ the process $\mathbb{P}_x$ -almost surely visits $\mathbb{H}_{0\downarrow} = \{z \in \mathbb{R}^d : z^{(1)} < 0\}$ and $\mathbb{H}_{0\uparrow} = \{z \in \mathbb{R}^d : z^{(1)} > 0\}$ the ‘lower’ and ‘upper’ half-spaces an infinite number of times.
- The process X started in $x = (x^{(1)}, \dots, x^{(d)})$ is regular for both the half spaces $(x^{(1)}, \infty) \times \mathbb{R}^{d-1}$ and $(-\infty, x^{(1)}) \times \mathbb{R}^{d-1}$. This is, for $\tau_{x^{(1)}}^\wedge := \inf\{t > 0 : X^{(1)} > x^{(1)}\}$ and $\tau_{x^{(1)}}^\vee := \inf\{t > 0 : X^{(1)} < x^{(1)}\}$, we necessarily have $\mathbb{P}_x(\tau_{x^{(1)}}^\wedge = 0) = \mathbb{P}_x(\tau_{x^{(1)}}^\vee = 0) = 1$.

- Crossing a hyperplane does not occur by creeping. This means, for $r \neq x^{(1)}$ and $\tau_r^\vee = \inf\{t > 0 : X_t^{(1)} < r\}$ and $\tau_r^\wedge = \inf\{t > 0 : X_t^{(1)} > r\}$

$$\mathbb{P}_x(X_{\tau_r^\vee}^{(1)} < r) = \mathbb{P}_x(X_{\tau_r^\wedge}^{(1)} > r) = 1.$$

With all these properties in mind we will develop a theory of excursions from a directional minimum/maximum. These are excursions ‘indexed’ on the set of times at which the process reaches a new minimum in the direction of $X^{(1)}$. By isotropy, these excursions work analogously for the maximum in the direction of $X^{(1)}$ and for extrema in any other given direction.

The excursion theory framework we are about to introduce is applicable to Markov Additive Processes. These processes encompass a variety of cases, including the isotropic α -stable Lévy process seen as $X = (X^{(1)}, X^{(2:d)})$ and the process underlying the skew-product representation of the isotropic α -stable process, known as the Lamperti-Kiu transform, among others.

0.4 Excursions of Markov Additive Processes

Before introducing the definition of Markov additive processes (MAPs) let us introduce the notion of Feller process. Consider E a locally compact Hausdorff space with countable base, such as $\mathbb{R}^d$, and $C_0(E)$ be the Banach space of real-valued continuous bounded functions on E which decay to 0 as $\|x\| \rightarrow \infty$ equipped with the supremum norm. Recall that an E -valued Markov process X defined in some completed, filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0})$ with laws $(\mathbb{P}_x : x \in E)$ is said to be a Feller process if its semigroup $\mathcal{P}_t[f](x) := \mathbb{E}_x[f(X_t)]$ satisfies the so-called Feller properties. That is, for any $f \in C_0(E)$,

1. for all $t \geq 0$ $\mathcal{P}_t[f] \in C_0(E)$, and
2. for all $x \in E$, $\lim_{t \downarrow 0} \mathcal{P}_t[f](x) = f(x)$.

An important consequence of X being Feller is that it is also strongly Markovian and admits a modification that is both càdlàg and quasi-left-continuous. The latter property means that for any increasing sequence of stopping times $(T_n)_{n \geq 0}$ with respect to the filtration $(\mathcal{F}_t)_{t \geq 0}$, converging to a stopping time T , we have $X(T_n) \rightarrow X(T)$ almost surely on the event $\{T < \infty\}$. We will refer to this modification as a *regular* Feller process. For more details the reader is referred to Section III.2 of [51].

We say that $(\Xi, \Upsilon) = ((\Xi_t, \Upsilon_t), t \geq 0)$ is a Markov additive process if it is a Feller

process on $\mathbb{R} \times \mathcal{S}$, with probabilities $(\mathbf{P}_{x,\theta} : x \in \mathbb{R}, \theta \in \mathcal{S})$, where $\mathcal{S}$ is a locally compact space with countable base, such that, for any $t \geq 0$, the conditional law of the process $((\Xi_{s+t} - \Xi_t, \Upsilon_{s+t}) : s \geq 0)$, given $((\Xi_u, \Upsilon_u), u \leq t)$ is that of (Ξ, Υ) under $\mathbf{P}_{0,\theta}$, with $\theta = \Upsilon_t$. For a MAP (Ξ, Υ) we call Ξ the *ordinate* and Υ the *modulator*.

For instance, the isotropic α -stable Lévy process X , split coordinate-wise as $(X^{(1)}, X^{(2:d)})$, constitutes a MAP with ordinate $X^{(1)}$ and modulator $X^{(2:d)}$. This is a simple consequence of the independence and stationarity of the increments of X .

The following example of a Markov additive process, related to isotropic α -stable processes through the so-called Lamperti-Kiu transform, will help us to understand more general excursions for MAPs and their consequences. We will present some of the results of [37], product of a theory of excursions from the radial minimum of X . These results are the inspiration for the development of excursions from directional extrema.

The Lamperti-Kiu transform

According to [1], there exists a MAP (ξ, Θ) with probabilities $(\mathbf{P}_{r,\theta} : r \in \mathbb{R}, \theta \in \mathbb{S}^{d-1})$ such that the isotropic α -stable Lévy process $X = (X_t : t \geq 0)$ issued at $x \in \mathbb{R}^d$, for $\alpha \in (0, 2]$ and $d \geq 2$, can be written as the following product, called the Lamperti-Kiu representation of X

$$X_t = \exp\{\xi_{\varphi(t)}\} \Theta_{\varphi(t)} \quad t \geq 0, \quad (0.10)$$

where $\varphi(t) = \inf\{s > 0 : \int_0^s e^{\alpha \xi_u} du > t\}$, and (ξ, Θ) is issued at $(\log|x|, \arg(x))$. Moreover, the radial process $|X| = (|X_t| : t \geq 0)$ can be expressed as $|X_t| = e^{\xi_{\varphi(t)}}$ $t \geq 0$, which implies that (0.10) is a skew-product representation of X in radius and position in the $(d-1)$ -dimensional sphere.

It was shown in [14] that, due to isotropy, the process ξ belongs to the class of so-called hypergeometric Lévy processes, a well-explored class with many known properties (cf. Chapter 4 of [35]). In particular, its Wiener-Hopf factorisation is known explicitly. That is, if its characteristic exponent $\Psi_\xi(\theta) = -\log \mathbf{E}_0[\exp(i\theta\xi_1)]$, $\theta \in \mathbb{R}$, then up to a multiplicative constant

$$\Psi_\xi(\theta) = \frac{\Gamma(\frac{1}{2}(-i\theta + \alpha))}{\Gamma(-\frac{1}{2}i\theta)} \times \frac{\Gamma(\frac{1}{2}(i\theta + d))}{\Gamma(\frac{1}{2}(i\theta + d - \alpha))}, \quad \theta \in \mathbb{R}, \quad (0.11)$$

where the two terms either side of the multiplication sign constitute the two Wiener-Hopf factors. These factors are known to characterise the running maximum and running infimum of $|X|$ and it has been proved that they belong to the class of so-

called beta subordinators (see [27, 35]). Based on the properties of beta subordinators one can check the regularity properties of ξ , which turns out to be regular for both the upper and lower half-lines. This means that any sphere of radius $r > 0$ is regular for both its interior and exterior for X .

For any $y \in \mathbb{R}$, let $\tau_y^- := \inf\{t > 0 : \xi_t < y\}$. We say that $((\xi, \Theta), \mathbf{P})$ is downwards regular if

$$\mathbf{P}_{0,\theta}(\tau_0^- = 0) = 1 \quad \forall \theta \in \mathbb{S}^{d-1}.$$

For $r > 0$, let $\tau_r^\oplus := \inf\{t > 0 : |X_t| < r\}$. Similarly, we say that $(X, \mathbb{P})$ is sphere-interior regular if and only if $\mathbb{P}_z(\tau_1^\oplus = 0) = 1$ for all $z \in \mathbb{R}^d$ with $|z| = 1$. As mentioned before the MAP in the Lamperti-Kiu representation of X , $((\xi, \Theta), \mathbf{P})$, is downwards regular. This fact is crucial for constructing what is called a local time at the minimum of ξ .

We will now develop an excursion theory framework from the minimum of the ordinate of general MAPs, nonetheless we will keep the notation (ξ, Θ) with laws $\mathbf{P}$, for a MAP in $\mathbb{R} \times \mathcal{S}$. The construction of excursions from different sets can be defined similarly. However, in this work we will only deal with excursions from the minimum/maximum.

Let $\xi_t := \inf_{s \leq t} \xi_s$ and $U_t := \xi_t - \xi_t$. Then under $\mathbf{P}_{0,\theta}$ the process $(\Theta_t, \xi_t, U_t)_{t \geq 0}$ is an $\mathcal{S} \times \mathbb{R} \times \mathbb{R}^+$ -valued right continuous process started at $(\theta, 0, 0)$, whose transition semigroup on $(0, +\infty)$ is given by

$$P_t f(v, x, u) := \mathbf{P}_{0,v} [f(\Theta_t, \xi_t + x, u \vee \bar{\xi}_t - \xi_t)],$$

for every $t > 0$ and every nonnegative measurable function $f : \mathcal{S} \times \mathbb{R} \times \mathbb{R}^+ \rightarrow \mathbb{R}^+$. We shall work with the canonical realisation of $(\Theta_t, \xi_t, U_t)_{t \geq 0}$ on the sample space of trajectories $\mathbb{D}_{\mathcal{S} \times \mathbb{R} \times \mathbb{R}^+}$, consisting of càdlàg mappings $\omega : [0, \infty) \rightarrow \mathcal{S} \times \mathbb{R} \times \mathbb{R}^+$.

We define $M := \{t \geq 0 : U_t = 0\}$ and $\bar{M}$ its closure in $\mathbb{R}^+$. The set $\mathbb{R}^+ \setminus \bar{M}$ is an open set and can be written as a union of intervals. We use $\mathbf{G}$ and $\mathbf{D}$, respectively, to denote the sets of left and right end points of such intervals. Define $\bar{R} := \inf\{t > 0 : t \in \bar{M}\}$. The downwards regularity implies that every point in $\mathcal{S}$ is regular for M in the sense that $\mathbf{P}_{0,\theta}(\bar{R} = 0) = 1$ for all $\theta \in \mathcal{S}$. Thus, by Theorem (4.1) of [41] there exist a continuous additive functional $t \mapsto L_t$ of $(\Theta_t, \xi_t, U_t)_{t \geq 0}$ which is carried by $\mathcal{S} \times \mathbb{R} \times \{0\}$ and a kernel $\mathfrak{P}$ from $\mathcal{S} \times \mathbb{R} \times \mathbb{R}^+$ into $\mathbb{D}_{\mathcal{S} \times \mathbb{R} \times \mathbb{R}^+}$ satisfying $\mathfrak{P}^{\theta,x,u}(\bar{R} = 0) = 0$ and

$\mathfrak{P}^{\theta,x,u} (1 - e^{-\bar{R}}) \leq 1$ such that

$$\mathbf{P}_{0,\theta} \left[\sum_{s \in \mathbb{G}} Z_s f \circ \theta_s \right] = \mathbf{P}_{0,\theta} \left[\int_0^\infty Z_s \mathfrak{P}^{\Theta_s, \xi_s, 0}(f) dL_s \right], \quad (0.12)$$

for θ_s the shift operator at time s , any nonnegative predictable process Z and any nonnegative function f which is measurable with respect to $\sigma((\Theta_t, \xi_t, U_t)_{t \geq 0})$. Moreover, by Theorem (5.1) of [41], $\mathfrak{P}^{\theta,x,0}((\Theta_0, \xi_0, U_0) \neq (\theta, x, 0)) = 0$, and under $\mathfrak{P}^{\theta,x,0}$, the process $(\Theta_t, \xi_t, U_t)_{t \geq 0}$ has the strong Markov property (as defined in Eq. (5.2) of [41]) with respect to P_t . In particular, if f is measurable with respect to $\sigma((\Theta_t, U_t)_{t \geq 0})$, then the right-hand side of (0.12) equals

$$\mathbf{P}_{0,\theta} \left[\int_0^\infty Z_s \mathfrak{P}^{\Theta_s, 0}(f) dL_s \right],$$

where $\mathfrak{P}$ denotes the kernel from $\mathcal{S} \times \mathbb{R}^+$ into $\mathbb{D}_{\mathcal{S} \times \mathbb{R}^+}$ for the process $(\Theta_t, U_t)_{t \geq 0}$ defined in the same way of [41]. It is known (see, for example, [24, Section 3]) that there is a nonnegative measurable function $\ell^+ : \mathcal{S} \rightarrow \mathbb{R}^+$ such that

$$\int_0^t \mathbf{1}_{\{s \in M\}} ds = \int_0^t \mathbf{1}_{\{s \in \bar{M}\}} ds = \int_0^t \ell^+(\Theta_s) dL_s \quad \forall t \geq 0 \quad \mathbf{P}_{0,\theta}\text{-a.s.} \quad (0.13)$$

Let L_t^{-1} be the right-continuous inverse process of L_t . Define $\xi_t^- := \xi_{L_t^{-1}}$ and $\Theta_t^- := \Theta_{L_t^{-1}}$ for all t such that $L_t^{-1} < +\infty$ and otherwise ξ_t^- and Θ_t^- are both assigned to be the cemetery state ∂ . One can verify by the strong Markov property that $(L_t^{-1}, \xi_t^-, \Theta_t^-)_{t \geq 0}$ defines a MAP, whose first two elements play the role of ordinates. Similarly, both $(\xi_t^-, \Theta_t^-)_{t \geq 0}$ and $(L_t^{-1}, \Theta_t^-)_{t \geq 0}$ are MAPs. These three processes are referred to as *descending ladder process*, *descending ladder height process* and *descending ladder time process*, respectively.

Split the set $\mathbb{R}^+ \setminus \bar{M}$ as a union of random intervals of the form (g, d) , if $d < \infty$. For such intervals, define

$$(\epsilon_s^{(g)}, \nu_s^{(g)}) := \begin{cases} (U_{g+s}, \Theta_{g+s}) & \text{if } 0 \leq s < d - g =: \zeta_g, \\ (\xi_{g^*} - \xi_g, \Theta_{g^*}) & \text{if } s \geq d - g. \end{cases} \quad (0.14)$$

where $g^* = \inf\{t > g : \xi_t < \xi_g\}$ is the next left endpoint of an excursion, $(\epsilon_s^{(g)}, \nu_s^{(g)})_{s \geq 0}$ is called an excursion from the minimum and ζ_g is called its lifetime. If $\zeta_g = \infty$, we just omit the second part of the definition (0.14). We use $\mathbb{G}$ to denote the set of all left

limits g , and $\mathcal{E}$ to denote the collection

$$\{(\epsilon_s^{(g)}(\omega), \nu_s^{(g)}(\omega))_{s \geq 0} : g \in \mathbf{G}(\omega), \omega \in \mathbb{D}_{\mathcal{S} \times \mathbb{R} \times \mathbb{R}^+}\},$$

which will be called the space of excursions. Let $\mathbf{n}_\theta^-$ be the image measure of $\mathfrak{P}^{\theta,0}$ under the mapping that stops the path of $(\Theta_t, U_t)_{t \geq 0}$ at time $\bar{R}$. A direct consequence of Equation (4.9) in [41] is that for any nonnegative measurable functionals $F : \mathbb{D}_{\mathbb{R}^+ \times \mathcal{S}} \rightarrow \mathbb{R}^+$ and $G : \mathbb{R}^+ \times \mathbb{D}_{\mathbb{R} \times \mathcal{S}} \rightarrow \mathbb{R}^+$,

$$\begin{aligned} \mathbf{P}_{y,\theta} & \left[\sum_{g \in \mathbf{G}} G(g, (\xi_t, \Theta_t)_{t < g}) F(\epsilon^{(g)}, \nu^{(g)}) \right] \\ &= \mathbf{P}_{y,\theta} \left[\int_0^\infty dL_s G(s, (\xi_t, \Theta_t)_{t < s}) \int_{\mathcal{E}} \mathbf{n}_{\Theta_s}^-(d\epsilon, d\nu) F(\epsilon, \nu) \right]. \end{aligned} \quad (0.15)$$

We call *exit formula* to the previous equation and $\{\mathbf{n}_\theta^- : \theta \in \mathcal{S}\}$ the *excursion measures at the minimum*.

For defining the excursions from the maximum it is not difficult to show that the pair $((\bar{\xi}_t - \xi_t, \Theta_t) : t \geq 0)$ forms a strong Markov process, where $\bar{\xi}_t := \sup_{s \leq t} \xi_s$, $t \geq 0$ is the running maximum of ξ . Suppose that the MAP (ξ, Θ) is upwards regular, then we can denote by $\bar{L}$ the local time at zero of $\bar{\xi} - \xi$, and introduce the following processes

$$\xi_t^+ = \xi_{\bar{L}_t^{-1}} \quad \text{and} \quad \Theta_{\bar{L}_t^{-1}}^+, \quad t \geq 0.$$

The local time $\bar{L}$ serves as a local time on the set $\{0\} \times \mathcal{S}$ of the Markov process $(\bar{\xi} - \xi, \Theta)$. It is not difficult to show that the pair (ξ^+, Θ^+) is a MAP and it will be called the ascending ladder height MAP.

Remark 0.6. Consider the Lamperti-Kiu transform (0.10) and notice that the excursions of (U, Θ) from $\{0\} \times \mathbb{S}^{d-1}$ translate into radial excursions for the process $Y_t := e^{\xi_t} \Theta_t$, $t \geq 0$. Ignoring the time change $\varphi(t)$ we see that the radial minima of the process Y agree with the radial minima of the stable process X .

Remark 0.7. The excursion framework constructed above can be used for the isotropic α -stable process $X = (X^{(1)}, X^{(2:d)})$ seen as a MAP. In this case $\mathcal{S} = \mathbb{R}^{d-1}$ and the excursions of X from the minimum/maximum of $X^{(1)}$ are called *directional excursions* on account that they can be generalised to any direction due to isotropy. The only requirement to be addressed is the regularity of $X^{(1)}$ with respect to the upper/lower half-lines, which is guaranteed by Theorem 0.3.

In following chapters we use excursions from the minimum/maximum of various MAPs. For all of them we will only specify the notation for descending/ascending processes, local time, and excursion measure. The excursion framework is crucial for all of the results in this work.

0.5 Results on excursions from the radial minimum of stable processes

In [37], the third paper in a series of papers [31, 38, 37] focused on studying Wiener-Hopf factorisations for the α -stable Lévy process, the authors consider the isotropic α -stable process $(X, \mathbb{P})$ in $\mathbb{R}^d$, $d \geq 2$, $\alpha \in (0, 2)$. They develop the excursion theory from the radial minimum we showed above, and proved the following series of results.

Theorem 0.8 (Theorem 1.1 of [37]). *Let $G(\infty) = \sup\{s > 0 : |X_s| = \inf_{u>0} |X_u|\}$ the instant of closest reach to the origin for X . Then for $|x| > 0$,*

$$\mathbb{P}_x(X_{G(\infty)} \in dy) = \frac{\pi^{-d/2} \Gamma(d/2)^2}{\Gamma((d-\alpha)/2) \Gamma(\alpha/2)} \frac{(|x|^2 - |y|^2)^{\alpha/2}}{|x-y|^d |y|^\alpha} dy, \quad 0 < |y| < |x|. \quad (0.16)$$

We will not show the proof of this theorem, however in the next section we will extend this result for transient Brownian Motion in $\mathbb{R}^d$ ($d \geq 3$). Although, it is adapted to one of the main differences between the two cases which is the continuity of the paths. The analysis of the point of closest approach to the origin is also relevant in the case $d = 1$ when $\alpha \in (0, 1)$, where the α -stable process exhibits transience. This scenario is addressed in [38].

The distribution of the point of closest reach is related to an important object in the theory of excursions from the radial minimum. Its density is, up to a simple change of variable, the potential measure of the descending ladder process (ξ^-, Θ^-) . To see this, we will use the excursion framework in the following way. Consider $x \in \mathbb{R}^d \setminus \{0\}$ and

a continuous bounded function $f : \mathbb{R}^d \rightarrow \mathbb{R}$, then by the exit formula (0.15)

$$\begin{aligned}
\mathbb{E}_x [f(X_{G(\infty)})] &= \mathbb{E}_{\log |x|, \arg(x)} \left[\sum_{g \in \mathbb{G}} f(e^{\xi_g} \Theta_g) \mathbf{1}_{\{\zeta_g = \infty\}} \right] \\
&= \mathbb{E}_{\log |x|, \arg(x)} \left[\int_0^\infty dL_t f(e^{\xi_t} \Theta_t) \mathbf{n}_{\Theta_t}(\zeta_t = \infty) \right] \\
&= \mathbb{E}_{\log |x|, \arg(x)} \left[\int_0^{L_\infty} dt f(e^{\xi_t^-} \Theta_t^-) \mathbf{n}_{\Theta_t^-}(\zeta_t = \infty) \right] \\
&= \int_{|z| < |x|} \mathcal{U}_x^-(dz) f(z) \mathbf{n}_{\arg(z)}^-(\zeta = \infty), \tag{0.17}
\end{aligned}$$

where

$$\mathcal{U}_x^-(dz) := \int_0^\infty \mathbf{P}_{\log |x|, \arg(x)}(e^{\xi_t^-} \Theta_t^- \in dz, t < L_\infty) dt, \quad |z| < |x| \tag{0.18}$$

is, up to the transformation $(r, \theta) \mapsto e^r \theta$, the potential of the descending ladder process (ξ_t^-, Θ^-) , and the choice of $(L, \mathbf{n}^-)$ can be taken so that the event $\{\zeta = \infty\}$ has $\mathbf{n}^-$ -measure equal to one for all $\theta \in \mathbb{S}^{d-1}$.

The second main result in [37] generalises the distribution of the point of closest reach to, roughly speaking, include not only information about the descending ladder process at a given time, but also, the process at the right-end of the excursion in course and an instant before. This result is called the triple law, alluding to those three instants.

Theorem 0.9 (Theorem 1.3 of [37]). *Fix $r > 0$ and define, for $x, z, y, v \in \mathbb{R}^d \setminus \{0\}$,*

$$\chi_x(z, y, v) := \pi^{-3/2} \frac{\Gamma((d + \alpha)/2) \Gamma(d/2)^2}{|\Gamma(-\alpha/2)| \Gamma(\alpha/2)^2} \frac{||z|^2 - |x|^2|^{\alpha/2} ||y|^2 - |z|^2|^{\alpha/2}}{|z|^\alpha |z - x|^d |z - y|^d |v - y|^{\alpha+d}}.$$

(i) *Write*

$$G(\tau_r^\oplus) = \sup\{s < \tau_r^\oplus : |X_s| = \inf_{u < s} |X_u|\},$$

for the instant of closest reach of the origin before first entry into $\mathbb{B}_r^d = \{x \in \mathbb{R}^d : |x| \leq r\}$. For $|x| > |z| > r$, $|y| > |z|$ and $|v| < r$,

$$\mathbb{P}_x \left(X_{G(\tau_r^\oplus)} \in dz, X_{\tau_r^\oplus -} \in dy, X_{\tau_r^\oplus} \in dv, \tau_r^\oplus < \infty \right) = \chi_x(z, y, v) dz dy dv.$$

(ii) *Define $\mathcal{G}(t) = \sup\{s < t : |X_s| = \sup_{u \leq s} |X_u|\}$, $t \geq 0$ and write, for $\tau_r^\ominus = \inf\{t > 0 : |X_t| > r\}$,*

$$\mathcal{G}(\tau_r^\ominus) = \sup\{s < \tau_r^\ominus : |X_s| = \sup_{u \leq s} |X_u|\},$$

for the instant of furthest reach from the origin immediately before first exit from $\mathbb{B}_r^d$. For $|x| < |z| < r$, $|y| < |z|$ and $|v| > r$,

$$\mathbb{P}_x \left(X_{\mathcal{G}(\tau_r^\ominus)} \in dz, X_{\tau_r^\ominus -} \in dy, X_{\tau_r^\ominus} \in dv, \tau_r^\oplus < \infty \right) = \chi_x(z, y, v) dz dy dv.$$

For the last main result, the so-called Deep Wiener-Hopf factorisation, we need to define a resolvent operator to be ‘factorised’ and a resolvent operator intrinsically related to the ascending/descending ladder processes, viz.

$$\mathbf{R}_z[f](\theta) := \mathbf{E}_{0,\theta} \left[\int_0^\infty e^{-z\xi_t} f(\Theta_t) dt \right], \quad \theta \in \mathbb{S}^{d-1}, z \in \mathbb{C}, \Re(z) \geq 0, \quad (0.19)$$

$$\rho_z[f](\theta) := \mathbf{E}_{0,\theta} \left[\int_0^\infty e^{-z\xi_t^+} f(\Theta_t^+) dt \right], \quad \theta \in \mathbb{S}^{d-1}, z \in \mathbb{C}, \Re(z) \geq 0, \quad (0.20)$$

for bounded measurable $f : \mathbb{S}^{d-1} \rightarrow [0, \infty)$. In fact, if we define the potential measure of the ascending ladder process analogously to (0.18), i.e.

$$\mathcal{U}_x^+(dz) := \int_0^\infty \mathbf{P}_x(e^{\xi_t^+} \Theta_t^+ \in dz) dt, \quad \theta \in \mathbb{S}^{d-1}, |x| < |z|, \quad (0.21)$$

it is easy to see that

$$\rho_z[f](\theta) = \int_{|z|>1} \mathcal{U}_x^+(dy) |y|^{-z} f(\arg(y)). \quad (0.22)$$

Thus, the deep Wiener-Hopf factorisation is stated in the following theorem.

Theorem 0.10 (Theorem 1.4 in [37]). *Suppose that $f : \mathbb{S}^{d-1} \rightarrow [0, \infty)$ is bounded and measurable. Then*

$$\mathbf{R}_{c-i\lambda}[f](\theta) = C_{\alpha,d} \rho_{i\lambda+d-\alpha-c}[\rho_{c-i\lambda}[f]](\theta), \quad \theta \in \mathbb{S}^{d-1}, \lambda \in \mathbb{R}, c \in (0, d-\alpha), \quad (0.23)$$

where $C_{\alpha,d} = 2^{-\alpha} \Gamma((d-\alpha)/2)^2 / \Gamma(d/2)^2$. Moreover,

$$\rho_z[f](\theta) = \frac{\pi^{-d/2} \Gamma(d/2)^2}{\Gamma((d-\alpha)/2) \Gamma(\alpha/2)} \int_{|y|>1} f(\arg(y)) \frac{||y|^2 - |\theta|^2|^{\alpha/2}}{|y|^{\alpha+z} |\theta - y|^d} dy, \quad \Re(z) > 0, \quad (0.24)$$

and

$$\mathbf{R}_z[f](\theta) = \frac{\Gamma((d-\alpha)/2)}{2^\alpha \pi^{d/2} \Gamma(\alpha/2)} \int_{\mathbb{R}^d} f(\arg(y)) |y|^{-z-\alpha} |y-\theta|^{\alpha-d} dy, \quad \Re(z) \in (0, d-\alpha). \quad (0.25)$$

Remark 0.11. Equation (0.23) ‘factorises’ the behaviour (seen as a resolvent) of the MAP underlying X into its ascending and descending components, where ascending is interpreted in terms of increasing radial distance. This is somewhat analogous to the real-valued Lévy process case (see Chapter 6 of [32]), where the characteristic exponent is factorised into two characteristic exponents, those of the ascending and descending ladder processes. We refer to this decomposition of operators as a deep factorisation.

Remark 0.12. The factorisation (0.23) is the first example of a Wiener-Hopf factorisation for a symmetric Markov Additive Process with a continuous state space ($\mathbb{S}^{d-1}$) for the modulator. Previous references to this type of factorisations, albeit for the case of modulators with finite state space, can be found in Theorem (3.28) of [25] or in Chapter IX of [3]. In particular for the MAPs associated to the Lamperti-Kiu transform of real self-similar Markov processes, [31] and [38] present the Deep Wiener-Hopf factorisation and applications.

Chapter 1

Contributions to the radial analysis of Isotropic stable Lévy processes

In the first part of this chapter we extend the results of [37] on radial excursions of the isotropic α -stable Lévy process to the case of Brownian Motion ($\alpha = 2$). Our main result will be a Deep Wiener-Hopf factorisation in the sense of Theorem 0.10. Many other identities will be obtained during the process of proving said factorisation, remarkably, explicit expressions for ascending and descending ladder potential measures. We will note, however, that not all the results from [37] make sense for Brownian motion, due to the continuity of its paths.

In the second part of the chapter, we establish and prove the Hunt-Nagasawa duality for isotropic α -stable processes and transient Brownian motion ($d \geq 3$), both conditioned on hitting a specified point. This duality is then used to time reverse the so-called triple law at first entry/exit of a ball, and finding a triple law at the last hit inside the ball.

1.1 Brownian motion's radial ladder potential measures

We will start by recalling that a Brownian motion $Y = (Y_t : t \geq 0)$ in $\mathbb{R}^d$, $d \geq 3$ with laws $(\mathbb{P}_x : x \in \mathbb{R}^d)$ (case $d = 2$ will be treated in Subsection 1.1.2) admits the

Lamperti-Kiu representation (0.10), i.e.

$$Y_t = e^{\xi_{\varphi(t)}} \Theta_{\varphi(t)} \quad t \geq 0,$$

where $\varphi(t) = \inf\{s > 0 : \int_0^s e^{2\xi_u} du \geq t\}$. In the particular case of Brownian motion ξ and Θ are independent processes; ξ is a Brownian motion with drift, the Lamperti transform of a Bessel(d), and Θ is a Brownian motion on $\mathbb{S}^{d-1}$ (cf. Chapter VII of [22] and [57]). It is well known that, for Y , points in $\mathbb{R}^d$ are polar and $\lim_{t \rightarrow \infty} |Y_t| = \infty$. This ensures that, when issued from $x \neq 0$, there exists a point of closest reach to the origin and the instant of closest reach can be denoted by

$$G(\infty) = \lim_{t \rightarrow \infty} G(t) = \lim_{t \rightarrow \infty} \sup\{s < t : |Y_s| = \inf_{u < t} |Y_u|\}.$$

The first result of this section is an extension of Theorem 0.8 to the case $\alpha = 2$, i.e. an explicit expression for the distribution of $Y_{G(\infty)}$.

Theorem 1.1 (Point of closest reach). *For $0 < |y| < |x|$*

$$\mathbb{P}_x(Y_{G(\infty)} \in dy) = \frac{\Gamma(d/2)(\frac{d}{2} - 1)}{\pi^{d/2}} \frac{|x|^2 - |y|^2}{|y|^2 |x - y|^d} dy, \quad |x| > |y|. \quad (1.1)$$

Our second main result is the deep Wiener-Hopf factorisation of Y which extends Theorem 0.10 to the case $\alpha = 2$. To do this we need to define the ascending and descending ladder potential measures. Recall that Y is regular for the interior and exterior of any sphere $\mathbb{S}_r^{d-1}$ for $r > 0$. Hence, a pair $(L, \mathbf{n}^-)$, where L is a local time at the radial minimum and $\mathbf{n}^-$ are the excursion measures, exists and can be defined in the same way of the previous section. Consequently, the MAPs (ξ^-, Θ^-) and (ξ^+, Θ^+) are defined in the obvious way, and one can consider the potential measures $\mathcal{U}_x^-(dy)$ and $\mathcal{U}^+(dy)$ as defined in (0.18) and (0.22) but for the Brownian case.

In the Brownian setting, the corresponding resolvent $\mathbf{R}$ defined as in (0.19) has the following explicit formula (cf. Section 3.1 of [45])

$$\mathbf{R}_z[f](\theta) = \frac{\Gamma((d-2)/2)}{4\pi^{d/2}} \int_{\mathbb{R}^d} f(\arg(y)) |y|^{-z-2} |y - \theta|^{2-d} dy, \quad \Re(z) \in (0, d-2).$$

Finally, let ρ be defined as in (0.20) for the Brownian ascending ladder MAP (ξ^+, Θ^+) . The abuse in the notation is intentional on account of the following theorem, which indicates that formulae in Theorem (0.10) can be extended to consider the case $\alpha = 2$.

Theorem 1.2 (Deep Wiener-Hopf factorisation for Brownian motion). *Suppose that $f : \mathbb{S}^{d-1} \rightarrow [0, \infty)$ is bounded and measurable. Then*

$$\mathbf{R}_{c-i\lambda}[f](\theta) = C_{2,d}\rho_{i\lambda+d-2-c}[\rho_{c-i\lambda}[f]](\theta), \quad \theta \in \mathbb{S}^{d-1}, \lambda \in \mathbb{R}, c \in (0, d-2), \quad (1.2)$$

where $C_{2,d} = \frac{1}{4(\frac{d}{2}-1)^2}$. Moreover,

$$\rho_z[f](\theta) = \frac{\Gamma(d/2-1)(d/2-1)}{\pi^{d/2}} \int_{|y|>1} f(\arg(y)) \frac{|y|^2 - |\theta|^2}{|y|^{2+z}|\theta-y|^d} dy, \quad \Re(z) > 0. \quad (1.3)$$

In this case, the Deep Wiener-Hopf factorisation decomposes the resolvent of the MAP underlying Y —expressed as an integral with respect to the Newtonian potential—into two resolvents corresponding to the ascending and descending ladder potentials, where ascending and descending refer to increases or decreases in radial distance. As in the α -stable case, the ascending and descending ladder MAPs are related by a duality principle. The specific duality relation employed is detailed in the proof.

1.1.1 Proof of the results

We will follow a similar path to that in [37], starting by noting that there is a relationship between the density of the point of closest reach to the origin and the descending ladder potential measure. Helped by the exit formula (0.15) and, by the same argument of in equation (0.17), we have that

$$\mathbb{E}_x [f(Y_{G(\infty)})] = \int_{|z|<|x|} \mathcal{U}_x^-(dz) f(z) \mathbf{n}_{\arg(z)}^-(\zeta = \infty), \quad (1.4)$$

where $\mathcal{U}_x^-(dz) = \int_0^\infty \mathbf{P}_{\log|x|, \arg(x)}(e^{\xi_t^-} \Theta_t^- \in dz, t < L_\infty) dt$, $|z| < |x|$. Noting that, by isotropy, $\mathbf{n}_{\arg(z)}^-(\zeta = \infty)$ is independent of z , we have that $\mathcal{U}_x^-(dy)$ is, up to a multiplicative constant, the density of the point of closest reach to the origin.

Now, to find the distribution of the point of closest reach we will use two ingredients: the point of closest reach to the origin of the radial process $|Y|$, and the joint distribution of $(|Y|, \arg(Y))$ at the moment of first hitting a sphere. Under a sensible conditioning, the latter will give us the information about the angular part at the point of closest reach. For the first one of these ingredients, we will obtain the distribution of the first hit to a sphere $\mathbb{S}_r^{d-1}$ from outside the ball of radius r .

By equation (1.5) in [9], just to mention one of many references in this regard, we know

that for $|x| < 1$, $z \in \mathbb{S}_1^{d-1}$ and $\tau_1^\circ = \inf\{t > 0 : |Y_t| = 1\}$

$$\mathbb{P}_x(X_{\tau_1^\circ} \in dz) = \frac{\Gamma(d/2)}{2\pi^{d/2}} \frac{1 - |x|^2}{|x - z|^d} \sigma_1(dz), \quad (1.5)$$

where σ_1 is the surface area on $\mathbb{S}_1^{d-1}$ normalized to have total measure one. This distribution is known as the Poisson kernel inside the unit ball.

We can easily obtain the distribution of the hitting on the sphere when the starting point is outside the sphere by appealing to inversion with respect to the sphere. This idea, originally used in [56] for Brownian motion was extended by Bogdan and Żak which in [8] to the isotropic α -stable process $\alpha \in (0, 2]$, establishes that if $K(x) = x/|x|^2$, $x \in \mathbb{R}^d \setminus \{0\}$, then $(K(X_{\eta(t)}), t \geq 0)$ under $\mathbb{P}_x$ is equal in law to $(X_t : t \geq 0)$ under $\mathbb{P}_{K(x)}^\circ$, where

$$\left. \frac{d\mathbb{P}_x^\circ}{d\mathbb{P}} \right|_{\sigma(X_s : s \leq t)} = \frac{|X_t|^{\alpha-d}}{|x|^{\alpha-d}}, \quad t \geq 0, \quad (1.6)$$

and $\eta(t) = \inf\{s > 0 : \int_0^s |X_u|^{-2\alpha} du > t\}$. This space-time transform of α -stable processes is known as the Riesz-Bogdan-Żak transform and is instrumental for many of the calculations, where roughly speaking, an inversion of space or time reversal is needed. In particular for our process of interest Y , we have for $|x| < 1$

$$\mathbb{P}_{K(x)}(Y_{\tau_1^\circ} \in dz, \tau_1^\circ < \infty) \left(\frac{|z|}{|x|} \right)^{2-d} = \mathbb{P}_x(Y_{\tau_1^\circ} \in dz).$$

Using that $K(K(x)) = x$ we have for $|x| > 1$ an explicit expression for the distribution of first hit to the unit sphere starting from $\mathbb{R}^d \setminus \mathbb{B}_1^d$,

$$\mathbb{P}_x(Y_{\tau_1^\circ} \in dz, \tau_1^\circ < \infty) = \frac{\Gamma(d/2)}{2\pi^{d/2}} \frac{|x|^2 - 1}{|x - z|^d} \sigma_1(dz).$$

Finally, we can use the scaling property to generalise this distribution to $\tau_r^\circ = \inf\{t > 0 : |Y_t| = r\}$, obtaining

$$\mathbb{P}_x(Y_{\tau_r^\circ} \in dz, \tau_r^\circ < \infty) = \frac{\Gamma(d/2)}{2\pi^{d/2}} \frac{1}{r} \frac{|x|^2 - r^2}{|x - z|^d} \sigma_r(dz), \quad |x| > r, z \in \mathbb{S}_r^{d-1}, \quad (1.7)$$

where σ_r is the surface area on $\mathbb{S}_r^{d-1} = \{z \in \mathbb{R}^d : |z| = r\}$, normalized to have total measure one.

Returning to the matter of the first ingredient in order to find the distribution of the

point of closest reach, let us integrate (1.7) on $\mathbb{S}_r^{d-1}$ to obtain an important insight in regards the radial part of the point of closest reach.

$$\begin{aligned}
\mathbb{P}_x(\tau_r^\circ < \infty) &= \int_{\mathbb{S}_r^{d-1}} \frac{\Gamma(d/2)}{2\pi^{d/2}} \frac{1}{r} \frac{|x|^2 - r^2}{|x - z|^d} \sigma_r(dz) \\
&= c \frac{(|x|^2 - r^2)}{r|x|^d} \int_0^\pi \frac{r^{d-1} \sin^{d-2}(\theta) d\theta}{|1 - 2r/|x| \cos(\theta) + (r/|x|)^2|^{d/2}} \\
&= \frac{r^{d-2}(|x|^2 - r^2)}{|x|^d} \left(1 - \frac{r^2}{|x|^2}\right)^{-1} \\
&= \left(\frac{r}{|x|}\right)^{d-2},
\end{aligned}$$

where c is an unimportant constant, and it is easy to verify that the last expression is a proper probability function for $|x| > r$.

By noting the equivalence of the events $\{|X_{G(\infty)}| > r\}$ and $\{\tau_r = \infty\}$, we conclude that for $0 < r < |x|$

$$\mathbb{P}_x(|Y_{G(\infty)}| \in dr) = (d-2) \left(\frac{r}{|x|}\right)^{d-3} \frac{1}{|x|} dr. \quad (1.8)$$

Remarkably, when $d = 3$, the radial position of the closest reach, $|Y_{G(\infty)}|$, started from $|x|$, follows a uniform distribution on $(0, |x|)$.

The second ingredient to find the joint distribution of $(|Y_{G(\infty)}|, \arg(Y_{G(\infty)}))$ is the explicit distribution (1.7), and to use them together we will define a key quantity which will limit to the joint distribution of $|Y_{G(\infty)}|$ and $\arg(Y_{G(\infty)})$. For $x \neq 0$, $|x| > r$, $\delta > 0$ and continuous, positive and bounded f on $\mathbb{R}^d$,

$$\Delta_r^\delta f(x) := \frac{1}{\delta} \mathbb{E}_x [f(\arg(Y_{G(\infty)})), |Y_{G(\infty)}| \in [r - \delta, r]]. \quad (1.9)$$

By conditioning on the first hit to the sphere of radius r we have

$$\Delta_r^\delta f(x) = \frac{1}{\delta} \int_{y \in \mathbb{S}_r^{d-1}} \mathbb{P}_x(Y_{\tau_r^\circ} \in dy, \tau_r^\circ < \infty) \mathbb{E}_y [f(\arg(Y_{G(\infty)})), |Y_{G(\infty)}| \in [r - \delta, r]].$$

Under the event $\{Y_{\tau_r^\circ} \in dy\}$ we will translate the expectation in the previous expression to the skew-product representation in the following way

$$\mathbb{E}_y [f(\arg(Y_{G(\infty)})), |Y_{G(\infty)}| \in [r - \delta, r]] = \mathbf{E}_{\log r, \arg(y)} [f(\Theta_{g_\infty}), e^{\xi_{g_\infty}} \in [r - \delta, r]],$$

where $g_\infty = \lim_{t \rightarrow \infty} \sup\{t > 0 : \xi_t = \inf_{s \leq t} \xi_s\}$ is the left endpoint of the last excursion from the minimum, or the instant of global minimum for ξ . Thus, because of the independence of ξ and Θ , and disintegration, we have

$$\begin{aligned} & \mathbf{E}_{\log r, \arg(y)} \left[f(\Theta_{g_\infty}), e^{\xi_{g_\infty}} \in [r - \delta, r] \right] \\ &= \int_{(t,s) \in [0, \infty) \times [r - \delta, r]} \mathbf{P}_{\log r}(g_\infty \in dt, e^{\xi_{g_\infty}} \in ds) \mathbf{E}_y[f(\Theta_t)]. \end{aligned}$$

In order to calculate the limit as $\delta \rightarrow 0$ in the previous expression recall that ξ is a Brownian motion with drift. Then, let us also call $\ell = (\ell_t : t \geq 0)$ to its local time at the minimum. This allows us to set

$$g_\infty = \ell_\infty^{-1}, \quad \text{and} \quad e^{\xi_{g_\infty}} = e^{-\hat{H}_\infty},$$

where $\hat{H}$ is the corresponding descending ladder height process.

Now note that, due to the continuity of ξ , any positive multiple of $-\inf_{s \leq t} \xi_s$ can be taken as a local time at the minimum for ξ , facilitating to write that for some $q > 0$

$$(\ell_\infty^{-1}, \xi_{\ell_\infty^{-1}}) \stackrel{d}{=} (T_{e_q}, -e_q), \quad (1.10)$$

where $T_z = \inf\{t > 0 : \xi_t = z\}$, for $z \in \mathbb{R}$. Therefore,

$$\begin{aligned} & \lim_{\delta \rightarrow 0} \frac{1}{\delta} \int_{(t,s) \in [0, \infty) \times [r - \delta, r]} \mathbf{P}_{\log r}(g_\infty \in dt, e^{\xi_{g_\infty}} \in ds) \mathbf{E}_y[f(\Theta_t)] \\ &= \lim_{\delta \rightarrow 0} \frac{1}{\delta} \int_{(t,u) \in [0, \infty) \times [1 - \delta/r, 1]} \mathbf{E}_{\log r} \left[\mathbf{1}_{\{\ell_\infty^{-1} \in ds, e^{\xi_{\ell_\infty^{-1}}} \in du\}} \right] \frac{1}{r} \mathbf{E}_y[f(\Theta_t)] \\ &= \lim_{\delta \rightarrow 0} \frac{1}{\delta} \int_{(t,u) \in [0, \infty) \times [1 - \delta/r, 1]} q e^{-qy} \mathbf{E}_0 \left[\mathbf{1}_{\{T_y \in ds, e^{-y} \in du\}} \right] \frac{1}{r} \mathbf{E}_y[f(\Theta_t)], \end{aligned}$$

where the second equality is given by equation (1.10). Finally, by a change of variables, we have

$$\begin{aligned} & \lim_{\delta \rightarrow 0} \frac{1}{\delta} \int_{(t,u) \in [0, \infty) \times [1 - \delta/r, 1]} q e^{-qy} \mathbf{E}_0 \left[\mathbf{1}_{\{T_y \in ds, e^{-y} \in du\}} \right] \frac{1}{r} \mathbf{E}_y[f(\Theta_t)] \\ &= \lim_{\delta \rightarrow 0} \frac{1}{\delta} \int_{(t,u) \in [0, \infty) \times [1 - \delta/r, 1]} q e^{-qy} \mathbf{E}_0 \left[\mathbf{1}_{\{T_{-\log u} \in ds\}} \right] \frac{du}{r} \mathbf{E}_y[f(\Theta_t)], \end{aligned}$$

where by the fundamental theorem of calculus, and recalling (cf. [57]) that $q = d - 2$

we obtain

$$\begin{aligned}
& \lim_{\delta \rightarrow 0} \frac{1}{\delta} \int_{(t,u) \in [0,\infty) \times [1-\delta/r, 1]} q e^{-qy} \mathbf{E}_0 \left[\mathbf{1}_{\{T_{-\log u} \in ds\}} \right] \frac{du}{r} \mathbf{E}_y [f(\Theta_t)] \\
&= \frac{q}{r} \mathbf{E}_{\arg y} [f(\theta_0)] \\
&= f(\arg(y))(d-2)r^{-1}.
\end{aligned}$$

Substituting this in the key quantity $\Delta_r^\delta f(x)$ we get

$$\lim_{\delta \rightarrow 0} \Delta_r^\delta f(x) = \left[\frac{\Gamma(d/2)}{2\pi^{d/2}} \frac{|x|^2 - r^2}{r} \right] \int_{y \in \mathbb{S}_r^{d-1}} \sigma_r(dy) |x - y|^{-d} f(\arg(y))(d-2)r^{-1}. \quad (1.11)$$

This finishes the proof of Theorem 1.1. $\square$

Notice that by (1.4) we have now an explicit expression for the descending ladder potential measure. For $|x| > |y| = r$,

$$\mathcal{U}_x^-(dy) = \frac{\Gamma(d/2)(\frac{d}{2} - 1)}{\pi^{d/2}} \frac{|x|^2 - r^2}{r^2 |x - y|^d} dy. \quad (1.12)$$

The same procedure above can be used to prove the following Lemma, which relates functionals from the theory of excursions from the radial minimum with known identities for Y .

Lemma 1.3. *For any bounded continuous function $f : \mathbb{R}^d \rightarrow 0$ and $r > 0$*

$$\lim_{\delta \rightarrow 0} \sup_{|x| \in (r-\delta, r]} \left| \frac{\int_{r-\delta \leq |y| \leq |x|} \mathcal{U}_x^-(dy) f(y)}{\int_{r-\delta \leq |y| \leq |x|} \mathcal{U}_x^-(dy)} - f(x) \right| = 0. \quad (1.13)$$

Ascending potential measure

Consider the change of measure of the Riesz-Bogdan-Żak transform (1.6). In terms of the underlying MAP (ξ, Θ) we see that $-i(\alpha - d)$ is a root of the exponent (0.11) and the change of measure corresponds to an Esscher transform of the Lévy process ξ , under which ξ drifts to $-\infty$. By an application of the optional stopping theorem, (1.6) is equivalent to the change of measure for ξ ,

$$\left. \frac{d\mathbf{P}_{x,\theta}^\circ}{d\mathbf{P}_{x,\theta}} \right|_{\sigma((\xi_s, \Theta_s) : s \leq t)} = e^{(2-d)(\xi_t - x)}, \quad t \geq 0. \quad (1.14)$$

It is not difficult to prove that the space-time transformed process $(K(Y_{\eta(t)}) : t \geq 0)$ is the Lamperti-Kiu transform of the MAP $(-\xi, \Theta)$, which implies at the level of MAPs, that (ξ, Θ) under the change of measure (1.14), when issued from $(\log |x|, \arg(x))$, $x \in \mathbb{R}^d$ is equal in law to $(-\xi, \Theta)$ when issued from $(-\log |x|, \arg(x))$. An important consequence of this ‘dual’ behaviour under the Riesz-Bogdan-Żak transform is that it offers the opportunity to study aspects of the process (ξ^+, Θ^+) . For instance, if we define $\mathcal{U}_x^+(dy)$, $|x| \leq |y|$ by

$$\mathcal{U}_x^+(dy) = \int_0^\infty \mathbf{P}_{\log |x|, \arg(x)} \left(e^{\xi_t^+} \Theta_t^+ \in dy \right) dt, \quad |y| \geq |x|. \quad (1.15)$$

Note that, unlike in the definition of $\mathcal{U}^-$, the condition $\{t < \bar{L}_\infty\}$ is omitted in the previous potential. This is justified by the fact that $\lim_{t \rightarrow \infty} |X_t| = \infty$, which in turn implies that $\lim_{t \rightarrow \infty} \xi_t = \infty$. Then, the Riesz-Bogdan-Żak transform ensures that for a Borel $A \subseteq \{z \in \mathbb{R}^d : |z| > |x|\}$

$$\frac{|z|^{2-d}}{|x|^{2-d}} \mathbf{P}_{\log |x|, \arg(x)} \mathbf{P} \left(e^{H_t^+} \Theta_t^+ \in A \right) = \mathbf{P}_{-\log |x|, \arg(x)} \left(e^{-H_t^-} \Theta_t^- \in K(A), t < \ell_\infty \right),$$

where $K(A) = \{K(z) : z \in A\}$. Hence, using that $dK(z) = \frac{dz}{|z|^{2d}}$ and $|K(x) - K(y)| = \frac{|x-y|}{|x||y|}$, for $|x| > 0$

$$\begin{aligned} \mathcal{U}_x^+(dz) &= \frac{|x|^{2-d}}{|z|^{2-d}} \mathcal{U}_{K(x)}^-(K(z)) \frac{dz}{|z|^{2d}} \\ &= \frac{\Gamma(d/2)(\frac{d}{2} - 1)}{\pi^{d/2}} \frac{|z|^2 - |x|^2}{|z|^2 |x - z|^d} dz, \end{aligned} \quad (1.16)$$

where the last equality is due to the explicit formula (1.12).

In order to prove the deep Wiener-Hopf factorisation we will relate the ascending ladder potential with a functional of the measure of excursions from the minimum $(\mathbf{n}_\theta^- : \theta \in \mathbb{S}^{d-1})$. The useful identity will be that of the explicit Green’s function outside the ball of radius r , i.e. the density of the following potential, for $|x| > r$,

$$\mathbb{E}_x \left[\int_0^{\tau_r^\circ} f(Y_t) dt \right] = \int_{|y| > r} f(y) h_r^\odot(x, y) dy. \quad (1.17)$$

Lemma 1.4. *For $|x|, |y| > r$ the potential measure in (1.17) has a density with respect to Lebesgue measure given by*

$$h_r^\odot(x, y) = \frac{\Gamma(d/2 - 1)}{4\pi^{d/2}} \left(|x - y|^{2-d} - \left| \frac{yr}{|y|} - \frac{x|y|}{r} \right|^{2-d} \right). \quad (1.18)$$

Proof. Let τ_1° denote the time at which the process first hits the unit sphere, and define the Green's function for the interior of the unit ball as the function $h_1^\odot(x, y)$, such that for $x, y \in \mathbb{B}_1^d$,

$$\mathbb{E}_x \left[\int_0^{\tau_1^\circ} \mathbf{1}_{\{Y_t \in dy\}} \right] = h_1^\odot(x, y) dy.$$

It is a classical result (see, for example, Chapter 2 of [20]) that this function is given by

$$h_1^\odot(x, y) = \frac{\Gamma(\frac{d}{2} - 1)}{4\pi^{d/2}} \left(|y - x|^{2-d} - \left| y|x| - \frac{x}{|x|} \right|^{2-d} \right). \quad (1.19)$$

As a corollary of Theorem 2 in [8], consequence of the Riesz-Bogdan-Żak transform, one gets

$$h_1^\odot(x, y) = |x|^{2-d} |y|^{2-d} h_1^\odot(K(x), K(y)), \quad |x|, |y| > 1.$$

Substituting (1.19) in the previous expression and noting that $|K(x) - K(y)| = \frac{|x-y|}{|x||y|}$ we obtain that

$$h_1^\odot(x, y) = \frac{\Gamma(\frac{d}{2} - 1)}{4\pi^{d/2}} \left(|y - x|^{2-d} - \left| x|y| - \frac{y}{|y|} \right|^{2-d} \right),$$

from where it is interesting to see that $h_1^\odot(x, y) = h_1^\odot(y, x)$, each one in its respective domain. Finally after using the scale property in our previous expression we obtain the desired result. $\square$

In virtue of Lemmas 1.3 and 1.4 we can now prove that there is a relationship between the occupation functional under the excursion measure $\mathbf{n}^-$ and integration under the ascending potential ladder measure $\mathcal{U}^+$. This ‘duality’ identity is well known to hold for Lévy processes in $\mathbb{R}$ (cf. Exercise VI.5 of [4]) and it is called Silverstein’s identity.

Proposition 1.5 (Radial Silverstein’s identity for Brownian Motion). *For $x \in \mathbb{R}^d \setminus \{0\}$ and continuous $f : \mathbb{R}^d \rightarrow \mathbb{R}$ whose support is compactly contained in the exterior of the ball of radius $|x|$*

$$\mathbf{n}_{\arg(x)}^- \left(\int_0^\zeta f(|x|e^{\epsilon(s)} \Theta_s^\epsilon) ds \right) = \frac{1}{4(\frac{d}{2} - 1)^2} \int_{|y| > |x|} \mathcal{U}_x^+(dy) f(y). \quad (1.20)$$

Proof. Let us start by pointing out that the occupation functional under the excursion measure appears naturally in the potential measure outside the sphere. Firstly, because the occupation for Brownian motion runs on a different speed than its Lamperti-Kiu transform, we use that $\int_0^{\varphi(t)} e^{2\xi_s} ds = t$, which after a change of variables yields

$$\mathbb{E}_x \left[\int_0^{\tau_r^\circ} f(Y_t) dt \right] = \mathbf{E}_{|x|, \arg(x)} \left[\int_0^\infty \mathbf{1}_{\{e^{\xi_t} \geq r\}} f(e^{\xi_t} \Theta_t) e^{2\xi_s} dt \right], \quad |x| > r.$$

Then, using the exit formula we obtain,

$$\mathbb{E}_x \left[\int_0^{\tau_r^\circ} f(Y_t) dt \right] = \int_{|x| > |z| > r} \mathcal{U}_x^-(dz) \mathbf{n}_{\arg(z)}^- \left(\int_0^\infty f(|z| e^{\epsilon(s)} \Theta_s^\epsilon) |z|^2 e^{2\epsilon(s)} ds \right). \quad (1.21)$$

We would like to use Lemma 1.3 for which we need to verify that, for f as in the statement of the proposition, the mapping $z \mapsto \mathbf{n}_{\arg(z)}^- \left(\int_0^\infty f(|z| e^{\epsilon(s)} \Theta_s^\epsilon) |z|^2 e^{2\epsilon(s)} ds \right)$ is continuous and bounded.

To prove finiteness we notice that the left-hand side of equation (1.21) is bounded by $\mathbb{E}_x \left[\int_0^\infty f(Y_t) dt \right]$, which is known to be finite under our assumptions. This necessarily implies boundedness of the occupation functional. Continuity is a consequence of the continuity of f and the continuity of the paths of $(\epsilon, \Theta^\epsilon)$ under $\mathbf{n}^-$.

Now, by Lemmas 1.3 and 1.4,

$$\begin{aligned} & \mathbf{n}_{\arg(z)}^- \left(\int_0^\infty f(|z| e^{\epsilon(s)} \Theta_s^\epsilon) |z|^2 e^{2\epsilon(s)} ds \right) \\ &= \lim_{r \uparrow |z|} \frac{\int_{|y| > r} f(y) h_r^\odot(z, y) dy}{\int_{r \leq |y| \leq |z|} \mathcal{U}_z^-(dy)} \\ &= \lim_{r \uparrow |z|} \frac{\int_{|y| > r} f(y) \frac{\Gamma(d/2-1)}{4\pi^{d/2}} \left(|z-y|^{2-d} - \left| \frac{yr}{|y|} - \frac{z|y|}{r} \right|^{2-d} \right) dy}{1 - \left(\frac{r}{|z|} \right)^{d-2}}. \end{aligned}$$

By studying the limits in the numerator and denominator, using dominated convergence in the former, we observe that both limits tend to zero, allowing to use L'Hopital's rule. Then, an elemental calculation yields

$$\mathbf{n}_{\arg(z)}^- \left(\int_0^\infty f(|z| e^{\epsilon(s)} \Theta_s^\epsilon) |z|^2 e^{2\epsilon(s)} ds \right) = \int_{|y| > r} f(y) \frac{\Gamma(d/2-1)}{4\pi^{d/2}} \frac{|y|^2 - |z|^2}{|z-y|^d} dy.$$

Finally, we just take $f^*(y) = f(y)|y|^{-2}$, and compare with equation (1.16) to conclude the proof. $\square$

Helped by Proposition 1.5, we can now prove the Deep Wiener-Hopf factorisation.

Proof of Theorem 1.2. By the exit formula (0.15), for bounded measurable $f : \mathbb{S}^{d-1} \rightarrow \mathbb{R}$, and $z \in \mathbb{C}$ such that $\Re(z) \in (0, d-2)$

$$\begin{aligned} \mathbf{R}_z[f](\theta) &= \mathbf{E}_{0,\theta} \left[\int_0^\infty e^{-z\xi_t} f(\Theta_t) \right] \\ &= \lim_{r \rightarrow 0} \int_{|x| > |y| > r} \mathcal{U}_x^-(dy) \mathbf{n}_{\arg(y)}^- \left(\int_0^\infty f(\Theta_s^\epsilon) (|y|e^{\epsilon(s)})^{z-2} ds \right) \\ &= \frac{1}{4(\frac{d}{2} - 1)^2} \int_{|x| > |y| > 0} \mathcal{U}_x^-(dy) \int_{|w| > |y|} \mathcal{U}_y^+(dw) f(\arg(w)) (|y||w|)^{z-d}, \end{aligned}$$

The result follows after using the definition of ρ and the Riesz-Bogdan-Żak transform, as in (1.16), on the descending potential measure $\mathcal{U}^-$. $\square$

So far we have only discussed the case $d \geq 3$, for which most of the techniques used in [37] are also applicable. However, the results for 2-dimensional, i.e. planar Brownian motion, can be stated as consequence of what we call *excursions from hyperplanes*, or excursions from directional minimum/maximum, see Remark 0.7. In the following subsection we will only present the results, while the proofs will be a particular case of those in the next chapter.

1.1.2 Deep Factorisation of Planar Brownian Motion

In the two-dimensional case, the Lamperti representation (0.10) of a planar Brownian motion Y with laws $(\mathbb{P}_x : x \in \mathbb{R}^2 \setminus \{0\})$ can be posed in the following way,

$$Y_t = e^{\xi_{\varphi(t)} + i\theta_{\varphi(t)}}, \quad t \geq 0,$$

where θ corresponds to the windings around the origin. The ordinate ξ is to be interpreted in the usual way, as the norm of the process and in this particular case, ξ and θ turn out to be independent Brownian motions (see Section 7.15 in [22]).

From this we have that $\limsup_{t \rightarrow \infty} \xi_t = -\liminf_{t \rightarrow \infty} \xi_t = \infty$, which implies that when started from $x \neq 0$ the point of closest reach to the origin is the origin itself.

However, helped by the independence of ξ and θ , we can find the potential measure of the ascending and descending ladder MAPs.

For simplicity in dealing with the deep Wiener-Hopf factorisation, we will not use the same definition of potential measures we used in the $d \geq 3$ case. It is enough to point out that excursions from the minimum/maximum radius of Y translate into excursions from the minimum/maximum in the direction of ξ of the planar BM (ξ, θ) .

It is well known (cf. Section 6.5 of [32]) that the inverse local time at the maximum for ξ , L^{-1} , is a $\frac{1}{2}$ -stable subordinator and ξ^+ and $-\xi^-$ are unit-rate linear drifts. Hence, given the independence of θ and L^{-1} , the ascending ladder MAP $(\xi_t^+, \theta_{L_t^{-1}})$ is equal in law to (t, S_t) where $(S_t)_{t \geq 0}$ is a Cauchy process.

Let us define the following potential ladder measures. Which in contrast to (1.15) and (1.12) are given in cartesian coordinates, viz.

$$U_{r,\gamma}^\pm(ds, d\phi) := \mathbf{E}_{r,\gamma} \left[\int_0^\infty \mathbf{1}_{\{\xi_t^\pm \in ds, \theta_t^\pm \in d\phi\}} dt \right]. \quad (1.22)$$

It is not difficult to see, given the distribution of (t, S_t) that the explicit expressions of the previous potentials are given by,

$$U_{r,\gamma}^\pm(ds, d\phi) = \frac{|r - s|}{\pi |(s, \phi) - (r, \gamma)|^2}. \quad (1.23)$$

We will pose the factorisation in the following way.

Theorem 1.6 (Deep Wiener-Hopf factorisation of planar Brownian Motion). *For $r > a$ and bounded and measurable $f : \mathbb{R} \rightarrow \mathbb{R}$, and $\tau_a^\xi = \inf\{t > 0 : \xi_t = a\}$*

$$\mathbf{E}_{r,\gamma} \left[\int_0^{\tau_a^\xi} f(\xi_t, \theta_t) dt \right] = \int_{r>s>a} U_{r,\gamma}^-(ds, d\phi) \int_{t>s} U_{s,\phi}^+(dt, d\iota) f(t, \iota). \quad (1.24)$$

1.2 Last hitting positions and duality

Let $(X_t : t \geq 0)$ be an isotropic α -stable Lévy process in $\mathbb{R}^d$, $d \geq 2$, $\alpha \in (0, 2)$, with laws $(\mathbb{P}_x : x \in \mathbb{R}^d)$. We are interested in the distribution of n -tuple laws related to the last hitting position inside the unit ball, i.e. if we define the last hitting time of the

ball by

$$\gamma_B := \begin{cases} \sup\{t > 0 : |X_t| \leq 1\} & \text{if } \{t : |X_t| \leq 1\} \neq \emptyset \\ 0 & \text{if } \{t : |X_t| \leq 1\} = \emptyset, \end{cases}$$

and, for $\gamma_B > 0$, we define the right endpoint of the subsequent excursion from the radial minimum

$$\vec{g}(\gamma_B) := \inf\{t > \gamma_B : |X_t| \leq \inf_{u \leq \gamma_B} |X_u|\},$$

then our objective is to find for any $x \in \mathbb{R}^d$

$$\mathbb{P}_x(X_{\gamma_B-} \in dw, X_{\gamma_B} \in dy, X_{\vec{g}(\gamma_B)} \in dz, \gamma_B > 0), \quad |w| < 1, |y| > |z| > 1. \quad (1.25)$$

Let us point out that for $\alpha \in (0, 2)$ and $d \geq 2$, $\lim_{t \rightarrow \infty} |X_t| = \infty$ $\mathbb{P}$ -almost surely, which ensures that $\gamma_B < \infty$ with probability one.

The distribution of X_{γ_B-} has been linked with concepts of potential theory, in particular with that of equilibrium measure, see for example [16] or [46]. In fact, that same distribution could be lifted up from Theorem 1.1 in [15] and equation (17) in [16]. In our case we will leverage on the theory of excursions from the radial minimum of X , specifically the triple law found in [37] and Hunt-Nagasawa duality, for generalising this result and give it a nice time-reversal interpretation.

The main result of this section, proved after presenting the Hunt-Nagasawa duality, is stated in the following theorem.

Theorem 1.7. *[Triple law at last hitting] Let $x \in \mathbb{R}^d$, $|v| < 1$ and $1 < |z| < |y|$*

$$\begin{aligned} & \mathbb{P}_x(X_{\gamma_B-} \in dv, X_{\gamma_B} \in dy, X_{\vec{g}(\gamma_B)} \in dz, \gamma_B > 0) \\ &= \pi^{-3/2} \frac{\Gamma((d+\alpha)/2)}{|\Gamma(-\alpha/2)|} \frac{\Gamma(d/2)^2}{\Gamma(\alpha/2)^2} |v-x|^{\alpha-d} \frac{||y|^{-2} - |z|^{-2}|^{\alpha/2}}{|z-y|^d |v-y|^{\alpha+d} |y|^{-\alpha}} dz dy dv. \end{aligned} \quad (1.26)$$

1.2.1 Hunt-Nagasawa duality and conditioned processes

We will start by introducing the isotropic α -stable process, $\alpha \in (0, 2)$, conditioned to continuously hit a point $\theta \in \mathbb{R}^d$. We will preemptively denote this process by $(X, \mathbb{P}^{\{\theta\}})$. This process was mentioned in [28], in the particular case $\alpha \in (0, 1]$. There, X is conditioned to continuously hit a patch S on the surface of the unit sphere. In the same paper it is said that $\mathbb{P}^{\{\theta\}}$ cannot be contextualised using the limiting description used for hitting patches on the sphere. However, we will show that the conditioning is simple, coincides with $\mathbb{P}^S$ when $S = \{\theta\}$ and, more importantly, is valid for $\alpha \in (0, 2)$.

Let us define the family of probabilities $(\mathbb{P}_x^{\{\theta\}} : x \in \mathbb{R} \setminus \{\theta\})$ by the following limit

$$\mathbb{P}_x^{\{\theta\}}(\Lambda) := \lim_{\epsilon \rightarrow 0} \mathbb{P}_x(\Lambda, t < \tau_{B_\epsilon(\theta)} | \tau_{B_\epsilon(\theta)} < \infty), \quad \Lambda \in \mathcal{F}_t, \quad (1.27)$$

where $B_\epsilon(\theta)$ is the ball of radius ϵ with center θ . The Markov property and shifting invariance yield

$$\begin{aligned} \mathbb{E}_x^{\{\theta\}} [\mathbf{1}_{\{\Lambda\}}] &= \lim_{\epsilon \rightarrow 0} \mathbb{E}_x \left[\mathbf{1}_{\{\Lambda, t < \tau_{B_\epsilon(\theta)}\}} \frac{\mathbb{P}_{X_t}(\tau_{B_\epsilon(\theta)} < \infty)}{\mathbb{P}_x(\tau_{B_\epsilon(\theta)} < \infty)} \right] \\ &= \lim_{\epsilon \rightarrow 0} \mathbb{E}_{x-\theta} \left[\mathbf{1}_{\{\Lambda^{\{\theta\}}, t < \tau_{B_\epsilon(0)}\}} \frac{\mathbb{P}_{X_t-\theta}(\tau_{B_\epsilon(0)} < \infty)}{\mathbb{P}_{x-\theta}(\tau_{B_\epsilon(0)} < \infty)} \right], \end{aligned} \quad (1.28)$$

where $\Lambda^{\{\theta\}}$ is defined as Λ but for the shifted process $(X_t + \theta)_{t \geq 0}$. We can now distinguish that the limiting expression in the last expectation is that of a process started at $x - \theta$ conditioned to be continuously absorbed at the origin. This conditional law is well known, and corresponds to the Doob's h transform $h(x) = |x|^{\alpha-d}$ (cf. Theorem 12.5 in [35]). Leading to the following result

$$\frac{d\mathbb{P}_x^{\{\theta\}}}{d\mathbb{P}_x} \Big|_{\mathcal{F}_t} = \frac{|X_t - \theta|^{\alpha-d}}{|x - \theta|^{\alpha-d}} \Big|_{\mathcal{F}_t}, \quad x \neq \theta.$$

In other words, we have proved that the process $(X, \mathbb{P}_x^{\{\theta\}})$ is a Doob h -transform of the process issued at x , with harmonic function $H(x) = |x - \theta|^{\alpha-d}$.

Our objective is now to prove that the Hunt-Nagasawa duality (cf. Appendix A.12 [35]) holds between the processes $(X, \mathbb{P}_x^{\{\theta\}})$ and $(X, \mathbb{P}_\theta^{\{x\}})$, for $x, \theta \in \mathbb{R}^d$ and given certain entrance law. Theorem 3.5 of [43] shows that, under suitable assumptions on the Markov process $(X, \mathbb{P}_x^{\{\theta\}})$, there is class of random times at which the pathwise time-reversal of the process is a Markov process whose transitions are given by those of $(X, \mathbb{P}_\theta^{\{x\}})$.

To be more precise about such class of random times suppose that $Y = (Y_t : t \leq \zeta)$ with probabilities $\mathbb{P}_x$, $x \in E$, is a regular Markov process on an open domain $E \subseteq \mathbb{R}^d$, with cemetery state Δ and killing time $\zeta = \inf\{t > 0 : Y_t = \Delta\}$. Let us additionally write $\mathbb{P}_\nu = \int_E \nu(da) \mathbb{P}_a$, for any probability measure ν on the state space of Y .

Suppose that $\mathcal{G}$ is the σ -algebra generated by Y and write $\mathcal{G}(\mathbb{P}_\nu)$ for its completion by the null sets of $\mathbb{P}_\nu$, for all possible probability distributions ν on E . Moreover, write $\bar{\mathcal{G}} = \bigcap_\nu \mathcal{G}(\mathbb{P}_\nu)$, where the intersection is taken over all probability measures on E . A random time $\mathbf{k}$ is called an L -time (generalised last exit time) if

(i) $\mathbf{k}$ is measurable in $\overline{\mathcal{G}}$, and $\mathbf{k} \leq \zeta$ almost surely with respect to $\mathbb{P}_\nu$, for all ν ,

(ii) $\{s < \mathbf{k} - t\} = \{t < \mathbf{k}\} \cap \{s < \mathbf{k} \circ (v_t)\}$ for all $t, s \geq 0$,

where v_t is the Markov shift operator. The class of L -times also contains classical last exit times like γ_B . The lifetime ζ is also an L -time.

Theorem 1.8. *On account of (A) and (B), listed and proved below, for any L -time $\mathbf{k}$, the time-reversed process $\overleftarrow{X}$, defined by $\overleftarrow{X}_t = X_{(\mathbf{k}-t)-}$, $t < \mathbf{k}$, under $\mathbb{P}_x^{\{\theta\}}$ is a time-homogeneous Markov process with transition probabilities*

$$\mathbb{P}_x^{\{\theta\}}(\overleftarrow{X}_t \in A | \overleftarrow{X}_s) = p_{t-s}^{\{\theta \rightarrow x\}}(\overleftarrow{X}_s, A), \quad \mathbb{P}_x^{\{\theta\}} - \text{almost surely}, \quad (1.29)$$

for all $0 < s < t$ and closed $A \in \mathbb{R}^d$, where $p_u^{\{\theta \rightarrow x\}}(z, A)$, $u \geq 0$, $z \in \mathbb{R}^d$, is the transition measure associated to the semigroup $\mathcal{P}^{\{\theta \rightarrow x\}}$ of $(X, \mathbb{P}_\theta^{\{x\}})$.

(A) The semigroup $\mathcal{P}^{\{x \rightarrow \theta\}}$ associated to $(X, \mathbb{P}_x^{\{\theta\}})$ is in weak duality with $\mathcal{P}^{\{\theta \rightarrow x\}}$ with respect to a measure μ which takes the following form.

$$\mu(dy) = \int_{\mathbb{R}^d} U^{\{\theta\}}(\omega, dy) \nu(d\omega)$$

where $U^{\{\theta\}}(w, dy)$ is the occupation measure $U^{\{\theta\}}(w, dy) = \mathbb{E}_w^{\{\theta\}} [\int_0^\infty \mathbf{1}_{\{X_t \in dy\}} dt]$ and the σ -finite measure $\nu(d\omega) \equiv 2^\alpha \pi^{d/2} \frac{\Gamma(\alpha/2)}{\Gamma((d-\alpha)/2)} \delta_x(\omega)$, i.e.

$$\mu(dy) = |x - y|^{\alpha-d} \frac{|y - \theta|^{\alpha-d}}{|x - \theta|^{\alpha-d}} dy.$$

Proof of (A). Let f and g be two compactly supported test functions and $p_t^{\{x \rightarrow \theta\}}$ and $p_t^{\{\theta \rightarrow x\}}$ represent the transition measure densities of $\mathcal{P}^{\{x \rightarrow \theta\}}$ and $\mathcal{P}^{\{\theta \rightarrow x\}}$ respectively,

then

$$\begin{aligned}
& \int_{\mathbb{R}^d} g(u) \mathcal{P}_t^{\{x \rightarrow \theta\}}[f](u) \mu(du) \\
&= \int_{\mathbb{R}^d} g(u) \int_{\mathbb{R}^d} p_t^{\{x \rightarrow \theta\}}(u, y) f(y) dy \mu(du) \\
&= \int_{\mathbb{R}^d} g(u) \int_{\mathbb{R}^d} p_t(u, y) \frac{|y - \theta|^{\alpha-d}}{|u - \theta|^{\alpha-d}} f(y) dy |x - u|^{\alpha-d} \frac{|u - \theta|^{\alpha-d}}{|x - \theta|^{\alpha-d}} du \\
&= \int_{\mathbb{R}^d} g(u) \int_{\mathbb{R}^d} p_t(y, u) \frac{|y - x|^{\alpha-d}}{|y - \theta|^{\alpha-d}} \frac{|y - \theta|^{\alpha-d}}{|u - \theta|^{\alpha-d}} f(y) dy |x - u|^{\alpha-d} \frac{|u - \theta|^{\alpha-d}}{|x - \theta|^{\alpha-d}} du \\
&= \int_{\mathbb{R}^d} f(y) \int_{\mathbb{R}^d} p_t^{\{\theta \rightarrow x\}}(y, u) g(u) du \mu(dy) \\
&= \int_{\mathbb{R}^d} f(y) \mathcal{P}_t^{\{\theta \rightarrow x\}}[g](y) \mu(dy).
\end{aligned}$$

□

(B) For any continuous compactly supported test function f and $a \in \mathbb{R}^d$, $\mathcal{P}_t^{\{x \rightarrow \theta\}}[f](a)$ is right-continuous in t for all $a \in \mathbb{R}^d$. For any $q > 0$, $U_{\hat{X}}^{(q)}[f](\bar{X}_t)$ the q -potential of $(\bar{X}, \mathbb{P}_\theta^{\{x\}})$ is right-continuous in t , where, for bounded and measurable $f : \mathbb{R}^d \rightarrow \mathbb{R}$

$$U_{\hat{X}}^{(q)}[f](a) := \int_0^\infty e^{-qt} \mathcal{P}_t^{\{\theta \rightarrow x\}}[f](a) dt, \quad a \in \mathbb{R}^d.$$

This condition is immediately satisfied due to the fact that $\mathcal{P}^{\{x \rightarrow \theta\}}$ and $\mathcal{P}^{\{\theta \rightarrow x\}}$ are right-continuous semigroups by virtue of their definition as a Doob h -transforms with respect to the Feller semigroup of the isotropic stable process.

Remark 1.9. The Riesz-Bogdan-Żak transform (1.6) under $\mathbb{P}_x$, which is equivalent in law to the process $(X, \mathbb{P}_{K(x)}^\circ)$, corresponds to the case $(X, \mathbb{P}_{K(x)}^{\{0\}})$.

The previous theorem will help us to prove the following proposition, which illustrates how the triple law at last hitting will be obtained.

Proposition 1.10. Let $x \in \mathbb{R}^d$ and $|y| < 1$

$$\mathbb{P}_x(X_{\gamma_B} \in dy, \gamma_B > 0) = C_{\alpha,d} |x - y|^{\alpha-d} (1 - |y|^2)^{-\alpha/2} dy.$$

Proof. Let us begin by recalling that the Riesz-Bogdan-Żak transform (1.6) inverts the space with respect the unit sphere $\mathbb{S}^{d-1}$ and puts in correspondence X_{γ_B} under $\mathbb{P}_x$ with $K(X_{\gamma_{B^c}})$ under $\mathbb{P}_{K(x)}^{\{0\}}$, where $\gamma_{B^c} = \sup\{t > 0 : |X_t| > 1\}$ and $K(x) = x/|x|^2$ for $x \neq 0$.

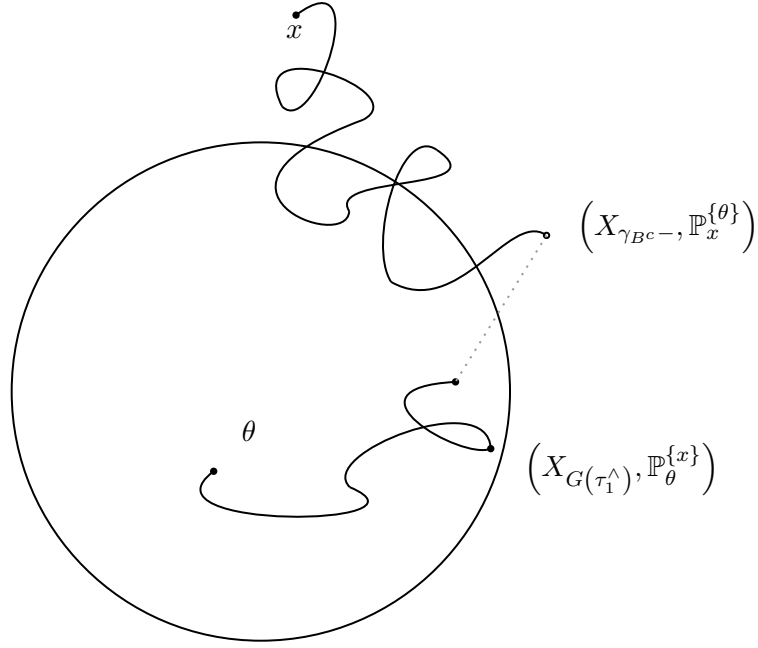


Figure 1-1: Duality between $\mathbb{P}_x^{\{\theta\}}$ and $\mathbb{P}_\theta^{\{x\}}$.

Therefore,

$$\begin{aligned} \mathbb{P}_x(X_{\gamma_B-} \in dy, \gamma_B > 0) &= \mathbb{P}_{K(x)}^{\{0\}}(X_{\gamma_{B^c}-} \in dK(y), \gamma_{B^c} > 0) \\ &= \mathbb{P}_0^{\{K(x)\}}(X_{\tau_1^\odot} \in dK(y), \tau_1^\odot < \infty), \end{aligned}$$

where $\tau_1^\odot = \inf\{t > 0 : |X_t| > 1\}$. The second equality above is given by Theorem 1.8, in this case, by time-reversing $(X, \mathbb{P}_{K(x)}^{\{0\}})$ from its lifetime (where it reaches zero). Thus we can identify the last hitting outside the unit ball with the reversed first hit outside the unit ball. See Figure 1-1.

Now we will use straightforward calculations on the last probability,

$$\begin{aligned}
\mathbb{P}_0^{\{K(x)\}}(X_{\tau_1^\odot} \in dK(y), \tau_1^\odot < \infty) &= \frac{H(K(y))}{H(0)} \mathbb{P}_0(X_{\tau_1^\odot} \in dK(y), \tau_1^\odot < \infty) \\
&= \frac{\left| \frac{y}{|y|^2} - \frac{x}{|x|^2} \right|^{\alpha-d}}{\left| 0 - \frac{x}{|x|^2} \right|^{\alpha-d}} \mathbb{P}_0(X_{\tau_1^\odot} \in dK(y), \tau_1^\odot < \infty) \\
&= |y-x|^{\alpha-d} |y|^{d-\alpha} C_{\alpha,d} (|y|^{-2} - 1)^{-\alpha/2} |y|^{-d} dy \\
&= C_{\alpha,d} |x-y|^{\alpha-d} (1-|y|^2)^{-\alpha/2} dy.
\end{aligned}$$

□

1.2.2 Proof of the triple law at last exit

We will now use the same idea from the previous proposition to characterise the joint law of three random variables at last hitting time. These random variables can be put in correspondence with the three random variables in the so-called triple law at the first exit of the unit ball (Theorem 1.3 in [37]). Define $G(\tau_1^\odot) = \sup\{t < \tau_1^\odot : |X_t| = \sup_{u \leq \tau_1^\odot} |X_u|\}$ the time of furthest reach from the origin before exiting the unit ball. If $\gamma_{B^c}-$ denotes the time of last visit to the complement of the unit ball, the counterpart of $G(\tau_1^\odot)$ under time reversal will be given by the time $\vec{G}(\gamma_{B^c}) := \inf\{t > \gamma_{B^c} : |X_t| = \sup_{u \geq \gamma_{B^c}} |X_u|\}$, which can be interpreted as the time of the next radial maximum after leaving for the last time the complement of the unit ball. Notice that due to $\vec{G}(\gamma_{B^c}) > \gamma_{B^c}$, such maximum cannot be bigger than one.

Proof of Theorem 1.7. Let $x \in \mathbb{R}^d$, $|y| > |z| > 1$ and $|w| < 1$. By the Riesz-Bogdan-Żak transform

$$\begin{aligned}
&\mathbb{P}_x(X_{\gamma_{B^-}} \in dv, X_{\gamma_B} \in dy, X_{\vec{g}(\gamma_B)} \in dz, \gamma_B > 0) \\
&= \mathbb{P}_{K(x)}^{\{0\}}(X_{\gamma_{B^c}-} \in dK(v), X_{\gamma_{B^c}} \in dK(y), X_{\vec{G}(\gamma_{B^c})} \in dK(z), \gamma_{B^c} > 0).
\end{aligned}$$

By Theorem 1.8 (see Figure 1-1 for reference)

$$\begin{aligned}
& \mathbb{P}_{K(x)}^{\{0\}}(X_{\gamma_{B^c}-} \in dK(v), X_{\gamma_{B^c}} \in dK(y), X_{\vec{G}(\gamma_{B^c})} \in dK(z), \gamma_{B^c} > 0) \\
&= \mathbb{P}_0^{\{K(x)\}}(X_{\tau_1^\otimes} \in dK(v), X_{\tau_1^\otimes-} \in dK(y), X_{G(\tau_1^\otimes)} \in dK(z), \tau_1^\otimes < \infty) \\
&= |v-x|^{\alpha-d} |v|^{d-\alpha} \chi_0(K(z), K(y), K(v)) |z|^{-2d} |y|^{-2d} |v|^{-2d} dz dy dv \\
&= |v-x|^{\alpha-d} |v|^{d-\alpha} \pi^{-3/2} \frac{\Gamma((d+\alpha)/2)}{|\Gamma(-\alpha/2)|} \frac{\Gamma(d/2)^2}{\Gamma(\alpha/2)^2} \frac{||y|^{-2} - |z|^{-2}|^{\alpha/2} |v|^{-2d}}{|z-y|^d |v-y|^{\alpha+d} |v|^{-\alpha-d} |y|^{-\alpha}} dz dy dv \\
&= |v-x|^{\alpha-d} \pi^{-3/2} \frac{\Gamma((d+\alpha)/2)}{|\Gamma(-\alpha/2)|} \frac{\Gamma(d/2)^2}{\Gamma(\alpha/2)^2} \frac{||y|^{-2} - |z|^{-2}|^{\alpha/2}}{|z-y|^d |v-y|^{\alpha+d} |y|^{-\alpha}} dz dy dv.
\end{aligned}$$

□

1.2.3 Last hit to $\mathbb{S}^{d-1}$ for transient Brownian motion

Let $(Y, \mathbb{P})$ be the d -dimensional Brownian motion with $d \geq 3$. It is known that Y is transient in the sense that $\lim_{t \rightarrow \infty} |Y_t| = \infty$. This makes possible to define the moment of last contact with the unit sphere in a similar fashion to what we did for the α -stable process. Define

$$\gamma_S = \begin{cases} \sup\{t > 0 : |Y_t| = 1\} & \text{if } \{t > 0 : |Y_t| = 1\} \neq \emptyset \\ 0 & \text{if } \{t > 0 : |Y_t| = 1\} = \emptyset. \end{cases}$$

Due to the continuity of the Brownian paths, the triple law does not longer make sense. However, we will obtain the distribution of the point of last contact using the same methodology used for the stable process.

We need to enunciate the following lemma, whose proof is verbatim of that of Theorem 1.8. In order to do this, we begin by noticing that if $(Y, \mathbb{P}_x^{\{\theta\}})$ is the Brownian motion issued at x and conditioned to hit $\theta \in \mathbb{R}^d$, defined analogously to the stable case (see (1.27)). Then, analogously to the stable case for $t < \inf\{t > 0 : |Y_t| = 0\}$

$$\left. \frac{d\mathbb{P}_x^{\{\theta\}}}{d\mathbb{P}_x} \right|_{\mathcal{F}_t} = \left. \frac{|Y_t - \theta|^{2-d}}{|x - \theta|^{2-d}} \right|_{\mathcal{F}_t},$$

where $(\mathcal{F}_t)_{t \geq 0}$ is the natural filtration of Y .

Lemma 1.11. *The consequences of Hunt-Nagasawa duality, for instance equation (1.29), hold between two conditioned transient Brownian motions $(Y, \mathbb{P}_x^{\{\theta\}})$ and $(Y, \mathbb{P}_\theta^{\{x\}})$ under the entrance laws δ_x and δ_θ respectively.*

The distribution of the last contact point with the unit sphere is given in the following proposition.

Proposition 1.12. *Let $x \in \mathbb{R}^d$,*

$$\mathbb{P}_x(Y_{\gamma_S} \in dy, \gamma_S > 0) = \frac{\Gamma(d/2)}{2\pi^{d/2}} |y - x|^{2-d} \sigma_1(dy), \quad y \in \mathbb{S}^{d-1}.$$

Proof. The steps are similar to those in the proof of Proposition 1.10, and we present them here in full to highlight modifications specific to this case: 1) the use of the last hitting time γ_S , 2) the h -transform of the Kelvin transform in the Brownian context, and 3) the application of the well-known explicit formula for the first hitting time of the unit sphere (see Theorem 3.43 in [42]). Recall that $\tau_1 = \inf\{t > 0 : |Y_t| = 1\}$, then

$$\begin{aligned} \mathbb{P}_x(Y_{\gamma_S} \in dy, \gamma_S > 0) &= \mathbb{P}_{K(x)}^{\{0\}}(Y_{\gamma_S} \in dy, \gamma_S > 0) \\ &= \mathbb{P}_0^{\{K(x)\}}(Y_{\tau_1} \in dy, \tau_1 < \infty) \\ &= \frac{|y - K(x)|^{2-d}}{|0 - K(x)|^{2-d}} \frac{\Gamma(d/2)}{2\pi^{d/2}} \sigma_1(dy) \\ &= \frac{\Gamma(d/2)}{2\pi^{d/2}} |x - y|^{2-d} \sigma_1(dy). \end{aligned}$$

□

The reader may identify that the measure in the last equation is exactly the potential measure of $(Y, \mathbb{P})$ on the sphere, i.e., the average occupation. This gives a nice and rather obvious interpretation to the distribution of the last contact position in the sphere.

Chapter 2

Directional excursions of stable processes

For the isotropic α -stable Lévy process in dimension $d \geq 2$ and $\alpha \in (0, 2)$, we present a theory of excursions from the minimum/maximum of the process with respect to a direction given by a normal vector v . For this construction, explicit expressions for the potential measures of the ladder processes are obtained, as well as other important excursion-related functionals. Applications of this excursion theory allow us to characterise the joint law of the process at three key random times, the instant of closest reach to a hyperplane before crossing it, the time of crossing the hyperplane and the moment right before the crossing. We call this result the triple law. An identity in the spirit of the so-called Deep Wiener-Hopf factorisation is also proved. At the end of the chapter we prove that after some light adjustments, we can follow the same methodology to find the ascending and descending potential measures and a Deep Wiener-Hopf-type factorisation for Brownian motion of dimension $d \geq 2$. The analysis of a point of closest reach to a hyperplane is not pertinent for Brownian motion, since it hits half-spaces continuously. Finally, by using the distribution of the point of closest reach to a hyperplane, we define and characterise the conditional law of the process to be attracted to a subset of a given hyperplane by one-side of the hyperplane, the conditional law of the process which never visits a half-space and the conditional law of a process which is repelled from a hyperplane.

2.1 Introduction

The study of fluctuation identities for α -stable Lévy processes was immensely benefited from the discoveries of Lamperti (cf. [40]) of a bijection between the class of positive self-similar Markov process and Lévy processes. Many articles use the so-called Lamperti transform to solve, in an intuitive probabilistic way, a variety of questions which in the past were often solved by purely analytic methods. See for instance [13] [39] [36]. More recently, the generalisation of the Lamperti transform, to a relationship between real-valued self-similar process and Markov Additive Processes (MAPs), a type of modulated Lévy processes, has given birth to various results on the study not only of the α -stable process on $\mathbb{R}$ but also the isotropic α -stable process in dimension $d \geq 2$.

In 2019, Kyprianou et al. [37], used the Lamperti-Kiu transform to develop a theory of excursions from the radial minimum, for the isotropic stable process in dimension $d \geq 2$. With this, they found the law of the point of radial closest reach to the origin, and exhibit the so-called deep factorisation for the Lamperti-Kiu-transform MAP (see Chapter 1). The characterisation of the point of closest reach allows, among other things, to make sense of the asymptotic conditioning of a process to continuously touch a set on the surface of the unitary sphere. This conditioning was studied in [34], and it opened the door to the question: can such conditioning be done if the sphere is replaced by a hyperplane? Simultaneously, in [28], the law of the process conditioned to touch a set on the surface of the sphere was also found. This time, in contrast to the conditioning in [34], the process is allowed to visit the interior of the unit ball an infinite number of times before touching said set, and did not need the theory of the radial point of closest reach. In this chapter the case of the hyperplane was also considered, contributing to the curiosity in characterising the dynamics of the stable process in relation to hyperplanes or half-spaces.

Other recent publications also highlight the relevance of investigations regarding Lévy processes in coordinate systems based on the projection onto an arbitrary hyperplane and its orthogonal direction. In [23], the authors investigate the decomposition of the trajectories of a Lévy process Y in $\mathbb{R}^d$ splitting into what happens before and after the process $\langle Y, v \rangle = (\langle Y_t, v \rangle : t \geq 0)$ reaches a minimum (here v is a fixed normal vector in $\mathbb{R}^d$). Another recent publication in regards stable processes and hyperplanes is [44], where the authors found a formula for the Laplace transform of the first hitting time to a hyperplane, and the distribution of the first hit on such hyperplane.

2.2 Results on excursions

The following results are consequences of the development of a theory of excursions from the minimum of X in the direction of $e_1 = (1, 0, \dots, 0)$. The research conducted in this direction has fundamentally followed the same steps of the theory of excursions from the radial minimum addressed in [37]. The main differences are the distributional properties of the radial process $|X| = (|X_t| : t \geq 0)$ versus those of $X^{(1)}$, a one-dimensional α -stable process. These differences present enough technical differences to dedicate a detailed treatment of the procedure.

The main results of this section for the isotropic α -stable process in $\mathbb{R}^d$ where $d \geq 2$ and $\alpha \in (0, 2)$ are Theorems 2.1, 2.3 and 2.7 presented below, and their proofs are provided in Subsection 2.2.1. In Subsection 2.2.2 we prove that some of the formulas for the α -stable process with $\alpha \in (0, 2)$, in particular the Deep Wiener-Hopf factorisation (Theorem 2.9), are also valid for the case of Brownian motion, although the methods used to prove this are somewhat different from those used in the first part.

Theorem 2.1 (Law of the one-sided point of closest reach to the hyperplane $\mathbb{H}_0$). *Let $x, y \in \mathbb{H}_{0\uparrow} = \{z \in \mathbb{R}^d : z^{(1)} > 0\}$, $G(t) := \sup\{s < t : X_s^{(1)} = \inf_{u < t} X_u^{(1)}\}$ denote the last instant before t where X reaches a minimum in its first coordinate, and $\tau_0^\vee := \inf\{s > 0 : X_s^{(1)} < 0\}$, the time of first passage below $\mathbb{H}_0 = \{z \in \mathbb{R}^d : z^{(1)} = 0\}$, then*

$$\mathbb{P}_x(X_{G(\tau_0^\vee)} \in dy) = C_{\alpha,d} \frac{|y^{(1)} - x^{(1)}|^{\alpha/2}}{|y^{(1)}|^{\alpha/2}} |x - y|^{-d} dy \quad \text{for } 0 < y^{(1)} < x^{(1)}, \quad (2.1)$$

where $C_{\alpha,d} = \pi^{-(d/2+1)} \Gamma(d/2) \sin(\alpha\pi/2)$.

The ‘one-sided’ qualification of this random variable refers to the fact that the directional minimum is recorded while the process remains in the upper half-space in which the hyperplane $\mathbb{H}_0$ separated the space. The concept of one-sided point of closest reach was born from the analogy to the point of closest distance to the origin, and it is the building block of the conditional law to approach a subset of the hyperplane $\mathbb{H}_0$ from one side. We will return to this conditional law at the end of this chapter.

Remark 2.2. *If we consider $x, y \in \mathbb{H}_{0\downarrow} = \{z \in \mathbb{R}^d : z^{(1)} < 0\}$ such that $x^{(1)} < y^{(1)} < 0$, replace $\tau_0^\vee$ by the stopping time $\tau_0^\wedge = \inf\{s > 0 : X_s^{(1)} > 0\}$ and define $\overline{G}(\tau_0^\wedge) := \sup\{s < \tau_0^\wedge : X_s^{(1)} = \sup_{u < t} X_u^{(1)}\}$ (the point of closest reach from below of $\mathbb{H}_0$) by symmetry, the kernel in (2.1) is also the density of $X_{\overline{G}(\tau_0^\wedge)}$.*

The use of an excursion framework to characterise directional minima, together with explicit expressions for functionals under the excursion measure, allow us to obtain joint laws of relevant quantities at first passage times in regards half-spaces. These n -tuple laws at first entry/exit of a half-space provide a non-trivial extension to the simple undershoot-overshoot laws.

Theorem 2.3 (Triple law at the first crossing of a hyperplane). *Fix $r \in \mathbb{R}$ and define for $w, x, y \in \mathbb{H}_{r\uparrow} = \{x \in \mathbb{R}^d : x^{(1)} > r\}$ and $z \in \mathbb{H}_{r\downarrow} = \{x \in \mathbb{R}^d : x^{(1)} < r\}$, the following density*

$$\Xi_w(x, y, z) := A_{\alpha, d} \frac{|w^{(1)} - x^{(1)}|^{\alpha/2}}{|w - x|^d} \frac{|x^{(1)} - y^{(1)}|^{\alpha/2}}{|x - y|^d} |y - z|^{-\alpha-d}$$

where $A_{\alpha, d} = \frac{2^\alpha \Gamma(d/2)^2 \Gamma(\frac{d+\alpha}{2})}{\pi^{3d/2} \Gamma(\alpha/2)^2 |\Gamma(-\alpha/2)|}$. Finally, let $G(\tau_r^\vee)$ denote the instant of closest reach to the hyperplane $\mathbb{H}_r$ before first entry into $\mathbb{H}_{r\downarrow}$. Then, for $w^{(1)} > x^{(1)} > r$, $y^{(1)} > x^{(1)}$ and $z^{(1)} < r$,

$$\mathbb{P}_w(X_{G(\tau_r^\vee)} \in dx, X_{\tau_r^\vee-} \in dy, X_{\tau_r^\vee} \in dz) = \Xi_w(x, y, z) dx dy dz. \quad (2.2)$$

Once again, by symmetry, the previous result can be easily extended to the case where the point of issue is in $\mathbb{H}_{r\downarrow}$ and the upward crossing occurs at $\tau_r^\wedge$. This is, for $w^{(1)} < x^{(1)} < r$, $y^{(1)} < x^{(1)}$ and $z^{(1)} > r$,

$$\mathbb{P}_w(X_{G(\tau_r^\wedge)} \in dx, X_{\tau_r^\wedge-} \in dy, X_{\tau_r^\wedge} \in dz) = \Xi_w(x, y, z) dx dy dz.$$

Technically speaking, the next two lemmas could be obtained by marginalising (2.2), and thus may be viewed as corollaries of Theorem 2.3. However, they can also be derived directly through a more straightforward approach. For this reason, we present them as standalone lemmas and provide direct proofs in Subsection 2.2.1.

Lemma 2.4 (Overshoot and undershoot at first crossing of a hyperplane). *Fix $r \in \mathbb{R}$ and define for $x, y \in \mathbb{H}_{r\uparrow}$ and $z \in \mathbb{H}_{r\downarrow}$*

$$\Xi_x(\bullet, y, z) := B_{\alpha, d} \left(|x - y|^{\alpha-d} \int_0^{\zeta_r^\vee(x, y)} (u+1)^{-d/2} u^{\alpha/2-1} du \right) |z - y|^{-(\alpha+d)}$$

where

$$\zeta_r^\vee(x, y) := \frac{4(x^{(1)} - r)(y^{(1)} - r)}{|x - y|^2} \quad \text{and} \quad B_{\alpha, d} = \frac{\Gamma((d+\alpha)/2) \Gamma(d/2)}{\pi^d |\Gamma(\alpha/2)^2 \Gamma(-\alpha/2)|}.$$

Then, for $y^{(1)} > r > z^{(1)}$

$$\mathbb{P}_x(X_{\tau_r^\vee-} \in dy, X_{\tau_r^\vee} \in dz) = \Xi_x(\bullet, y, z) dy dz. \quad (2.3)$$

Finally, in a second lemma below we give the law of just the overshoot distribution. However, we note that its proof again avoids performing a marginalisation of, this time, the density (2.3), and we use a third approach for its proof.

Lemma 2.5 (Overshoot at first crossing of a hyperplane). *Fix $r \in \mathbb{R}$ and define for $x \in \mathbb{H}_{r\uparrow}$ and $z \in \mathbb{H}_{r\downarrow}$*

$$\Xi_x(\bullet, \bullet, z) := C_{\alpha,d} |x - z|^{-d} \frac{|r - x^{(1)}|^{\alpha/2}}{|r - z^{(1)}|^{\alpha/2}},$$

where $C_{\alpha,d} = \pi^{-(d/2+1)} \Gamma(d/2) \sin(\alpha\pi/2)$. Then

$$\mathbb{P}_x(X_{\tau_r^\vee} \in dz) = \Xi_x(\bullet, \bullet, z) dz \quad z^{(1)} < r < x^{(1)}. \quad (2.4)$$

Remark 2.6. Lemma 2.5 is known already from [12] and Lemma 2.4 is also implicitly known from the same reference, albeit stated up to multiplication by the stable jump measure as the Green function for X killed on exiting $\mathbb{H}_{r\uparrow}$. The approach taken by [12] uses potential analysis rather than the fluctuation theory approach that we use here.

In the same spirit of the so-called deep Wiener-Hopf factorisation for the radial MAP (ξ, Θ) in the Lamperti-Kiu representation of X , we exhibit a deep factorisation for the orthogonal decomposition $(\langle X, e_1 \rangle, X_{e_1^\perp}) = (X^{(1)}, X^{(2:d)})$, which, as pointed out in Remark 0.7, turns out to be a MAP. An analogous relationship to the Deep Wiener-Hopf factorisation in the radial case (in Theorem 0.10) turns out to be true for this decomposition. It is worth mentioning that in [53], the potential measure of a Lévy process killed in a half-space is *factorised* in a pair of measures associated to a path decomposition at the time of a directional minimum. We show that for the isotropic α -stable process, those measures are in fact the ascending and descending potential ladder measures, giving them a simple probabilistic meaning. Our derivation of the potentials also follows a different approach, based on theory of excursions.

To state the deep Wiener-Hopf factorisation result we require to define the (d -dimensional) ascending and descending ladder processes in the direction of $X^{(1)}$. Recall the framework of excursions in Section 0.4, in the particular case of Remark 0.7, consider $L = (L_t : t \geq 0)$ the local time at $\{0\} \times \mathbb{R}^{d-1}$ for the process X reflected on its

directional minimum $(X_t^{(1)} - \underline{X}_t^{(1)}, X_t^{(2:d)})_{t \geq 0}$. Consider the d -dimensional descending ladder height process $(H^-, X^{(2:d)-})$ given by the following relationship

$$(H_t^-, X_t^{(2:d)-}) = (X_{L_t^{-1}}^{(1)}, X_{L_t^{-1}}^{(2:d)}) \quad t \geq 0, \quad (2.5)$$

where $L_t^{-1} = \inf\{s > 0 : L_s > t\}$ is the right-continuous inverse of the local time L . Notice that by the definition of L , and the fact that under $\mathbb{P}_x$, $X^{(1)}$ is a symmetric α -stable process, $-H^-$ is the usual one-dimensional descending ladder process as in equation (0.5), and hence an $\alpha/2$ -stable subordinator. Analogously, we can define the ascending ladder height process, and denote it by $(H^+, X^{(2:d)+})$.

Let us introduce the following potential operators, for f a continuous function with compact support on $\mathbb{R}^{d-1}$, $\theta \in \mathbb{R}$, and $x^{(2:d)} \in \mathbb{R}^{d-1}$

$$\mathbf{R}_\theta[f](x) := \mathbb{E}_{0, x^{(2:d)}} \left[\int_0^\infty e^{i\theta X_t^{(1)}} f(X_t^{(2:d)}) dt \right],$$

now define the potentials of the descending and ascending ladder processes, $(H^-, X^{(2:d)-})$ and $(H^+, X^{(2:d)+})$ respectively, by

$$\hat{\rho}_\theta[f](x) := \mathbb{E}_{0, x^{(2:d)}} \left[\int_0^\infty e^{i\theta H_t^-} f(X_t^{(2:d)-}) dt \right],$$

and

$$\rho_\theta[f](x) := \mathbb{E}_{0, x^{(2:d)}} \left[\int_0^\infty e^{i\theta H_t^+} f(X_t^{(2:d)+}) dt \right].$$

Notice that

$$\hat{\rho}_\theta[f](x) = \int_{(-\infty, 0) \times \mathbb{R}^d} e^{i\theta y^{(1)}} f(y^{(2:d)}) U_{0, x^{(2:d)}}^-(dy),$$

where $U_x^-(dy) = \int_0^\infty \mathbb{P}_x \left((H_t^-, X_t^{(2:d)-}) \in dy \right) dt$ for $x^{(1)} > y^{(1)}$. Similarly one can define $U_x^+(dy)$. The measures U^+ and U^- are called the ascending and descending ladder measures.

The density of the potential of a d -dimensional stable process is a well known object (cf. Chapter 3 of [35]), called the Riesz potential, and allows us to write the operator $\mathbf{R}_\theta[f](x)$ as follows,

$$\mathbf{R}_\theta[f](x) = 2^{-\alpha} \pi^{-d/2} \frac{\Gamma((d-\alpha)/2)}{\Gamma(\alpha/2)} \int_{\mathbb{R}^d} e^{i\theta y^{(1)}} f(y^{(2:d)}) |x - y|^{\alpha-d} dy.$$

Theorem 2.7 (Deep Wiener-Hopf planar factorisation). *For a continuous compactly*

supported function $f : \mathbb{R}^{d-1} \rightarrow \mathbb{R}$, $\theta \in \mathbb{R}$ and $x = (0, x^{(2:d)})$, $x^{(2:d)} \in \mathbb{R}^{d-1}$ the potential operator $\mathbf{R}$ corresponding to the isotropic α -stable process can be represented as the composition of ascending and descending potential operators as follows

$$\mathbf{R}_\theta[f](x) = \hat{\rho}_\theta[\rho_\theta[f]](x). \quad (2.6)$$

Moreover,

$$\rho_\theta[f](x) = \int_{y^{(1)} > 0} e^{i\theta y^{(1)}} f(y^{(2:d)}) \frac{\Gamma(d/2)}{\pi^{d/2} \Gamma(\alpha/2)} \frac{(y^{(1)})^{\alpha/2}}{|x - y|^d} dy$$

and

$$\hat{\rho}_\theta[f](x) = \rho_{-\theta}[f](x).$$

Results for Brownian Motion

Consider Y a Brownian motion in $d \geq 2$ with laws $(\mathbb{P}_x : x \in \mathbb{R}^d)$. It is known that $Y^{(1)}$ under $\mathbb{P} := \mathbb{P}_0$ is a real-valued Brownian motion independent of $Y^{(2:d)}$, another Brownian motion in $\mathbb{R}^{d-1}$. In an abuse of notation we will consider $L = (L_t : t \geq 0)$ the local time at the minimum for $Y^{(1)}$. If we define, analogously to what we did for the α -stable process, $(H^-, Y^{(2:d)-})$ and $(H^+, Y^{(2:d)+})$ the descending and ascending ladder height processes respectively and we denote by $U_x^+(dy)$ and $U_x^-(dy)$ the ascending and descending potential measures, i.e.

$$U_x^+(dy) := \int_0^\infty \mathbb{P}_x \left((H_t^+, Y_t^{(2:d)+}) \in dy \right) dt, \quad \text{for } x^{(1)} < y^{(1)}$$

and

$$U_x^-(dy) := \int_0^\infty \mathbb{P}_x \left((H_t^-, Y_t^{(2:d)-}) \in dy \right) dt, \quad \text{for } x^{(1)} > y^{(1)}$$

then we have the following result, combining the two cases of α -stable processes, $\alpha \in (0, 2)$ and $\alpha = 2$.

Proposition 2.8. *The explicit expression for the ascending/descending potential measures of an α -stable process with $\alpha \in (0, 2]$ are given by*

$$U_x^+(dy) = \frac{\Gamma(d/2)}{\pi^{d/2} \Gamma(\alpha/2)} \frac{(y^{(1)} - x^{(1)})^{\alpha/2}}{|x - y|^d} dy, \quad y^{(1)} > x^{(1)} \quad (2.7)$$

$$U_x^-(dy) = \frac{\Gamma(d/2)}{\pi^{d/2} \Gamma(\alpha/2)} \frac{(y^{(1)} - x^{(1)})^{\alpha/2}}{|x - y|^d} dy, \quad y^{(1)} < x^{(1)}. \quad (2.8)$$

Moreover, we can write the following ‘factorisation’ for the so-called Newtonian poten-

tial.

Theorem 2.9 (Deep Wiener-Hopf factorisation for Brownian motion). *Consider $f : \mathbb{R}^{d-1} \rightarrow \mathbb{R}$ a compactly supported continuous function, $\theta \in \mathbb{R}$, $x^{(2:d)} \in \mathbb{R}^{d-1}$ and let us extend our definitions of $\mathbf{R}$ to consider the Brownian case, i.e. also if $\alpha = 2$*

$$\mathbf{R}_\theta[f](x) = \mathbb{E}_{0, x^{(2:d)}} \left[\int_0^\infty e^{i\theta Y^{(1)}} f(Y^{(2:d)}) \right] = \frac{\Gamma(\frac{d-2}{2})}{4\pi^{d/2}} \int_{\mathbb{R}^d} e^{i\theta y^{(1)}} f(y^{(2:d)}) |x - y|^{2-d} dy,$$

then

$$\mathbf{R}_\theta[f](x) = \hat{\rho}_\theta[\rho_\theta[f]](x). \quad (2.9)$$

Moreover,

$$\rho_\theta[f](x) = \int_{y^{(1)} > 0} e^{i\theta y^{(1)}} f(y^{(2:d)}) \frac{\Gamma(d/2)}{\pi^{d/2}} \frac{y^{(1)}}{|x - y|^d} dy$$

and

$$\hat{\rho}_\theta[f](x) = \rho_{-\theta}[f](x).$$

2.2.1 Proofs of results for $\alpha \in (0, 2)$

Proofs of Lemmas 2.4 and 2.5

In order to prove Lemmas 2.4 and 2.5 we use a method of converting results for first entry into a ball into first entry into a half space by expanding the radius of the sphere asymptotically so that, with re-centring, the interior of the ball becomes a half-space. We call this the *flat earth theory*, in the sense that, from an infinitesimally localised perspective, the surface of a sphere appears to be that of a flat surface.

The lemma we present below combines a geometric argument with the Lévy and scaling properties of the stable process $(X, \mathbb{P})$.

Lemma 2.10 (Flat earth approximation). *Let $\tau_r^\oplus = \inf\{t > 0 : |X_t| < r\}$ and suppose $f : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ is bounded and measurable. Then*

$$\mathbb{E}_x [f(X_{\tau_r^\vee}, X_{\tau_r^\vee -})] = \lim_{R \rightarrow \infty} \mathbb{E}_{x + R\mathbf{e}_1} \left[f(X_{\tau_{R+r}^\oplus} - R\mathbf{e}_1, X_{\tau_{R+r}^\oplus} - R\mathbf{e}_1), \tau_R^\oplus < \infty \right], \quad (2.10)$$

where $\mathbf{e}_1 = (1, 0, \dots, 0) \in \mathbb{R}^d$.

Proof. We start by approximating $\mathbb{H}_{r\downarrow} = \{z \in \mathbb{R}^d : z^{(1)} < r\}$, $r > 0$, by an increasing sequence of balls $\mathbb{B}_{(-R\mathbf{e}_1, R+r)} = \{z \in \mathbb{R}^d : |z + R\mathbf{e}_1| < R + r\} \subseteq \mathbb{H}_{r\downarrow}$ where $R > 0$. We

have

$$\lim_{R \rightarrow \infty} \mathbb{B}_{(-R\mathbf{e}_1, R+r)} = \mathbb{H}_{r\downarrow}.$$

To see why, we only have to show that $\mathbb{H}_{r\downarrow} \subseteq \lim_{R \rightarrow \infty} \mathbb{B}_{(-R\mathbf{e}_1, R+r)}$. Indeed, take an arbitrary point in $\mathbb{H}_{r\downarrow}$ written $y = (r - r_0, y^{(2:d)})$. An elementary calculation shows that if $R > (2r_0)^{-1}(r_0^2 + |y^{(2:d)}|^2) - r$, then $y \in \mathbb{B}_{(-R\mathbf{e}_1, R+r)}$.

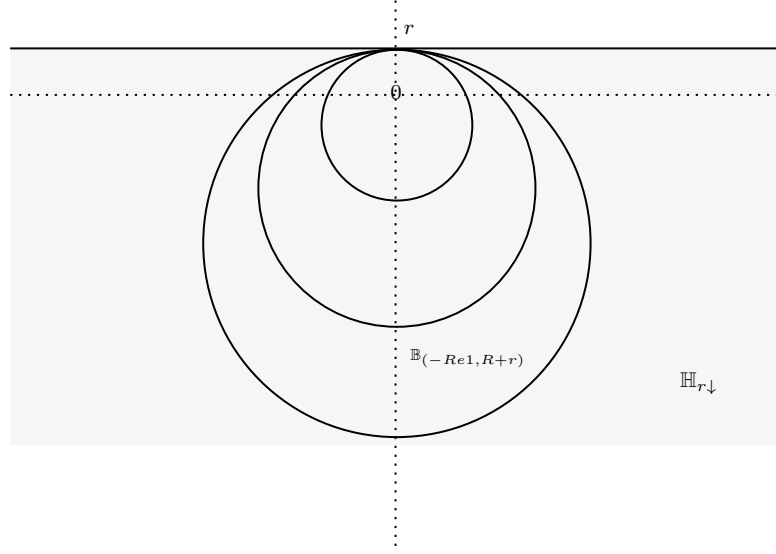


Figure 2-1: Approximation of $\mathbb{H}_{r\downarrow}$ by $\mathbb{B}_{(-R\mathbf{e}_1, R+r)}$.

The next assertion is that we can define a decreasing sequence of stopping times which will limit to $\tau_r^\vee$. If $\tau_{\mathbb{B}_{R+r}} = \inf\{t > 0 : X_t \in \mathbb{B}_{(-R\mathbf{e}_1, R+r)}\}$, we claim that, for all $x \in \mathbb{R}^d$ with $x^{(1)} > r$, $\mathbb{P}_x$ -almost surely it holds that

$$\tau_{\mathbb{B}_{R+r}} \downarrow \tau_r^\vee,$$

as $R \rightarrow \infty$.

The first thing we need to point out is that $(\tau_{\mathbb{B}_{R+r}}, R > 0)$ is a decreasing sequence of stopping times, hence its limit is a stopping time. For $\mathbb{P}_x$ -almost every ω let $y(\omega) = X_{\tau_r^\vee(\omega)}(\omega)$. Then there exists a $R_0(\omega) > 0$ such that $y(\omega) \in \mathbb{B}_{(-R_0(\omega)\mathbf{e}_1, R_0(\omega)+r)}$, implying by convexity that $\tau_{\mathbb{B}_{R+r}}(\omega) = \tau_r^\vee(\omega)$ for all $R > R_0$. In which case $X_{\tau_r^\vee(\omega)}(\omega) = X_{\tau_{\mathbb{B}_{R+r}}(\omega)}(\omega)$ and $X_{\tau_r^\vee(\omega)-}(\omega) = X_{\tau_{\mathbb{B}_{R+r}}(\omega)-}(\omega)$.

Hence, for any $x \in \mathbb{H}_{r\uparrow}$ appealing to dominated convergence (as f is bounded),

$$\begin{aligned}\mathbb{E}_x[f(X_{\tau_r^\vee}, X_{\tau_r^\vee-})] &= \mathbb{E}_x \left[\lim_{R \rightarrow \infty} f(X_{\tau_{\mathbb{B}_{R+r}}}, X_{\tau_{\mathbb{B}_{R+r}}-}) \right] \\ &= \lim_{R \rightarrow \infty} \mathbb{E}_x \left[f(X_{\tau_{\mathbb{B}_{R+r}}}, X_{\tau_{\mathbb{B}_{R+r}}-}), \tau_{\mathbb{B}_{R+r}} < \infty \right] \\ &= \lim_{R \rightarrow \infty} \mathbb{E}_{x+R\mathbf{e}_1} \left[f(X_{\tau_{R+r}^\oplus} - R\mathbf{e}_1, X_{\tau_{R+r}^\oplus-} - R\mathbf{e}_1), \tau_{R+r}^\oplus < \infty \right],\end{aligned}$$

where the last equality is given by stationary and independent increments. $\square$

Next we give the proof of Lemma 2.5 which is almost trivial in light of Lemma 2.10. We leave the proof of Lemma 2.4 to the reader, pointing to Corollary 1.4 of [37] as the starting expression from which limits can be taken.

Proof. Applying Lemma 2.10 using the explicit formula for the distribution of the first hit inside a ball found in [6], we have that

$$\begin{aligned}\mathbb{E}_x[f(X_{\tau_r^\vee})] &= \lim_{R \rightarrow \infty} \mathbb{E}_{x+R\mathbf{e}_1} \left[f(X_{\tau_{R+r}^\oplus} - R\mathbf{e}_1); \tau_{R+r}^\oplus < \infty \right] \\ &= \lim_{R \rightarrow \infty} C_{\alpha,d} \int_{|y| < R+r} f(y - R\mathbf{e}_1) \frac{|(R+r)^2 - |x + R\mathbf{e}_1|^2|^{\alpha/2}}{|(R+r)^2 - |y|^2|^{\alpha/2} |(x + R\mathbf{e}_1) - y|^d} dy \\ &= \lim_{R \rightarrow \infty} C_{\alpha,d} \int_{|z+R\mathbf{e}_1| < R+r} f(z) \frac{|(R+r)^2 - |x + R\mathbf{e}_1|^2|^{\alpha/2}}{|(R+r)^2 - |y + R\mathbf{e}_1|^2|^{\alpha/2}} |x - z|^{-d} dy \\ &= C_{\alpha,d} \int_{\mathbb{H}_{r\downarrow}} f(z) |x - y|^{-d} \lim_{R \rightarrow \infty} \frac{|2R(r - x^{(1)}) + r^2 - |x|^2|^{\alpha/2}}{|2R(r - y^{(1)}) + r^2 - |y|^2|^{\alpha/2}} dz \\ &= C_{\alpha,d} \int_{\mathbb{H}_{r\downarrow}} f(z) |x - y|^{-d} \frac{|r - x^{(1)}|^{\alpha/2}}{|r - y^{(1)}|^{\alpha/2}} dy,\end{aligned}$$

which completes the proof. $\square$

Remark 2.11. Before starting the proof of the fundamental results in this chapter, we address the rather obvious question as to why flat Earth theory cannot be used to address Theorem 2.1. In theory this may well be possible but we could not see a way to ensure that the point of closest reach to the origin preceding first entry to the ball $\mathbb{B}_{(-R\mathbf{e}_1, R+r)}$ asymptotically converges to $X_{G(\tau_r^\vee)}$. For this reason, we take a different approach to proving Theorem 2.1, which uses the so-called directional excursion as defined in Remark 0.7.

In order to calculate the distribution of the one-sided closest reach to $\mathbb{H}_0$ let us begin by defining the remaining elements from the theory of excursions from the min-

imum/maximum of $X^{(1)}$. To define the excursion process we start by expressing the endpoints of the intervals of excursions. Let $\mathbf{G} = \cup_{t>0} \mathbf{g}_t$ and $\mathbf{D} = \cup_{t>0} \mathbf{d}_t$ where $\mathbf{g}_t := \sup\{s < t : X_s^{(1)} = \underline{X}_s^{(1)}\}$ and $\mathbf{d}_t := \inf\{s > t : X_s^{(1)} = \underline{X}_s^{(1)}\}$. Hence, for all $t > 0$ such that $\mathbf{d}_t > \mathbf{g}_t$, the excursions of X from its directional minimum in the first axis, or equivalently the excursions of $(X^{(1)} - \underline{X}^{(1)}, X^{(2:d)})$ from the set $\{0\} \times \mathbb{R}^{d-1}$ are encoded in the following process

$$(\epsilon_{\mathbf{g}_t}^{(1)}(s), \epsilon_{\mathbf{g}_t}^{(2:d)}(s)) := (X_{\mathbf{g}_t+s}^{(1)} - X_{\mathbf{g}_t}^{(1)}, X_{\mathbf{g}_t+s}^{(2:d)}) \quad 0 \leq s \leq \mathbf{d}_t - \mathbf{g}_t =: \zeta_{\mathbf{g}_t}. \quad (2.11)$$

Such excursions live in the space $\mathbb{D}_{\mathbb{R}^+ \times \mathbb{R}^{d-1}}$ of càdlàg paths in $\mathbb{R}^+ \times \mathbb{R}^{d-1}$, such that

$$(\epsilon^{(1)}, \epsilon^{(2:d)}) = \left((\epsilon^{(1)}(t), \epsilon^{(2:d)}(t)) : t \leq \zeta \right) \text{ with lifetime } \zeta = \inf\{s > 0 : \epsilon^{(1)}(s) < 0\}$$

such that $(\epsilon^{(1)}(0), \epsilon^{(2:d)}(0)) \in \{0\} \times \mathbb{R}^{d-1}$, $(\epsilon^{(1)}(s), \epsilon^{(2:d)}(s)) \in (0, \infty) \times \mathbb{R}^{d-1}$ and $(\epsilon^{(1)}(\zeta), \epsilon^{(2:d)}(\zeta)) \in (-\infty, 0) \times \mathbb{R}^{d-1}$.

The exit formula (0.15) can be reinterpreted for the previous process as

$$\begin{aligned} \mathbb{E}_{(x^{(1)}, x^{(2:d)})} \left[\sum_{g \in \mathbf{G}} F(X_s, s < g) H\left((\epsilon_g^{(1)}, \epsilon_g^{(2:d)})\right) \right] \\ = \mathbb{E}_{(x^{(1)}, x^{(2:d)})} \left[\int_0^\infty F(X_s, s < t) \mathbb{N}_{X_t^{(2:d)}}^- \left(H(\epsilon^{(1)}, \epsilon^{(2:d)}) \right) dL_t \right], \end{aligned} \quad (2.12)$$

where F is non-negative and bounded on the space of càdlàg paths on $\mathbb{R}^d$ and H is measurable on $\mathbb{D}(\mathbb{R}^+ \times \mathbb{R}^{d-1})$. Under the measures $(\mathbb{N}_{x^{(2:d)}}^-, x^{(2:d)} \in \mathbb{R}^{d-1})$, the process $(\epsilon^{(1)}(s), \epsilon^{(2:d)}(s); s < \zeta)$ is Markovian with the same semigroup as $X = (X^{(1)}, X^{(2:d)})$ killed at its first hitting time of $(-\infty, 0] \times \mathbb{R}^{d-1}$.

The exit system $(L, \mathbb{N}^-)$ is not unique, but once L is chosen, the measures, $(\mathbb{N}_{x^{(2:d)}}^-, x^{(2:d)} \in \mathbb{R}^{d-1})$ are determined but for a L -neglectable set, i.e. a set A such that

$$\mathbb{E}_{x, \theta} \left[\int_0^\infty \mathbf{1} \left((X_s^{(1)}, X_s^{(2:d)}) \in A \right) dL_s \right] = 0.$$

Analogously to the definition of the elements of an excursion theory from the directional minima of X , we can similarly define excursions based on the directional maxima. These are described as excursions of the process $(\bar{X}^{(1)} - X^{(1)}, X^{(2:d)})$ from the set $\{0\} \times \mathbb{R}^{d-1}$. In the isotropic case, the primary difference from the earlier construction lies in the convention for defining the excursion process: here, its first coordinate is

nonnegative. Specifically, we define:

$$\left(\bar{\epsilon}_g^{(1)}(s), \bar{\epsilon}_g^{(2:d)}(s)\right) := (X_g^{(1)} - X_{g+s}^{(1)}, X_{g+s}^{(2:d)}), \quad 0 \leq s \leq \bar{\zeta}, \quad (2.13)$$

where g represents the canonical left endpoint of an excursion from the maximum, and $\bar{\zeta}$ denotes the lifetime of the excursion.

This allows us to define $(\bar{L}, \mathbb{N}^+)$, an appropriate exit system for these excursions. By isotropy, for any $z \in \mathbb{R}^{d-1}$ and any nonnegative measurable function f on $\mathbb{R}^{d-1}$, the following equality holds:

$$\mathbb{N}_z^+ \left(f \left(\bar{\epsilon}^{(1)}(t), \bar{\epsilon}^{(2:d)}(t) \right) : t \leq \bar{\zeta} \right) = \mathbb{N}_z^- \left(f \left(\epsilon^{(1)}(t), \epsilon^{(2:d)}(t) \right) : t \leq \zeta \right).$$

Proof of Theorem 2.1

Proof of Theorem 2.1. Let $x \in \mathbb{H}_{0\uparrow}$ and $G(\tau_0^\vee)$ as defined in Theorem 2.1, by the exit formula (2.12)

$$\begin{aligned} \mathbb{E}_x \left[f(X_{G(\tau_0^\vee)}) \right] &= \mathbb{E}_x \left[\sum_{t \in \mathbb{G}} f(X_t^{(1)}, X_t^{(2:d)}) \mathbf{1}(X_t^{(1)} + \epsilon_t(\zeta) < 0) \right] \\ &= \mathbb{E}_x \left[\int_0^\infty f(X_t^{(1)}, X_t^{(2:d)}) \mathbb{N}_{X_t^{(2:d)}}^-(X_t^{(1)} + \epsilon_t(\zeta) < 0) dL_t \right]. \end{aligned}$$

In the above expression, the event in the excursion measure depends only on the first coordinate of the excursion process. Hence it is independent of the excursion measure index $X_t^{(2:d)}$ and can be written in terms of the jump measure, say $\Pi_-^{(1)}$, of the descending ladder process associated to $X^{(1)}$, which is known to be an $\alpha/2$ -stable subordinator. More precisely,

$$\mathbb{N}_{x^{(2:d)}}^-(\epsilon_t^{(1)}(\zeta) < -y) = \Pi_-^{(1)}(y, \infty) = \frac{1}{\Gamma(1 - (\alpha/2))} y^{-\alpha/2}, \quad y > 0.$$

As such, we have

$$\begin{aligned}
\mathbb{E}_x \left[g(X_{G(\tau_0^\vee)}) \right] &= \mathbb{E}_x \left[\int_0^\infty g(X_t) \frac{1}{\Gamma(1 - (\alpha/2))} (X_t^{(1)})^{-\alpha/2} dL_t \right] \\
&= \frac{1}{\Gamma(1 - (\alpha/2))} \mathbb{E}_x \left[\int_0^\infty g(H_t^{(1)-}, X_t^{(2:d)-}) (H_t^{(1)-})^{-\alpha/2} dt \right] \\
&= \frac{1}{\Gamma(1 - (\alpha/2))} \int_{\{y^{(1)} < x^{(1)}\}} g(y^{(1)}, y^{(2:d)}) (y^{(1)})^{-\alpha/2} U_x^-(dy), \quad (2.14)
\end{aligned}$$

where $H_t^{(1)-} := X_{L_t^{-1}}^{(1)}$ and $X_t^{(2:d)-} := X_{L_t^{-1}}^{(2:d)}$, $t \geq 0$, and we define the descending ladder potential measure

$$U_x^-(dy) := \int_0^\infty \mathbb{P}_x \left((H_t^{(1)-}, X_t^{(2:d)-}) \in dy \right) dt \quad y^{(1)} < x^{(1)}, y^{(2:d)} \in \mathbb{R}^{d-1}. \quad (2.15)$$

We note that the integral on the right-hand side of (2.14) is over the whole of $\mathbb{R}^d$ with the restriction that $\{y^{(1)} < x^{(1)}\}$. In the sequel, we will see many such integrals where it is implicit that we are integrating over $\mathbb{R}^d$ and only the restriction over this space is indicated in the delimiter.

In conclusion, we showed that the law of $X_{G(\tau_0^\vee)}$ under $\mathbb{P}_x$ (up to a constant) is nothing but $U_x^-(dy)(y^{(1)})^{-\alpha/2}$. We therefore focus on identifying $U_x^-(dy)$. The proof we follow is guided by the calculations leading to Theorem 1.1 in [37].

Let us define, for $x \in \mathbb{R}^d$ such that $x^{(1)} > r > 0$, $\delta > 0$, and continuous, positive and bounded f on $\mathbb{R}^+ \times \mathbb{R}^{d-1}$,

$$\Delta_r^\delta f(x) = \frac{1}{\delta} \mathbb{E}_x [f(r, X_{G(\tau_0^\vee)}^{(2:d)}), X_{G(\tau_0^\vee)}^{(1)} \in [r - \delta, r]]. \quad (2.16)$$

We want to establish the limit as $\delta \rightarrow 0$. In order to do so, we start by conditioning on the first entry into the half-space $\mathbb{H}_{r\downarrow}$, this can be done by appealing to Lemma 2.5,

which we have already proved. We have

$$\begin{aligned}
\Delta_r^\delta f(x) &= \frac{1}{\delta} \int_{\{r-\delta \leq y^{(1)} \leq r\}} \mathbb{P}_x(X_{\tau_r^\vee} \in dy) \mathbb{E}_y[f(r, X_{G(\tau_0^\vee)}^{(2:d)}), X_{G(\tau_0^\vee)}^{(1)} \in [r-\delta, y^{(1)}]] \\
&= \frac{1}{\delta} C_{\alpha,d} \int_{\{r-\delta \leq y^{(1)} \leq r\}} dy |x-y|^{-d} \frac{|r-x^{(1)}|^{\alpha/2}}{|r-y^{(1)}|^{\alpha/2}} \mathbb{E}_y[f(r, X_{G(\tau_0^\vee)}^{(2:d)}), X_{G(\tau_0^\vee)}^{(1)} \in [r-\delta, y^{(1)}]] \\
&= \frac{1}{\delta} C_{\alpha,d} \int_{\{r-\delta \leq y^{(1)} \leq r\}} dy |x-y|^{-d} \frac{|r-x^{(1)}|^{\alpha/2}}{|r-y^{(1)}|^{\alpha/2}} \\
&\quad \times \frac{1}{\Gamma(1-(\alpha/2))} \int_{\{r-\delta < z^{(1)} \leq y^{(1)}\}} f(r, z^{(2:d)}) (z^{(1)})^{-\alpha/2} U_y^-(dz), \quad (2.17)
\end{aligned}$$

where the third equality is given by equation (2.14).

Next we provide a limiting approximation for the inner integral of (2.17) as $\delta \rightarrow 0$. From now on we will write $\tilde{U}_x^-(dz) = (z^{(1)})^{-\alpha/2} U_x^-(dz)$, $z \in \mathbb{R}^d$ and recall that this is proportional to the law of $X_{G(\tau_0^\vee)}$ under $\mathbb{P}_x$. We claim that

$$\lim_{\delta \rightarrow 0} \sup_{\{y \in \mathbb{R}^d : y^{(1)} \in (r-\delta, r]\}} \left| \frac{\int_{\{r-\delta < z^{(1)} \leq y^{(1)}\}} f(r, z^{(2:d)}) \tilde{U}_y^-(dz)}{\int_{\{r-\delta < z^{(1)} \leq y^{(1)}\}} \tilde{U}_y^-(dz)} - f(r, y^{(2:d)}) \right| = 0. \quad (2.18)$$

Suppose that $H_{r,\delta,\varepsilon}(y) \subset \mathbb{R}^d$ is the solid cylinder defined by $\{x \in \mathbb{R}^d : |x^{(2:d)} - y^{(2:d)}| \leq \varepsilon \text{ and } r-\delta \leq x^{(1)} \leq r\}$, denote by $H_\varepsilon(y) \subseteq \mathbb{R}^d$ the section of the cylinder given by $\{x^{(2:d)} \in \mathbb{R}^{d-1} : |x^{(2:d)} - y^{(2:d)}| \leq \varepsilon\}$.

Thanks to the continuity of f , we can choose ε such that

$$\sup_{z \in H_\varepsilon(y)} |f(r, y^{(2:d)}) - f(r, z^{(2:d)})| < \varepsilon',$$

for some $\varepsilon' \ll 1$. Then we have

$$\begin{aligned}
&\sup_{\{y \in \mathbb{R}^d : y^{(1)} \in (r-\delta, r]\}} \left| \frac{\int_{\{r-\delta < z^{(1)} \leq y^{(1)}\}} f(r, z^{(2:d)}) \tilde{U}_y^-(dz)}{\int_{\{r-\delta < z^{(1)} \leq y^{(1)}\}} \tilde{U}_y^-(dz)} - f(r, y^{(2:d)}) \right| \\
&\leq \varepsilon' + 2\|f\|_\infty \sup_{\{y \in \mathbb{R}^d : y^{(1)} \in (r-\delta, r]\}} \frac{\int_{\{r-\delta < z^{(1)} \leq y^{(1)}\}} \tilde{U}_y^-(dz) \mathbf{1}_{\{z^{(2:d)} \notin H_\varepsilon(y)\}}}{\int_{\{r-\delta < z^{(1)} \leq y^{(1)}\}} \tilde{U}_y^-(dz)}. \quad (2.19)
\end{aligned}$$

Analysing the fraction on the right-hand side of (2.19), we notice that

$$\begin{aligned}
& \frac{\int_{\{r-\delta < z^{(1)} \leq y^{(1)}\}} \tilde{U}_y^-(dz) \mathbf{1}_{\{z^{(2:d)} \notin H_\varepsilon(y)\}}}{\int_{\{r-\delta < z^{(1)} \leq y^{(1)}\}} \tilde{U}_y^-(dz)} \\
&= \frac{\int_{\{r-\delta < z^{(1)} \leq y^{(1)}\}} U_y^-(dz) (z^{(1)})^{-\alpha/2} \mathbf{1}_{\{z^{(2:d)} \notin H_\varepsilon(y)\}}}{\int_{\{r-\delta < z^{(1)} \leq y^{(1)}\}} U_y^-(dz) (z^{(1)})^{-\alpha/2}} \\
&= c_\alpha \frac{\int_{\{r-\delta < z^{(1)} \leq y^{(1)}\}} U_y^-(dz) \Pi_-^{(1)}(z^{(1)} - (r - \delta), \infty) \left(1 - \frac{(r-\delta)}{z^{(1)}}\right)^{\alpha/2} \mathbf{1}_{\{z^{(2:d)} \notin H_\varepsilon(y)\}}}{\int_{\{r-\delta < z^{(1)} \leq y^{(1)}\}} U_y^-(dz) (z^{(1)})^{-\alpha/2}} \\
&\leq c_\alpha \frac{\int_{\{r-\delta < z^{(1)} \leq y^{(1)}\}} U_y^-(dz) \Pi_-^{(1)}(z^{(1)} - (r - \delta), \infty) \left(1 - \frac{(r-\delta)}{y^{(1)}}\right)^{\alpha/2} \mathbf{1}_{\{z^{(2:d)} \notin H_\varepsilon(y)\}}}{\int_{\{r-\delta < z^{(1)} \leq y^{(1)}\}} U_y^-(dz) (y^{(1)})^{-\alpha/2}} \\
&= c_\alpha \frac{\mathbb{P}_y(H_{\sigma_{r-\delta}^\vee}^{(2:d)-} \notin H_\varepsilon(y))}{\left(1 - \frac{(r-\delta)}{y^{(1)}}\right)^{-\alpha/2} (y^{(1)})^{-\alpha/2} U_{y^{(1)}}^{(1)-}(r - \delta, y^{(1)})}, \tag{2.20}
\end{aligned}$$

where $c_\alpha = \Gamma(1 - (\alpha/2))$, $\sigma_{r-\delta}^\vee = \inf\{t > 0 : H_t^{(1)-} < r - \delta\}$ and, recalling $y = (y^{(1)}, y^{(2:d)})$,

$$U_{y^{(1)}}^{(1)-}(dz^{(1)}) := \int_0^\infty \mathbb{P}_y(H_t^{(1)-} \in dz^{(1)}) dt \quad z^{(1)} < y^{(1)}.$$

As $y \in [r - \delta, r] \times H_\varepsilon(y)$, if we denote by $\nu_\varepsilon = \inf\{t > 0 : X_t^{(2:d)-} \notin H_\varepsilon(y)\}$, which only depends on y through $y^{(2:d)}$, by the right-continuity of the paths of X we conclude that, for the numerator in (2.20),

$$\lim_{\delta \rightarrow 0} \sup_{\{y \in \mathbb{R}^d : y^{(1)} \in (r-\delta, r]\}} \mathbb{P}_y(X_{\sigma_{r-\delta}^\vee}^{(2:d)-} \notin H_\varepsilon(y)) \leq \lim_{\delta \rightarrow 0} \sup_{\{y \in \mathbb{R}^d : y^{(1)} \in (r-\delta, r]\}} \mathbb{P}_y(\nu_\varepsilon < \sigma_{r-\delta}^\vee) = 0.$$

On the other hand, recalling (cf. Chapter 3 of [35]) that the known descending ladder height occupation measure of a one-dimensional symmetric stable process (cf. Chapter 3 of [35]) gives us

$$U_{y^{(1)}}^{(1)-}(dv) = \frac{1}{\Gamma(\alpha/2)} (y^{(1)} - v)^{(\alpha/2)-1}, \quad v \leq y^{(1)}, \tag{2.21}$$

for the denominator of (2.20) we have

$$\begin{aligned}
& \left(1 - \frac{(r - \delta)}{y^{(1)}}\right)^{-\alpha/2} (y^{(1)})^{-\alpha/2} U_{y^{(1)}}^{(1)-}(r - \delta, y^{(1)}) \\
&= \left(1 - \frac{(r - \delta)}{y^{(1)}}\right)^{-\alpha/2} (y^{(1)})^{-\alpha/2} \int_0^{y^{(1)} - (r - \delta)} \frac{u^{(\alpha/2)-1}}{\Gamma(\alpha/2)} du \\
&= \frac{1}{\Gamma(1 + (\alpha/2))}.
\end{aligned}$$

Hence the denominator of (2.20) is nothing but a constant. The right-hand side of (2.20) therefore goes to zero uniformly for $y^{(1)} \in (r - \delta, r]$ as $\delta \rightarrow 0$, which, from (2.19), means that we have proved the claim (2.18).

Now returning to (2.17), using (2.18), for small δ we have

$$\begin{aligned}
\Delta_r^\delta f(x) &= \frac{1}{\delta} C_{\alpha,d} \int_{\{r - \delta \leq y^{(1)} \leq r\}} dy |x - y|^{-d} \frac{|r - x^{(1)}|^{\alpha/2}}{|r - y^{(1)}|^{\alpha/2}} \\
&\quad \frac{1}{\Gamma(1 - (\alpha/2))} \int_{\{r - \delta < z^{(1)} \leq y^{(1)}\}} \left(f(r, z^{(2:d)}) - f(r, y^{(2:d)}) + f(r, y^{(2:d)}) \right) \tilde{U}_y^-(dz) \\
&= D(\varepsilon) \Delta_r^\delta 1(x) \\
&\quad + \frac{1}{\delta} C_{\alpha,d} \int_{\{r - \delta \leq y^{(1)} \leq r\}} dy |x - y|^{-d} \frac{|r - x^{(1)}|^{\alpha/2}}{|r - y^{(1)}|^{\alpha/2}} f(r, y^{(2:d)}) \mathbb{P}_y(X_{G(\tau_0^\vee)}^{(1)} > r - \delta),
\end{aligned} \tag{2.22}$$

where $|D(\varepsilon)| < \varepsilon$.

We deal with the two terms on the right-hand side of (2.22) sequentially. First, we note from the original definition in (2.16) that

$$\begin{aligned}
\lim_{\delta \rightarrow 0} \Delta_r^\delta 1(x) &= \frac{1}{\delta} \mathbb{P}_x(X_{G(\tau_0^\vee)}^{(1)} \in [r - \delta, r]) \\
&= \lim_{\delta \rightarrow 0} \frac{1}{\delta} \int_{r - \delta}^r \frac{(x^{(1)} - u)^{(\alpha/2)-1} u^{-\alpha/2}}{\Gamma(\alpha/2) \Gamma(1 - (\alpha/2))} du \\
&= \frac{(x^{(1)} - r)^{(\alpha/2)-1} r^{-\alpha/2}}{\Gamma(\alpha/2) \Gamma(1 - (\alpha/2))}.
\end{aligned} \tag{2.23}$$

where, we have appealed to the one-dimensional analogue of (2.14); using $U_{y_0}^{(1)-}$ in place of U_y^- . The limit in (2.23) tells us that the first term on the right-hand side of (2.22) can be ignored as $\delta \rightarrow 0$.

For the second term on the right-hand side of (2.22), we note that its limit as $\delta \rightarrow 0$ is

equal to

$$\begin{aligned} \lim_{\delta \rightarrow 0} \Delta_r^\delta f(x) &= C_{\alpha,d} \int_{\mathbb{R}^{d-1}} dy^{(2:d)} \frac{|r - x^{(1)}|^{\alpha/2} f(r, y^{(2:d)})}{|(x^{(1)} - r)^2 + (x^{(2:d)} - y^{(2:d)})|^{d/2}} \\ &\quad \times \lim_{\delta \rightarrow 0} \frac{1}{\delta} \int_{r-\delta}^r dy^{(1)} \frac{\mathbb{P}_y(X_{G(\tau_0^\vee)}^{(1)} > r - \delta)}{(r - y^{(1)})^{\alpha/2}}. \end{aligned} \quad (2.24)$$

For the limit on the right-hand of (2.24), we can compute

$$\begin{aligned} &\lim_{\delta \rightarrow 0} \frac{1}{\delta} \int_{r-\delta}^r dy^{(1)} \frac{\mathbb{P}_y(X_{G(\tau_0^\vee)}^{(1)} > r - \delta)}{(r - y^{(1)})^{\alpha/2}} \\ &= \lim_{\delta \rightarrow 0} \frac{1}{\delta} \int_{r-\delta}^r dy^{(1)} (r - y^{(1)})^{-\alpha/2} \int_{r-\delta}^{y^{(1)}} \frac{(y^{(1)} - u)^{(\alpha/2)-1} u^{-\alpha/2}}{\Gamma(\alpha/2)\Gamma(1 - (\alpha/2))} du \\ &= \frac{r^{-\alpha/2}}{(\alpha/2)\Gamma(\alpha/2)\Gamma(1 - (\alpha/2))} \lim_{\delta \rightarrow 0} \frac{1}{\delta} \int_{r-\delta}^r dy^{(1)} (r - y^{(1)})^{-\alpha/2} (y^{(1)} - r + \delta)^{\alpha/2} \\ &= \frac{r^{-\alpha/2}}{(\alpha/2)\Gamma(\alpha/2)\Gamma(1 - (\alpha/2))} \lim_{\delta \rightarrow 0} \frac{1}{\delta} \int_0^\delta v^{-\alpha/2} (\delta - v)^{\alpha/2} dv \\ &= \frac{r^{-\alpha/2}}{\Gamma(1 + (\alpha/2))\Gamma(1 - (\alpha/2))} \int_0^1 w^{-\alpha/2} (1 - w)^{\alpha/2} dw \\ &= r^{-\alpha/2}. \end{aligned} \quad (2.25)$$

where the first equality uses a calculation similar to (2.23).

Putting (2.25) back into (2.24), we have from (2.16) that

$$\begin{aligned} &\mathbb{E}_x \left[f(X_{G(\tau_0^\vee)}^{(1)}, X_{G(\tau_0^\vee)}^{(2:d)}) \right] \\ &= \int_0^{x^{(1)}} \lim_{\delta \rightarrow 0} \Delta_r^\delta f(x) dr \\ &= C_{\alpha,d} \int_0^{x^{(1)}} dr \int_{\mathbb{R}^{d-1}} dy^{(2:d)} \frac{|r - x^{(1)}|^{\alpha/2} f(r, y^{(2:d)})}{r^{\alpha/2} |(x^{(1)} - r)^2 + (x^{(2:d)} - y^{(2:d)})|^{d/2}} \end{aligned} \quad (2.26)$$

which completes the proof of Theorem 2.1. $\square$

Proof of Theorem 2.3

There are two formulae we will need to first derive in order to prove Theorem 2.3. For the first, just as we have constructed the directional excursions from the minimum in the first axis, we can construct directional excursions from the maximum in the first axis resulting in an ascending ladder process, say $(H^{(1)+}, X^{(2:d)+})$. If we define the

renewal function of the ascending ladder process of the latter, say $U_x^+(\mathrm{d}y)$, analogously to equation (2.15), then it is immediate from Theorem 2.1 and symmetry that

$$U_x^-(\mathrm{d}z) = \frac{\Gamma(d/2)}{\pi^{d/2}\Gamma(\alpha/2)} \frac{(x^{(1)} - z^{(1)})^{\alpha/2}}{|x - z|^d} \mathrm{d}z \quad \text{for } x^{(1)} > z^{(1)}, z^{(2:d)} \in \mathbb{R}^{d-1}. \quad (2.27)$$

and

$$U_x^+(\mathrm{d}z) = \frac{\Gamma(d/2)}{\pi^{d/2}\Gamma(\alpha/2)} \frac{(z^{(1)} - x^{(1)})^{\alpha/2}}{|x - z|^d} \mathrm{d}z \quad \text{for } x^{(1)} < z^{(1)}, z^{(2:d)} \in \mathbb{R}^{d-1}. \quad (2.28)$$

These two expressions prove Proposition 2.8 in the case $\alpha \in (0, 2)$.

The second formula we need to prove Theorem 2.3 is provided by the following lemma, which gives us an occupation functional under the excursion measure.

Lemma 2.12. *For $x \in \mathbb{R}^d$ and continuous $g : \mathbb{R}^d \rightarrow [0, \infty)$ whose support is compactly embedded in $\mathbb{H}_{r\uparrow}$*

$$\begin{aligned} \mathbb{N}_{x^{(2:d)}}^- \left(\int_0^\zeta g(x^{(1)} + \epsilon^{(1)}(u), \epsilon^{(2:d)}(u)) \mathrm{d}u \right) &= \int_{\{x^{(1)} < z^{(1)}\}} g(z) \frac{\Gamma(d/2)}{\pi^{d/2}\Gamma(\alpha/2)} \frac{(z^{(1)} - x^{(1)})^{\alpha/2}}{|x - z|^d} \mathrm{d}z, \\ &= \int_{\{x^{(1)} < z^{(1)}\}} g(z) U_x^+(\mathrm{d}z). \end{aligned} \quad (2.29)$$

Proof. By the last exit formula (2.12) we note that

$$\begin{aligned} &\mathbb{E}_x \left[\int_0^{\tau_r^\vee} g(X_s) \mathrm{d}s \right] \\ &= \mathbb{E}_x \left[\sum_{g \in \mathbb{G}} \mathbf{1}_{\{\underline{X}_g^{(1)} > r\}} \int_g^d g(\underline{X}_g^{(1)} + \epsilon_g^{(1)}(s), \epsilon_g^{(2:d)}(s)) \mathrm{d}s \right] \\ &= \mathbb{E}_x \left[\int_0^\infty \mathbf{1}_{\{\underline{X}_s^{(1)} > r\}} \mathbb{N}_{X_s^{(2:d)}}^- \left(\int_0^\zeta g(y + \epsilon^{(1)}(u), \epsilon^{(2:d)}(u)) \mathrm{d}u \right) \Big|_{y=X_s^{(1)}} \mathrm{d}L_s \right] \\ &= \mathbb{E}_x \left[\int_0^\infty \mathbf{1}_{\{H_s^{(1)-} > r\}} \mathbb{N}_{H_s^{(2:d)-}}^- \left(\int_0^\zeta g(y + \epsilon^{(1)}(u), \epsilon^{(2:d)}(u)) \mathrm{d}u \right) \Big|_{y=H_s^{(1)-}} \mathrm{d}s \right] \\ &= \int_{\{r < z^{(1)} < x^{(1)}\}} U_x^-(\mathrm{d}z) \mathbb{N}_{z^{(2:d)}}^- \left(\int_0^\zeta g(z^{(1)} + \epsilon^{(1)}(u), \epsilon^{(2:d)}(u)) \mathrm{d}u \right) \\ &= \int_{\{r < z^{(1)} < x^{(1)}\}} \tilde{U}_x^-(\mathrm{d}z) (z^{(1)})^{\alpha/2} \mathbb{N}_{z^{(2:d)}}^- \left(\int_0^\zeta g(z^{(1)} + \epsilon^{(1)}(u), \epsilon^{(2:d)}(u)) \mathrm{d}u \right). \end{aligned} \quad (2.30)$$

Next, we want to make use of an argument very similar to (2.18) replacing the role of

f there by the function

$$\kappa(z) := (z^{(1)})^{\alpha/2} \mathbb{N}_{z^{(2:d)}}^- \left(\int_0^\zeta g(z^{(1)} + \epsilon(u), \epsilon^{(2:d)}(u)) \right).$$

Note that we cannot use (2.18) directly as the function f there cannot take the form of κ . Nonetheless, the result in (2.18) still holds when we make the aforesaid exchange of functions, providing we can demonstrate that $\kappa(z)$ is bounded and continuous in z .

Boundedness of κ is ensured by the compact support property of g . Continuity of κ is ensured by dominated convergence, using the compact embedding of the support of g , the boundedness and continuity of g and the fact that

$$\mathbb{N}_{z^{(2:d)}}^- \left(\int_0^\zeta g(z^{(1)} + \epsilon(u), \epsilon^{(2:d)}(u)) \right) = \mathbb{N}_0^- \left(\int_0^\zeta g(z^{(1)} + \epsilon(u), z^{(2:d)} + \epsilon^{(2:d)}(u)) \right).$$

Hence using κ in place of f in (2.18) we have on the one hand that

$$(x^{(1)})^{\alpha/2} \mathbb{N}_{x^{(2:d)}}^- \left(\int_0^\zeta g(x^{(1)} + \epsilon(u), \epsilon^{(2:d)}(u)) du \right) = \lim_{r \uparrow x^{(1)}} \frac{\mathbb{E}_x \left[\int_0^{\tau_r^\vee} g(X_s) ds \right]}{\int_{\{r < z^{(1)} < x^{(1)}\}} \tilde{U}_x^-(dz)}. \quad (2.31)$$

Although we have not calculated an expression for the numerator on the right-hand side of (2.31), it is easily derived from Lemma 2.4. Indeed, it is easy to derive from the underlying Lévy system of X that, for $x, y \in \mathbb{H}_{r\uparrow}$ and $z \in \mathbb{H}_{r\downarrow}$,

$$\Xi_x(\bullet, y, z) dy dz = \mathbb{E}_x \left[\int_0^{\tau_r^\vee} \mathbf{1}_{\{X_s \in dy\}} ds \right] \Pi(dz - y).$$

As such, we have, for $x, y \in \mathbb{H}_{r\uparrow}$ and $z \in \mathbb{H}_{r\downarrow}$,

$$\begin{aligned} h_r^\vee(x, y) &:= \mathbb{E}_x \left[\int_0^{\tau_r^\vee} \mathbf{1}_{\{X_s \in dy\}} ds \right] \\ &= E_{\alpha, d} |x - y|^{\alpha-d} \int_0^{\zeta_r^\vee(x, y)} (u + 1)^{-d/2} u^{\alpha/2-1} du, \end{aligned} \quad (2.32)$$

where $E_{\alpha, d} = \Gamma(d/2)/2^\alpha \pi^{d/2} \Gamma(\alpha/2)^2$ and we recall

$$\zeta_r^\vee(x, y) := \frac{4(x^{(1)} - r)(y^{(1)} - r)}{|x - y|^2}.$$

Therefore, on the other hand we have that

$$\begin{aligned}
& \lim_{r \uparrow x^{(1)}} \frac{\mathbb{E}_x \left[\int_0^{\tau_r^\vee} g(X_s) ds \right]}{\int_{\{r < z^{(1)} < x^{(1)}\}} \tilde{U}_x^-(dz)} \\
&= \lim_{r \uparrow x^{(1)}} \frac{\int_{\{x^{(1)} < z^{(1)}\}} \mathbf{1}_{\{r < z^{(1)}\}} g(z) h_r^\vee(x, z) dz}{\int_{\{r < z^{(1)} < x^{(1)}\}} (z^{(1)})^{-\alpha/2} U_x^{(1)-}(dz^{(1)})} \\
&= \lim_{r \uparrow x^{(1)}} \frac{\int_{\{x^{(1)} < z^{(1)}\}} \mathbf{1}_{\{r < z^{(1)}\}} g(z) E_{\alpha,d} |x - z|^{\alpha-d} \int_0^{\zeta_r^\vee(x,z)} (u+1)^{-d/2} u^{\alpha/2-1} du dz}{\Gamma(\alpha/2)^{-1} \int_r^{x^{(1)}} u^{-\alpha/2} (x^{(1)} - u)^{\alpha/2-1} du} \\
&= E_{\alpha,d} \Gamma(\alpha/2) \int_{\{x^{(1)} < z^{(1)}\}} g(z) |x - z|^{\alpha-d} \lim_{r \uparrow x^{(1)}} \frac{\int_0^{\zeta_r^\vee(x,z)} (u+1)^{-d/2} u^{\alpha/2-1} du}{\int_r^{x^{(1)}} u^{-\alpha/2} (x^{(1)} - u)^{\alpha/2-1} du} dz \\
&= E_{\alpha,d} \Gamma(\alpha/2) (x^{(1)})^{\alpha/2} \int_{\{x^{(1)} < z^{(1)}\}} g(z) |x - z|^{\alpha-d} \left(\frac{4(z^{(1)} - x^{(1)})}{|x - z|^2} \right)^{\alpha/2} dz \\
&= 2^\alpha E_{\alpha,d} \Gamma(\alpha/2) (x^{(1)})^{\alpha/2} \int_{\{x^{(1)} < z^{(1)}\}} g(z) \frac{(z^{(1)} - x^{(1)})^{\alpha/2}}{|x - z|^d} dz. \tag{2.33}
\end{aligned}$$

where we have used (2.21) and, in the last equality, we used L'Hôpital rule. Equating (2.31) with (2.33) and noting that

$$2^\alpha E_{\alpha,d} \Gamma(\alpha/2) = \frac{\Gamma(d/2)}{\pi^{d/2} \Gamma(\alpha/2)}$$

completes the proof. $\square$

Remark 2.13. We can remove the assumption that g is compactly supported in the previous Lemma, leaving only the requirement that g is continuous and bounded by appealing to a sequence of increasing functions $g_n \uparrow g$ and monotone convergence across the equality (2.29) applied to the sequence g_n .

Proof of Theorem 2.3. Using the fact that $X^{(1)}$ does not creep downwards, from the underlying Lévy system we get that

$$\mathbb{E}_w \left[f(X_{G(\tau_r^\vee)}) g(X_{\tau_r^\vee-}) h(X_{\tau_r^\vee}) \right] = \mathbb{E}_w \left[\int_0^{\tau_r^\vee} f(X_{G(t)}) g(X_t) k(X_t) dt \right], \tag{2.34}$$

where $k(y) = \int_{\{y^{(1)} + u^{(1)} < r\}} \Pi(du) h(y + u)$.

Putting together identity (2.34) with Lemma 2.12, and rearranging the integrals we

obtain

$$\begin{aligned}
& \mathbb{E}_w \left[\int_0^{\tau_r^\vee} f(X_{G(t)}) g(X_t) k(X_t) dt \right] \\
&= \int_{\{r < x^{(1)} < w^{(1)}\}} f(x) U_w^-(dx) \mathbb{N}_{x^{(2:d)}}^- \left(\int_0^\infty (g \cdot k)(x^{(1)} + \epsilon^{(1)}(s), \epsilon^{(2:d)}(s)) ds \right) \\
&= \int_{\{r < x^{(1)} < w^{(1)}\}} f(x) U_w^-(dx) \int_{\{x^{(1)} < y^{(1)}\}} g(y) U_x^+(dy) \int_{\{y^{(1)} + u^{(1)} < r\}} \Pi(du) h(y + u) \\
&\hspace{25em} (2.35) \\
&= \int_{\{r < x^{(1)} < w^{(1)}\}} \int_{\{x^{(1)} < y^{(1)}\}} \int_{\{z^{(1)} < r\}} f(x) g(y) h(z) U_w^-(dx) U_x^+(dy) K_{\alpha,d} |z - y|^{-(\alpha+d)}.
\end{aligned}$$

where $K_{\alpha,d} = 2^\alpha \pi^{-d/2} \Gamma((d + \alpha)/2) / |\Gamma(-\alpha/2)|$ and we have used (0.9), from which the result follows. $\square$

Proof of the Deep Wiener-Hopf factorisation

By (2.14) it is possible to rewrite the descending and ascending ladder potentials ρ and $\hat{\rho}$ as follows, for $z = i\theta$, $\theta \in \mathbb{R}$

$$\hat{\rho}_z[f](x) = \int_{y^{(1)} < x^{(1)}} e^{zy^{(1)}} f(y^{(2:d)}) U_x^-(dy)$$

and

$$\rho_z[f](x) = \int_{y^{(1)} > x^{(1)}} e^{zy^{(1)}} f(y^{(2:d)}) U_x^+(dy),$$

where the explicit expressions for the potentials are given in equations (2.27) and (2.28), respectively.

Taking limits when $r \rightarrow -\infty$ in (2.30) and making $f^*(x) = e^{-zx^{(1)}} f(x^{(2:d)})$ we have that

$$\begin{aligned}
& \mathbf{R}_z[f](x) \\
&= \lim_{r \downarrow -\infty} \mathbb{E}_{0,x^{(2:d)}} \left[\int_0^{\tau_r^\vee} f^*(X_t) dt \right] \tag{2.36} \\
&= \lim_{r \downarrow -\infty} \int_{r < w^{(1)} < 0} U_{0,x^{(2:d)}}^-(dw) \mathbb{N}_{w^{(2:d)}}^- \left(\int_0^\zeta e^{z(w^{(1)} + \epsilon(s))} f(\epsilon^{(2:d)}(s)) ds \right) \\
&= \int_{w^{(1)} < 0} U_{0,x^{(2:d)}}^-(dw) \int_{w^{(1)} < y^{(1)}} e^{zy^{(1)}} f(y^{(2:d)}) U_{w^{(1)},w^{(2:d)}}^+(dy) \\
&= \int_{w^{(1)} < 0} U_{0,x^{(2:d)}}^-(dw) \int_{v^{(1)} > 0} e^{z(v^{(1)} + w^{(1)})} f(y^{(2:d)}) U_{w^{(1)},w^{(2:d)}}^+ \left((dv^{(1)} + w^{(1)}) \times dy^{(2:d)} \right) \\
&= \int_{w^{(1)} < 0} U_{0,x^{(2:d)}}^-(dw) e^{zw^{(1)}} \int_{v^{(1)} > 0} e^{zv^{(1)}} f(y^{(2:d)}) U_{0,w^{(2:d)}}^+ \left(dv^{(1)} \times dy^{(2:d)} \right) \\
&= \int_{w^{(1)} < 0} e^{zw^{(1)}} \left[\int_{v^{(1)} > 0} e^{zv^{(1)}} f(y^{(2:d)}) U_{0,w^{(2:d)}}^+ \left(dv^{(1)} \times dy^{(2:d)} \right) \right] U_{0,x^{(2:d)}}^-(dw) \\
&= \hat{\rho}_z [\rho_z[f]](x). \tag{2.37}
\end{aligned}$$

□

2.2.2 Proofs for Brownian motion

Consider a Brownian motion $Y = (Y_t : t \geq 0)$ in dimension $d \geq 3$ with laws $(\mathbb{P}_x : x \in \mathbb{R}^d)$.

Due to the regularity of 0 with respect to $(0, \infty)$ and $(-\infty, 0)$ for $Y^{(1)}$, the construction of a local time at a directional minimum/maximum is analogous to that one done in Section 0.4. It is clear then that an adequate choice of local time for indexing excursions from the directional minimum is the local time at zero for $Y^{(1)} - \underline{Y}^{(1)}$, denoted by $L = (L_t : t \geq 0)$. Let L^{-1} be its right-continuous inverse. By the independence of $Y^{(1)}$ and $Y^{(2:d)}$, L and L^{-1} are also independent of $Y^{(2:d)}$.

There are additional properties of Brownian local times that we can take advantage of. In order to state them let us first define the ascending/descending ladder height processes. We define $(H^-, Y^{(2:d)-}) = (H_t^-, Y_t^{(2:d)-} : t \geq 0)$ by the following relations

$$H_t^- = Y_{L_t^{-1}}^{(1)}, \quad Y_t^{(2:d)-} = Y_{L_t^{-1}}^{(2:d)}, \quad t \geq 0. \tag{2.38}$$

If $g_t = \sup\{s < t : Y_s^{(1)} = \underline{Y}_t^{(1)}\}$ and $d_t = \inf\{s > t : Y_s = \underline{Y}_t^{(1)}\}$, we define the

excursions from the minimum as the process

$$\epsilon_t(s) := (Y_{g_t+s}^{(1)} - Y_{g_t}^{(1)}, Y_{g_t+s}^{(2:d)}), \quad 0 \leq s \leq \zeta_{\epsilon_t} := d_t - g_t. \quad (2.39)$$

By the standard excursion argument, there exists a measure $\mathbb{N}_{x^{(2:d)}}^-$ such that (2.12) holds. The elements of an excursion framework from the minimum of Y in the direction of $Y^{(1)}$ are defined in the analogous way, with $H_t^+ = Y_{L_t^{-1}}^{(1)}$ and $Y_t^{(2:d)+} = Y_{L_t^{-1}}^{(2:d)}$, $t \geq 0$.

We will use the independence between the coordinates of Y to find useful identities of the excursion measure and the ascending and descending ladder potentials. We will start by doing the latter, defining the ascending and descending ladder potential measures as follows

$$U_x^+(dy) = \mathbb{E}_x \left[\int_0^\infty \mathbf{1}_{\{H_t^+ \in dy^{(1)}, Y_t^{(2:d)+} \in dy^{(2:d)}\}} dt \right], \quad x^{(1)} < y^{(1)}.$$

$$U_x^-(dy) = \mathbb{E}_x \left[\int_0^\infty \mathbf{1}_{\{H_t^- \in dy^{(1)}, Y_t^{(2:d)-} \in dy^{(2:d)}\}} dt \right], \quad x^{(1)} > y^{(1)}.$$

Proof of Proposition 2.8, case $\alpha = 2$

Proof. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous and bounded function. It is well known (cf. Chapter 6 of [32]) that for Brownian motion, up to a constant $\bar{L} = (\bar{L}_t : t \geq 0)$ given by $\bar{L}_t := \bar{Y}_t^{(1)}$ is a local time at the maximum for $Y^{(1)}$. Without loss of generality let us choose $(\bar{L}, \mathbb{N}^+)$ such that $\mathbb{P}_0$ -a.s. $H_t^+ = t$ for $t \geq 0$

$$\mathbb{E} \left[\int_0^\infty f(H_t^+, Y_t^{(2:d)+}) dt \right] = \mathbb{E} \left[\int_0^\infty f(t, Y_{L_t^{-1}}^{(2:d)}) dt \right].$$

Thus, by the independence of $Y^{(1)}$ and $Y^{(2:d)}$ and using that

$$\bar{L}_t^{-1} = \inf\{s > 0 : Y_s^{(1)} = t\}, \quad t > 0$$

is a $\frac{1}{2}$ -stable process, by subordination we can use the transition kernel of the Cauchy process

$$\begin{aligned}\mathbb{E} \left[\int_0^\infty f \left(t, Y_{L_t^{-1}}^{(2:d)} \right) dt \right] &= \int_{y^{(2:d)} \in \mathbb{R}^{d-1}} dy^{(2:d)} \int_0^\infty dt f(t, y^{(2:d)}) p_t^{\alpha=1, d-1}(y^{(2:d)}) \\ &= \int_{y^{(2:d)} \in \mathbb{R}^{d-1}} dy^{(2:d)} \int_0^\infty dt f(t, y^{(2:d)}) \frac{\Gamma(d/2)}{\pi^{d/2}} \frac{t}{(t^2 + |y^{(2:d)}|^2)^{d/2}} \\ &= \int_{y \in \mathbb{H}_{0+}} \frac{\Gamma(d/2)}{\pi^{d/2}} \frac{y^{(1)}}{|y|^d} f(y).\end{aligned}$$

the result for $U_x^+(dy)$, $x^{(1)} > y^{(1)}$ follows by shifting. For $U_x^-(dy)$, the proof is analogous due to symmetry. Finally, comparing the expression with that of the ladder potential measures (2.27) and (2.28) we conclude that Proposition 2.8 for $\alpha = 2$. $\square$

Proof of Theorem 2.9

Return to Equation (2.36) and note that the same holds in the Brownian case, with the slight difference that $\tau_r^\vee$ needs to be replaced by $\tau_r^{(1)} = \inf\{t > 0 : Y_t^{(1)} = r\}$. Then, if in an abuse of notation, $\mathbb{N}^-$ is the excursion measure in the the excursions from the minimum of $Y^{(1)}$, we have

$$\mathbf{R}_z[f](x) = \lim_{r \downarrow -\infty} \int_{r < w^{(1)} < 0} U_{0,x^{(2:d)}}^-(dw) \mathbb{N}_{w^{(2:d)}}^- \left(\int_0^\zeta e^{z(w^{(1)} + \epsilon^{(1)}(s))} f(\epsilon^{(2:d)}(s)) ds \right). \quad (2.40)$$

Hence, to verify the statement of the Deep Wiener-Hopf factorisation we need to prove that Lemma 2.12 holds in the Brownian case, i.e. to prove that

$$\mathbb{N}_x^- \left(\int_0^\zeta f(x^{(1)} + \epsilon^{(1)}(s), \epsilon^{(2:d)}(s)) ds \right) = \int_{y^{(1)} > x^{(1)}} f(y) U_x^+(dy).$$

In order to prove this we will require two ingredients, the first one is analogous to the approximation relationship (2.18) for the Brownian case. The extension of this equation is not immediate from the stable case and we will give an adequate proof of it. The second ingredient is an expression for the potential measure on the upper half-space. More specifically the density $h_r^{(1)}(x, y)$ of the measure

$$\mathbb{E}_x \left[\int_0^{\tau_r^{(1)}} \mathbf{1}_{\{Y_t \in dy\}} dt \right], \quad x^{(1)}, y^{(1)} > r.$$

For obtaining this density we will separate the cases $d = 2$ and $d \geq 3$. For the first case we lift the formula directly from [20], whilst for the second case we use a flat-earth argument on the result of Lemma 1.4.

Let us start developing the first ingredient, in the form of the following lemma

Lemma 2.14. *For a bounded continuous function $f : \mathbb{H}_{x^{(1)}} \rightarrow \mathbb{R}$, $d \geq 2$, and $U_x^-(dy)$ the descending ladder potential measure of Y*

$$\lim_{r \uparrow x^{(1)}} \frac{\int_{y^{(1)} \in [r, x^{(1)}]} U_x^-(dy) f(x^{(1)}, y^{(2:d)})}{(r - x^{(1)})} = f(x). \quad (2.41)$$

Proof. We will leverage on the explicit formula we have for $U_x^-(dy)$.

$$\begin{aligned} \lim_{r \uparrow x^{(1)}} \frac{\int_{y^{(1)} \in [r, x^{(1)}]} U_x^-(dy) f(x^{(1)}, y^{(2:d)})}{(x^{(1)} - r)} \\ = \lim_{r \uparrow x^{(1)}} \frac{\int_{y^{(1)} \in [r, x^{(1)}]} \frac{\Gamma(d/2)}{\pi^{d/2}} \frac{x^{(1)} - y^{(1)}}{|x - y|^d} f(x^{(1)}, y^{(2:d)})}{(x^{(1)} - r)} dy \\ \stackrel{L'H}{=} \lim_{r \uparrow x^{(1)}} \int_{y^{(2:d)} \in \mathbb{R}^{d-1}} \frac{\Gamma(d/2)}{\pi^{d/2}} \frac{x^{(1)} - r^{(1)}}{|x - (r, y^{(2:d)})|^d} f(x^{(1)}, y^{(2:d)}) dy^{(2:d)}. \end{aligned} \quad (2.42)$$

It turns out that the kernel in the last expression is the density of the first hitting of a hyperplane of height r . We can see this by appealing to the harmonic measure on the exterior of a sphere and a flat-earth argument. Let $\tau_{\partial \mathbb{B}_{R+r}} = \inf\{t > 0 : Y_t \in \partial \mathbb{B}_{(-R\mathbf{e}_1, R+r)}\}$, then, in order to extend the flat-earth argument to Brownian motion it is enough to observe that due to the continuity of the paths of Y , for any $x \in \mathbb{R}^d$ with $x^{(1)} > r$, $\mathbb{P}_x$ -almost surely it holds that

$$\tau_{\partial \mathbb{B}_{R+r}} \downarrow \tau_r^{(1)}.$$

From Equation (1.5) in [9] we have that for $|x| < r$ and $y \in \mathbb{S}_a^{d-1}$

$$\mathbb{P}_x(Y_{\tau_r^\oplus} \in dy) = \frac{\Gamma(d/2)}{2\pi^{d/2}} \frac{1}{r} \frac{r^2 - |x|^2}{|x - y|^d} dy.$$

Then we obtain the distribution of $Y_{\tau_r^{(1)}}$ by taking the following limit. For $x \in \mathbb{H}_{r \uparrow}$ and

$y \in \mathbb{H}_r$,

$$\begin{aligned}
\mathbb{P}_x(Y_{\tau_r^{(1)}} \in dy) &= \lim_{R \rightarrow \infty} \mathbb{P}_{x+Re_1} \left(Y_{\tau_{R+r}^{\oplus}} \in dy + Re_1 \right) \\
&= \lim_{R \rightarrow \infty} \frac{\Gamma(d/2)}{2\pi^{d/2}} \frac{1}{|x-y|^d} \frac{r^2 + 2rR + R^2 - |x|^2 - 2Rx^{(1)} - R^2}{(r+R)} dy \\
&= \frac{\Gamma(d/2)}{\pi^{d/2}} \frac{x^{(1)} - r}{|x - (r, y^{(2:d)})|^d} dy^{(2:d)}. \tag{2.43}
\end{aligned}$$

Hence, identifying the last expression in (2.42) yields

$$\begin{aligned}
\lim_{r \uparrow x^{(1)}} \frac{\int_{y^{(1)} \in [r, x^{(1)}]} U_x^-(dy) f(x^{(1)}, y^{(2:d)})}{(r - x^{(1)})} &= \lim_{r \uparrow x^{(1)}} \mathbb{E}_x \left[f(x^{(1)}, Y_{\tau_r^{(1)}}^{(2:d)}) \right] \\
&= f(x),
\end{aligned}$$

where the last equality is consequence of the quasi-left continuity of Y . $\square$

Now we will find the expressions for the density of the potential on $\mathbb{H}_{r\uparrow}$ and prove in each case the so-called Silverstein's identity.

Case $d = 2$. From Chapter 2 of [20], in the case $d = 2$ we know an expression for the Green's function on a half-space. For $(s, \gamma), (u, \phi) \in \mathbb{R}^2$, such that $s, u > 0$, this is given by

$$h_0^{(1)}((s, \gamma), (u, \phi)) = \Phi((u, \phi) - (s, \gamma)) - \Phi((u, \phi) - (-s, \gamma)), \tag{2.44}$$

where $\Phi((s, \phi)) = \frac{-1}{2\pi} \log |(s, \phi)|$.

By shifting we obtain the expression for the density of the potential on the half-space $\{(u, \phi) : u > r, \phi \in \mathbb{R}\}$. It must be given by

$$\begin{aligned}
h_r^\vee((s, \gamma), (u, \phi)) &= \frac{-1}{2\pi} (\log |(u - s, \phi - \gamma)| - \log |(u + s - 2r, \phi - \gamma)|) \\
&= \frac{1}{2\pi} \log \frac{|(u + s - 2r, \phi - \gamma)|}{|(u - s, \phi - \gamma)|},
\end{aligned}$$

for $u, s > r$.

Using a standard excursion argument and Lemma 2.14

$$\begin{aligned}
\mathbb{N}_\gamma^- \left(\int_0^\zeta f(s + \epsilon_t^{(1)}, \epsilon_t^{(2)}) dt \right) &= \lim_{r \uparrow s} \frac{\mathbb{E}_{s,\gamma} \left[\int_0^{\tau_r^{(1)}} f(Y_t) dt \right]}{\int_{\{r < u < s\} \times \mathbb{R}} U_{s,\gamma}^-(du, d\phi)} \\
&= \lim_{r \uparrow s} \frac{\int_{u>r} h_r^{(1)}((s, \gamma), (u, \phi)) f(u, \phi) du d\phi}{s - r} \\
&= \lim_{r \uparrow s} \frac{\int_{u>r} \frac{1}{2\pi} \log \frac{|(u+s-2r, \phi-\gamma)|}{|(u-s, \phi-\gamma)|} f(u, \phi) du d\phi}{s - r}.
\end{aligned}$$

Analysing the limit of the expression inside the last integral we have, after using L'Hôpital's rule,

$$\lim_{r \uparrow s} \frac{\log \frac{|(u+s-2r, \phi-\gamma)|}{|(u-s, \phi-\gamma)|}}{s - r} = \frac{2(u - s)}{|(u, \phi) - (s, \gamma)|^2}. \quad (2.45)$$

Hence, recalling the explicit expression for U^+ , we have just proved Silverstein's identity in the case $d = 2$, which reads

$$\mathbb{N}_\gamma^- \left(\int_0^\zeta f(s + \epsilon_t^{(1)}, \epsilon_t^{(2)}) dt \right) = \int_{\{u>s\} \times \mathbb{R}} U_{s,\gamma}^+(du, d\phi) f(u, \phi). \quad (2.46)$$

Lemma 2.15. *In the case $d \geq 3$, the potential $h_r^{(1)}(x, y)$, given by*

$$h_r^{(1)}(x, y) = \mathbb{E}_x \left[\int_0^{\tau_r^{(1)}} \mathbf{1}_{\{Y_t \in dy\}} \right], \quad x^{(1)}, y^{(1)} < r,$$

can be obtained by the following limiting procedure. If $x^{(1)}, y^{(1)} > r$ and $e_1 := (1, 0, \dots, 0)$,

$$\begin{aligned}
h_r^{(1)}(x, y) &= \lim_{R \rightarrow \infty} h_{r+R}^\odot(x + Re_1, y + Re_1) \\
&= \frac{\Gamma(\frac{d}{2} - 1)}{4\pi^{d/2}} |x - y|^{2-d} \left(1 - \left(1 + \frac{4(x^{(1)} - r)(y^{(1)} - r)}{|x - y|^2} \right)^{\frac{2-d}{2}} \right). \quad (2.47)
\end{aligned}$$

Proof. Let us calculate the limit as $R \rightarrow \infty$ directly on (1.18).

$$\begin{aligned}
&\lim_{R \rightarrow \infty} h_{r+R}^\odot(x + Re_1, y + Re_1) \\
&= \frac{\Gamma(\frac{d}{2} - 1)}{4\pi^{d/2}} \left(|x - y|^{2-d} - \lim_{R \rightarrow \infty} \left| \frac{(y + Re_1)|x + Re_1|}{r + R} - \frac{(r + R)(x + Re_1)}{|x + Re_1|} \right|^{2-d} \right).
\end{aligned}$$

Analysing the limit in the right-hand side

$$\begin{aligned} & \lim_{R \rightarrow \infty} \left| \frac{(y + Re_1)|x + Re_1|}{r + R} - \frac{(r + R)(x + Re_1)}{|x + Re_1|} \right|^{2-d} \\ &= \lim_{R \rightarrow \infty} \left| \frac{|y + Re_1|^2 |x + Re_1|^2 - 2(r + R)^2 \langle x + Re_1, y + Re_1 \rangle + (r + R)^4}{(r + R)^2} \right|^{\frac{2-d}{2}}. \end{aligned}$$

In the numerator of the previous expression all the terms of order R^4 and R^3 cancel before taking the limit. Then,

$$\begin{aligned} & \lim_{R \rightarrow \infty} \left| \frac{(y + Re_1)|x + Re_1|}{r + R} - \frac{(r + R)(x + Re_1)}{|x + Re_1|} \right|^{2-d} \\ &= \left| |y - x|^2 + 4 \left(r^2 - x^{(1)} y^{(1)} - r(x^{(1)} + y^{(1)}) \right) \right|^{\frac{2-d}{2}} \\ &= \left| |y - x|^2 + 4 \left(x^{(1)} - r \right) \left(y^{(1)} - r \right) \right|^{\frac{2-d}{2}} \\ &= |y - x|^{2-d} \left| 1 + \frac{4 \left(x^{(1)} - r \right) \left(y^{(1)} - r \right)}{|x - y|^2} \right|^{\frac{2-d}{2}}. \end{aligned}$$

In conclusion

$$h_r^{(1)}(x, y) = \frac{\Gamma(\frac{d}{2} - 1)}{4\pi^{d/2}} |x - y|^{2-d} \left(1 - \left(1 + \frac{4(x^{(1)} - r)(y^{(1)} - r)}{|x - y|^2} \right)^{\frac{2-d}{2}} \right), \quad (2.48)$$

which in fact coincides with the expression for $h_r^\vee(x, y)$ in equation (2.32) when $\alpha = 2$.

□

Then, by Lemma 2.14 we can claim that

$$\begin{aligned} \mathbb{N}_{x^{(2:d)}}^- \left(\int_0^\zeta f \left(x^{(1)} + \epsilon(s), \epsilon^{(2:d)} \right) ds \right) &= \lim_{r \uparrow x^{(1)}} \frac{\int_{y^{(1)} \in [r, x^{(1)}]} U_x^-(dy) f(x^{(1)}, y^{(2:d)})}{(r - x^{(1)})} \\ &= \lim_{r \uparrow x^{(1)}} \frac{\mathbb{E}_x \left[\int_0^{\tau_r^{(1)}} f(Y_t) dt \right]}{(x^{(1)} - r)} \\ &= \lim_{r \uparrow x^{(1)}} \frac{\int_{y^{(1)} > r} f(y) h_r^{(1)}(x, y) dy}{(x^{(1)} - r)}, \end{aligned}$$

and using the expression we obtained for $h_r^{(1)}(x, y)$ in Lemma 2.15, we have

$$\begin{aligned} \mathbb{N}_{x^{(2:d)}}^- \left(\int_0^\zeta f \left(x^{(1)} + \epsilon(s), \epsilon^{(2:d)} \right) ds \right) \\ = \lim_{r \uparrow x^{(1)}} \frac{\int_{y^{(1)} > r} f(y) \frac{\Gamma(\frac{d}{2}-1)}{4\pi^{d/2}} |x-y|^{2-d} \left(1 - \left(1 + \frac{4(x^{(1)}-r)(y^{(1)}-r)}{|x-y|^2} \right)^{\frac{2-d}{2}} \right)}{(x^{(1)} - r)}. \end{aligned}$$

Hence, after an application of L'Hopital we conclude that

$$\begin{aligned} \mathbb{N}_{x^{(2:d)}}^- \left(\int_0^\zeta f \left(x^{(1)} + \epsilon(s), \epsilon^{(2:d)} \right) ds \right) &= \int_{y^{(1)} > x^{(1)}} \frac{\Gamma(d/2)}{\pi^{d/2}} \frac{(y^{(1)} - x^{(1)})}{|x-y|^d} f(y) dy \\ &= \int_{y^{(1)} > x^{(1)}} U_x^+(dy) f(y). \end{aligned}$$

In other words, we have proved Silverstein's identity for $d \geq 2$. To finally prove the statement of Theorem 2.9 we just repeat the steps (2.36)-(2.37), which briefly put say that for $z = i\theta$ and $x^{(2:d)} \in \mathbb{R}^{d-1}$

$$\begin{aligned} \mathbb{E}_{0, x^{(2:d)}} \left[\int_0^\infty e^{zY_t^{(1)}} f(Y_t^{(2:d)}) dt \right] \\ = \int_{\{y^{(1)} < 0\}} U_{0, x^{(2:d)}}^-(dy) \mathbb{N}_{y^{(2:d)}}^- \left(\int_0^\zeta e^{z(y^{(1)} + \epsilon(s))} f(\epsilon^{(2:d)}(s)) ds \right) \\ = \int_{\{y^{(1)} < 0\}} U_{0, x^{(2:d)}}^-(dy) \int_{\{w^{(1)} > y^{(1)}\}} e^{zw^{(1)}} f(w^{(2:d)}) U_y^+(dw) \\ = \int_{y^{(1)} < 0} e^{zy^{(1)}} \left[\int_{v^{(1)} > 0} e^{zv^{(1)}} f(w^{(2:d)}) U_{0, y^{(2:d)}}^+(dv^{(1)} \times dw^{(2:d)}) \right] U_{0, x^{(2:d)}}^-(dy) \\ = \hat{\rho}_z [\rho_z[f]](x) \end{aligned}$$

This proves the Deep Wiener-Hopf factorisation for the Brownian case for $d \geq 2$.

2.3 Results on conditional laws

2.3.1 Touching the hyperplane from one side

The following results are an extension of those in [34], where the law of the point of closest reach to the origin $X_{G(\infty)}$ is used to condition the isotropic α -stable Lévy process $(X, \mathbb{P})$ to continuously approach a set on the unit sphere. In this case we will use the law of the *point of closest reach to a hyperplane before crossing such hyperplane*. The explicit expression of the law of $X_{G(\tau_0^\vee)}$ was explicitly obtained in Subsection 2.2.

Consider S , a subset of $\mathbb{H}_0 = \{z \in \mathbb{R}^d : z^{(1)} = 0\}$, such that it has strictly positive Lebesgue measure ℓ_{d-1} on $\mathbb{H}_0$ or it is a point. We aim to make sense of the law of X conditioned to approach S continuously from the half-space $\mathbb{H}_{0\uparrow}$. We will give a probabilistic construction of such conditional probability via a limiting procedure. Moreover, this approach allows us to identify this law as a Doob h -transform of the stable process killed when first entering $\mathbb{H}_{0\downarrow}$.

Let us formalise the idea of conditioning the process X to be attracted to S from $\mathbb{H}_{0\uparrow}$. When S is not a point, let us define for $\epsilon > 0$ $S_\epsilon = (0, \epsilon) \times \pi_{(2:d)}(S)$, where $\pi_{(2:d)}$ refers to the projection function from $\mathbb{R}^d$ to $\mathbb{R}^{d-1}$ that suppresses the first coordinate component. Consider the event $C_\epsilon^\vee = \{X_{G(\tau_0^\vee)} \in S_\epsilon\}$ where $G(\tau_0^\vee)$ is as previously defined and $\tau_0^\vee = \inf\{t > 0 : X_t^{(1)} < 0\}$ denotes the first hitting time of $\mathbb{H}_{0\downarrow}$.

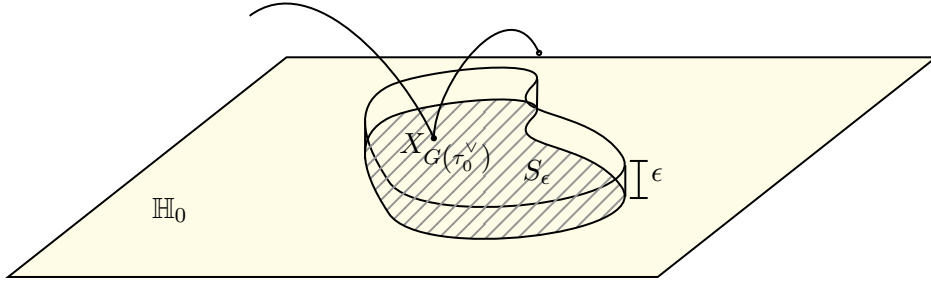


Figure 2-2: Representation of the sets S (dashed), S_ϵ and the event $C_\epsilon^\vee$.

We are interested in the asymptotic conditioning, for an event $A \in \mathcal{F}_t$

$$\mathbb{P}_x^\vee(A, t < \zeta) := \lim_{\beta \rightarrow 0} \lim_{\epsilon \rightarrow 0} \mathbb{P}_x(A, t < \tau_\beta^\vee | C_\epsilon^\vee), \quad x \in \mathbb{H}_{0\uparrow}, \quad (2.49)$$

where $\tau_\beta^\vee = \inf\{t > 0 : X_t^{(1)} < \beta\}$, $\beta > 0$. This conditioning, intuitively forces the probability measure to consider only those paths in which the point of closest reach to $\mathbb{H}_0$ is in S_ϵ , whilst $t < \tau_\beta^\vee$ controls that the event does not occur at the moment of closest reach.

To condition to approach S *from below* (from the lower half-space $\mathbb{H}_{0\downarrow}$) we can naturally define $\mathbb{P}^\wedge$ using $S_{-\epsilon} = (-\epsilon, 0) \times \pi_{(2:d)}(S)$, $\tau_0^\wedge$ and $C_\epsilon^\wedge = \{X_{G(\tau_0^\wedge)} \in S_{-\epsilon}\}$. In any case, if $S = \{v\} \in \mathbb{H}_0$, the sets $S_{\pm\epsilon}$ are adapted to converge to S on $\mathbb{H}_{0\uparrow}$ or $\mathbb{H}_{0\downarrow}$. For instance one could take $S_\epsilon = (0, \epsilon) \times \mathbb{B}^{(d-1)}(v, \epsilon)$.

It is therefore clear that the family of probabilities $(\mathbb{P}_x^\vee : x \in \mathbb{H}_{0\uparrow})$ have the property

that

$$\mathbb{P}_x^\vee(X_s \in \mathbb{H}_{0\uparrow} \text{ for all } s < \zeta \text{ and } X_{\zeta-} \in S) = 1, \quad x \in \mathbb{H}_{0\uparrow},$$

with an analogous property for $(\mathbb{P}_x^\wedge : x \in \mathbb{H}_{0\downarrow})$.

In order to state our main result let us define the following functions for $x \notin \mathbb{H}_0$,

$$H_S(x) = \begin{cases} \int_{\pi_{(2:d)}(S)} |x^{(1)}|^{\alpha/2} \left| (x^{(1)})^2 + |x^{(2:d)} - y^{(2:d)}|^2 \right|^{-d/2} dy^{(2:d)} & \text{if } \ell_{d-1}(S) > 0 \\ |x^{(1)}|^{\alpha/2} \left| (x^{(1)})^2 + |x^{(2:d)} - v^{(2:d)}|^2 \right|^{-d/2} & \text{if } S = \{v\}. \end{cases} \quad (2.50)$$

Note that H_S is well defined when $S = \mathbb{H}_0$.

Theorem 2.16. *Let $S \subset \mathbb{H}_0$ such that $\ell_{d-1}(S) > 0$ or $S = \{v\}$ a point on $\mathbb{H}_0$. Then for all points of issue $x \notin \mathbb{H}_0$ we have*

$$\left. \frac{d\mathbb{P}_x^\vee}{d\mathbb{P}_x} \right|_{\mathcal{F}_t} = \mathbf{1}_{(t < \tau_0^\vee)} \frac{H_S(X_t)}{H_S(x)}, \quad \text{if } x \in \mathbb{H}_{0\uparrow}, \quad (2.51)$$

and otherwise

$$\left. \frac{d\mathbb{P}_x^\wedge}{d\mathbb{P}_x} \right|_{\mathcal{F}_t} = \mathbf{1}_{(t < \tau_0^\wedge)} \frac{H_S(X_t)}{H_S(x)}, \quad \text{if } x \in \mathbb{H}_{0\downarrow}, \quad (2.52)$$

where the h -transform $H_S(x)$ is given in (2.50). In particular, $(\mathbb{P}_x^\vee : x \in \mathbb{H}_{0\uparrow})$ and $(\mathbb{P}_x^\wedge : x \in \mathbb{H}_{0\downarrow})$ are Markovian families.

The following result extends the previous change of measure, to be well-defined when the point of issue x is in $\mathbb{H}_0 \setminus S$. To do this, we need to make sense of the h -transform on $H(x)$ when $x^{(1)} \downarrow 0$.

Theorem 2.17. *Let $S \subset \mathbb{H}_0$ be a closed set with $\ell_{d-1}(S) > 0$ or $S = \{v\}$ for a fixed point $v \in \mathbb{H}_0$. The processes $(X, \mathbb{P}^\vee)$ and $(X, \mathbb{P}^\wedge)$ can be extended in a consistent way to include points of issue $x \in \mathbb{H}_0 \setminus S$ with path-wise continuous entry, via*

$$\mathbb{P}_x^\vee(X_t \in dy, t < \zeta) := \kappa_{\alpha,d} \frac{H_S(y)}{h(x)} \mathbb{N}_{x^{(2:d)}}^- \left(\left(\epsilon^{(1)}(t), \epsilon^{(2:d)}(t) \right) \in dy, t < \zeta \right) \quad \text{for } y \in \mathbb{H}_{0\uparrow} \quad (2.53)$$

and

$$\mathbb{P}_x^\wedge(X_t \in dy, t < \zeta) := \kappa_{\alpha,d} \frac{H_S(y)}{h(x)} \mathbb{N}_{x^{(2:d)}}^+ \left(\left(-\epsilon^{(1)}(t), \epsilon^{(2:d)}(t) \right) \in dy, t < \zeta \right) \quad \text{for } y \in \mathbb{H}_{0\downarrow}, \quad (2.54)$$

where, for $x \in \mathbb{H}_0 \setminus S$

$$h(x) = \int_{\pi_{(2:d)}(S)} |u^{(2:d)} - x^{(2:d)}|^{-d} du^{(2:d)} \quad \text{and} \quad \kappa_{\alpha,d} = -\Gamma(-\alpha/2) \frac{\sin(\alpha\pi/2)}{\pi}.$$

Here, path-wise continuous entry means that for all $x \in \mathbb{H}_0 \setminus S$

$$\mathbb{P}_x^\vee \left(\lim_{t \downarrow 0} X_t = x \right) = \mathbb{P}_x^\wedge \left(\lim_{t \downarrow 0} X_t = x \right) = 1.$$

Notice that the notation $\mathbb{N}^-$ refers to the excursion measure for the excursions of X from its directional infimum (in the direction of the $x^{(1)}$ -axis), as defined in (2.11) and $\mathbb{N}^+$ refer to the excursion measure of excursions from the directional maximum, see equation (2.13) and the discussion below. Moreover, due to isotropy, for $y \in \mathbb{H}_{0\downarrow}$

$$\mathbb{N}_{x^{(2:d)}}^+ \left(\left(\epsilon^{(1)}(t), \epsilon^{(2:d)}(t) \right) \in dy, t < \zeta \right) = \mathbb{N}_{x^{(2:d)}}^- \left(\left(\epsilon^{(1)}(t), \epsilon^{(2:d)}(t) \right) \in dy, t < \zeta \right).$$

Proofs of conditional laws from one side

Proof of Theorem 2.16

We start by recalling that we have an explicit expression for the law of the one-sided point of closest reach to $\mathbb{H}_{0\downarrow}$ given in Theorem 2.1.

Let us define $\tau_\beta^\vee = \inf\{t > 0 : X_t^{(1)} < \beta\}$, then for any $A \in \mathcal{F}_t$ by the Markov property

$$\mathbb{P}_x(A, t < \tau_\beta^\vee | C_\epsilon^\vee) = \mathbb{E}_x \left[\mathbf{1}_{\{A, t < \tau_\beta^\vee\}} \frac{\mathbb{P}_{X_t}(C_\epsilon^\vee)}{\mathbb{P}_x(C_\epsilon^\vee)} \right]. \quad (2.55)$$

In order to prove Theorem 2.16, it is enough to show that, for all $\beta > 0$, (2.51) is true for sets of the form $A \cap \{t < \tau_\beta^\vee\} \in \mathcal{F}_t$, in which case the full statement follows by monotone convergence as we take $\beta \downarrow 0$.

We note from Theorem 2.1 that if S is not a point on $\mathbb{H}_0$ and $\ell_{d-1}(S) > 0$

$$\begin{aligned}
& \mathbb{P}_x(X_{G(\tau_0^\vee)} \in S_\epsilon) \\
&= \int_{S_\epsilon} C_{\alpha,d} \frac{|x^{(1)} - y^{(1)}|^{\alpha/2}}{|y^{(1)}|^{\alpha/2}} |x - y|^{-d} dy \\
&= \int_{\pi_{(2:d)}(S)} \int_{(0,\epsilon)} C_{\alpha,d} \frac{|x^{(1)} - y^{(1)}|^{\alpha/2}}{|y^{(1)}|^{\alpha/2}} |x - y|^{-d} dy^{(1)} dy^{(2:d)}. \\
&= \int_{\pi_{(2:d)}(S)} \int_{(0,\epsilon)} C_{\alpha,d} \frac{|x^{(1)} - y^{(1)}|^{\alpha/2}}{|y^{(1)}|^{\alpha/2}} \left| (x^{(1)} - y^{(1)})^2 + |x^{(2:d)} - y^{(2:d)}|^2 \right|^{-d/2} dy^{(1)} dy^{(2:d)}.
\end{aligned}$$

To capture the leading order of the limit as $\epsilon \rightarrow 0$ we notice that, separating the previous integrand in $(y^{(1)})^{-\alpha/2}$ and the rest of the expression, the latter is a continuous function of $y^{(1)}$, and

$$\int_0^\epsilon (y^{(1)})^{-\alpha/2} dy^{(1)} = 2/(2 - \alpha) \epsilon^{1-\alpha/2},$$

thus

$$\begin{aligned}
& \lim_{\epsilon \rightarrow 0} \epsilon^{\alpha/2-1} \mathbb{P}_x(X_{G(\tau_0^\vee)} \in S_\epsilon) \\
&= \int_{\pi_{(2:d)}(S)} C_{\alpha,d} \left(x^{(1)}\right)^{\alpha/2} \left| (x^{(1)} - 0)^2 + |x^{(2:d)} - y^{(2:d)}|^2 \right|^{-d/2} dy^{(2:d)}. \quad (2.56)
\end{aligned}$$

It is not too difficult to notice that the last integral is bounded by the same integral over all $\mathbb{H}_0$, which for any $x \in \mathbb{H}_{0\uparrow}$ is given by

$$\begin{aligned}
& \int_{\mathbb{R}^{d-1}} C_{\alpha,d} \left(x^{(1)}\right)^{\alpha/2} \left| (x^{(1)})^2 + |x^{(2:d)} - y^{(2:d)}|^2 \right|^{-d/2} dy^{(2:d)} \\
&= C_{\alpha,d} \frac{\pi^{d/2}}{\Gamma(d/2)} \left(x^{(1)}\right)^{\alpha/2-1} < \infty.
\end{aligned}$$

Hence, by dominated convergence using that for any $\beta > 0$

$$\sup_{x^{(1)} > \beta} \frac{\int_{\pi_{(2:d)}(S)} \left(x^{(1)}\right)^{\alpha/2} \left| (x^{(1)} - 0)^2 + |x^{(2:d)} - y^{(2:d)}|^2 \right|^{-d/2} dy^{(2:d)}}{(x^{(1)})^{\alpha/2-1}} < \infty,$$

we take the limit inside (2.55), as long as $x \notin \mathbb{H}_0$ and $t < \tau_\beta^\vee$, we get

$$\lim_{\epsilon \rightarrow 0} \frac{\mathbb{P}_{X_t}(C_\epsilon^\vee)}{\mathbb{P}_x(C_\epsilon^\vee)} = \frac{\int_{\pi_{(2:d)}(S)} \left(X_t^{(1)} \right)^{\alpha/2} \left| \left(X_t^{(1)} \right)^2 + |X_t^{(2:d)} - y^{(2:d)}|^2 \right|^{-d/2} dy^{(2:d)}}{\int_{\pi_{(2:d)}(S)} \left(x^{(1)} \right)^{\alpha/2} \left| \left(x^{(1)} \right)^2 + |x^{(2:d)} - y^{(2:d)}|^2 \right|^{-d/2} dy^{(2:d)}}. \quad (2.57)$$

Taking into account the previous limit, it is sensible to define the h -transform by the following function

$$H_S(x) = \begin{cases} \int_{\pi_{(2:d)}(S)} \left(x^{(1)} \right)^{\alpha/2} \left| \left(x^{(1)} \right)^2 + |x^{(2:d)} - y^{(2:d)}|^2 \right|^{-d/2} dy^{(2:d)} & \text{if } S \neq \mathbb{H}_0 \\ \left(x^{(1)} \right)^{\alpha/2-1} & \text{if } S = \mathbb{H}_0. \end{cases} \quad (2.58)$$

To finish the proof of (2.51), we just need to notice that when $S = \{v\} \in \mathbb{H}_0$, due to the continuity of $|x^{(1)} - y^{(1)}|^{\alpha/2} \left| \left(x^{(1)} - y^{(1)} \right)^2 + |x^{(2:d)} - y^{(2:d)}|^2 \right|^{-d/2}$, when integrating on the ball $\mathbb{B}^{(d-1)}(v, \epsilon)$ and taking the limit when $\epsilon \rightarrow 0$, we get that the following limit holds

$$\lim_{\epsilon \rightarrow 0} \frac{\mathbb{P}_{X_t}(C_\epsilon^\vee)}{\mathbb{P}_x(C_\epsilon^\vee)} = \frac{\left(X_t^{(1)} \right)^{\alpha/2} \left| \left(X_t^{(1)} \right)^2 + |X_t^{(2:d)} - v^{(2:d)}|^2 \right|^{-d/2}}{\left(x^{(1)} \right)^{\alpha/2} \left| \left(x^{(1)} \right)^2 + |x^{(2:d)} - v^{(2:d)}|^2 \right|^{-d/2}},$$

concluding that in this case

$$H_{\{v\}}(x) = \left(x^{(1)} \right)^{\alpha/2} \left| \left(x^{(1)} \right)^2 + |x^{(2:d)} - v^{(2:d)}|^2 \right|^{-d/2}. \quad (2.59)$$

Finally, the proof of (2.52) is analogous (all $x^{(1)}$ must be replaced by $|x^{(1)}|$), given that by symmetry, the law of the respective point of closest reach is

$$\mathbb{P}_x(X_{G(\tau_0^\wedge)} \in dy) = C_{\alpha,d} \frac{|x^{(1)} - y^{(1)}|^{\alpha/2}}{|y^{(1)}|^{\alpha/2}} |x - y|^{-d} dy \quad x^{(1)} < y^{(1)} < 0. \quad (2.60)$$

This proves Theorem 2.16. $\square$

The proof of Theorem 2.17 is less straightforward due to a series of consistency checks. However, the main idea of the extension of the conditional law to include points of issue lying on the hyperplane, comes by setting a limiting equation between the potential in the upper half-space $h_0^\vee(x, y)$, $x, y \in \mathbb{H}_{0^\uparrow}$, and the h -transformed resolvent of the

process $(X, \mathbb{P}_x^\vee)$ as $x^{(1)} \downarrow 0$. We start by guessing the form of $\kappa_{\alpha,d}$ and h . Assuming that (2.53) holds let $x \in \mathbb{H}_0 \setminus S$ and $y \in \mathbb{H}_{0\uparrow}$ and define

$$\begin{aligned} \rho^\vee(x, dy) &:= \int_0^\infty \mathbb{P}_x^\vee(X_t \in dy, t < \zeta) dt \\ &= \kappa_{\alpha,d} \frac{H_S(y)}{h(x)} \int_0^\infty \mathbb{N}_{x^{(2:d)}}^- \left(\left(\epsilon^{(1)}(u), \epsilon^{(2:d)}(u) \right) \in dy, u < \zeta \right) du \\ &= \kappa_{\alpha,d} \frac{H_S(y)}{h(x)} U_{0,x^{(2:d)}}^+(dy), \end{aligned}$$

where we used Silverstein's identity (Lemma 2.12). Substituting the explicit expression for the potential in the previous equation we obtain

$$\begin{aligned} \rho^\vee(x, dy) &= \kappa_{\alpha,d} \frac{\Gamma(d/2)}{\pi^{d/2} \Gamma(\alpha/2)} \frac{H_S(y)}{h(x)} (y^{(1)})^{\alpha/2} |y - x|^{-d} dy \\ &= \kappa_{\alpha,d} \frac{\Gamma(d/2)}{\pi^{d/2} \Gamma(\alpha/2)} \frac{H_S(y) H_{\{x\}}(y)}{h(x)}. \end{aligned} \quad (2.61)$$

On the other hand, we can extend the definition of $\rho^\vee$ in a natural way to consider $x \in \mathbb{H}_{0\uparrow}$, and use, pre-emptively, that $\mathbb{P}^\vee$ is a Doob's h -transform of $\mathbb{P}$, with harmonic function H . Hence, it is possible to establish for $x, y \in \mathbb{H}_{0\uparrow}$

$$\rho^\vee(x, dy) = \frac{H_S(y)}{H_S(x)} h_0^\vee(x, y) dy, \quad (2.62)$$

where, by equation (2.32), $h_0^\vee(x, y) = E_{\alpha,d} |x - y|^{\alpha-d} \int_0^{\zeta_0^\vee(x,y)} (u+1)^{-d/2} u^{\alpha/2-1} du$ and $\zeta_0^\vee(x, y) = \frac{4x^{(1)}y^{(1)}}{|x-y|^2}$.

Taking the limit when $x^{(1)} \downarrow 0$

$$\begin{aligned} &\lim_{x^{(1)} \downarrow 0} \rho^\vee(x, dy) \\ &= \frac{H_S(y) E_{\alpha,d} \left| (y^{(1)})^2 + |x^{(2:d)} - y^{(2:d)}|^2 \right|^{(\alpha-d)/2}}{\int_{\pi_{(2:d)}(S)} \left| (0)^2 + |u^{(2:d)} - x^{(2:d)}|^2 \right|^{-d/2} du^{(2:d)}} \lim_{x^{(1)} \downarrow 0} \frac{\int_0^{\zeta_0^\vee(x,y)} (u+1)^{-d/2} u^{\alpha/2-1} du}{(x^{(1)})^{\alpha/2}} \\ &= E_{\alpha,d} \frac{2}{\alpha} 2^\alpha \frac{H_S(y) H_{\{x\}}(y)}{\int_{\pi_{2:d}(S)} |u^{(2:d)} - x^{(2:d)}|^{-d} du^{(2:d)}}, \end{aligned} \quad (2.63)$$

where we have used L'Hopital rule in the last equality and confirmed the correct defi-

inition of h . In conclusion, setting equation (2.61) equal to equation (2.63) implies that $\kappa_{\alpha,d} = -\Gamma(-\alpha/2) \frac{\sin(\alpha\pi/2)}{\pi}$ and $h(x) = \int_{\pi^{(2:d)}(S)} |u^{(2:d)} - x^{(2:d)}|^{-d} du^{(2:d)}$.

Checking the consistency of $(X, \mathbb{P}_x^\vee)$ when $x \in \mathbb{H}_0 \setminus S$ means verifying a series of ‘good behaviour’ properties for this extension. The first one would be proving that the right hand side of (2.53) defines a finite measure.

To prove finiteness we must prove that for any $x^{(2:d)} \in \mathbb{H}_0 \setminus S$ and $t > 0$,

$$\mathbb{N}_{x^{(2:d)}}^- \left(H_S \left(\epsilon^{(1)}(t), \epsilon^{(2:d)}(t) \right), t < \zeta \right) < \infty.$$

Given that $H_S(y) \leq H_{\mathbb{H}_0}(y) = (y^{(1)})^{\alpha/2-1}$, which tends to 0 as $x^{(1)} \rightarrow 0$, it suffices to prove for any $R > 0$ and $t \geq 0$ that

$$\mathbb{N}_{x^{(2:d)}}^- \left(H_S \left(\epsilon^{(1)}(t), \epsilon^{(2:d)}(t) \right), \epsilon^{(1)}(t) < R, t < \zeta \right) + \mathbb{N}_{x^{(2:d)}}^- (\epsilon^{(1)}(t) > R, t < \zeta) < \infty.$$

The second term is bounded by $\mathbb{N}_{x^{(2:d)}}^- (T_R^\wedge \leq t < \zeta)$, where $T_R^\wedge = \inf\{t > 0 : \epsilon^{(1)}(t) > R\}$. Such bound must be finite, otherwise there would be an infinite rate of arbitrary tall excursions, which is known to be impossible (cf. Chapter IV of [4]).

In order to prove that the first term is finite, let us define, for a given $R > 0$, the auxiliary integral $I_S(x^{(2:d)}) := \int_0^\infty \mathbb{N}_{x^{(2:d)}}^- (H_S(\epsilon^{(1)}(t), \epsilon^{(2:d)}(t)), \epsilon^{(1)}(t) < R, t < \zeta) dt$. If we prove that this integral is finite, then it is also the case (or at least Lebesgue-a.e.) for its integrand for any $t > 0$. This trick will allow us to use the occupation functional under $\mathbb{N}^-$, for which we know the explicit expression. By Lemma 2.12 and using the definition of H_S we have

$$\begin{aligned} I_S(x^{(2:d)}) &= \frac{C}{h(x)} \int_{\{y^{(1)} > 0\}} U_{(0, x^{(2:d)})}^+(dy) H_S(y) \\ &= \frac{C}{h(x)} \int_{\{y^{(1)} > 0\}} \frac{(y^{(1)})^{\alpha/2} dy}{|(y^{(1)})^2 + |x^{(2:d)} - y^{(2:d)}|^2|^{d/2}} \int_{\pi^{(2:d)}(S)} \frac{(y^{(1)})^{\alpha/2} du^{(2:d)}}{|(y^{(1)})^2 + |u^{(2:d)} - y^{(2:d)}|^2|^{d/2}} \end{aligned}$$

where C is an unimportant constant that may change between lines. Solving the integral with respect to $dy^{(2:d)}$, which we can do by completing the convolution of two independent $(d-1)$ -dimensional Cauchy densities, the expression turns into

$$\frac{C}{h(x)} \int_0^R \int_{\pi^{(2:d)}(S)} \frac{(y^{(1)})^{\alpha/2-1} du^{(2:d)}}{(2y^{(1)})^2 + |x^{(2:d)} - u^{(2:d)}|^2|^{d/2}} dy^{(1)},$$

hence, by bounding the denominator, we obtain

$$I_S(x^{(2:d)}) \leq \frac{C}{h(x)^2} \int_0^R (y^{(1)})^{\alpha/2-1} dy^{(1)} < \infty.$$

There are still two other verifications to do, starting by showing that the related semi-group is well defined. By setting $\mathcal{P}_t[g](x) := \mathbb{E}_x^\vee [g(X_t)]$ for g a continuous, bounded and measurable function. We will see this is a consequence of the Markov property of the excursion measure $\mathbb{N}_{x^{(2:d)}}^-$ and the fact that it possesses the same semigroup as $(X, \mathbb{P}_x)$ killed at $\tau_0^\vee$. Let $x \in \mathbb{H}_0 \setminus S$, and $\underline{\epsilon}(t) := (\epsilon^{(1)}(t), \epsilon^{(2:d)}(t))$ for $t \geq 0$ be a short notation for the d -dimensional excursion process from the minimum in the direction of $X^{(1)}$. Then

$$\begin{aligned} \mathbb{E}_x^\vee [g(X_{t+s})] &= \kappa_{\alpha,d} h(x)^{-1} \mathbb{N}_{x^{(2:d)}}^- (H_S(\underline{\epsilon}(t+s)) g(\underline{\epsilon}(t+s)) \mathbf{1}(t+s < \zeta)) \\ &= \kappa_{\alpha,d} h(x)^{-1} \mathbb{N}_{x^{(2:d)}}^- \left(H_S(\underline{\epsilon}(t)) \mathbf{1}(t < \zeta) \mathbb{E}_{\underline{\epsilon}(t)} \left[\frac{H_S(X_s)}{H_S(\underline{\epsilon}(t))} g(X_s) \mathbf{1}(s < \tau_0^\vee) \right] \right) \\ &= \kappa_{\alpha,d} h(x)^{-1} \mathbb{N}_{x^{(2:d)}}^- \left(H_S(\underline{\epsilon}(t)) \mathbf{1}(t < \zeta) \mathbb{E}_{\underline{\epsilon}(t)}^\vee [g(X_s)] \right) \\ &= \mathbb{E}_x^\vee \left[\mathbb{E}_{\underline{\epsilon}(t)}^\vee [g(X_s)] \right] \\ &= \mathcal{P}_t^\vee [\mathcal{P}_s^\vee [g]](x). \end{aligned}$$

The pathwise continuity property for $(X, \mathbb{P}^\vee)$, our last consistency check, means that $\mathbb{P}_x^\vee(X_{0+} = x) = 1$ for $x \in \mathbb{H}_0 \setminus S$. This is a consequence of the classical excursion theory for Lévy process, which ensure such excursions begin continuously, i.e.

$$\mathbb{N}_{x^{(2:d)}}^- \left(\lim_{t \downarrow 0} (\epsilon(t), \epsilon^{(2:d)}(t)) \neq (0, x^{(2:d)}) \right) = 0.$$

The case for the extension of $(X, \mathbb{P}^\vee)$ follows analogously, as the expression for the resolvent density $h_0^\wedge(x, y)$ of $\mathbb{E}_x \left[\int_0^{\tau_0^\wedge} \mathbf{1}(X_t \in dy) dt \right]$, $x, y \in \mathbb{H}_{0\downarrow}$, is symmetric to that of $h_0^\vee(x, y)$.

Proposition 2.18 (Distribution of the hitting location). *Suppose that $S \subset \mathbb{H}_0$ is a closed set with $\ell_{d-1}(S) > 0$, and let S' be a closed subset of S . Then for any $x \notin \mathbb{H}_0$ we have*

$$\mathbb{P}_x^\vee(X_{\zeta-} \in S') = \frac{\int_{\pi_{(2:d)}(S')} |y - x^{(2:d)}|^{-d} dy}{\int_{\pi_{(2:d)}(S)} |y - x^{(2:d)}|^{-d} dy} \quad \text{if } x \in \mathbb{H}_{0\uparrow}$$

and

$$\mathbb{P}_x^\wedge(X_{\zeta-} \in S') = \frac{\int_{\pi_{(2:d)}(S')} |y - x^{(2:d)}|^{-d} dy}{\int_{\pi_{(2:d)}(S)} |y - x^{(2:d)}|^{-d} dy} \quad \text{if } x \in \mathbb{H}_{0\downarrow}.$$

Proof. Let us begin by defining the set $S'_\epsilon = (0, \epsilon) \times \pi_{(2:d)}(S')$ and retake the limiting procedure, considering that $\mathbb{P}_x^\vee(X_\zeta \in S') = \lim_{\epsilon \rightarrow 0} \mathbb{P}_x^\vee(X_{G(\tau_0^\vee)} \in S'_\epsilon | X_{G(\tau_0^\vee)} \in S_\epsilon)$. Then

$$\begin{aligned} \mathbb{P}_x^\vee(X_{\zeta-} \in S') &= \lim_{\epsilon \rightarrow 0} \mathbb{P}_x^\vee(X_{G(\tau_0^\vee)} \in S'_\epsilon | X_{G(\tau_0^\vee)} \in S_\epsilon) \\ &= \lim_{\epsilon \rightarrow 0} \frac{\mathbb{P}_x^\vee(X_{G(\tau_0^\vee)} \in S'_\epsilon)}{\mathbb{P}_x^\vee(X_{G(\tau_0^\vee)} \in S_\epsilon)} \\ &= \frac{\int_{\pi_{(2:d)}(S')} |y - x^{(2:d)}|^{-d} dy}{\int_{\pi_{(2:d)}(S)} |y - x^{(2:d)}|^{-d} dy}. \end{aligned}$$

Once again, the procedure for $\mathbb{P}^\wedge$ is analogous. □

2.3.2 Oscillatory attraction/repulsion to a hyperplane

In [28], the isotropic d -dimensional α -stable process with $\alpha \in (0, 1]$ and $d \geq 2$ (which would not hit a hyperplane) is conditioned to continuously approach a subset D of the hyperplane $\mathbb{H}_0$. It is shown in Theorem 1 therein that this conditioning corresponds to a Doob h -transform, where the h function is $M_D(x) = \int_S |x - y|^{\alpha-d} \ell_{d-1}(dy)$ for $x \notin \mathbb{H}_0$. Moreover, taking $D = \mathbb{H}_0$ then

$$M_{\mathbb{H}_0}(x) \propto |x^{(1)}|^{\alpha-1}. \quad (2.64)$$

For X a one-dimensional symmetric α -stable with $\alpha \in (0, 1]$, the same Doob h -transform corresponds to the process conditioned to be absorbed at the origin. However, the same h -transform has an interpretation when $\alpha \in (1, 2)$, in that case it is equivalent to conditioning the process to be repelled from zero. We will show that when $d \geq 2$ and $\alpha \in (1, 2)$, the h -transform (2.64) is equivalent to condition the process to be repelled from the hyperplane $\mathbb{H}_0$. By repelling we mean the family of probabilities $(\mathbb{P}_x^\succ : x \in \mathbb{R}^d \setminus \mathbb{H}_0)$ be a family of probabilities such that

$$\mathbb{P}_x^\succ(A) := \lim_{a \rightarrow \infty} \mathbb{P}_x(A \cap \{t < \tau^{B_a^c}\} | \tau^{B_a^c} < \tau_{\{0\}}) \quad x \notin \mathbb{H}_0, A \in \mathcal{F}_t,$$

where $\tau_{\{0\}} = \inf\{t : X_t^{(1)} = 0\}$ and $\tau^{B_a^c} = \inf\{t : |X_t^{(1)}| > a\}$. Thus our assertions is the characterisation of this law by the following change of measure.

Theorem 2.19. *Let X be a d -dimensional isotropic α -stable process with $\alpha \in (1, 2)$, then for $x \notin \mathbb{H}_0$ and $A \in \mathcal{F}_t$,*

$$\left. \frac{d\mathbb{P}_x^\asymp}{d\mathbb{P}_x} \right|_{\mathcal{F}_t} = \frac{|X_t^{(1)}|^{\alpha-1}}{|x^{(1)}|^{\alpha-1}} \mathbf{1}_{\{t < \zeta\}}. \quad (2.65)$$

The proof uses the fact that under $\mathbb{P}$, $X^{(1)}$ is a symmetric α -stable Lévy process, then by Theorem 6.7 in [35],

$$\mathbb{P}_x(\tau^{B_a^c} < \tau_{\{0\}}) = (\alpha - 1) \left(\frac{x^{(1)}}{a} \right)^{\alpha-1} \int_1^{a/x^{(1)}} (t^2 - 1)^{\alpha/2-1} dt.$$

This identity helps us finding the leading order behaviour as $a \rightarrow \infty$, the rest of the proof consists in using the Markov property and dominated convergence.

Summarising the two results, the h -transform $|x^{(1)}|^{\alpha-1}$ changes the behaviour of the process according to the regime of α , so that when $\alpha \in (0, 1]$ it will be attracted to $\mathbb{H}_0$ whilst it will be repelled from it when $\alpha \in (1, 2)$. Such a behaviour is what one would expect after *inverting* the space on the first coordinate, this is, the mapping $K^{(1)}(x) := (\frac{x^{(1)}}{|x^{(1)}|}, x^{(2:d)})$. Inversion of space has been treated radially, by the well known Kelvin transform. In one dimension, such a transform is related with the previous change of measure via the Riesz-Bogdan-Żak Theorem (cf. [8]). This motivates our future research, aiming to prove that there exists a time change and a spatial transformation such that, the transformed process is equal in law to $(X, \mathbb{P}^\asymp)$.

2.3.3 Condition to never visit the half-space

The conditional law of the process X to stay in one of the half-spaces $\mathbb{H}_{0\uparrow}$ or $\mathbb{H}_{0\downarrow}$ is a question we addressed in the previous section. This conditional behaviour depends only on the behaviour of $X^{(1)}$, which under $\mathbb{P}$ is a real-valued symmetric α -stable process. Then, for $x \in \mathbb{H}_{\uparrow}$, we define the family of probabilities $\mathbb{P}^{\uparrow\mathbb{H}_0} = (\mathbb{P}_x^{\uparrow\mathbb{H}_0} : x \in \mathbb{H}_{0\uparrow})$ by the following limit,

$$\mathbb{P}_x^{\uparrow\mathbb{H}_0}(A) = \lim_{q \downarrow 0} \mathbb{P}_x(A, t < \mathbf{e}_q | \tau_0^\vee > \mathbf{e}_q) \quad A \in \mathcal{F}_t, t < \tau_0^\vee,$$

where $\mathbf{e}_q$ is an exponential random variable of parameter q , independent of X . The limiting parameter agrees with the usual convention that an exponential variable with parameter zero is infinite with probability one.

Proposition 2.20. *Let X be a d -dimensional isotropic α -stable process and $x \in \mathbb{H}_{0\uparrow}$,*

then

$$\left. \frac{d\mathbb{P}_x^{\uparrow\mathbb{H}_0}}{d\mathbb{P}_x} \right|_{\mathcal{F}_t} = \frac{|X_t^{(1)}|^{\alpha/2}}{|x^{(1)}|^{\alpha/2}} \mathbf{1}_{\{t < \zeta\}}. \quad (2.66)$$

Proof. Let $\mathbf{e}_q$ be an exponential random variable of rate q independent of X and $A \in \mathcal{F}_t$.

$$\begin{aligned} \mathbb{P}_x^{\uparrow\mathbb{H}_0}(A, t < \zeta) &= \lim_{q \downarrow 0} \mathbb{P}_x(A, t < \mathbf{e}_q | \tau_0^\vee > \mathbf{e}_q) \\ &= \lim_{q \downarrow 0} \mathbb{E}_x \left(\mathbf{1}_{(A \cap \{t < \mathbf{e}_q \wedge \tau_0^\vee\})} \frac{\mathbb{P}_{X_t}(\tau_0^\vee > \mathbf{e}_q)}{\mathbb{P}_x(\tau_0^\vee > \mathbf{e}_q)} \right) \\ &= \lim_{q \downarrow 0} \mathbb{E}_x \left(\mathbf{1}_{(A \cap \{t < \mathbf{e}_q \wedge \tau_0^\vee\})} \frac{\widehat{U}_q(X_t^{(1)})}{\widehat{U}_q(x^{(1)})} \right) \end{aligned}$$

where in the last equality we used the Wiener-Hopf factorisation (cf. 13.2.1 [32]) of the real-valued symmetric α -stable processes and $\widehat{U}_q(x)$ is the density of the q -potential of the descending ladder process, given by $\widehat{U}_q(x^{(1)}) dx^{(1)} = \mathbb{E} \left[\int_0^\infty e^{-qt} \mathbf{1}_{(H_t^- \in dx^{(1)})} dt \right]$. Using that $\widehat{U}_q(x) \uparrow \widehat{U}_0(x)$ and that we know an explicit expression for $\widehat{U}_0(x) = \widehat{U}(x)$, given in (0.6), this yields

$$\begin{aligned} \mathbb{P}_x^{\uparrow\mathbb{H}_0}(A, t < \zeta) &= \mathbb{E}_x \left(\mathbf{1}_{(A \cap \{t < \tau_0^\vee\})} \frac{\widehat{U}(X_t^{(1)})}{\widehat{U}(x^{(1)})} \right) \\ &= \mathbb{E}_x \left(\mathbf{1}_{(A \cap \{t < \tau_0^\vee\})} \frac{(X_t^{(1)})^{\alpha/2}}{(x^{(1)})^{\alpha/2}} \right). \end{aligned}$$

Notice that the crux of these calculations is the fact that the event $\{\tau_0 > \mathbf{e}_q\}$ depends only on the first coordinate of X , which under $\mathbb{P}$ is a real-valued symmetric α -stable process. The details of the use of the dominated convergence theorem follow exactly as in Chapter 13 of [32]. The case when $x \in \mathbb{H}_{0\downarrow}$ is treated analogously, but using $U_q(X_t^{(1)})$ instead of $\widehat{U}_q(X_t^{(1)})$. It turns out that by symmetry, both potentials are the same. $\square$

Chapter 3

Self-similar Markov processes and Cylindrical MAPs

In this chapter we introduce a new representation of self-similar Markov processes on the space $E := \mathbb{R}^d \setminus \mathbb{H}_0 = \{z \in \mathbb{R}^d : z^{(1)} \neq 0\}$ in terms of Markov Additive Processes (MAPs). If $X = (X^{(1)}, X^{(2:d)})$ is a self-similar Markov process on E , this representation consists of the classical Lamperti-Kiu transformation applied to $X^{(1)}$ (cf. [38]), but with an extended modulator which now accounts not only for the sign of $X^{(1)}$ but also for $X^{(2:d)}$. We refer to this representation as the cylindrical Lamperti-Kiu transform, and the corresponding transformed process as a cylindrical MAP.

We will explore various examples of self-similar Markov processes on $\mathbb{R}^d \setminus \mathbb{H}_0$. In the first section, we focus on the isotropic α -stable process killed upon its first hit into $\mathbb{H}_0$. For this process, we will identify the corresponding MAP and describe it through its Lévy system. Leveraging this, we will show a space-time transformation of the stable process whose cylindrical MAP serves as a dual MAP, akin to the Riesz-Bogdan-Żak transform. Additionally, we will develop an excursion theory and apply it to express certain fluctuations of the stable process relative to hyperplanes and slabs in terms of ladder processes associated with their cylindrical MAP transforms.

The second section is devoted to examples of ssMPs on $\mathbb{H}_{0\uparrow} = \{z \in \mathbb{R}^d : z^{(1)} > 0\}$, which we refer to as cylindrically positive self-similar Markov processes (cpssMps). These processes will be constructed from the isotropic α -stable process through killing, censoring, reflecting, and conditioning, thereby generalising the notion of positive self-similar Markov processes to higher dimensions. For each example, we will present their cylindrical MAP and employ excursion theory to derive alternative formulations of in-

interesting fluctuation identities for their self-similar counterpart. In some cases, explicit calculations will be provided, particularly for certain cpssMps related to Brownian motion.

The benefits of using the cylindrical Lamperti-Kiu transform stem from its ability to facilitate the study of ssMP fluctuations within the framework of MAPs, a theory that has seen significant development in recent years (cf. [30], [29]).

3.1 Introduction

The construction of positive self-similar Markov processes (pssMps) by killing, censoring, reflecting or conditioning an α -stable Lévy process on $\mathbb{R}$ is undoubtedly an effective technique used in understanding the behaviour of the stable process (cf. Chapter V of [35]). Roughly speaking, this technique consists of finding an appropriate embedded pssMp and ‘translating’ a question for the one-dimensional stable process which is difficult to solve, (e.g. the distribution of a first hit into a given set, calculation of potentials, conditional laws of the process or entrance laws) into a simpler question regarding the Lamperti transform of the identified pssMp. A good example of this is found in [36], where the authors find an explicit expression for the distribution of the first hit into the interval $(0, 1)$, by showing that this corresponds to the distribution of the first passage below 1 for a ‘censored’ α -stable Lévy process, which conveniently erases the negative parts of its paths. After proving that the censored process is indeed a pssMp, it is possible to solve the distribution of the first passage below 1 by relating it to the first passage below zero of its Lamperti transform.

With the generalisation of the Lamperti transform to represent real-valued self-similar Markov processes or d -dimensional self-similar processes in terms of MAPs (cf. [1]), the previous technique has been extended to consider problems in these more general spaces. For instance, it was used in [37] to answer questions of interest in Potential theory which have been addressed in the previous chapters of this thesis.

Our cylindrical transform builds on MAP representations of self-similar Markov processes and appears to be more suitable for studying questions related to stable processes and domains defined by $x^{(1)}$, such as half-spaces, hyperplanes, or infinite strips. Later in this chapter, we will explore fluctuations of α -stable Lévy processes concerning these sets, using various examples of path transformations that lead us to work with cylindrically positive self-similar Markov processes.

3.2 The Cylindrical Lamperti-Kiu transform

Before presenting the main result of this chapter let us define precisely the notion of self-similar Markov process X in E . A regular Feller process X (see Section 0.4 for the definition) with laws $\mathbb{P}$ as described above will be called a self-similar Markov process (of index α) if there exists a constant $\alpha > 0$ such that, for any $x \in \mathbb{R}^d \setminus \mathbb{H}_0$ and $c > 0$,

$$\text{the law of } (cX_{c^{-\alpha}t} : t \geq 0) \text{ under } \mathbb{P}_x \text{ is } \mathbb{P}_{cx},$$

where $\mathbb{P}_x$ is the law of X when issued from x .

We will generally omit to mention $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0})$ and in what follows, a process satisfying the properties above will be denoted by $(X, \mathbb{P})$. An additional consideration to be made is that we will allow the process X to be sent to an additional absorbing state $\dagger$ when X hits $\mathbb{H}_0$. The time $\zeta_c = \inf\{t > 0 : X_t \in \mathbb{H}_0\}$ will be called the lifetime of the process and we must notice that absorption may or may not hold with positive probability.

Also, see Section 0.4 for the definition of a Markov Additive Process (ξ, Θ) . In the same way we did consider possibly killed self-similar Markov processes, we will now consider the possibility for a MAP of being sent to an absorbing state $\delta = (\delta_\xi, \delta_\Theta)$ and will denote this lifetime by ζ_{MAP} .

In contrast to the radial Lamperti-Kiu transform (0.10) where the ordinate represents, up to a time-change, the behaviour of $\log |X|$ and the modulator corresponds $\arg(X) = X/|X|$, in our main result below we present a MAP representation of X in which the ordinate corresponds a time change of $\log |X^{(1)}|$, and the correlation structure between $X^{(1)}$ and $X^{(2:d)}$ is encoded in a modulator which in some sense represents, additionally to the sign of $X^{(1)}$, the position of the $(2 : d)$ coordinates over the plane $\mathbb{R}^{d-1}$ under a scaling by $|X^{(1)}|$.

Theorem 3.1 (Cylindrical Lamperti-Kiu transform). *Let $\alpha > 0$ and $\{(\xi, (J, \Xi)), \mathbf{P}_{y,j,\theta}\}$ be a MAP in $\mathbb{R} \times (\{-1, 1\} \times \mathbb{R}^{d-1})$, with ordinate ξ and modulator (J, Ξ) , lifetime ζ_{MAP} and absorbing state δ . Define the process X by*

$$X_t = \begin{cases} (J_{\varphi(t)} e^{\xi_{\varphi(t)}}, e^{\xi_{\varphi(t)}} \Xi_{\varphi(t)}), & \text{if } t < \int_0^{\zeta_{MAP}} \exp(\alpha \xi_s) ds, \\ \delta & \text{if } t \geq \int_0^{\zeta_{MAP}} \exp(\alpha \xi_s) ds, \end{cases} \quad (3.1)$$

where $\varphi(t) = \inf\{s > 0 : \int_0^s e^{\alpha \xi_u} du > t\}$, for $t < \int_0^{\zeta_{MAP}} \exp(\alpha \xi_s) ds$. Define the

probability measure $\mathbb{P}_x = \mathbf{P}_{\log |x^{(1)}|, \text{sgn}(x^{(1)}), \frac{x^{(2:d)}}{|x^{(1)}|}}$, for $x \in E$. Then the process $(X, \mathbb{P})$ is an E -valued ssMp, with index α and lifetime $\int_0^{\zeta_{MAP}} e^{\alpha \xi_s} ds$.

Conversely, let $(X, \mathbb{P})$ be a self-similar Markov process on E with index α and lifetime ζ_c . Define the process $(\xi, (J, \Xi))$ taking values in $(0, \infty) \times (\{-1, 1\} \times \mathbb{R}^{d-1})$ by

$$(\xi_t, (J_t, \Xi_t)) = \begin{cases} \left(\log |X_{A(t)}^{(1)}|, \left(\text{sgn}(X_{A(t)}^{(1)}), X_{A(t)}^{(2:d)} / |X_{A(t)}^{(1)}| \right) \right) & \text{if } t < \int_0^{\zeta_c} |X_s^{(1)}|^{-\alpha} ds, \\ \dagger & \text{if } t \geq \int_0^{\zeta_c} |X_s^{(1)}|^{-\alpha} ds, \end{cases} \quad (3.2)$$

where $\dagger$ is a cemetery state and $A(t) = \inf\{s > 0 : \int_0^s |X_u^{(1)}|^{-\alpha} du > t\}$ for $t < \int_0^{\zeta_c} |X_s^{(1)}|^{-\alpha} ds$. Define the probability measure $\mathbf{P}_{y,j,\eta} = \mathbb{P}_{je^y, \theta e^{-y}}$. Then $(\xi, (J, \Xi))$ with laws $\mathbf{P}_{y,j,\eta}$ is a MAP in $\mathbb{R} \times \{-1, 1\} \times \mathbb{R}^{d-1}$ with lifetime $\int_0^{\zeta_c} |X_u^{(1)}|^{-\alpha} du$.

Remark 3.2. If X is one-dimensional, the representation above reduces to that in [38], where the modulator is given by $J \in \{-1, 1\}$. Furthermore, if X is positive, the representation is the Lamperti transform and ξ is a Lévy process.

Proof of Theorem 3.1. Let us begin considering a (possibly) killed MAP $(\xi, (J, \Xi))$ with laws $(\mathbf{P}_{y,j,\eta} : y \in \mathbb{R}, j \in \{-1, 1\}, \eta \in \mathbb{R}^{d-1})$ as in the statement. Let $(\mathcal{G}_t : t \geq 0)$ be a filtration for the MAP. The process $Y_t = (J_t e^{\xi_t}, e^{\xi_t} \Xi_t)$, with probabilities $\mathbb{P}$ as described in the theorem, is the image of the MAP through a one to one measurable mapping. Hence it is a regular Feller process in the filtration $(\mathcal{G}_t : t \geq 0)$. The time change $\varphi(t)$ is the right inverse of a continuous additive functional of Y and it is strictly increasing on $(0, \zeta_{MAP})$ then, by (2.11) in Chapter V of [7], X is Feller under the filtration $(\mathcal{G}_{\varphi(t)} : t \geq 0)$.

We finish this part by proving the scaling property of $(X, \mathbb{P})$. Let $c > 0$ and $x \in \mathbb{R}^d \setminus \mathbb{H}_0$, we want to prove that for $t < \int_0^{\zeta_{MAP}} e^{\alpha \xi_s} ds$

$$\mathbb{E}_x \left[f \left(X_t^{(c)} \right) \right] = \mathbb{E}_{cx} [f(X_t)],$$

where $X^{(c)}$ is the process defined by $X_t^{(c)} = cX_{c^{-\alpha}t}$ for $t \geq 0$. Notice that the conditionally independent increments of the ordinate ξ guarantee that for $y \in \mathbb{R}$, $j \in \{-1, 1\}$, $\theta \in \mathbb{R}^{d-1}$ and $t < \zeta_{MAP}$

$$\mathbf{E}_{y,j,\theta} [f(\log c + \xi_t, J_t, \Xi_t)] = \mathbf{E}_{y+\log c, j, \theta} [f(\xi_t, J_t, \Xi_t)]. \quad (3.3)$$

The time-change φ in the theorem evaluated in the time $c^{-\alpha}t$ can also be expressed in

terms of a time-change $\varphi^{(c)}$ for the process $\xi_t^{(c)} := \log c + \xi_t$, noting that

$$\begin{aligned}\varphi(c^{-\alpha}t) &= \inf\{s : \int_0^s e^{\alpha\xi_u} du > c^{-\alpha}t\} \\ &= \inf\{s : \int_0^s e^{\alpha(\log c + \xi_u)} du > t\} \\ &=: \varphi^{(c)}(t).\end{aligned}$$

Hence, using the previous notation, for $x \in \mathbb{R}^d \setminus \mathbb{H}_0$ and $t < \int_0^{\zeta_{MAP}} e^{\alpha\xi_s} ds$

$$\begin{aligned}\mathbb{E}_x \left[f \left(X_t^{(c)} \right) \right] &= \mathbf{E}_{\log |x^{(1)}|, \text{sgn} x^{(1)}, x^{(2:d)} / |x^{(1)}|} \left[f(cJ_{\varphi(c^{-\alpha}t)} e^{\xi_{\varphi(c^{-\alpha}t)}}, c e^{\xi_{\varphi(c^{-\alpha}t)}} \Xi_{\varphi(c^{-\alpha}t)}) \right] \\ &= \mathbf{E}_{\log |x^{(1)}|, \text{sgn} x^{(1)}, x^{(2:d)} / |x^{(1)}|} \left[f(J_{\varphi^{(c)}(t)} e^{\xi_{\varphi^{(c)}(t)}}, e^{\xi_{\varphi^{(c)}(t)}} \Xi_{\varphi^{(c)}(t)}) \right] \\ &= \mathbf{E}_{\log c |x^{(1)}|, \text{sgn} x^{(1)}, x^{(2:d)} / |x^{(1)}|} \left[f(J_{\varphi(t)} e^{\xi_{\varphi(t)}}, e^{\xi_{\varphi(t)}} \Xi_{\varphi(t)}) \right] \\ &= \mathbb{E}_{cx} [f(X_t)].\end{aligned}$$

This proves the first implication in the theorem.

Consider now $(X, \mathbb{P})$ a ssMp in E with index α . The proof that $(\xi, (J, \Xi))$ is a Feller process follows analogously to its converse in the first part. In fact, if we consider $A(t)$ as in the statement if $t < \int_0^{\zeta_c} |X_s^{(1)}|^{-\alpha} ds$, and set $A(t) = \infty$ if $t \geq \int_0^{\zeta_c} |X_s^{(1)}|^{-\alpha} ds$, then $A(t)$ is an $\mathcal{F}_t$ -stopping time for all t , hence $\{(\xi, (J, \Xi)), \mathbf{P}\}$ is a Feller process in the filtration $\mathcal{G}_t = \mathcal{F}_{A(t)}$. It just remains to verify that this process possesses conditionally independent increments.

In the following, θ_t will denote the Markov shift operator for $(X, \mathbb{P}, (\mathcal{F}_t : t \geq 0))$, so that for all $s, t \geq 0$

$$A(t+s) = A(t) + \theta_{A(t)} A(s).$$

Let f be a bounded measurable function, $x \in \mathbb{R}^d \setminus \mathbb{H}_0$ and $y = \log |x^{(1)}|$, $j = \text{sgn}(x^{(1)})$ and $\eta = x^{(2:d)} / |x^{(1)}|$, then simply using the definition in the statement we have

$$\begin{aligned}\mathbf{E}_{y,j,\eta} [f(\xi_{t+s} - \xi_t, J_{t+s}, \Xi_{t+s}) \mathbf{1}_{(t+s < \zeta_{MAP})} | \mathcal{G}_t] \\ = \mathbb{E}_x \left[f \left(\log \frac{|X_{A(t+s)}^{(1)}|}{|X_{A(t)}^{(1)}|}, \text{sgn} \left(X_{A(t+s)}^{(1)} \right), \frac{X_{A(t+s)}^{(2:d)}}{|X_{A(t+s)}^{(1)}|} \right) \mathbf{1}_{(A(t+s) < \zeta_c)} \middle| \mathcal{F}_{A(t)} \right].\end{aligned}$$

Using the shift operator and the Markov property of X on the previous expression yields

$$\begin{aligned} & \mathbb{E}_x \left[f \left(\log \frac{\theta_{A(t)} |X_{A(s)}^{(1)}|}{|X_{A(t)}^{(1)}|}, \operatorname{sgn} \left(\theta_{A(t)} X_{A(s)}^{(1)} \right), \theta_{A(t)} \frac{X_{A(s)}^{(2:d)}}{|X_{A(s)}^{(1)}|} \right) \right. \\ & \quad \left. \times \mathbf{1}_{(A(t) + \theta_{A(t)} A(s) < \zeta_c)} \middle| \mathcal{F}_{A(t)} \right] \\ &= \mathbb{E}_{X_{A(t)}} \left[f \left(\log \frac{|X_{A(s)}^{(1)}|}{|X_{A(t)}^{(1)}|}, \operatorname{sgn} \left(X_{A(s)}^{(1)} \right), \frac{X_{A(s)}^{(2:d)}}{|X_{A(s)}^{(1)}|} \right) \mathbf{1}_{(A(s) < \zeta_c)} \right] \mathbf{1}_{(A(t) < \zeta_c)}. \end{aligned}$$

Finally, we will use the scaling property of $(X, \mathbb{P})$ with $c = |X_{A(t)}^{(1)}| > 0$, recalling that $X_t^{(c)} = cX_{c^{-\alpha}t}$ and $A^{(c)}(t)$ is the same time change $A(t)$ but defined in terms of $X^{(c)}$ instead of X . The scaling implies the following,

$$\begin{aligned} & \mathbf{E}_{y,j,\theta} [f(\xi_{t+s} - \xi_t, J_{t+s}, \Xi_{t+s}) \mathbf{1}_{(t+s < \zeta_{MAP})} | \mathcal{G}_t] \\ &= \mathbb{E}_{\frac{X_{A(t)}}{c}} \left[f \left(\log \frac{|cX_{c^{-\alpha}A^{(c)}(s)}^{(1)}|}{c}, \operatorname{sgn} \left(cX_{c^{-\alpha}A^{(c)}(s)}^{(1)} \right), \frac{cX_{c^{-\alpha}A^{(c)}(s)}^{(2:d)}}{|cX_{c^{-\alpha}A^{(c)}(s)}^{(1)}|} \right) \right. \\ & \quad \left. \times \mathbf{1}_{(A^{(c)}(s) < \zeta_c^{(c)})} \right] \mathbf{1}_{(A(t) < \zeta_c)} \\ &= \mathbb{E}_{\operatorname{sgn}(x^{(1)}), x^{(2:d)}/|x^{(1)}|} \left[f \left(\log |X_{A(s)}^{(1)}|, \operatorname{sgn} \left(X_{A(s)}^{(1)} \right), \frac{X_{A(s)}^{(2:d)}}{|X_{A(s)}^{(1)}|} \right) \mathbf{1}_{(A(s) < \zeta_c)} \right] \mathbf{1}_{(A(t) < \zeta_c)} \\ &= \mathbf{E}_{0, J_t, \Xi_t} [f(\xi_s, J_s, \Xi_s) \mathbf{1}_{(s < \zeta_{MAP})}] \mathbf{1}_{(t < \zeta_{MAP})}, \end{aligned}$$

where in the second equality we used that

$$\begin{aligned} c^{-\alpha} A^{(c)}(t) &= c^{-\alpha} \inf \{s > 0 : \int_0^s |cX_{c^{-\alpha}u}^{(1)}|^{-\alpha} du > t\} \\ &= \inf \{c^{-\alpha} s > 0 : \int_0^{c^{-\alpha} s} |X_w^{(1)}|^{-\alpha} dw > t\} \\ &= A(t). \end{aligned}$$

This completes the proof of the theorem. $\square$

3.2.1 Lévy systems

We will now outline how certain MAPs derived from the cylindrical Lamperti-Kiu of self-similar Markov processes on E can be characterised by examining their jump rates. The key tool for this characterisation is the concept of Lévy System, represented by the pair (H, Π) . Here, H is an additive functional that tracks the dynamics of the modulator, while Π is a transition kernel that, conditionally on the current state of the modulator, describes the instantaneous jump rate of the ordinate.

We need to assume that the modulator $(J_t, \Xi_t)_{t \geq 0}$ in the cylindrical Lamperti-Kiu transform is a Hunt process, this is, among other conditions, asking that (J, Ξ) is quasi-left continuous on $[0, \infty)$ (see Chapter 1 of [7] for a precise definition of Hunt processes), and note that by definition $(\xi_t)_{t \geq 0}$ is quasi-left continuous on $[0, \zeta_{MAP})$. It was shown in [18] that for MAPs with these conditions there exists a continuous increasing additive functional $t \mapsto H_t$ of (J, Ξ) and a transition kernel Π from $\{-1, 1\} \times \mathbb{R}^{d-1}$ to $(\{-1, 1\} \times \mathbb{R}^{d-1}) \times \mathbb{R}$, called the jump measure of the MAP, satisfying for all $(j, \theta) \in \{-1, 1\} \times \mathbb{R}^{d-1}$

$$\Pi((j, \theta), \{(j, \theta), 0\}) = 0, \quad \int_{\mathbb{R}} (1 \wedge |y|^2) \Pi((j, \theta), \{(j, \theta) \times dy\}) < \infty,$$

and such that for every non-negative predictable process Z and non-negative measurable function $g : \{-1, 1\} \times \mathbb{R}^{d-1} \times \{-1, 1\} \times \mathbb{R}^{d-1} \times \mathbb{R} \rightarrow \mathbb{R}^+$,

$$\begin{aligned} \mathbf{E}_{0,j,\theta} \left[\sum_{s \leq t} Z_s g(J_{s-}, \Xi_{s-}, J_s, \Xi_s, \xi_s - \xi_{s-}) \mathbf{1}_{\{(J_{s-}, \Xi_{s-}) \neq (J_s, \Xi_s) \text{ or } \xi_{s-} \neq \xi_s\}} \right] \\ = \mathbf{E}_{0,j,\theta} \left[\int_0^t dH_s Z_s \int_{\{-1,1\} \times \mathbb{R}^{d-1} \times \mathbb{R}} \Pi(J_s, \Xi_s, i, dv, dy) g(J_s, \Xi_s, i, v, y) \right], \quad (3.4) \end{aligned}$$

for all $(j, \theta) \in \{-1, 1\} \times \mathbb{R}^{d-1}$ and $t \geq 0$.

It was proved in Theorem 2.23 of [19] that similar to the Lévy-Itô decomposition of Lévy processes we can decompose ξ as the addition of three types of conditionally independent processes: an additive functional, a Gaussian process and a purely discontinuous process. If all the dynamics of ξ are explained by its purely discontinuous part then it is clear that the jump measure in the Lévy system characterises the MAP. That is the case of $(\xi, (J, \Xi))$, the MAP corresponding to the cylindrical Lamperti-Kiu transform of the isotropic α -stable process, for which we can calculate Π explicitly. Some references on the study of MAPs and Lévy systems are [18, 17, 24, 3, 30] however, the

reader must be aware that most of the examples of MAPs in these texts consider the modulator to be a Markov chain.

The Lamperti-stable cylindrical MAP

Let us consider $X = (X_t : t \geq 0)$, an isotropic α -stable Lévy process with laws $(\mathbb{P}_x : x \in \mathbb{R}^d \setminus \mathbb{H}_0)$. This process is self-similar of index $\alpha \in (0, 2)$. And defined up to its lifetime $\tau_{\{0\}} := \inf\{t > 0 : X_t \in \mathbb{H}_0\}$, by Theorem 3.1 there exists a MAP $(\xi, (J, \Xi))$ with laws $(\mathbf{P}_{(y,j,\theta)} : y \in \mathbb{R}, j \in \{-1, 1\}, \theta \in \mathbb{R}^{d-1})$ such that equation (3.2) holds. We will call the process $(\xi, (J, \Xi))$ the Lamperti-stable cylindrical MAP.

Notice that the killing time $\tau_{\{0\}}$ depends only on the behaviour of $(X_t^{(1)} : t \geq 0)$ as a marginal of X , and by the hitting-points properties of real-valued stable process, this implies that the stable cylindrical MAP is killed in finite time only if $\alpha \in (1, 2)$. Conversely, when $\alpha \in (0, 1]$ the MAP does not experience killing.

The pure-jump behaviour of the α -stable process causes the Lamperti-stable cylindrical MAP to possess a pure-jump behaviour, i.e., for any $t > 0$ the whole dynamics of the process will be governed by the jump measure of the MAP. Notice from the expression of the Lévy measure of X , equation (0.9), that co-planar jumps do not occur $\mathbb{P}$ -a.s. This implies that the modulator (J, Ξ) and the ordinate ξ jump together. In the case of (J, Ξ) , the jump may be in Ξ only or in both (J, Ξ) but never of J solely.

The jump measure of the stable cylindrical MAP is calculated in the following proposition. The MAP property is reflected in the fact that the jump measure is independent of the current position of the ordinate ξ , and just depends on the state of the modulator.

Proposition 3.3 (Lévy system of the stable cylindrical MAP). *Let $(X, \mathbb{P})$ be the isotropic α -stable process in $\mathbb{R}^d$ killed when it first hits $\mathbb{H}_0$, and $(\xi, (J, \Xi))$ its associated cylindrical MAP. Then, for a bounded measurable function $f : [0, \infty) \times \mathbb{R} \times \mathbb{R} \times \{-1, 1\} \times \{-1, 1\} \times \mathbb{R}^{d-1} \times \mathbb{R}^{d-1} \rightarrow \mathbb{R}$ such that $f(\cdot, \cdot, 0, \cdot, \cdot, \cdot, \cdot) = 0$ and i, θ in $\{-1, 1\}$ and $\mathbb{R}^{d-1}$ respectively,*

$$\begin{aligned} \mathbf{E}_{0,i,\theta} \left[\sum_{t>0} f(t, \xi_{t-}, \Delta \xi_t, J_{t-}, J_t, \Xi_{t-}, \Xi_t) \right] \\ = \int_{t \in (0, \infty)} \int_{x \in \mathbb{R}} \int_{y \in \mathbb{R}} \sum_{j \in \{-1, 1\}} \sum_{\phi \in \{-1, 1\}} \int_{w \in \mathbb{R}^{d-1}} \int_{z \in \mathbb{R}^{d-1}} \\ \mathbf{V}_{i,\theta}(dt, dx, j, dw) \mathcal{K}_{j,w}(dy, \phi, dz) f(t, x, y, j, \phi, w, z), \end{aligned}$$

where

$$\begin{aligned}\mathbf{V}_{i,\theta}(\mathrm{d}t, \mathrm{d}x, j, \mathrm{d}w) &= \mathbf{P}_{0,i,\theta}(\xi_t \in \mathrm{d}x, J_t = j, \Xi_t \in \mathrm{d}w) \mathrm{d}t, \\ \mathcal{K}_{j,w}(\mathrm{d}y, \phi, \mathrm{d}z) &= \frac{1}{2} K_{\alpha,d} e^{\mathrm{d}y} ((e^y \phi - j)^2 + |e^y z - w|^2)^{-\frac{\alpha+d}{2}} \mathrm{d}y \mathrm{d}z,\end{aligned}$$

and

$$K_{\alpha,d} = \frac{2^\alpha \Gamma((d+\alpha)/2)}{\pi^{d/2} |\Gamma(-\alpha/2)|}.$$

The proof consists in applying a series of changes of variables between ‘cylindrical’ and cartesian coordinates, and incorporating the time-change of the cylindrical Lamperti-Kiu transform. The procedure is not illuminating and hence it is omitted.

3.2.2 The Cylindrical Riesz-Bogdan-Żak transform

Let us introduce a time-space transform of the α -stable process which inverts the space in the first coordinate. Let $K_1 x = \left(\frac{x^{(1)}}{|x^{(1)}|^2}, \frac{x^{(2:d)}}{|x^{(1)}|^2} \right)$ for any $x \in \mathbb{R}^d \setminus \mathbb{H}_0$, and consider the following time change

$$\eta(t) = \inf \left\{ s : \int_0^s |X_u^{(1)}|^{-2\alpha} \mathrm{d}u > t \right\}, \quad t \geq 0.$$

We define the cylindrical Riesz-Bogdan-Żak (RBZ) transform of X killed when first hitting $\mathbb{H}_0$, as the process $(Y, \mathbb{P})$ given by $Y_t = K_1 X_{\eta(t)}$ for $t < \tau_{\mathbb{H}_0}$ and sent to a cemetery state if $t \geq \tau_{\mathbb{H}_0}$.

Proposition 3.4. *The cylindrical RBZ transform $(Y, \mathbb{P})$ is a ssMp in $\mathbb{R}^d \setminus \mathbb{H}_0$ and its associated cylindrical MAP $(\xi', (J', \Xi'))$ under $\mathbf{P}'_0$ is equal in law to $(-\xi, (J, \Xi))$ under $\mathbf{P}_0$.*

Proof. Let us begin by proving a ‘scaling’ property for the time change η . Define, for $c > 0$ and $t \geq 0$ $X_t^{(c)} := cX_{c^{-\alpha}t}$. To refer to the k -th coordinate of $X^{(c)}$ we will use $X^{(c,k)}$. Then

$$\begin{aligned}\eta^{(c)}(t) &= \inf \left\{ s : \int_0^s |X_u^{(c,1)}|^{-2\alpha} \mathrm{d}u > t \right\} \\ &= \inf \left\{ s : \int_0^s |cX_{c^{-\alpha}u}^{(1)}|^{-2\alpha} \mathrm{d}u > t \right\} \\ &= \inf \left\{ s : \int_0^{c^{-\alpha}s} |X_w^{(1)}|^{-2\alpha} \mathrm{d}w > c^\alpha t \right\} \\ &= c^\alpha \eta(c^\alpha t).\end{aligned}$$

Now we will prove that Y possesses the scaling property with index α on $\mathbb{R}^d \setminus \mathbb{H}_0$. For $c > 0$, $x \in \mathbb{R}^d \setminus \mathbb{H}_0$ and a bounded measurable function $f : \mathbb{R}^d \rightarrow \mathbb{R}$,

$$\begin{aligned} \mathbb{E}_x [f(cY_{c^{-\alpha}t})] &= \mathbb{E}_{x/|x^{(1)}|^2} [f(cK_1 X_{\eta(c^{-\alpha}t)})] \\ &= \mathbb{E}_{x/|x^{(1)}|^2} \left[f \left(\frac{\frac{1}{c} X_{c^\alpha \eta^{(1/c)}(t)}}{|\frac{1}{c} X_{c^\alpha \eta^{(1/c)}(t)}|^2} \right) \right] \\ &= \mathbb{E}_{x/c|x^{(1)}|^2} \left[f \left(\frac{X_{\eta(t)}}{|X_{\eta(t)}^{(1)}|^2} \right) \right] \\ &= \mathbb{E}_{cx} [f(Y_t)], \end{aligned}$$

where in the penultimate equation we used the self-similarity property of X with $1/c > 0$. The fact that Y is Markov follows from two facts: first, the transform K_1 is bijective on $\mathbb{R}^d \setminus \mathbb{H}_0$ and second, the time-change η is a ‘perfect continuous homogeneous additive functional of X ’ which by Section III.21 of [51] implies Y is a strong Markov process. The Feller properties follow from the right-continuity and Feller properties of X . Now that it is well-established that Y is a ssMp, we will exhibit its Lévy system to claim the equality in law. Let f be as in Proposition 3.3, then

$$\begin{aligned} &\mathbf{E}'_{0,j,\theta} \left[\sum_{t>0} f(t, \xi'_{t-}, \Delta \xi'_t, J'_{t-}, J'_t, \Xi'_{t-}, \Xi'_t) \right] \\ &= \mathbb{E}_{(1,\theta)} \left[\sum_{t>0} f \left(t, \log |Y_{A(t)-}^{(1)}|, \log \frac{|Y_{A(t)}^{(1)}|}{|Y_{A(t)-}^{(1)}|}, \text{sgn} Y_{A(t)-}^{(1)}, \text{sgn} Y_{A(t)}^{(1)}, \frac{Y_{A(t)-}^{(2:d)}}{|Y_{A(t)-}^{(1)}|}, \frac{Y_{A(t)}^{(2:d)}}{|Y_{A(t)}^{(1)}|} \right) \right], \end{aligned}$$

where $A(t) = \inf\{s : \int_0^s |Y_u^{(1)}|^{-\alpha} du > t\}$. Notice that $Y_{A(t)} = K_1 X_{\eta(A(t))}$ which implies

$$|Y_{A(t)}^{(1)}| = |X_{\eta(A(t))}^{(1)}|^{-1}; \quad Y_{A(t)}^{(2:d)} = \frac{X_{\eta(A(t))}^{(2:d)}}{|X_{\eta(A(t))}^{(1)}|^2}, \quad (3.5)$$

and $|X_{\eta(A(t))}^{(1)}| > 0$ implies that $\text{sgn} Y_{A(t)}^{(1)} = \text{sgn} X_{\eta(A(t))}^{(1)}$. Moreover,

$$\begin{aligned}
\eta(A(t)) &= \eta \left(\inf \{ r : \int_0^r |Y_v^{(1)}|^{-\alpha} dv > t \} \right) \\
&= \eta \left(\inf \{ r : \int_0^r |X_{\eta(v)}^{(1)}|^{-\alpha} dv > t \} \right) \\
&\quad (\text{Using that } d\eta^{-1}(w) = |X_w^{(1)}|^{2\alpha} dw) = \eta \left(\inf \{ r : \int_0^{\eta(r)} |X_w^{(1)}|^{-\alpha} dw > t \} \right) \\
&\quad (\text{By the monotonicity of } \eta) = \inf \{ s : \int_0^s |X_w^{(1)}|^{-\alpha} dw > t \} \\
&= A_X(t), \tag{3.6}
\end{aligned}$$

where $A_X(t)$ is the time-change in the cylindrical Lamperti-Kiu transform of X . Using (3.5) and (3.6) in the Lévy system, we obtain

$$\begin{aligned}
&\mathbf{E}'_{0,j,\theta} \left[\sum_{t>0} f(t, \xi'_{t-}, \Delta \xi'_t, J'_{t-}, J'_t, \Xi'_{t-}, \Xi'_t) \right] \\
&= \mathbb{E}_{(1,\theta)} \left[\sum_{t>0} f \left(t, -\log |X_{A_X(t)-}^{(1)}|, \log \frac{|X_{A_X(t)-}^{(1)}|}{|X_{A_X(t)}^{(1)}|}, \text{sgn} X_{A_X(t)-}^{(1)}, \right. \right. \\
&\quad \left. \left. \text{sgn} X_{A_X(t)}^{(1)}, \frac{X_{A_X(t)-}^{(2:d)}}{|X_{A_X(t)-}^{(1)}|}, \frac{X_{A_X(t)}^{(2:d)}}{|X_{A_X(t)}^{(1)}|} \right) \right] \\
&= \mathbf{E}_{0,j,\theta} \left[\sum_{t>0} f(t, -\xi_{t-}, -\Delta \xi_t, J_{t-}, J_t, \Xi_{t-}, \Xi_t) \right].
\end{aligned}$$

This is what we wanted to prove. $\square$

3.2.3 Fluctuation identities for the stable cylindrical MAP

There are a series of fluctuation identities for the α -stable process which emerge naturally from the cylindrical Lamperti-Kiu representation. These fluctuations are higher-dimensional analogues of identities investigated in [36, 38, 52], and, to the best of our knowledge, they do not have an explicit formula. However, in this section we will reinterpret them in terms of the cylindrical stable MAP and finally express them in terms of potential ladder measures and excursion functionals which provide insight into their analytic structure.

The excursions will be a particular case of those developed in Section 0.4. In the

radial MAP (0.10) the MAP modulator represents either the sign of the stable process when $d = 1$, or the angular position in polar coordinates when $d \geq 2$. In our case the modulator contains the information of the sign of $X^{(1)}$ and the position of the remaining $(2 : d)$ scaled by $|X^{(1)}|$, hence the excursions from the minimum or maximum of ξ will represent, for the stable process, excursions from the minimal/maximal distance to $\mathbb{H}_0$ (see Figure 3-1).

To work under a framework of excursions from the minimum of ξ for the MAP $(\xi, (J, \Xi))$ we appeal to Section 0.4 just pointing out that 1 is regular for $(1, \infty)$ and $(-\infty, 1)$ for $X^{(1)}$, hence 0 is regular for both $(-\infty, 0)$ and $(0, \infty)$ for ξ , and we can consider a continuous additive functional $L = (L_t : t \geq 0)$ to be the local time at zero for the ordinate reflected at its minimum $\xi - \xi_-$. Therefore, it is possible to define the descending ladder MAP $(H_t^-, (J^-, \Xi^-)_{t \geq 0})$ by the relationship

$$(H_t^-, (J^-, \Xi^-)_t) = \left(-\xi_{L_t^{-1}}, (J, \Xi)_{L_t^{-1}} \right), \quad t \geq 0. \quad (3.7)$$

For the purpose of defining the excursions, consider $g_t = \sup\{s < t : \xi_s = \inf_{u < s} \xi_u\}$ and $d_t = \inf\{s > t : \xi_s = \inf_{u < s} \xi_u\}$, the starting and ending instants of the excursion that contains $(\xi_t, (J_t, \Xi_t))$. Thus, the excursions are the processes defined by

$$(\epsilon_t(s), (J_t^\epsilon(s), \Xi_t^\epsilon(s))) = (\xi_{g_t+s} - \xi_{g_t}, (J_{g_t+s}, \Xi_{g_t+s})), \quad \text{for } 0 \leq s \leq d_t - g_t =: \zeta_t. \quad (3.8)$$

It is pertinent to notice that these excursions can be thought as excursions from running minimal distance to $\mathbb{H}_0$ (see Fig 3-1). We will consider $(\mathbf{n}_{j,\theta}^- : j \in \{-1, 1\}, \theta \in \mathbb{R}^{d-1})$ to be the family of excursion measures such that the exit formula (0.15) holds.

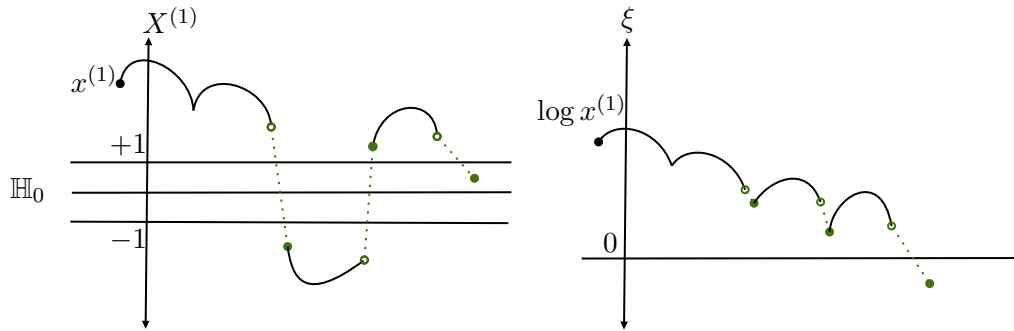


Figure 3-1: Lateral view of the excursions from the minimum of $|X^{(1)}|$ and the relationship, up to a time-change, with excursions from the minimum of ξ . The horizontal axes represent the position on the hyperplane $\mathbb{H}_0$ for the left-side plot, and $\{-1, 1\} \times \mathbb{R}^{d-1}$ for the plot on the right.

We now rewrite in the following Corollaries (of Theorem 3.1 and the excursion framework) some fluctuation identities for the stable process in terms of excursion theory for the cylindrical Lamperti-Kiu. The first one is a rather simple identity relating first passage problems of these two processes.

Corollary 3.5. *By defining the stopping times $\tau_{(-1,1)} = \inf\{t > 0 : X_t \in (-1, 1) \times \mathbb{R}^{d-1}\}$ and $T_0^- = \inf\{t > 0 : \xi_t < 0\}$ it is easy to see that for continuous bounded $f : \mathbb{R}^d \rightarrow \mathbb{R}$ and $x \notin (-1, 1) \times \mathbb{R}^{d-1}$*

$$\mathbb{E}_x \left[f \left(X_{\tau_{(-1,1)}} \right); \tau_{(-1,1)} < \infty \right] = \mathbf{E}_{x_c} \left[f \left(J_{T_0^-} e^{\xi_{T_0^-}}, e^{\xi_{T_0^-}} \Xi_{T_0^-} \right); T_0^- < \infty \right], \quad (3.9)$$

where x_c is a convenient short notation for $(\log |x^{(1)}|, \text{sgn}(x^{(1)}), x^{(2:d)} / |x^{(1)}|) \in \mathbb{R} \times \{-1, 1\} \times \mathbb{R}^{d-1}$.

The second identity is the excursion-theory development of the right-hand side of equation (3.9). This will give place to an Ikeda-Watanabe-type expression (cf. [21]) for the distribution of the first hit into $(-1, 1) \times \mathbb{R}^{d-1}$.

Corollary 3.6. *for continuous bounded $f : \mathbb{R}^d \rightarrow \mathbb{R}$ and $x \notin (-1, 1) \times \mathbb{R}^{d-1}$,*

$$\begin{aligned} \mathbb{E}_x [f(X_{\tau_{(-1,1)}}); \tau_{(-1,1)} < \infty] &= \int_{\{-\log |x^{(1)}| < r < 0\} \times \mathcal{S}} U_{x_c}^-(dr, di, d\theta) \\ &\quad \times \int_{(-\infty, r) \times \mathcal{S}} f(je^{-r+s}, e^{-r+s}\phi) \mathbf{n}_{i,\theta}^-(\mathcal{E}(\zeta) \in ds \times dj \times d\phi; \zeta < \infty) \end{aligned}$$

Proof. Extending the right-hand side of (3.9), summing over all the left limits of the excursion intervals we get

$$\begin{aligned} \mathbb{E}_x [f(X_{\tau_{(-1,1)}}); \tau_{(-1,1)} < \infty] &= \mathbf{E}_{x_c} \left[f \left(J_{T_0^-} e^{\xi_{T_0^-}}, e^{\xi_{T_0^-}} \Xi_{T_0^-} \right); T_0^- < \infty \right] \\ &= \mathbf{E}_{x_c} \left[\sum_{g \in G} \mathbf{1}_{\{0 < \xi_g < \log |x^{(1)}|, d < \infty\}} f(J_d e^{\xi_d}, e^{\xi_d} \Xi_d) \right] \\ &= \mathbf{E}_{x_c} \left[\int_0^\infty dL_t \mathbf{1}_{\{0 < \xi_t < \log |x^{(1)}|\}} \mathbf{n}_{J_t, \Xi_t}^- \left(f(J_{t+\zeta} e^{\xi_{t+\zeta}}, e^{\xi_{t+\zeta}} \Xi_{t+\zeta}); \xi_{t+\zeta} < 0, \zeta < \infty \right) \right], \end{aligned}$$

where we recall that $G = \cup_{t \geq 0} g_t$. By a change of variable and equations (3.7) and

(3.8), the previous expression is equal to

$$\begin{aligned}
& \mathbf{E}_{x_c} \left[\int_0^{L_\infty} dt \mathbf{1}_{\{0 < -H_t^- < \log |x^{(1)}|\}} \right. \\
& \quad \left. \mathbf{n}_{J_t^-, \Xi_t^-}^- \left(f(J_\zeta^\epsilon e^{-H_t^- + \epsilon(\zeta)}, e^{-H_t^- + \epsilon(\zeta)} \Xi_\zeta^\epsilon); \epsilon(\zeta) < H_t^-, \zeta < \infty \right) \right] \\
&= \int_{\{-\log |x^{(1)}| < r < 0\} \times \mathcal{S}} U_{x_c}^-(dr, di, d\theta) \\
& \quad \int_{(-\infty, r) \times \mathcal{S}} f(je^{-r+s}, e^{-r+s}\phi) \mathbf{n}_{i, \theta}^-(\mathcal{E}(\zeta) \in ds \times dj \times d\phi; \zeta < \infty)
\end{aligned}$$

where

$$U_{x_c}^-(dr, di, d\theta) := \int_0^\infty \mathbf{P}_{x_c}((H_t^-, J_t^-, \Xi_t^-) \in dr \times di \times d\theta; t < L_\infty) dt,$$

$\mathcal{S} := \{-1, 1\} \times \mathbb{R}^{d-1}$ and we call $U_{x_c}^-(dr, di, d\theta)$ the descending ladder potential measure. The factor $\mathbf{n}_{i, \theta}^-(\epsilon(\zeta), (J^\epsilon(\zeta), \Xi^\epsilon(\zeta))) \in ds \times dj \times d\phi; \zeta < \infty$ is called the jump measure of the descending ladder MAP $(H_t^-, (J_t^-, \Xi_t^-))$. $\square$

In the case $\alpha \in (0, 1)$, for which $\tau_{\{0\}} = \infty$ almost surely, we can define a non-trivial point of closest reach to the hyperplane $\mathbb{H}_0$. Let us call the instant of closest reach to the time

$$G(\infty) = \lim_{t \rightarrow \infty} \sup\{s < t : |X_s^{(1)}| = \inf_{u < s} |X_u^{(1)}|\}.$$

Also, if $\alpha \in (0, 1)$ the MAP experiences no killing and $\lim_{t \rightarrow \infty} \xi_t = \infty$, then it makes sense to talk about $g_\infty = \lim_{t \rightarrow \infty} g_t$, which corresponds to the global minimum of ξ and hence, $X_{G(\infty)}$ and $(\xi_{g_\infty}, (J_{g_\infty}, \Xi_{g_\infty}))$ can be put in correspondence. With this in mind we will show that the point of closest reach to the hyperplane is related to the descending ladder potential measure, in the sense that the latter is, up to a change of variable, the density of the former.

Corollary 3.7. *For $\alpha \in (0, 1)$ and $x \notin \mathbb{H}_0$ and $f : \mathbb{R}^d \rightarrow \mathbb{R}$,*

$$\mathbb{E}_x[f(X_{G(\infty)})] = \int_{\{-\infty < r < \log |x^{(1)}|\} \times \mathcal{S}} U_{x_c}^-(dr, di, d\theta) f(ie^r, e^r \theta).$$

Proof.

$$\begin{aligned}
\mathbb{E}_x [f(X_{G(\infty)})] &= \mathbf{E}_{x_c} \left[f(J_{g_\infty} e^{\xi_{g_\infty}}, e^{\xi_{g_\infty}} \Xi_{g_\infty}) \right] \\
&= \mathbf{E}_{x_c} \left[\sum_{g \in G} \mathbf{1}(\zeta_g = \infty) f(J_g e^{\xi_g}, e^{\xi_g} \Xi_g) \right] \\
&= \mathbf{E}_{x_c} \left[\int_0^\infty dL_t f(J_t e^{\xi_t}, e^{\xi_t} \Xi_t) \mathbf{n}_{J_t, \Xi_t}^-(\zeta = \infty) \right] \\
&= \mathbf{E}_{x_c} \left[\int_0^{L_\infty} dt f(J_t^- e^{-H_t}, e^{-H_t} \Xi_t^-) \mathbf{n}^-(\zeta = \infty) \right] \\
&= \int_{\{-\infty < r < \log |x^{(1)}|\} \times \mathcal{S}} U_{x_c}^-(dr, di, d\theta) f(i e^r, e^r \theta).
\end{aligned}$$

In the fourth equality we used that the lifetime of the excursion depends only of ξ , hence the event $\{\zeta = \infty\}$ is independent of (J_t, Ξ_t) and finally, that the pair $(L, \mathbf{n}^-)$ can be taken so that $\mathbf{n}^-(\zeta = \infty) = 1$. $\square$

In the one-dimensional case, for $\alpha \in (0, 1)$, when the modulator is only given by J , an explicit expression for U^- can be obtained from the density of the point of closest reach, which was found in Proposition 1.1 of [38], which in the symmetric case ($\rho = 1/2$ in the formula therein) has the following density,

$$\mathbb{P}_x(X_{G(\infty)} \in dz) = \frac{2^{-\alpha} \Gamma(1 - \alpha/2)}{\Gamma(1 - \alpha) \Gamma(\alpha/2)} \frac{(x^2 - |z|^2)^{\alpha/2}}{|z|^\alpha} |x - z|^{-1} dz \quad |z| < |x|.$$

The previously described point of closest reach to $\mathbb{H}_0$ also coincides with the radial point of closest reach to the origin in Theorem 0.8 when $d = 1$.

3.3 Cylindrically positive self-similar Markov processes

In this section we will present examples of self-similar processes in $\mathbb{H}_{0\uparrow}$ and their respective cylindrical Lamperti-Kiu transform. The restriction to the upper half-space, in terms of their cylindrical Lamperti-Kiu transform, implies that $J \equiv 1$, reason why we will omit J in the notation for the modulator of the MAP. We call these processes *cylindrically positive self-similar Markov processes* and they generalise the notion of positive ssMps to greater dimensions. In terms of their transforms, they pass from being a Lévy process to be a Markov Additive Process with a modulator space $\mathbb{R}^{d-1}$.

The purpose of this section is to define specific cpssMps derived from both the isotropic

α -stable Lévy process and Brownian motion, and demonstrate how their cylindrical Lamperti-Kiu transform enables the application of excursion theory for MAPs. This approach allows us to reinterpret interesting fluctuations of stable Lévy processes using ladder MAPs and potential measures. The Lamperti transform has already been instrumental in uncovering new fluctuation results for the α -stable Lévy process, and it is expected to be similarly useful here, provided the challenges of dealing with MAPs instead of Lévy processes can be addressed.

In a final example of cpssMps, we will explore in depth how the MAP framework can be applied to address a specific fluctuation of stable processes: the potential measure for the isotropic α -stable process Lévy killed upon entering the infinite strip $(0, 1) \times \mathbb{R}^{d-1}$. We will highlight the key challenges, which primarily involve handling complicated integrals, decomposing the paths of MAP as a resurrected MAP with jumps in both its ordinate and modulator, and understanding the duality of cylindrical MAPs. The aim with this final example is to provide a preliminary skeleton for future work, to be refined once these challenges are resolved.

The cylindrically pssMps we will treat are:

1. The ascending ladder process in the direction of $X^{(1)}$, $(H_t, X_t^{(2:d)+})_{t \geq 0}$, given in equation (2.5).
2. The isotropic α -stable ($\alpha \in (0, 2)$ and $\alpha = 2$) killed on exiting $\mathbb{H}_{0\uparrow}$.
3. The isotropic α -stable α -stable ($\alpha \in (0, 2)$ and $\alpha = 2$) conditioned to stay in the upper half-space $\mathbb{H}_{0\uparrow}$.
4. The isotropic α -stable α -stable ($\alpha \in (0, 2)$ and $\alpha = 2$) conditioned to hit $\mathbb{H}_0$ from one side.
5. The isotropic α -stable reflected on $\mathbb{H}_0$ ($\alpha \in (0, 2)$).
6. The censored isotropic α -stable ($\alpha \in (0, 2)$).

In all the examples the self-similar processes are considered up to the first hit of $\mathbb{H}_0$. This is only relevant if $\alpha \in (1, 2]$.

3.3.1 Ascending ladder process in the direction of $X^{(1)}$

Definition

In Chapter 2 we defined $(H, X^{(2:d)+})$, a process on $\mathbb{H}_{0\uparrow}$ with probabilities $(\mathbb{P}_x : x \in \mathbb{H}_{0\uparrow})$, by the relation

$$(H_t, X_t^{(2:d)+}) = (X_{\bar{L}_t^{-1}}^{(1)}, X_{\bar{L}_t^{-1}}^{(2:d)}), \quad t \geq 0,$$

where $\bar{L}_t^{-1}$ is the inverse local time at the maximum of $X^{(1)}$. The first coordinate of this process is the well-known ascending ladder height of the symmetric α -stable Lévy process, hence an $\alpha/2$ -stable subordinator.

By Lemma 6.8 of [32], for any $t \geq 0$, $\bar{L}_t^{-1}$ is an stopping time with respect to the natural filtration of X , then the strong Markov property of X guarantees that $(H, X^{(2:d)+})$ is Markovian. Moreover, by the definition of local time, as a continuous process, $(\bar{L}_t^{-1} : t \geq 0)$ is right-continuous. Hence, the right-continuity of X now ensures that $(H, X^{(2:d)+})$ is Feller. In order to prove that this process is in fact a cpssMps we only need to show that it satisfies the scaling property.

Let us recall that the process $(\bar{L}_t^{-1} : t \geq 0)$ is a $1/2$ -stable subordinator (cf. Subsection 6.5.3 in [32]). Consider $c, d > 0$, $x \in \mathbb{H}_{0\uparrow}$ and continuous and bounded $f : \mathbb{R}^d \rightarrow \mathbb{R}$, then

$$\begin{aligned} & \mathbb{E} \left[f(H_t, X_t^{(2:d)+}) \middle| \bar{L}_0^{-1} = 0, (H_0, X_0^{(2:d)}) = x \right] \\ &= \mathbb{E} \left[f(X_{\bar{L}_t^{-1}}^{(1)}, X_{\bar{L}_t^{-1}}^{(2:d)}) \middle| \bar{L}_0^{-1} = 0, X_0 = x \right] \\ &= \mathbb{E} \left[f(X_{d\bar{L}_{d^{-1}/2}^{-1}}^{(1)}, X_{d\bar{L}_{d^{-1}/2}^{-1}}^{(2:d)}) \middle| \bar{L}_0^{-1} = 0, X_0 = x \right] \\ (c^{-\alpha} = d^{-1}) &= \mathbb{E} \left[f(cX_{\bar{L}_{c^{-\alpha}/2}^{-1}}^{(1)}, cX_{\bar{L}_{c^{-\alpha}/2}^{-1}}^{(2:d)}) \middle| \bar{L}_0^{-1} = 0, X_0 = x/d \right] \\ &= \mathbb{E} \left[f(cX_{\bar{L}_{c^{-\alpha}/2t}^{-1}}^{(1)}, cX_{\bar{L}_{c^{-\alpha}/2t}^{-1}}^{(2:d)}) \middle| \bar{L}_0^{-1} = 0, (H_0, X_0^{(2:d)}) = x/d \right]. \end{aligned} \quad (3.10)$$

This proves that $(H, X^{(2:d)+})$ is a self-similar Markov process on $\mathbb{H}_{0\uparrow}$ of index $\alpha/2$, and by Theorem 3.1, there exists a cylindrical MAP (ξ^+, Ξ^+) such that for $t \geq 0$

$$(H_t, X_t^{(2:d)+}) = e^{\xi_{\varphi^+(t)}^+}(1, \Xi_{\varphi^+(t)}^+), \quad (3.11)$$

where in this case $\varphi^+(t) = \inf\{s > 0 : \int_0^s e^{\alpha/2 \xi_u^+} du > t\}$.

3.3.2 The stable killed on exiting $\mathbb{H}_{0\uparrow}$

Definition

Let us define the isotropic α -stable Lévy process killed at its first passage below $\mathbb{H}_0$. This is a process $X^* = (X_t^* : t \geq 0)$ with laws $(\mathbb{P}_x : x \in \mathbb{H}_{0\uparrow})$, given by

$$X_t^* = X_t \mathbf{1}_{\{t < \tau_0^\vee\}}, \quad t \geq 0,$$

where X is the isotropic α -stable process and $\tau_0^\vee = \inf\{t > 0 : X_t^{(1)} < 0\}$. From the construction of X^* it is clear that this process is Markovian and possesses the Feller properties. It just remains to show that this process satisfies the scaling property. Let $c > 0$, define $X^{[c]} := cX_{c^{-\alpha}t}$ for $t \geq 0$, and $\tau_0^{\vee[c]} := \inf\{t > 0 : X_t^{[c](1)} < 0\}$, then

$$\begin{aligned} \tau_0^{\vee[c]} &= \inf\{t > 0 : cX_{c^{-\alpha}t}^{(1)} < 0\} \\ &= c^\alpha \inf\{c^{-\alpha}t : X_{c^{-\alpha}t}^{(1)} < 0\} \\ &= c^\alpha \tau_0^\vee, \end{aligned}$$

and for a bounded continuous function $f : \mathbb{R}^d \rightarrow \mathbb{R}$,

$$\begin{aligned} \mathbb{E}_x \left[f(cX_{c^{-\alpha}t}^*) \right] &= \mathbb{E}_x \left[f(cX_{c^{-\alpha}t} \mathbf{1}_{\{c^{-\alpha}t < \tau_0^\vee\}}) \right] \\ &= \mathbb{E}_x \left[f(X_t^{[c]} \mathbf{1}_{\{t < \tau_0^{\vee[c]}\}}) \right] \\ &= \mathbb{E}_{cx} \left[f(X_t \mathbf{1}_{\{t < \tau_0^\vee\}}) \right] \\ &= \mathbb{E}_{cx} [f(X_t^*)], \end{aligned}$$

where the penultimate equality is given by the self-similarity of X . Therefore, by the cylindrical Lamperti-Kiu transform (Theorem 3.1) we have that

$$X_t^* = (X_t^{(1)*}, X_t^{(2:d)*}) = e^{\xi_{\varphi^*(t)}^*} \left(1, \Xi_{\varphi^*(t)}^* \right) \quad (3.12)$$

where (ξ^*, Ξ^*) is governed by laws $(\mathbf{P}_{x,\theta}^* : x \in \mathbb{R}, \theta \in \mathbb{R}^{d-1})$, and we call this process the killed cylindrical MAP. In virtue of $X^{(1)}$ being a symmetric α -stable Lévy process, section 5.3 of [35] guarantees that the MAP (ξ^*, Ξ^*) is killed at an exponential time of parameter $q^* = \Gamma(\alpha) \frac{\sin(\pi\alpha/2)}{\alpha} > 0$. We denote the lifetime of this MAP by ζ_{MAP}^* .

Fluctuation identities

All the following fluctuations are Corollaries of the cylindrical MAP representation of the α -stable Lévy process killed on exiting $\mathbb{H}_{0\uparrow}$.

For the first fluctuation we can spatially match the stable process an instant before crossing below $\mathbb{H}_0$ with the MAP, an instant before being killed.

Corollary 3.8. *For any continuous and bounded $f : \mathbb{R}^d \rightarrow \mathbb{R}$, $x \in (0, 1) \times \mathbb{R}^{d-1}$, and taking $x_c = (\log x^{(1)}, x^{(2:d)}/x^{(1)})$*

$$\mathbb{E}_x \left[f(X_{\tau_0^\vee -}) \right] = \mathbf{E}_{x_c}^* \left[f \left(e^{\xi_{\zeta_{MAP}^*}^*}, e^{\xi_{\zeta_{MAP}^*}^* - \Xi_{\zeta_{MAP}^*}^*} \right) \right]. \quad (3.13)$$

Remark 3.9. *For any continuous bounded $f : \mathbb{R} \times \mathbb{R}^{d-1} \rightarrow \mathbb{R}$ the potential of the killed cylindrical MAP is related to the potential of the stable before entering a half-space, viz.*

$$\begin{aligned} \mathbf{E}_{x_c}^* \left[\int_0^{\zeta_{MAP}^*} f(\xi_t^*, \Xi_t^*) dt \right] &= \mathbb{E}_x \left[\int_0^{A(\zeta_{MAP}^*)} f \left(\log |X_{A(t)}^{(1)}|, \frac{X_{A(t)}^{(2:d)}}{|X_{A(t)}^{(1)}|} \right) dt \right] \\ &= \mathbb{E}_x \left[\int_0^{A(\zeta_{MAP}^*)} f \left(\log |X_{A(t)}^{(1)}|, \frac{X_{A(t)}^{(2:d)}}{|X_{A(t)}^{(1)}|} \right) |X_{A(t)}^{(1)}|^{-\alpha} dA(t) \right] \\ &= \mathbb{E}_x \left[\int_0^{\tau_0^\vee} f \left(\log |X_s^{(1)}|, \frac{X_s^{(2:d)}}{|X_s^{(1)}|} \right) |X_s^{(1)}|^{-\alpha} ds \right] \\ &= \int_{(0, \infty)} \int_{\mathbb{R}^{d-1}} f \left(y^{(1)}, \frac{y^{(2:d)}}{y^{(1)}} \right) \left(y^{(1)} \right)^{-\alpha} h_0^\vee(x, y) dy^{(2:d)} dy^{(1)}, \end{aligned}$$

where $h_0^\vee(x, y)$ is given explicitly in Lemma (2.32).

Similarly to how it was done in [39] to obtain the distribution of the first hit outside the interval $(0, 1)$, using the killed cylindrical MAP it is possible to relate the distribution of the first exit of the strip $(0, 1) \times \mathbb{R}^{d-1}$ with the distribution of first passage above zero (in the ordinate).

Corollary 3.10 (First exit from the strip). *Let $x \in (0, 1) \times \mathbb{R}^{d-1}$, $x_c = (\log x^{(1)}, x^{(2:d)}/x^{(1)})$ and $T_0^+ = \inf\{t > 0 : \xi_t^* > 0\}$,*

$$\mathbb{E}_x \left[f(X_{\tau_1^\wedge}); \tau_1^\wedge < \tau_0^\vee \right] = \mathbf{E}_{x_c}^* \left[f \left(e^{\xi_{T_0^+}^*}, e^{\xi_{T_0^+}^* - \Xi_{T_0^+}^*} \right); T_0^+ < \zeta_{MAP}^* \right]$$

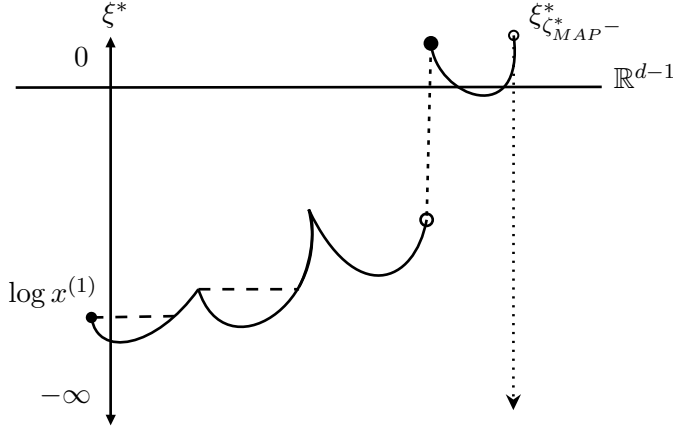


Figure 3-2: Excursions from the maximum until the first passage above 0.

We can use ladder processes and potentials to deep dive on Corollary 3.10. To do this, let us work under the framework of excursions from the maximum of ξ^* . Figure 3-2 shows excursions from the maximum are used to characterise the distribution in Corollary 3.10. The elements of such excursion theory are listed below.

- $\overline{G}$ denotes the set of left endpoints of excursions from the maximum of ξ^* .
- ℓ is the local time at the maximum of ξ^* , $\ell_\infty < \infty$ $\mathbf{P}^*$ -a.s. on account of the exponential killing time of parameter $q^* > 0$. The ascending ladder MAP is defined by $(H^{*+}, \Xi^{*+}) := (\xi_{\ell_t^-}^*, \Xi_{\ell_t^-}^*), t < \ell_\infty$.
- The excursion process is $(\epsilon_g(s), \Xi_s^{*\epsilon_g})_{0 \leq s \leq \zeta_g}$, and its excursion measure is given by $\mathbf{n}_\theta^{*+}$ for $\theta \in \mathbb{R}^{d-1}$

To simplify the notation in the following equations, let us define the function $f_e(y, \theta) = f(e^y, e^y \theta)$, for $y \in \mathbb{R}$ and $\theta \in \mathbb{R}^{d-1}$. Then,

$$\mathbb{E}_x \left[f(X_{\tau_1^\wedge}); \tau_1^\wedge < \tau_0^\vee \right] = \mathbf{E}_{x_c}^* \left[f_e \left(\xi_{T_0^+}^*, \Xi_{T_0^+}^* \right); T_0^+ < \zeta_{MAP}^* \right].$$

Now, by the exit formula (0.15), we can identify the random variables $(\xi_{T_0^+}^*, \Xi_{T_0^+}^*)$ with

the right endpoint of an excursion from the maximum.

$$\begin{aligned}
& \mathbf{E}_{x_c}^* \left[f_e \left(\xi_{T_0^+}^*, \Xi_{T_0^+}^* \right); T_0^+ < \zeta_{MAP}^* \right] \\
&= \mathbf{E}_{x_c}^* \left[\sum_{g \in \overline{G}} \mathbf{1}_{\{\xi_g^* < 0, \xi_{g+\zeta_g}^* > 0, g+\zeta_g < \zeta_{MAP}^*\}} f_e \left(\xi_{g+\zeta_g}^*, \Xi_{g+\zeta_g}^* \right) \right] \\
&= \mathbf{E}_{x_c}^* \left[\sum_{g \in \overline{G}} \mathbf{1}_{\{\xi_g^* < 0, \xi_g^* > \epsilon_g(\zeta_g), g+\zeta_g < \zeta_{MAP}^*\}} f_e \left(\xi_g^* - \epsilon_g(\zeta_g), \Xi_{\zeta_g}^{*\epsilon} \right) \right].
\end{aligned}$$

Using, in the last expression, the independent increments of the MAP at time g , and the exit system $(\ell, \mathbf{n}^{*+})$ we obtain

$$\begin{aligned}
& \mathbf{E}_{x_c}^* \left[\int_0^\infty d\ell \mathbf{1}_{\{\xi_t^* < 0, t < \zeta_{MAP}^*\}} \mathbf{n}_{\Xi_t^*}^+ \left(f_e \left(\xi_t^* - \epsilon_t(\zeta_t), \Xi_{\zeta_t}^{*\epsilon} \right); \xi_t^* > \epsilon_t(\zeta_t), \zeta_t < \zeta_{MAP}^* \right) \right] \\
&= \mathbf{E}_{x_c}^* \left[\int_0^{\ell_\infty} ds \mathbf{1}_{\{H_s^{*+} < 0\}} \mathbf{n}_{\Xi_s^{*+}}^+ \left(f_e \left(H_s^{*+} - \epsilon(\zeta), \Xi_\zeta^{*\epsilon} \right); H_s^{*+} > \epsilon(\zeta), \zeta < \zeta_{MAP}^* \right) \right] \\
&= \int_{(\log x^{(1)}, 0)} \int_{\mathbb{R}^{d-1}} U_{x_c}^{*+}(dy, d\phi) \mathbf{n}_\phi^+ \left(f_e(y - \epsilon(\zeta), \Xi_\zeta^{*\epsilon}); y > \epsilon(\zeta), \zeta < \zeta_{MAP}^* \right), \quad (3.14)
\end{aligned}$$

where

$$U_{x_c}^{*+}(dy, d\phi) = \int_0^\infty \mathbf{P}_{x_c}^*(H_t^{*+} \in dy, \Xi_t^{*+} \in d\phi, t < \ell_\infty) dt.$$

Finally, we can define for $z \in (0, \infty)$ and $\phi, \theta \in \mathbb{R}^{d-1}$,

$$\Pi_\phi^{*+}(dz, d\theta) := \mathbf{n}_\phi^{*+}(-\epsilon(\zeta) \in dz, \Xi_\zeta^{*\epsilon} \in d\theta, \zeta < \zeta_{MAP}^*),$$

the jump measure of the ascending ladder MAP, and substitute in (3.14) to conclude with the following corollary.

Corollary 3.11. *For a bounded continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ and $x \in (0, 1) \times \mathbb{R}^{d-1}$, the first exit of the strip (when it occurs by above) has the following Ikeda-Watanabe-type expression,*

$$\begin{aligned}
& \mathbb{E}_x \left[f(X_{\tau_1^\wedge}); \tau_1^\wedge < \tau_0^\vee \right] \\
&= \int_{(\log x^{(1)}, 0)} \int_{\mathbb{R}^{d-1}} U_{x_c}^{*+}(dy, d\phi) \int_{(-y, \infty) \times \mathbb{R}^{d-1}} \Pi_\phi^{*+}(dz, d\theta) f_e(y + z, \theta). \quad (3.15)
\end{aligned}$$

Developing a theory of excursions from the minimum of ξ^* it is possible to relate the one-sided point of closest reach in Theorem 2.1 to the global minimum of (ξ^*, Ξ^*) , and

hence to the descending potential ladder measure of the killed cylindrical MAP.

Similarly, we can identify the instant of overall maximum before the exponential clock $\mathbf{e}_{q^*}, \bar{g}_\infty$. This random time can be put in correspondence with the instant of directional maximum before crossing below $\mathbb{H}_0$, or in proper notation

$$\bar{G}(\tau_0^\vee) = \sup\{s \leq \tau_0^\vee : X_s^{(1)} = \sup_{u \leq t} X_u^{(1)}\}.$$

Corollary 3.12. *Consider $x \in \mathbb{R}^d$ such that $x^{(1)} > 0$ and bounded continuous $f : \mathbb{R}^d \rightarrow \mathbb{R}$, then let $g_\infty^* = \sup\{s \leq \zeta_{MAP}^* : \xi_s^* = \xi_{\zeta_{MAP}^*}^*\}$,*

$$\mathbb{E}_x \left[f(X_{G(\tau_0^\vee)}) \right] = \mathbf{E}_{x_c}^* \left[f(e^{\xi_{g_\infty^*}^*}, e^{\xi_{g_\infty^*}^*} \Xi_{g_\infty^*}^*) \right]. \quad (3.16)$$

Remark 3.13. *The random variable $X_{\bar{G}(\infty)}$ is related to the ascending ladder potential of the MAP as follows,*

$$\begin{aligned} \mathbb{E}_x^* \left[f(X_{\bar{G}(\infty)}^*) \right] &= \mathbf{E}_{x_c}^* \left[f(e^{\xi_{g_\infty^*}^*}, e^{\xi_{g_\infty^*}^*} \Xi_{g_\infty^*}^*) \right] \\ &= \mathbf{E}_{x_c}^* \left[\sum_{g \in \bar{G}} \mathbf{1}_{\{\zeta_g = \infty\}} f(e^{\xi_g^*}, e^{\xi_g^*} \Xi_g^*) \right] \\ &= \mathbf{E}_{x_c}^* \left[\int_0^\infty d\ell_t f(e^{\xi_{gt}^*}, e^{\xi_t^*} \Xi_t^*) \mathbf{n}_{\Xi_t^*}^{*+}(\zeta = \infty) \right] \\ &= \int_{\{x_c^{(1)} < y^{(1)}\} \times \mathbb{R}^{d-1}} U_{x_c}^{*+}(dz, d\phi) f(e^z, e^z \phi) \mathbf{n}_\phi^{*+}(\zeta = \infty). \end{aligned}$$

We complement this example of cylindrically positive ssMps by studying the case of Brownian motion, for which many of the expressions included in the fluctuations above are explicitly known, or easily obtained from the literature.

Killed Brownian Motion as a Cylindrical MAP

Consider $X = (X_t : t \geq 0)$ with laws $\mathbb{P}$, a d -dimensional Brownian motion. Let us define, similarly as before, a Brownian motion killed on exiting $\mathbb{H}_{0\uparrow}$ as the process Y given by

$$Y_t = X_t \mathbf{1}_{\{t < \tau_{\mathbb{H}_0}\}}, \quad \text{for } t \geq 0,$$

where $\tau_{\mathbb{H}_0} = \inf\{t > 0 : X^{(1)} = 0\}$. It is not difficult to see that Y is a self-similar Markov process on $\mathbb{H}_{0\uparrow}$. The proof is verbatim to the proof for stable processes at the

beginning of Subsection 3.3.2.

Recycling our notation from the previous section, and by virtue of Theorem 3.2, let us denote by (ξ^*, Ξ^*) with laws $(\mathbf{P}_{x,\theta}^* : x \in \mathbb{R}, \theta \in \mathbb{R}^{d-1})$ the cylindrical Lamperti-Kiu transform MAP of Y . This is,

$$Y_t = e^{\xi_{\varphi(t)}^*} (1, \Xi_{\varphi(t)}^*), \quad 0 \leq t < \tau_{\mathbb{H}_0}.$$

Note that as Y approaches $\mathbb{H}_0$ continuously, the theory of the one-dimensional Lamperti transform (cf. Section 13.3 of [32]), tells us that $\zeta_{MAP}^* = \infty$ and $\lim_{t \rightarrow \infty} \xi_t^* = -\infty$. In fact, this agrees with the fact that $q^* = \Gamma(\alpha) \frac{\sin(\pi\alpha/2)}{\alpha}$ is zero if $\alpha = 2$, and corresponds to the convention that an exponential random variable of parameter 0 is defined as $+\infty$.

In the one-dimensional case it is not difficult to show that the Lamperti transform ξ^* of $Y^{(1)}$ is a Brownian motion with drift coefficient $-\frac{1}{2}$. To see this, consider the function $f(t, x) = e^{-\frac{1}{2}t+x}$, hence a straightforward application of Itô's formula tells us that the process $(f(t, B_t) : t \geq 0)$, where $(B_t)_{t \geq 0}$ is a real valued Brownian motion, is a martingale. Therefore, by the Dambis-Dubins-Schwarz Theorem we have that, for $t < \tau_{\mathbb{H}_0}$, the process $Y^{(1)}$ can be written as $Y_t^{(1)} = e^{-\frac{1}{2}\varphi(t)+B_{\varphi(t)}}$, where $\varphi(t) = \inf\{s > 0 : \int_0^s e^{-\frac{1}{2}u+B_u} du > t\}$.

Explicit ascending/descending ladder potential measures

Helped by known fluctuations of Brownian Motion and formulae in the literature, we will present the explicit expressions of the potential measures of ascending and descending ladder MAPs. To do this, let us define local times and ladder processes.

Let $\ell = (\ell_t : t \geq 0)$ be the local time at the maximum of ξ^* . The ascending ladder MAP is defined by,

$$(\xi_t^{*+}, \Xi_t^{*+}) := (\xi_{\ell_t^{-1}}^*, \Xi_{\ell_t^{-1}}^*), \quad t \geq 0. \quad (3.17)$$

Analogously, if $\underline{\ell} = (\underline{\ell}_t : t \geq 0)$ is the local time at the minimum of ξ^* , the descending ladder MAP is given by

$$(\xi_t^{*-}, \Xi_t^{*-}) := (\xi_{\underline{\ell}_t^{-1}}^*, \Xi_{\underline{\ell}_t^{-1}}^*), \quad t \geq 0. \quad (3.18)$$

Finally, let $L = (L_t : t \geq 0)$ be the local time at the maximum of $X^{(1)}$ and $\underline{L} = (\underline{L}_t : t \geq 0)$ is the local time at the minimum of $X^{(1)}$.

For calculating the potential measures we will relate the ladder MAPs in (3.17) and (3.18) with the ladder processes of Y , or with those of X considering the additional condition of not being ‘killed’ at $\mathbb{H}_0$. We will start by proving a relationship between local times (at the maximum or minimum) of the cylindrical MAP, and local times at the directional maxima/minima of X .

Lemma 3.14 (Identification of local times). *Suppose $t < \tau_{\mathbb{H}_0}$ and $A(t)$ be the additive functional given in the cylindrical Lamperti-Kiu transform of killed Brownian motion Y , i.e., for $t < \tau_{\mathbb{H}_0}$ $A(t) = \inf\{s > 0 : \int_0^s |Y_u^{(1)}|^{-2} du > t\}$, then*

$$L_t^{-1} = A(\ell_{\log t}^{-1}), \quad \underline{L}_t^{-1} = A(\underline{\ell}_{\log t}^{-1}).$$

Proof. The first thing to notice is that $Y^{(1)}$ before its killing time $\tau_{\mathbb{H}_0}$, and ξ^* are continuous Lévy processes of unbounded variation with continuous paths, for which according to Section 6.1 of [32], the running supremum processes, given by $\bar{Y}_t^{(1)} := \sup_{s \leq t} Y_s^{(1)}$ and $\bar{\xi}_t^* := \sup_{s \leq t} \xi_s^*$, serve as local times at the maximum for its respective process. In other words, the inverse local time at the maximum for $Y^{(1)}$ is given by

$$L_t^{-1} = \inf\{u > 0 : \bar{Y}_u^{(1)} > t\}, \quad \text{for } t < \tau_{\mathbb{H}_0}.$$

Using the Lamperti transform of $Y^{(1)}$ and the fact that $A(t) = \varphi^{-1}(t)$, we obtain for $t < \tau_{\mathbb{H}_0}$ that

$$\begin{aligned} L_t^{-1} &= \inf\{u > 0 : e^{\bar{\xi}_{\varphi(u)}^*} > t\} \\ &= \inf\{A(u) > 0 : e^{\bar{\xi}_u^*} > t\} \\ &= A\left(\inf\{u > 0 : \bar{\xi}_u^* > \log t\}\right) \\ &= A(\ell_{\log t}^{-1}), \end{aligned}$$

where the two last equalities are given by the monotonicity of A and the same observation that $\bar{\xi}^*$ is an adequate choice of local time at the maximum of ξ^* . The proof for the local time at the minimum follows analogously given that a local time at the minimum of $Y^{(1)}$ and ξ^* is a local time at the maximum of the processes $-Y^{(1)}$ and $-\xi^*$ respectively, which are continuous and of unbounded variation. $\square$

Proposition 3.15 (Explicit expression for descending potential). *For $\theta \in \mathbb{R}^{d-1}$ and*

$f : (-\infty, 0) \times \mathbb{R}^{d-1} \rightarrow \mathbb{R}$ continuous and bounded

$$\int_0^\infty dt \mathbf{E}_{0,\theta}^* [f(\xi_t^{*-}, \Xi_t^{*-})] = \int_{-\infty}^0 \int_{\mathbb{R}^{d-1}} \frac{\Gamma(d/2)}{\pi^{d/2}} \frac{e^{-2z}}{|e^{-4z} + |y - e^{-z}\theta|^2|^{d/2}} f(z, y) dy dz. \quad (3.19)$$

Proof. Consider θ and f as in the statement, hence by developing the left-hand side we have

$$\begin{aligned} \int_0^\infty dt \mathbf{E}_{0,\theta}^* [f(\xi_t^{*-}, \Xi_t^{*-})] &= \int_0^\infty dt \mathbf{E}_{0,\theta}^* [f(\xi_{\underline{\ell}_t}^{*-}, \Xi_{\underline{\ell}_t}^{*-})] \\ &= \int_0^\infty dt \mathbb{E}_{(1,\theta)} \left[f \left(\log |Y_{A(\underline{\ell}_t)}^{(1)}|, \frac{Y_{A(\underline{\ell}_t)}^{(2:d)}}{|Y_{A(\underline{\ell}_t)}^{(1)}|} \right) \right] \\ (\text{By Lemma 3.14}) &= \int_0^\infty dt \mathbb{E}_{(1,\theta)} \left[f \left(\log |Y_{\underline{L}_{e^{-t}}}^{(1)}|, \frac{Y_{\underline{L}_{e^{-t}}}^{(2:d)}}{|Y_{\underline{L}_{e^{-t}}}^{(1)}|} \right) \right] \\ &= \int_0^\infty dt \mathbb{E}_{(1,\theta)} \left[f \left(-t, \frac{Y_{\underline{L}_{e^{-t}}}^{(2:d)}}{e^{-t}} \right) \right]. \end{aligned} \quad (3.20)$$

In the last equality we used that $Y_t^{(1)} = X_t^{(1)}$ for $t \leq \tau_{\mathbb{H}_0}$, thus we invoked the well known result that the descending ladder height process of a one-dimensional Brownian motion, behaves as $-t$. Then, doing $t = \log s$ and using that $\underline{L}_s^{-1}$ is a Lévy subordinator, we rewrite the expression (3.20) as

$$\begin{aligned} &\int_1^\infty \frac{ds}{s} \mathbb{E}_{(1,\theta)} \left[f \left(-\log s, s Y_{\underline{L}_s^{-1}}^{(2:d)} \right) \right] \\ &= \int_1^\infty \frac{ds}{s} \int_{y \in \mathbb{R}^{d-1}} p_s^{1,d-1}(\theta, y) f(-\log s, sy) dy, \end{aligned}$$

where $p_s^{1,d-1}$ is the transition kernel density of a $(d-1)$ -dimensional Cauchy process, and so is given by

$$p_s^{1,d-1}(x, y) = \frac{\Gamma(d/2)}{\pi^{d/2}} \frac{s}{(s^2 + |x - y|^2)^{d/2}}, \quad x, y \in \mathbb{R}^{d-1}.$$

Substituting and doing basic changes of variable, we get the following

$$\begin{aligned}
& \int_1^\infty \int_{y \in \mathbb{R}^{d-1}} \frac{\Gamma(d/2)}{\pi^{d/2}} \frac{s}{|s^2 + |y - \theta|^2|^{d/2}} f(-\log s, sy) dy \frac{ds}{s} \\
&= \int_1^\infty \int_{u \in \mathbb{R}^{d-1}} \frac{\Gamma(d/2)}{\pi^{d/2}} \frac{1}{|s^2 + |\frac{u}{s} - \theta|^2|^{d/2}} f(-\log s, u) s^{1-d} du ds \\
&= \int_{-\infty}^0 \int_{u \in \mathbb{R}^{d-1}} \frac{\Gamma(d/2)}{\pi^{d/2}} \frac{e^{-z(2-d)}}{|e^{-2z} + |e^z u - \theta|^2|^{d/2}} f(z, u) du dz.
\end{aligned}$$

This concludes the proof of the proposition. $\square$

It is important to note that the effect that Y being killed has on the potential of (ξ^{*-}, Ξ^{*-}) in the previous calculation, is only that of allowing the logarithm of the spatial transform to be well defined until its lifetime. Intuitively speaking, the descending ladder of $X^{(1)}$ is the same of $Y^{(1)}$, as long as both processes are positive. However, the killing plays a more complex role in the ascending ladder potential.

Before giving an explicit formula for the potential of (ξ^{*+}, Ξ^{*+}) , we present an example of how said potential can be used in unveiling fluctuations of X .

Proposition 3.16 (Point of vertical furthest reach before killing). *Let us define $\bar{G}(\tau_0^{(1)}) = \sup\{t < \tau_0^{(1)} : X_t^{(1)} = \bar{X}_{\tau_0^{(1)}}^{(1)}\}$, consider $x \in \mathbb{H}_{0\uparrow}$, $x_c = (\log x^{(1)}, x^{(2:d)}/x^{(1)})$ and $f : (0, \infty) \times \mathbb{R}^{d-1} \rightarrow \mathbb{R}$ continuous and bounded, then, up to a multiplicative constant,*

$$\mathbb{E}_x \left[f \left(X_{\bar{G}(\tau_0^{(1)})} \right) \right] = \int_{\{r > \log x^{(1)}\} \times \mathbb{R}^{d-1}} \mathcal{U}_{x_c}^{*+}(dr, d\phi) f(e^r, e^r \phi), \quad (3.21)$$

where $\mathcal{U}_{x_c}^{*+}(dr, d\phi) := \int_0^\infty \mathbf{P}_{x_c}^*(\xi_t^{*+} \in dr, \Xi_t^{*+} \in d\phi, t < \ell_\infty) dt$.

Proof. Let $x \in \mathbb{H}_{0\uparrow}$ and $\bar{g}_\infty = \lim_{t \rightarrow \infty} \sup\{u < t : \xi_u^* = \bar{\xi}_t^*\}$ the time of global maximum of ξ^* . We know $\bar{g}_\infty$ exists because $\lim_{t \rightarrow \infty} \xi_t^* = -\infty$, and this point will allow us to identify the point of furthest reach in a framework of excursions for the MAP. Consider $(\mathbf{n}_\theta^{*+} : \theta \in \mathbb{R}^{d-1})$ the measure of excursions from the maximum of ξ^* , according to

Section 0.4. Then, by the exit formula (0.15),

$$\begin{aligned}
\mathbb{E}_x \left[f \left(X_{\overline{G}(\tau_0^{(1)})} \right) \right] &= \mathbf{E}_{x_c}^* \left[f \left(e^{\xi_{\overline{g}\infty}^*}, e^{\xi_{\overline{g}\infty}^*} \Xi_{\overline{g}\infty}^* \right) \right] \\
&= \mathbf{E}_{x_c}^* \left[\sum_{g \in \overline{G}} f \left(e^{\xi_g^*}, e^{\xi_g^*} \Xi_g^* \right) \mathbf{1}_{\{\zeta_g = \infty\}} \right] \\
&= \mathbf{E}_{x_c}^* \left[\int_0^\infty d\ell_t f \left(e^{\xi_t^*}, e^{\xi_t^*} \Xi_t^* \right) \mathbf{n}_{\Xi_t^*}^{*+}(\zeta_t = \infty) \right] \\
&= \mathbf{E}_{x_c}^* \left[\int_0^{\ell_\infty} dt f \left(e^{\xi_{\ell_t^{-1}}^*}, e^{\xi_{\ell_t^{-1}}^*} \Xi_{\ell_t^{-1}}^* \right) \mathbf{n}^{*+}(\zeta = \infty) \right] \\
&= \mathbf{E}_{x_c}^* \left[\int_0^{\ell_\infty} dt f \left(e^{\xi_t^{*+}}, e^{\xi_t^{*+}} \Xi_t^{*+} \right) \mathbf{n}^{*+}(\zeta = \infty) \right] \\
&= \int_{\{r > \log x^{(1)}\} \times \mathbb{R}^{d-1}} \mathcal{U}_{x_c}^{*+}(dr, d\phi) f(e^r, e^r \phi), \tag{3.22}
\end{aligned}$$

where we are using the independence of the event $\{\zeta_t = \infty\}$ with the modulator Ξ_t^* , and excursion theory guarantees that $\mathbf{n}^{*+}(\zeta = \infty)$ is a constant, which we ‘absorb’ in the expression for the potential measure. $\square$

Now we will give an explicit expression for the ascending potential ladder measure of the killed cylindrical MAP (ξ^*, Ξ^*) .

Proposition 3.17. *Let $\theta \in \mathbb{R}^{d-1}$, and $f : (0, \infty) \times \mathbb{R}^{d-1} \rightarrow \mathbb{R}$ continuous and bounded. The potential measure of the ascending ladder MAP (ξ^{*+}, Ξ^{*+}) possesses a density with respect to Lebesgue measure $u_{0,\theta}^{*+}(r, \phi)$, $r \in (0, \infty)$, $\phi \in \mathbb{R}^{d-1}$, given in the following expression*

$$\begin{aligned}
&\int_0^\infty dt \mathbf{E}_{0,\theta}^* [f(\xi_t^{*+}, \Xi_t^{*+}); t < \ell_\infty] \\
&= \int_0^\infty \int_{\mathbb{R}^{d-1}} dr d\phi u_{0,\theta}^{*+}(r, \phi) f(r, \phi) \\
&= \int_0^\infty \int_{\mathbb{R}^{d-1}} dr d\phi \left[e^{r(d-1)} \frac{\Gamma(d/2)}{\pi^{d/2}} \sum_{k=-\infty}^\infty \frac{e^r - 1 + 2ke^r}{(|e^r \phi - \theta|^2 + (e^r - 1 + 2ke^r)^2)^{d/2}} \right] f(r, \phi).
\end{aligned}$$

Proof. We start by using the definition of (ξ^{*+}, Ξ^{*+}) , to then use the cylindrical

Lamperti-Kiu transform, obtaining

$$\begin{aligned} & \int_0^\infty dt \mathbf{E}_{0,\theta}^* \left[f(\xi_{\ell_t^{-1}}^*, \Xi_{\ell_t^{-1}}^*); t < \ell_\infty \right] \\ &= \int_0^\infty dt \mathbb{E}_{(1,\theta)} \left[f \left(\log |Y_{A(\ell_t^{-1})}^{(1)}|, \frac{Y_{A(\ell_t^{-1})}^{(2:d)}}{|Y_{A(\ell_t^{-1})}^{(1)}|} \right); A(\ell_t^{-1}) < \zeta \right]. \end{aligned}$$

Then, from Proposition 3.14 and the the last equation, this is

$$\int_0^\infty dt \mathbb{E}_{(1,\theta)} \left[f \left(\log |Y_{L_{e^t}^{-1}}^{(1)}|, \frac{Y_{L_{e^t}^{-1}}^{(2:d)}}{|Y_{L_{e^t}^{-1}}^{(1)}|} \right); L_{e^t}^{-1} < \zeta \right]. \quad (3.23)$$

Notice that the fluctuation above occurs before the killing of Y , so we will refer equivalently to the same fluctuation in terms of X . Now, we use that the event $\{L_{e^t}^{-1} < \zeta\}$ is equivalent to $\{\tau_{e^t}^{(1)} < \tau_{\mathbb{H}_0}\}$, where $\tau_{e^t}^{(1)} = \inf\{t > 0 : X_t^{(1)} = e^t\}$, and recalling that $\tau_{\mathbb{H}_0}$ depends only on $X^{(1)}$, we can develop (3.23) as follows

$$\begin{aligned} & \int_0^\infty dt \mathbb{E}_{(1,\theta)} \left[f \left(t, \frac{X_{L_{e^t}^{-1}}^{(2:d)}}{e^t} \right); \tau_{e^t}^{(1)} < \tau_0^{(1)} \right] \\ &= \int_0^\infty dt \mathbb{E}_{(1,\theta)} \left[f \left(t, \frac{X_{\tau_{e^t}^{(1)}}^{(2:d)}}{e^t} \right); \tau_{e^t}^{(1)} < \tau_0^{(1)} \right] \\ &= \int_0^\infty dt \int_0^\infty ds \mathbb{P}_1 \left(\tau_{e^t}^{(1)} \in ds; \tau_{e^t}^{(1)} < \tau_0^{(1)} \right) \mathbb{E}_\theta \left[f \left(t, \frac{X_s^{(2:d)}}{e^t} \right) \right], \end{aligned}$$

with the last equality given by the independence of $X^{(1)}$ and $X^{(2:d)}$. Now, using $p_t^{(d-1)}(y)$ to denote the transition kernel density of $(d-1)$ -dimensional Brownian motion, we have, by the explicit expression of $\mathbb{P}_1 \left(\tau_{e^t}^{(1)} \in ds; \tau_{e^t}^{(1)} < \tau_0^{(1)} \right)$ (Formula 3.0.6 of Part II. Chapter 1 of [10]),

$$\begin{aligned} & \int_0^\infty dt \int_0^\infty ds \int_{\mathbb{R}^{d-1}} dy \mathbb{P}_1 \left(\tau_{e^t}^{(1)} \in ds; \tau_{e^t}^{(1)} < \tau_0^{(1)} \right) p_s^{(d-1)}(y - \theta) f \left(t, \frac{y}{e^t} \right) \\ &= \int_0^\infty dt \int_0^\infty ds \int_{\mathbb{R}^{d-1}} dy s s_s(1, e^t - 0) p_s^{(d-1)}(y - \theta) f \left(t, \frac{y}{e^t} \right), \end{aligned}$$

where

$$ss_s(u, v) = \sum_{k=-\infty}^{\infty} \frac{v - u + 2kv}{\sqrt{2\pi}s^{3/2}} e^{-(v-u+2kv)^2/2s}, \quad u < v, s > 0.$$

Therefore, straightforward calculations lead to

$$\begin{aligned} & \int_0^\infty dt \int_{\mathbb{R}^{d-1}} d\phi e^{t(d-1)} \left[\int_0^\infty ds ss_s(1, e^t) p_s^{(d-1)}(e^t \phi - \theta) \right] f(t, \phi) \\ &= \int_0^\infty dt \int_{\mathbb{R}^{d-1}} d\phi e^{t(d-1)} \\ & \quad \left[\frac{\Gamma(d/2)}{\pi^{d/2}} \sum_{k=-\infty}^{\infty} \frac{e^t - 1 + 2ke^t}{(|e^t \phi - \theta|^2 + (e^t - 1 + 2ke^t)^2)^{d/2}} \right] f(t, \phi). \end{aligned} \quad (3.24)$$

This finishes the proof of the theorem. $\square$

3.3.3 Stable process conditioned to live in the upper half-space

Definition

From Subsection 2.3.3, we know that the h -conditioned stable process to stay in the upper half-space $\mathbb{H}_{0\uparrow}$, $(X, \mathbb{P}^\uparrow)$, is given by the change of measure

$$\left. \frac{d\mathbb{P}_x^\uparrow}{d\mathbb{P}_x} \right|_{\mathcal{F}_t} = \frac{(X_t^{(1)})^{\alpha/2}}{(x^{(1)})^{\alpha/2}} \mathbf{1}_{\{t < \tau_0^\vee\}}, \quad x \in \mathbb{H}_{0\uparrow}, t \geq 0, \quad (3.25)$$

It is not difficult to prove that $(X, \mathbb{P}^\uparrow)$ is a self-similar Markov process on $\mathbb{H}_{0\uparrow}$ and that there exists a cylindrical MAP $(\xi^\uparrow, \Xi^\uparrow)$ with laws $\mathbf{P}^\uparrow$, such that

$$\begin{aligned} \mathbb{E}_{(1, x^{(2:d)})}^\uparrow [f(X_t)] &= \mathbf{E}_{0, x^{(2:d)}}^\uparrow \left[f(e^{\xi_{\varphi(t)}^\uparrow}, e^{\xi_{\varphi(t)}^\uparrow} \Xi_{\varphi(t)}^\uparrow) \right] \\ &= \mathbf{E}_{0, x^{(2:d)}}^* \left[e^{\alpha/2 \xi_{\varphi(t)}^*} f(e^{\xi_{\varphi(t)}^*}, e^{\xi_{\varphi(t)}^*} \Xi_{\varphi(t)}^*) \mathbf{1}_{\{t < \zeta_{MAP}^*\}} \right]. \end{aligned} \quad (3.26)$$

Brownian motion conditioned to live in $\mathbb{H}_{0\uparrow}$

Let $(Y_t)_{t \geq 0} = (Y_t^{(1)}, Y_t^{(2:d)})_{t \geq 0}$, with laws $(\mathbb{P}_x : x \in \mathbb{R}^d)$, be a Brownian motion in $\mathbb{R}^d$, and $\tau_{\{0\}} = \inf\{t > 0 : Y_t^{(1)} = 0\}$.

Defining Brownian motion conditioned to live in the upper half-space is easy due to the independence of its coordinates. It is well known that the measure $\mathbb{P}^\uparrow$ given by

$$\left. \frac{d\mathbb{P}_{x^{(1)}}^\uparrow}{d\mathbb{P}_{x^{(1)}}} \right|_{\mathcal{F}_t} = \frac{Y_t^{(1)}}{x^{(1)}} \mathbf{1}_{\{t < \tau_{\{0\}}\}}, \quad x^{(1)} \in (0, \infty), t \geq 0, \quad (3.27)$$

is that of the Brownian motion $Y^{(1)}$ conditioned to be positive. Hence the change of measure (3.25) extends to $\alpha \in (0, 2]$. Moreover, it is known that $(Y^{(1)}, \mathbb{P}^\uparrow)$ has the law of a Bessel(3) (cf. Chapter VI of [49]).

3.3.4 Stable process conditioned to hit $\mathbb{H}_0$ from above

Definition

Consider $(X, \mathbb{P})$ the isotropic α -stable process. We proved in Subsection 2.3.1 that it is possible to condition the process to hit a set on $\mathbb{H}_0$ without visiting $\mathbb{H}_{0\downarrow}$, by equivalently conditioning on the event where the one-sided point of closest reach hits said set. The h -transform there is given by (2.58), hence the change of measure is

$$\frac{d\mathbb{P}_x^\downarrow}{d\mathbb{P}_x} \Big|_{\mathcal{F}_t} = \frac{(X_t^{(1)})^{\alpha/2-1}}{(x^{(1)})^{\alpha/2-1}} \mathbf{1}_{\{t < \tau_0^\vee\}}, \quad x \in \mathbb{H}_{0\uparrow}, t \geq 0. \quad (3.28)$$

Notice that the previous change of measure is trivial in the case $\alpha = 2$ and take us back to Brownian Motion killed on exiting $\mathbb{H}_{0\uparrow}$. This occurs because if started in $\mathbb{H}_{0\uparrow}$, Brownian Motion almost surely hits $\mathbb{H}_0$ after staying all its lifetime in $\mathbb{H}_{0\uparrow}$.

It is not difficult to prove that in the case $\alpha \in (0, 2)$, the h -transformed process (3.28) is a cylindrically positive self-similar Markov process and that it has a cylindrical Lamperti-Kiu representation.

On the cylindrical Williams' decomposition

Defining $\mathbb{P}^\uparrow$ and $\mathbb{P}^\downarrow$ suggests the possibility of applying a Williams' decomposition to the path of $(X, \mathbb{P})$, similar to the approach in [29]. Specifically, $\mathbb{P}$ can be described up to its (random) directional minimum in the vertical direction as $\mathbb{P}^\downarrow$, and as $\mathbb{P}^\uparrow$ after moving away from this global minimum. However, for the isotropic α -stable process in $\mathbb{H}_{0\uparrow}$, no such global minimum exists.

Nevertheless, if we lift the restriction to a positive first coordinate and return to the general cylindrical MAP $(\xi, (J, \Xi))$, it was noted in Corollary 3.7 that for $\alpha \in (0, 1)$, there exists a point of minimal distance to $\mathbb{H}_0$. This observation hints that the measure introduced in Section 2.3.2, which is adapted from [28], for X conditioned to hit $\mathbb{H}_0$, might be one of the key components needed to describe $\mathbb{P}$ via a Williams' decomposition.

3.3.5 Stable process reflected on $\mathbb{H}_0$

Definition

Let us define $Z = (Z_t : t \geq 0)$ the stable process reflected on $\mathbb{H}_0$, by the relationship

$$Z_t = (Z_t^{(1)}, Z_t^{(2:d)}) = (|X_t^{(1)}|, X_t^{(2:d)}) =: \mathbf{R}(X_t), \quad t < \tau_{\{0\}},$$

and $Z_t = \dagger$ for $t \geq \tau_{\{0\}}$, and equip this process with the laws of the isotropic α -stable Lévy process $(\mathbb{P}_x : x \in \mathbb{H}_{0\dagger})$.

Proving that Z has the scaling property is easy and similar to how we proved it for the killed process (Subsec. 3.3.2). One can show that if $Z^{[c]}$ is the process given by $Z_t^{[c]} = cZ_{c^{-\alpha}t}$ for $t \geq 0$ and some $c > 0$, then $\tau_{\{0\}}^{[c]} = \inf\{t > 0 : Z_t^{[c](1)} = 0\} = c^\alpha \tau_{\{0\}}$, and the proof follows by the scaling property of X .

For proving the Markov property we will appeal to the isotropy of the transition kernel of X , $p_t(x, y)$, which implies symmetry with respect to $\mathbb{H}_0$, i.e., for $x, y \in \mathbb{R}^d$

$$p_t((x^{(1)}, x^{(2:d)}), (y^{(1)}, y^{(2:d)})) = p_t((-x^{(1)}, x^{(2:d)}), (-y^{(1)}, y^{(2:d)})),$$

and notice that for $x \in \mathbb{H}_{0\dagger}$, $t \geq 0$ and any continuous bounded $f : \mathbb{H}_{0\dagger} \rightarrow \mathbb{R}$ we can define

$$\mathcal{Q}_t[f](x) := \int_{\mathbb{H}_{0\dagger}} [p_t(x, (y^{(1)}, y^{(2:d)})) + p_t(x, (-y^{(1)}, y^{(2:d)}))] f(y) dy,$$

and easily verify that

$$\mathcal{Q}_t[f](\mathbf{R}(x)) = \mathcal{P}_t[f \circ \mathbf{R}](x), \quad (3.29)$$

where $(\mathcal{P}_t)_{t \geq 0}$ denotes the semigroup of $(X, \mathbb{P})$. Hence, on account of $\mathbf{R}$ being a surjective function and by equation (3.29), Lemma 14.1 of [50] affirms that Z is a Markov process with semigroup $(\mathcal{Q}_t)_{t \geq 0}$. Moreover, by the continuity of $\mathbf{R}$, the right-continuity of X , and dominated convergence Z is a Feller process.

Therefore, as Z is a ssMp on E , there exists a cylindrical MAP (ξ^R, Ξ^R) such that for $t < \tau_{\{0\}}$

$$Z_t = e^{\xi_{\varphi(t)}^R} \times (1, \Xi_{\varphi(t)}^R).$$

Remark 3.18. *In the case $d = 1$, the process $(\xi_t^R)_{t \geq 0}$ is a Lévy process and by Proposition 5.20 of [35] it can be decomposed as follows*

- *It is equal in law to the sum of two independent Lévy processes ξ^L and ξ^{CR} ,*

- The Lévy process ξ^L has characteristic exponent

$$\Psi^*(z) - q^*, \quad z \in \mathbb{R},$$

where Ψ^* is given by

$$\Psi^*(z) = \frac{\Gamma(\alpha - iz)}{\Gamma(\alpha/2 - iz)} \frac{\Gamma(1 + iz)}{\Gamma(1 - \alpha/2 + iz)}, \quad z \in \mathbb{R},$$

and q^* is the killing rate of ξ^* , which by Subsection 3.3.2 is equal to $\Gamma(\alpha)/(\Gamma(\alpha/2)\Gamma(1 - \alpha/2))$. Said in another way, ξ^L is formed by removing the effect of independent killing from ξ^* .

- The process ξ^{C_R} is a compound Poisson process with jump rate q^* and jump distribution F^{C_R} on $\mathbb{R}$, given by its characteristic function,

$$\int_{\mathbb{R}} e^{i\theta x} F^{C_R}(dx) = \frac{\Gamma(1 + i\theta)\Gamma(\alpha - i\theta)}{\Gamma(\alpha)}.$$

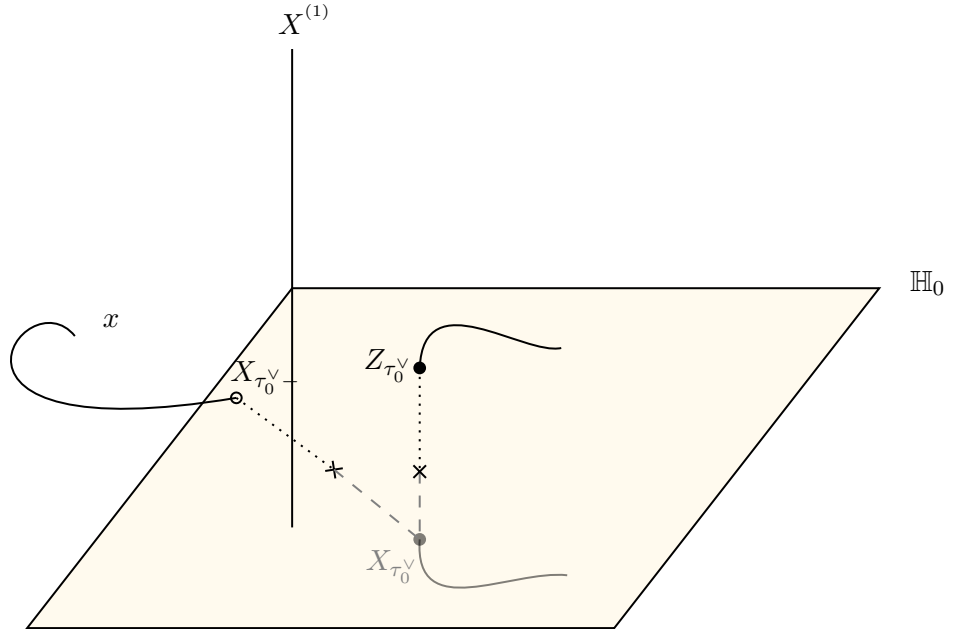


Figure 3-3: First moment of reflection for Z and its relationship with $(X_{\tau_0^\vee-}, X_{\tau_0^\vee})$

Characterisation

We can give a description of the paths of (ξ^R, Ξ^R) in a similar spirit to the decomposition in the previous remark, in terms of the killed MAP (ξ^*, Ξ^*) and additional jumps that occur exponentially often. Recall that $\tau_0^\vee = \inf\{t > 0 : X_t^{(1)} < 0\}$ is almost surely finite, then by the strong Markov property at $\tau_0^\vee$, one can deduce that it is sufficient to study the process $(Z_t : t \leq \tau_0^\vee)$. It is clear that the path section $(Z_t : t < \tau_0^\vee)$ agrees with $(X_t : t < \tau_0^\vee)$ and their cylindrical Lamperti-Kiu transforms coincide as well. Given that (ξ^*, Ξ^*) is the cylindrical MAP which describes X killed on exiting $\mathbb{H}_{0\uparrow}$, the dynamics of (ξ^R, Ξ^R) agree with those of (ξ^*, Ξ^*) up to, but not including, the moment of killing of the latter. Instead of being killed at rate q^* , (ξ^R, Ξ^R) experiences an additional jump $(\Delta_\xi^R, \Delta_\Xi^R)$. Let T be the first time the MAP experiences said jump, i.e. $(\Delta_\xi^R, \Delta_\Xi^R) = (\xi_T - \xi_{T-}, \Xi_T - \Xi_{T-})$, then we will

1. Prove that, conditionally on Ξ_{T-} , $(\Delta_\xi^R, \Delta_\Xi^R)$ is independent of $((\xi_t^R, \Xi_t^R) : t \leq T)$.
2. Find an explicit expression for the distribution of $(\Delta_\xi^R, \Delta_\Xi^R)$ by using the joint distribution of $(X_{\tau_0^\vee-}, X_{\tau_0^\vee})$ under $\mathbb{P}_x$, that we know explicitly.
3. Embed the previous jump into a killing-resurrecting representation of the Lévy system of (ξ^R, Ξ^R) .

In order to prove this we will require, the joint law of $(Z_{\tau_0^\vee-}, Z_{\tau_0^\vee})$ which can easily be identified from that of $(X_{\tau_0^\vee-}, X_{\tau_0^\vee})$, noting that $Z_{\tau_0^\vee}^{(1)} = -X_{\tau_0^\vee}^{(1)}$ and that $Z_{\tau_0^\vee}^{(2:d)} = X_{\tau_0^\vee}^{(2:d)}$, see Fig. 3-3. By the triple law (Thm.2.3), we have for $x, w, y \in \mathbb{H}_{0\uparrow}$ such that $w^{(1)} < x^{(1)}, w^{(1)} < y^{(1)}$ and $z \in \mathbb{H}_{0\downarrow}$

$$\begin{aligned}
& \mathbb{P}_x(X_{\tau_0^\vee-} \in dy, X_{\tau_0^\vee} \in dz) \\
&= dy dz \int_{\{w: 0 < w^{(1)} < y^{(1)} \wedge x^{(1)}\}} C \frac{|x^{(1)} - w^{(1)}|^{\alpha/2} |w^{(1)} - y^{(1)}|^{\alpha/2}}{|x - w|^d |w - y|^d} |y - z|^{-(\alpha+d)} dw \\
&=: f_0(y^{(1:d)}, z^{(1:d)}) dy dz.
\end{aligned} \tag{3.30}$$

Notice that for a test function and $h : \mathbb{R} \times \mathbb{R}^{d-1} \rightarrow \mathbb{R}$

$$\mathbb{E}_{x_c} [h(\Delta_\xi^R, \Delta_\Xi^R)] = \mathbb{E}_x \left[h \left(\log \frac{-X_{\tau_0^\vee}^{(1)}}{X_{\tau_0^\vee-}^{(1)}}, \frac{X_{\tau_0^\vee}^{(2:d)}}{-X_{\tau_0^\vee}^{(1)}} - \frac{X_{\tau_0^\vee-}^{(2:d)}}{X_{\tau_0^\vee-}^{(1)}} \right) \right],$$

hence, for $g : \mathbb{R} \times \mathbb{R}^{d-1} \rightarrow \mathbb{R}$ another test function, we are interested in proving that, conditionally on the value of $\frac{X_{\tau_0^\vee-}^{(2:d)}}{X_{\tau_0^\vee-}^{(1)}}$, the argument of h is independent of that of g in

the following expectation

$$\mathbb{E}_x \left[g(X_{\tau_0^\vee -}) h \left(\frac{-X_{\tau_0^\vee}^{(1)}}{X_{\tau_0^\vee -}^{(1)}}, \frac{X_{\tau_0^\vee}^{(2:d)}}{-X_{\tau_0^\vee}^{(1)}} - \frac{X_{\tau_0^\vee -}^{(2:d)}}{X_{\tau_0^\vee -}^{(1)}} \right) \right].$$

Thus, by equation (3.30),

$$\begin{aligned} & \mathbb{E}_x \left[g(X_{\tau_0^\vee -}) h \left(\frac{-X_{\tau_0^\vee}^{(1)}}{X_{\tau_0^\vee -}^{(1)}}, \frac{X_{\tau_0^\vee}^{(2:d)}}{-X_{\tau_0^\vee}^{(1)}} - \frac{X_{\tau_0^\vee -}^{(2:d)}}{X_{\tau_0^\vee -}^{(1)}} \right) \right] \\ &= \int_{z \in \mathbb{H}_{0\downarrow}} \int_{y \in \mathbb{H}_{0\uparrow}} g(y^{(1:d)}) h \left(\frac{-z^{(1)}}{y^{(1)}}, \frac{z^{(2:d)}}{-z^{(1)}} - \frac{y^{(2:d)}}{-y^{(1)}} \right) f_0(y^{(1:d)}, z^{(1:d)}) dy dz \\ &= \int_{s \in (0, \infty), \theta^{(2:d)} \in \mathbb{R}^{d-1}} \int_{y \in \mathbb{H}_{0\uparrow}} g(y^{(1:d)}) h(s, \theta^{(2:d)}) f_0(y^{(1:d)}, z^{(1:d)}) dy dz, \end{aligned}$$

with $s = -z^{(1)}/y^{(1)}$ and $\theta^{(2:d)} = \frac{z^{(2:d)}}{-z^{(1)}} - \frac{y^{(2:d)}}{y^{(1)}}$ and the integrals in the previous expression are taken as follow: $s \in \mathbb{R}$, $\theta \in \mathbb{R}^{d-1}$ and $y \in \mathbb{H}_{0\uparrow}$. Now notice that $dz^{(1)} = -y^{(1)}ds$ and $dz^{(2:d)} = (sy^{(1)})^{d-1}d\theta^{(2:d)}$, which all together imply

$$\begin{aligned} & \mathbb{E}_x \left[g(X_{\tau_0^\vee -}) h \left(\frac{-X_{\tau_0^\vee}^{(1)}}{X_{\tau_0^\vee -}^{(1)}}, \frac{X_{\tau_0^\vee}^{(2:d)}}{-X_{\tau_0^\vee}^{(1)}} - \frac{X_{\tau_0^\vee -}^{(2:d)}}{X_{\tau_0^\vee -}^{(1)}} \right) \right] \\ &= \int_{s \in (0, \infty), \theta^{(2:d)} \in \mathbb{R}^{d-1}} \int_{y \in \mathbb{H}_{0\uparrow}} g(y^{(1:d)}) h(s, \theta^{(2:d)}) \\ & \quad f_0 \left(y^{(1:d)}, y^{(1)}s \left(-1, \theta^{(2:d)} + \frac{y^{(2:d)}}{y^{(1)}} \right) \right) s^{d-1} (y^{(1)})^{d-1} (-y^{(1)}) dy ds d\theta^{(2:d)}, \end{aligned}$$

By analysing the expression (3.30), we observe that the only term depending on z is $|y - z|^{-(\alpha+d)}$, then substituting $z = y^{(1)}s \left(-1, \theta^{(2:d)} + \frac{y^{(2:d)}}{y^{(1)}} \right)$ we get

$$|y - z|^{-(\alpha+d)} = |y^{(1)}|^{-\alpha-d} \left| s \left(1, \theta^{(2:d)} + \frac{y^{(2:d)}}{y^{(1)}} \right) - \left(1, \frac{y^{(2:d)}}{y^{(1)}} \right) \right|^{-(\alpha+d)}.$$

This helps us to conclude that $\left(\frac{-X_{\tau_0^\vee}^{(1)}}{X_{\tau_0^\vee -}^{(1)}}, \frac{X_{\tau_0^\vee}^{(2:d)}}{-X_{\tau_0^\vee}^{(1)}} - \frac{X_{\tau_0^\vee -}^{(2:d)}}{X_{\tau_0^\vee -}^{(1)}} \right)$ depends on $(X_t : 0 \leq t < \tau_0^\vee)$

only through $\frac{X_{\tau_0^-}^{(2:d)}}{X_{\tau_0^-}^{(1)}}$, or in MAP terms, only through Ξ_{T-} . Moreover,

$$\begin{aligned} \mathbf{P}_{x_c} \left(\Delta_\xi^R \in ds, \Delta_\Xi^R \in d\theta^{(2:d)} \middle| \Xi_{T-} = \gamma^{(2:d)} \right) \\ \propto s^{d-1} |s(1, \theta^{(2:d)} + \gamma^{(2:d)}) - (1, \gamma^{(2:d)})|^{-(\alpha+d)} ds d\theta^{(2:d)}, \end{aligned} \quad (3.31)$$

and we will call resurrection jump kernel of the reflected MAP to the distribution

$$\mathcal{R}_{\gamma^{(2:d)}}^R(dy, d\theta^{(2:d)}) = \mathbf{P}_0 \left(\Delta_\xi^R \in dy, \Delta_\Xi^R \in d\theta^{(2:d)} \middle| \Xi_{T-} = \gamma^{(2:d)} \right), \quad \text{for } \gamma^{(2:d)} \in \mathbb{R}^{d-1}.$$

Lévy system in terms of killing and resurrecting

We will identify the elements of the Lévy system of (ξ^R, Ξ^R) in terms of the Lévy system of (ξ^*, Ξ^*) and the additional jumps studied in the previous part. Let us define $\mathbf{V}^*$ an additive functional of Ξ^* and $\mathcal{K}^*$ a jump measure such that for any bounded continuous function $f : (0, \infty) \times \mathbb{R} \times \mathbb{R}^{d-1} \times \mathbb{R}^{d-1} \rightarrow \mathbb{R}$, $f(\cdot, 0, \cdot, \cdot) = 0$,

$$\begin{aligned} \mathbf{E}_{x_c} \left[\sum_{t>0} f(t, \Delta \xi_t^*, \Xi_{t-}^*, \Xi_t^*) \mathbf{1}_{\{t < \zeta_{MAP}^*\}} \right] \\ = \mathbf{E}_{x_c} \left[\int_{(0, e_q^*) \times \mathbb{R}^{d-1}} \mathbf{V}_{x_c}^*(dt, d\theta) \int_{\mathbb{R} \times \mathbb{R}^{d-1}} \mathcal{K}_\theta^*(dy, d\phi) f(t, y, \theta, \phi) \right]. \end{aligned} \quad (3.32)$$

The existence of the pair $\mathbf{V}^*, \mathcal{K}^*$ is guaranteed by a similar calculation to that in Proposition 3.3.

Define $e_0^q = 0$, and let us label the times between the resurrection jumps of the MAP (ξ^R, Ξ^R) by $\{e_1^q, e_2^q, \dots\}$, i.e., e_1 corresponds to τ_0 in terms of the isotropic α -stable process and e_k^q to the difference between the times of k -th and $(k-1)$ -th crossings through $\mathbb{H}_0$. Due to the symmetry of $X^{(1)}$ we have that $\mathbf{P}_{x_c}(e_k < \infty) = 1$ for any $x_c \in \mathbb{R} \times \mathbb{R}^{d-1}$. Consider $x_c \in \mathbb{R} \times \mathbb{R}^{d-1}$ and a continuous and bounded function $f : [0, \infty) \times \mathbb{R} \times \mathbb{R}^{d-1} \times \mathbb{R}^{d-1} \rightarrow \mathbb{R}$, s.t. $f(\cdot, 0, \cdot, \cdot) = 0$, then by the Markov property and the conditional independence of the resurrecting jumps we can split the trajectories of

(ξ^R, Ξ^R) as follows

$$\begin{aligned}
& \mathbf{E}_{x_c} \left[\sum_{t>0} f(t, \Delta \xi_t^R, \Xi_{t-}^R, \Xi_t^R) \right] \\
&= \mathbf{E}_{x_c} \left[\sum_{k \geq 1} \int_{e_{k-1}^q}^{e_k^q} \int_{\mathbb{R}^{d-1}} \mathbf{V}_{\Xi_{e_{k-1}^q}^R}^* (ds, dx) \int_{\mathbb{R} \times \mathbb{R}^{d-1}} \mathcal{K}_x^* (dy, dz) f(s, y, x, z) \right. \\
&\quad \left. + \int_{\mathbb{R} \times \mathbb{R}^{d-1}} f(e_k^q, y_r, \Xi_{e_k^q}^R, \theta^{(2:d)}) \mathcal{R}_{\Xi_{e_k^q}^R}^R (dy_r, d\theta^{(2:d)}) \right]. \quad (3.33)
\end{aligned}$$

3.4 Censored stable and the first entry to the strip

Definition

We are interested in the half-space censored process, meaning that we want to re-sample the isotropic α -stable Lévy process $(X_t : t \geq 0)$ according to the clock

$$I(t) = \int_0^t \mathbf{1}_{\{\text{sgn}(X_s^{(1)})=1\}} ds = \int_0^t \mathbf{1}_{\{J_{\varphi(s)}=1\}} ds, \quad t \geq 0,$$

so that

$$\vec{\tilde{X}}_t = (\vec{\tilde{X}}_t^{(1)}, \vec{\tilde{X}}_t^{(2:d)}) := X_{\gamma(t)} = (X_{\gamma(t)}^{(1)}, X_{\gamma(t)}^{(2:d)}), \quad t \geq 0,$$

where $\{\gamma(t)\}_{t \geq 0}$ is the right-inverse of $\{I(t)\}_{t \geq 0}$, i.e., $\gamma(t) = \inf\{s > 0 : I(s) > t\}$ for $0 \leq t < \tau_{\mathbb{H}_0}$. In the following we will consider $\alpha \in (0, 1)$, so that $\tau_{\mathbb{H}_0} = \infty$ a.s.

We can see that $\vec{\tilde{X}}$ has the scaling property by noting that for $c > 0$ and setting $X_t^{[c]} = cX_{c^{-\alpha}t}$ and $\vec{\tilde{X}}_t^{[c]} = c \vec{\tilde{X}}_{c^{-\alpha}t}$ for $t \geq 0$,

$$\begin{aligned}
\gamma^{[c]}(t) &:= \inf\{s > 0 : \int_0^s \mathbf{1}_{\{cX_{c^{-\alpha}u}^{(1)} > 0\}} du > t\} \\
&= c^\alpha \inf\{c^{-\alpha}s > 0 : \int_0^{c^{-\alpha}s} \mathbf{1}_{\{X_w^{(1)} > 0\}} dw > c^{-\alpha}t\} \\
&= c^\alpha \gamma(c^{-\alpha}t).
\end{aligned}$$

Hence, for bounded and measurable $f : \mathbb{H}_{0\uparrow} \rightarrow \mathbb{R}$,

$$\begin{aligned}
\mathbb{E}_{cx} \left[f(\tilde{X}_t) \right] &= \mathbb{E}_{cx} \left[f(X_{\gamma(t)}) \right] \\
&= \mathbb{E}_x \left[f(X_{\gamma^{[c]}(t)}^{[c]}) \right] \\
&= \mathbb{E}_x \left[f(cX_{c^{-\alpha}\gamma^{[c]}(t)}) \right] \\
&= \mathbb{E}_x \left[f(cX_{\gamma(c^{-\alpha}t)}) \right] \\
&= \mathbb{E}_x \left[f(c\tilde{X}_{c^{-\alpha}t}) \right] \\
&= \mathbb{E}_x \left[f(\tilde{X}_t^{[c]}) \right].
\end{aligned}$$

Also notice that the Markov property of $\tilde{X}$ follows by Section III.21 of [50] and noting that $\{I(t)\}_{t \geq 0}$ is a perfect continuous homogeneous additive functional. Moreover, due to $\mathbb{P}_x(\lim_{t \downarrow 0} \gamma(t) = 0) = 1$ for any $x \in \mathbb{H}_{0\uparrow}$, the right continuity of X and dominated convergence, $\tilde{X}$ is a Feller process. Therefore, by Theorem 3.1 we can find a MAP $(\tilde{\xi}, \tilde{\Xi})$, for which

$$\tilde{X}_t = (\tilde{X}_t^{(1)}, \tilde{X}_t^{(2:d)}) = e^{\tilde{\xi} \tilde{\varphi}_t} (1, \tilde{\Xi}_{\tilde{\varphi}_t}), \quad t \geq 0, \quad (3.34)$$

where $\tilde{\varphi}_t = \inf\{s > 0 : \int_0^s e^{\alpha \tilde{\xi}_u} du > t\}$. We call the process $(\tilde{\xi}, \tilde{\Xi})$ the censored cylindrical MAP.

Remark 3.19. *In the case $d = 1$, Theorem 5.16 in [35] guarantees that $\tilde{\xi}$ is a Lévy process equal in law to the sum of two independent Lévy processes ξ^L and ξ^C .*

1. *The Lévy process ξ^L is that of Remark 3.18, i.e., the process that results from stripping away the killing of the Lamperti transform of $X^{(1)}$ killed on exiting $(0, \infty)$.*
2. *The process ξ^C is a compound Poisson process with jump rate $q^* = \Gamma(\alpha)/(\Gamma(\alpha/2)\Gamma(1-\alpha/2))$ and jump distribution F^c on $\mathbb{R}$, given by its characteristic function,*

$$\int_{\mathbb{R}} e^{i\theta x} F^C(dx) = \frac{\sin(\pi\alpha/2)}{\pi\Gamma(\alpha)} \Gamma(1-\alpha/2+i\theta) \gamma(\alpha/2-i\theta) \Gamma(1+i\theta) \Gamma(\alpha-i\theta), \quad \theta \in \mathbb{R}.$$

Characterisation

Unlike the one-dimensional censored process studied in [36], we cannot characterise the whole censored MAP $(\tilde{\xi}, \tilde{\Xi})$ with a single characteristic exponent. Nonetheless, we

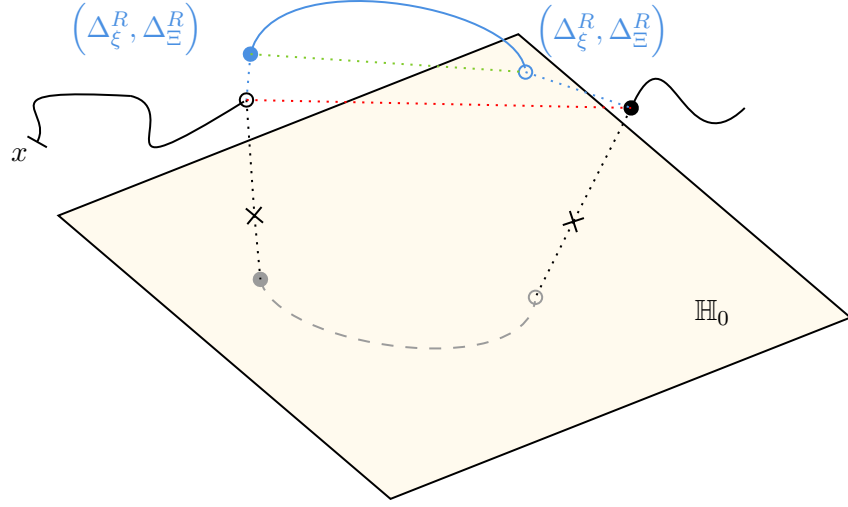


Figure 3-4: In solid black line, a typical trajectory of the censored process $\tilde{X}$. The first resurrection jump is denoted by the dotted red line. In dashed grey line, the ‘censored’ part of the trajectory and in solid blue its reflection. The dotted blue lines represent, up to a change of variables, the resurrection jumps of the reflected process. In dotted green line the difference whilst the reflected process does not experiment a resurrection jump.

can describe this censored process pathwise as we did for the stable process reflected on $\mathbb{H}_0$. Applying the strong Markov property at $\tau_0^\vee$, it is sufficient to study the process $(\tilde{X}_t : t \leq \tau_0^\vee)$. It is clear that the path section $(\tilde{X}_t : t < \tau_0^\vee)$ agrees with $(X_t : t < \tau_0^\vee)$; however, rather than being killed at time $\tau_0^\vee$, the process $\tilde{X}$ experiences a jump to a state in $\mathbb{H}_{0\uparrow}$. Recalling that (ξ^*, Ξ^*) is the cylindrical MAP that describes the process X killed on exiting $\mathbb{H}_{0\uparrow}$, it follows that the dynamics of $(\tilde{\xi}, \tilde{\Xi})$ agree with those of (ξ^*, Ξ^*) up to, but not including, the moment of killing of the latter. Instead of being killed at rate q^* , $(\tilde{\xi}, \tilde{\Xi})$ experiences an additional jump $(\Delta_{\tilde{\xi}}, \Delta_{\tilde{\Xi}})$. Let T be the first time the MAP experiences said jump, i.e. $(\Delta_{\tilde{\xi}}, \Delta_{\tilde{\Xi}}) = (\tilde{\xi}_T - \tilde{\xi}_{T-}, \tilde{\Xi}_T - \tilde{\Xi}_{T-})$. Similarly to the reflected case, we have for $x \in \mathbb{H}_{0\uparrow}$ and $x_c = (\log x^{(1)}, x^{(2:d)}/x^{(1)})$,

$$\mathbf{E}_{x_c} \left[f(\Delta_{\tilde{\xi}}, \Delta_{\tilde{\Xi}}) \right] = \mathbb{E}_x \left[f \left(\log \left(\frac{X_{\sigma_0^\wedge}^{(1)}}{X_{\tau_0^\vee-}^{(1)}} \right), \frac{X_{\sigma_0^\wedge}^{(2:d)}}{X_{\tau_0^\vee-}^{(1)}} - \frac{X_{\tau_0^\vee-}^{(2:d)}}{X_{\tau_0^\vee-}^{(1)}} \right) \right], \quad (3.35)$$

where $\sigma_0^\wedge = \inf\{s > \tau_0^\vee : X^{(1)} > 0\}$. The explicit expression for the joint distribution of $X_{\tau_0^\vee-}$ and $X_{\sigma_0^\wedge}$ could be obtained by using the Markov property and the triple law

(Theorem 2.3), viz.

$$\mathbb{P}_x(X_{\tau_0^\vee-} \in dy, X_{\sigma_0^\wedge} \in dw) = \int_{z \in \mathbb{H}_{0\downarrow}} \mathbb{P}_x(X_{\tau_0^\vee-} \in dy, X_{\tau_0^\vee} \in dz) \mathbb{P}_z(X_{\tau_0^\wedge} \in dw). \quad (3.36)$$

However, to prove that these jumps are conditionally (on $\tilde{\Xi}_{T-}$) independent of the past, and to fully characterise the distribution of the jumps is more complicated than in the reflected MAP case.

To elaborate on the conditional independence, notice that the jump $(\Delta_{\tilde{\xi}}, \Delta_{\tilde{\Xi}})$ is equal in distribution to the addition of three jumps. See Figure 3-4 for reference. The first jump is $(\Delta_{\xi}^R, \Delta_{\Xi}^R)$, the second is equal in distribution to $(\xi_{e_{q^*}}^*, \Xi_{e_{q^*}}^*)$ under $\mathbf{P}_0$, and the last one is, by symmetry, again $(\Delta_{\xi}^R, \Delta_{\Xi}^R)$. Given $\tilde{\Xi}_{T-}$, the addition of these jumps is conditionally independent of the starting point x_c .

In principle, an explicit expression for the distribution of the resurrection jump $(\Delta_{\tilde{\xi}}, \Delta_{\tilde{\Xi}})$ could be obtained by substituting the known formulas into equation (3.36), and it could be possible to exhibit the conditional independence previously mentioned. Yet, this d -dimensional integral takes the following form, which we were not able to solve if $d \geq 2$,

$$dw f(x, y) (w^{(1)})^{-\alpha/2} \int_{z \in \mathbb{H}_{0\downarrow}} |y - z|^{-\alpha-d} (-z^{(1)})^{\alpha/2} |z - w|^{-d} dz, \quad (3.37)$$

where, from Lemma 2.4,

$$f(x, y) = B_{\alpha,d} C_{\alpha,d} \left(\left| \left(1, \frac{x^{(2;d)}}{x^{(1)}} \right) - y \right|^{\alpha-d} \int_0^{\frac{4y^{(1)}}{\left| \left(1, \frac{x^{(2;d)}}{x^{(1)}} \right) - y \right|^2}} (u+1)^{-d/2} u^{\alpha/2-1} du \right),$$

$B_{\alpha,d} = \frac{\Gamma((d+\alpha)/2)\Gamma(d/2)}{\pi^d |\Gamma(\alpha/2)^2 \Gamma(-\alpha/2)|}$ and $C_{\alpha,d} = \pi^{-(d/2+1)} \Gamma(d/2) \sin(\alpha\pi/2)$. Notice, however, that the integral in expression (3.37) is finite as it is bounded by

$$\begin{aligned} & \int_{\{z^{(1)} < 0\}} |y^{(1)} - z^{(1)}|^{-\alpha-1} (-z^{(1)})^{\alpha/2} |z^{(1)} - w^{(1)}|^{-1} dz^{(1)} \\ & \propto (y^{(1)})^{-\alpha/2-1} {}_2F_1\left(1, 1 + \alpha/2; 2 + \alpha; 1 - w^{(1)}/y^{(1)}\right) < \infty, \end{aligned}$$

where we recall that ${}_2F_1$ is the Gauss-hypergeometric function.

Fluctuation identities

Analogously to the Lamperti-stable cylindrical MAP and the killed cylindrical MAP, we can develop an excursion theory from the minimum of $\tilde{\xi}$ to exhibit some connections

between this and fluctuation identities of the isotropic α -stable Lévy process. If $\tilde{L} = (\tilde{L}_t : t \geq 0)$ represents the local time at the minimum for $\tilde{\xi}$ we will define the descending ladder MAP as

$$(\tilde{\xi}_t^-, \tilde{\Xi}_t^-) = (\tilde{\xi}_{\tilde{L}_t^{-1}}, \tilde{\Xi}_{\tilde{L}_t^{-1}}), \quad t \geq 0, \quad (3.38)$$

and by

$$(\epsilon(s), \tilde{\Xi}^\epsilon(s) : s \in [0, \zeta])$$

the canonical excursion from the minimum of $\tilde{\xi}$, with lifetime ζ and excursion measures $(\tilde{\mathbf{n}}_\phi^- : \phi \in \mathbb{R}^{d-1})$. See Section 0.4 for reference.

Corollary 3.20 (First entry to the strip). *For continuous and bounded f , $f_e(x) = f(e^{x^{(1)}}, e^{x^{(1)}} x^{(2:d)})$, $x \in (1, \infty) \times \mathbb{R}^{d-1}$ and $x_c = (\log x^{(1)}, x^{(2:d)}/x^{(1)})$*

$$\begin{aligned} \mathbb{E}_x[f(\tilde{X}_{\tau_{(0,1)}}); \tau_{(0,1)} < \infty] &= \mathbb{E}_x[f(X_{\tau_{(0,1)}}); \tau_{(0,1)} < \infty] \\ &= \mathbf{E}_{x_c} \left[f \left(e^{\tilde{\xi}_{T_0^-}}, e^{\tilde{\xi}_{T_0^-}} \tilde{\Xi}_{T_0^-} \right); T_0^- < \infty \right] \\ &= \int_{(y,z) \in (-\infty, -r) \times \mathbb{R}^{d-1}} \int_{(r,\phi) \in (0, \log x^{(1)}) \times \mathbb{R}^{d-1}} \tilde{U}_{x_c}^-(dr, d\phi) \tilde{\Pi}_\phi^-(dy, dz) f_e(r+y, z), \end{aligned}$$

where

$$\tilde{U}_{x_c}^-(dr, d\phi) = \int_0^\infty \mathbf{P}_{x_c}(\tilde{\xi}_t^- \in dr, \tilde{\Xi}_t^- \in d\phi, t < \tilde{L}_\infty), \quad r < \log x^{(1)}, \phi \in \mathbb{R}^{d-1}$$

and

$$\tilde{\Pi}_\phi^-(dy, dz) = \tilde{\mathbf{n}}_\phi^-(\epsilon(\zeta) \in dy, \tilde{\Xi}^\epsilon(\zeta) \in dz, \zeta < \infty).$$

Corollary 3.21 (Positive point of closest reach). *Due to the transience of $\tilde{X}^{(1)}$ when $\alpha \in (0, 1)$, there exists a time of global minimum of the censored MAP. Let us denote this time by $\tilde{g}_\infty$ and its respective ‘instant of minimal vertical distance’ for $\tilde{X}$, $\tilde{G}(\infty)$.*

There is a relationship between the point of closest reach to $\mathbb{H}_0$ of the uncensored isotropic α -stable process X , and the global minimum $\tilde{g}_\infty$. In fact $X_{G(\infty)} = \tilde{X}_{\tilde{G}(\infty)} = e^{\tilde{\xi}_{\tilde{g}_\infty}}(1, \tilde{\Xi}_{\tilde{g}_\infty})$ only if $X_{G(\infty)}^{(1)} > 0$. We call $\tilde{X}_{\tilde{G}(\infty)}$ the positive point of closest reach to the hyperplane $\mathbb{H}_0$.

Remark 3.22. We have the following fluctuation identity, for $x \in (0, \infty) \times \mathbb{R}^{d-1}$,

$$\begin{aligned}
\mathbb{E}_x \left[f \left(\tilde{X}_{\tilde{G}(\infty)} \right) \right] &= \mathbf{E}_{x_c} \left[f(e^{\tilde{\xi}_{\tilde{g}_\infty}}, e^{\tilde{\xi}_{\tilde{g}_\infty}} \tilde{\Xi}_{\tilde{g}_\infty}) \right] \\
&= \mathbf{E}_{x_c} \left[\sum_{g \in G} \mathbf{1}(\tilde{\zeta}_g = \infty) f(e^{\tilde{\xi}_g}, e^{\tilde{\xi}_g} \tilde{\Xi}_g) \right] \\
&= \mathbf{E}_{x_c} \left[\int_0^\infty d\tilde{L}_t f(e^{\tilde{\xi}_t}, e^{\tilde{\xi}_t} \tilde{\Xi}_t) \tilde{\mathbf{n}}_{\tilde{\Xi}_t}^- (\tilde{\zeta} = \infty) \right] \\
&= \mathbf{E}_{x_c} \left[\int_0^{\tilde{L}^\infty} dt f(e^{-\tilde{H}_t}, e^{-\tilde{H}_t} \tilde{\Xi}_t^-) \tilde{\mathbf{n}}^- (\tilde{\zeta} = \infty) \right] \\
&\propto \int_{-\infty < r < \log |x^{(1)}| \times \mathcal{S}} \tilde{U}_{x_c}^- (dr, d\theta) f(e^r, e^r \theta).
\end{aligned}$$

Corollary 3.23 (Potential killed at entering the strip). For $x \in (1, \infty) \times \mathbb{R}^{d-1}$ and f a continuous bounded function supported on $(0, \infty) \times \mathbb{R}^{d-1}$

$$\begin{aligned}
\mathbb{E}_x \left[\int_0^{\tau_{(0,1)}} f(\tilde{X}_t) dt \right] &= \mathbf{E}_{x_c} \left[\int_0^{T_0^-} f(e^{\tilde{\xi}_t}, e^{\tilde{\xi}_t} \tilde{\Xi}_t) dt \right] \\
&= \int_{(r, \phi) \in (0, \log x^{(1)}) \times \mathbb{R}^{d-1}} \tilde{U}_{x_c}^- (dr, d\phi) \tilde{\mathbf{n}}_\phi^- \left(\int_0^\zeta f_e(r + \epsilon(s), \tilde{\Xi}^\epsilon(s)) ds \right). \quad (3.39)
\end{aligned}$$

We will deep dive in how we would attempt to use the recently developed theory of MAPs (cf. [29]) for finding explicit expressions to the right-hand side of equation (3.39). There are a number of important facts that will help us to investigate $\tilde{X}$. The first fact is that the ascending ladder MAP properties of $\tilde{X}$ agree with those of X . The second fact of interest is that the behaviour of the occupation of the excursions from the ascending ladder process tells us something about the descending ladder MAP via duality.

3.4.1 Ascending behaviour of the censored MAP

The ascending ladder invariant measure $\tilde{\pi}^+$

Consider $\tilde{L}^*$ the local time at the maximum of $(\tilde{\xi}, \tilde{\Xi})$ and let us denote by $(\tilde{\xi}^+, \tilde{\Xi}^+)$ to the process given by

$$(\tilde{\xi}^+, \tilde{\Xi}^+) = (\tilde{\xi}_{\tilde{L}_t^{*-1}}, \tilde{\Xi}_{\tilde{L}_t^{*-1}}), \quad t \geq 0.$$

We call this process the ascending ladder MAP for $\tilde{X}$, and recall that it has the same range as the ascending ladder MAP of X , also written (ξ^+, Ξ^+) and given in equation (3.7). We are interested in the invariant measure of the modulator of the former, $\tilde{\pi}^+$, which is the same as the invariant measure of the latter, π^+ . We note that both of these probability measures are supported on $\mathbb{R}^{d-1}$.

Theorem 3.24. *If we parameterise the support space of $\tilde{\pi}^+$ in polar coordinates so that $\theta \in \text{supp } \tilde{\pi}^+$ is written $\theta = y\phi$, then*

$$\tilde{\pi}^+(dy, d\phi) = \frac{2\Gamma(d/2)}{\Gamma((d-1)/2)} \frac{y^{d-2}}{(1+y^2)^{d/2}} \sigma_1^{(d-2)}(d\phi) dy, \quad y > 0, \phi \in \mathbb{S}^{d-2}$$

which is otherwise written as

$$\tilde{\pi}^+(dx) = \pi^{-(\frac{d-1}{2})} \Gamma(d/2) \frac{dx}{(1+|x|^2)^{d/2}}, \quad x \in \mathbb{R}^{d-1}$$

where $\sigma_1^{(d-2)}$ is the uniform measure on $\mathbb{S}^{d-2}$ normalised to have unit mass.

Proof. To calculate π^+ and hence $\tilde{\pi}^+$, let us start with the following observations. Let us denote by $\tau_z^\wedge = \inf\{t > 0 : X_t^{(1)} > z\}$, for $z > 0$. Self-similarity tells us that

$$\frac{X_{\tau_z^\wedge}^{(2:d)}}{X_{\tau_z^\wedge}^{(1)}} \stackrel{d}{=} \frac{X_{\tau_1^\wedge}^{(2:d)}}{X_{\tau_1^\wedge}^{(1)}} \quad \text{for all } z > 0.$$

As such, it is quite clear that the ratio above on the left converges in distribution to the ratio above on the right.

Next we observe that for any $x \in \mathbb{R}^d$ such that $x^{(1)} > 0$, we have for $z > x^{(1)}$ and non-negative, bounded measurable f , that

$$\lim_{|x| \rightarrow 0} \mathbb{E}_x \left[f \left(\frac{X_{\tau_1^\wedge}^{(2:d)}}{X_{\tau_1^\wedge}^{(1)}} \right) \right] = \lim_{|x| \rightarrow 0} \mathbf{E}_{0, x^{(2:d)}/x^{(1)}} \left[f(\Xi_{T_{-\log x^{(1)}}}^+) \right]$$

where $T_z^+ = \inf\{t > 0 : \xi_t^+ > z\}$. Taking into account the fact that the process $\Xi_{T_{-\log x^{(1)}}}^+$, $x > 0$, is the same as taking the limit of $f(\Xi_x)$, $x > 0$, we conclude that

$$\mathbb{E}_0 \left[f \left(\frac{X_{\tau_1^\wedge}^{(2:d)}}{X_{\tau_1^\wedge}^{(1)}} \right) \right] = \int_{\mathbb{R}^{d-1}} f(\theta) \pi^+(d\theta).$$

From Lemma 2.5 we know that

$$\mathbb{P}_0(X_{\tau_1^\wedge} \in dx) = C_{\alpha,d} \frac{|x|^{-d}}{(x^{(1)} - 1)^{\alpha/2}} dx, \quad x^{(1)} > 1,$$

where $C_{\alpha,d} = \pi^{-(d/2+1)} \Gamma(d/2) \sin(\alpha\pi/2)$. Hence

$$\mathbb{E}_0 \left[f \left(\frac{X_{\tau_1^\wedge}^{(2:d)}}{X_{\tau_1^\wedge}^{(1)}} \right) \right] = C_{\alpha,d} \int_1^\infty \int_{\mathbb{R}^{d-1}} f(x^{(2:d)}/x^{(1)}) \frac{|(x^{(1)})^2 + |x^{(2:d)}|^2|^{-d/2}}{(x^{(1)} - 1)^{\alpha/2}} dx^{(1)} dx^{(2:d)}.$$

Now write $z^{(2:d)} = x^{(2:d)}/x^{(1)}$ and $dx^{(2:d)} = (x^{(1)})^{d-1} dz^{(2:d)}$. We now have that

$$\begin{aligned} \mathbb{E}_0 \left[f \left(\frac{X_{\tau_1^\wedge}^{(2:d)}}{X_{\tau_1^\wedge}^{(1)}} \right) \right] &= C_{\alpha,d} \int_{\mathbb{R}^{d-1}} f(z^{(2:d)}) |1 + |z^{(2:d)}|^2|^{-d/2} dz^{(2:d)} \int_1^\infty \frac{dx^{(1)}}{(x^{(1)})(x^{(1)} - 1)^{\alpha/2}} \\ &= C_{\alpha,d}/C_{\alpha,1} \int_{\mathbb{R}^{d-1}} f(z^{(2:d)}) |1 + |z^{(2:d)}|^2|^{-d/2} dz^{(2:d)}, \end{aligned}$$

from which the statement of the theorem follows. $\square$

The invariant measure $\tilde{\pi}$

Theorem 3.25. *If we parameterise the support space of $\tilde{\pi}$ in polar coordinates so that $\theta \in \text{supp } \tilde{\pi}$ is written $\theta = y\phi$, then*

$$\tilde{\pi}(dy, d\phi) = \frac{2\Gamma(d/2)}{\Gamma((d-1)/2)} \frac{y^{d-2}}{(1+y^2)^{d/2}} \sigma_1^{(d-2)}(d\phi) dy, \quad y \geq 0, \phi \in \mathbb{S}^{d-2},$$

which is also written as

$$\tilde{\pi}(dx) = \pi^{-(\frac{d-1}{2})} \Gamma(d/2) \frac{dx}{(1+|x|^2)^{d/2}}, \quad x \in \mathbb{R}^{d-1}.$$

Proof. We appeal again to self-similarity and note that

$$\frac{\tilde{X}_t^{(2:d)}}{\tilde{X}_t^{(1)}} \stackrel{d}{=} \frac{\tilde{X}_1^{(2:d)}}{\tilde{X}_1^{(1)}}$$

which tells us that there is a distributional limit for $\tilde{X}_t^{(2:d)}/\tilde{X}_t^{(1)}$ as $t \rightarrow \infty$. On the

other hand, we also recognise that for $t \geq 0$

$$\frac{X_{\gamma_t}^{(2:d)}}{|X_{\gamma_t}^{(1)}|} \stackrel{d}{=} \frac{\tilde{X}_t^{(2:d)}}{\tilde{X}_t^{(1)}}.$$

Hence the strategy of our proof is to relate the desired stationary distribution to that of $X_t^{(2:d)}/|X_t^{(1)}|$, appealing to isotropy and time change with the help of Revuz measures (cf. [54]).

To this end, let us start by considering the stationary distribution of $X_t^{(2:d)}/|X_t^{(1)}|$ on the upper half-space under $\mathbb{P}_0$. With a now familiar argument, we have that self-similarity implies that the $X_t^{(2:d)}/|X_t^{(1)}|$ is equal in distribution to $X_1^{(2:d)}/|X_1^{(1)}|$ for all $t \geq 0$ and hence the law of the latter represents the stationary distribution π of the former. Isotropy dictates that $\arg(X_1)$ must be uniformly distributed on $\mathbb{S}^{d-1}$. As such, if we call φ the angle to the $x^{(1)}$ -axis and write in spherical polar coordinates in $X_1^{(2:d)} = Y\Theta$, where $Y \geq 0$ and $\Theta \in \mathbb{S}^{d-2}$, then, in addition Θ being uniformly distributed, the spherical polar coordinates (φ, Θ) represent $\arg(X_1)$. Moreover,

$$\text{sgn}(X_1^{(1)}) = 2\mathbf{1}_{\{\varphi \in [0, \pi/2)\}} - 1 \quad \text{and} \quad \tan \varphi = \frac{Y_t}{X_t^{(1)}}.$$

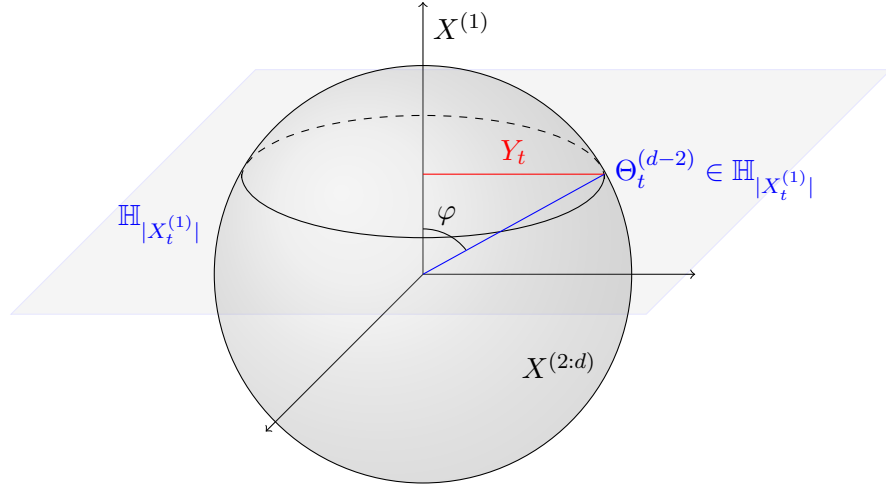


Figure 3-5: Polar decomposition on the hyperplane.

Classical calculations for spherical polar coordinates, cf. [5], tells us that

$$\mathbb{P}_0((\varphi, \Theta) \in (d\vartheta, d\phi)) = \left(\frac{2\pi^{(d-1)/2}}{\Gamma((d-1)/2)} \right) \sin^{d-2} \vartheta \sigma_1^{(d-2)}(d\phi) d\vartheta, \quad \vartheta \in [0, \pi), \phi \in \mathbb{S}^{d-2}.$$

Hence, we have that the stationary distribution of $X_1^{(2:d)}/|X_1^{(1)}|$, seen as $\left(\frac{Y}{|X_1^{(1)}|}, \Theta\right)$, is given by

$$\begin{aligned} & \int_0^\infty \int_{\mathbb{S}^{d-2}} f(y, \phi) \pi(j, dy, d\phi) \\ &= \mathbb{E}_0[f(|\tan \varphi|, \Theta)] \\ &= \int_0^\pi \int_{\mathbb{S}^{d-2}} f(\tan \vartheta, \phi) \left(\frac{2\pi^{(d-1)/2}}{\Gamma((d-1)/2)} \right) \sin^{d-2} \vartheta \sigma_1^{(d-2)}(d\phi) d\vartheta \\ &= \int_0^\infty \int_{\mathbb{S}^{d-2}} f(y, \phi) \left(\frac{2\pi^{(d-1)/2}}{\Gamma((d-1)/2)} \right) \frac{y^{d-2}}{(1+y^2)^{d/2}} \sigma_1^{(d-2)}(d\phi) dy, \end{aligned}$$

where we have written $y = \tan \vartheta$ and hence $d\vartheta = dy/(1+y^2)$, moreover, we have used $\sin \vartheta = y \cos \vartheta$ and $\cos \vartheta^2 = 1/(1 + \tan^2 \vartheta) = 1/(1 + y^2)$.

Technically speaking, we can extend the definition of π to give us the stationary distribution of the pair $(J_1, X_1^{(2:d)}/|X_1^{(1)}|)$ appealing to the earlier observation that $J_1 = \text{sgn}(X_1^{(1)}) = 2\mathbf{1}_{\{\varphi \in [0, \pi/2)\}} - 1$. Indeed, using obvious notation and symmetry, this tells us that

$$\begin{aligned} & \sum_{j=\pm 1} \int_0^\infty \int_{\mathbb{S}^{d-2}} f(j, y, \phi) \pi(j, dy, d\phi) \\ &= \int_0^\infty \int_{\mathbb{S}^{d-2}} [f(1, y, \phi) + f(-1, y, \phi)] \frac{C_{d-1} y^{d-2}}{2(1+y^2)^{d/2}} \sigma_1^{(d-2)}(d\phi) dy, \end{aligned}$$

where $C_{d-1} = \frac{2\pi^{(d-1)/2}}{\Gamma((d-1)/2)}$. Next, we can appeal to pp. 240-241 of [54] which gives the form of invariant measures for time changed Markov processes by appealing to Revuz measures. The invariant measure of $\tilde{\tilde{X}}_t^{(2:d)}/\tilde{\tilde{X}}_t^{(1)}$ is given by

$$\begin{aligned} \int_0^\infty \int_{\mathbb{S}^{(d-2)}} f(y, \phi) \tilde{\tilde{\pi}}'(dy, d\phi) &:= \lim_{t \rightarrow 0} \frac{1}{t} \mathbb{E}_\pi \left[\int_0^t f(Y_s, \Theta_s) \mathbf{1}_{\{J_s=1\}} ds \right] \\ &= \int_0^\infty \int_{\mathbb{S}^{d-2}} f(y, \phi) \frac{C_{d-1} y^{d-2}}{2(1+y^2)^{d/2}} \sigma_1^{(d-2)}(d\phi) dy \end{aligned}$$

where we are writing $X_t^{(2:d)} = (Y_t, \Theta_t) \in [0, \infty) \times \mathbb{S}^{(d-2)}$ in spherical polar coordinates for $\mathbb{R}^{d-1}$.

Finally, we need to move from the stationary distribution of $\tilde{\tilde{X}}_t^{(2:d)}/\tilde{\tilde{X}}_t^{(1)}$ to that of $\tilde{\tilde{\Xi}}_t$, which we will do by again appealing to an associated Revuz measure since $\tilde{\tilde{\Xi}}_t =$

$\tilde{X}_{\tilde{\varphi}_t^{-1}}^{(2:d)} / | \tilde{X}_{\tilde{\varphi}_t^{-1}}^{(1)} |$. Note, that from the definition of $\tilde{\varphi}$ we can easily deduce that

$$\tilde{\varphi}_t^{-1} = \inf\{s > 0 : \int_0^s | \tilde{X}_u^{(1)} |^{-\alpha} du > t\}, \quad t \geq 0. \quad (3.40)$$

To do this we should note that $| \tilde{X}_t^{(1)} | = \exp(\tilde{\xi}_{\tilde{\varphi}_t})$, $t \geq 0$ and from the earlier work on censored one-dimensional stable processes, cf. [36], we know that $\tilde{\xi}$ is a Lévy process. In particular, we know that, like all Lévy processes, $\tilde{\xi}$ is invariant with respect to Lebesgue measure. Appealing to the Revuz measure associated to $|X_t^{(1)}|$ via the Lamperti transform of ξ , it is easy to deduce that $| \tilde{X}_t^{(1)} |$ is invariant with respect to $\ell(dx) := |x^{(1)}|^\alpha dx$ on $(0, \infty)$. To understand the invariant measure on Ξ , we note that for non-negative, bounded measurable f on $(0, \infty) \times (0, \infty) \times \mathbb{S}^{(d-2)}$, recalling the definition for $\tilde{\varphi}_t$,

$$\begin{aligned} & \int_{\mathbb{R}} \int_0^\infty \int_{\mathbb{S}^{d-2}} f(x, y, \phi) \tilde{\pi}(dy, d\phi) dx \\ &= \lim_{t \rightarrow 0} \frac{1}{t} \mathbb{E}_{\ell \times \tilde{\pi}'} \left[\int_0^t f(| \tilde{X}_s^{(1)} |, Y_s, \Theta_s) | \tilde{X}_s^{(1)} |^{-\alpha} ds \right] \\ &= \int_{(0, \infty)} \int_0^\infty \int_{\mathbb{S}^{d-2}} f(x, y, \phi) \frac{C_{d-1} y^{d-2}}{(1+y^2)^{d/2}} \sigma_1^{(d-2)}(d\phi) dy dx. \end{aligned}$$

This finishes the proof of Theorem 3.25. $\square$

Ascending potential of the censored MAP

The ascending ladder MAP $(\tilde{\xi}^+, \tilde{\Xi}^+)$ has potential given by

$$\tilde{U}_{(r, \theta)}^+(dx, d\theta) = \int_0^\infty \mathbf{P}_{r, \theta}(\tilde{\xi}_t^+ \in dx, \tilde{\Xi}_t^+ \in d\theta) dt, \quad x > r, \theta \in \mathbb{R}^{d-1}.$$

Theorem 3.26. *In cartesian coordinates, the potential for the ascending ladder MAP $(\tilde{\xi}^+, \tilde{\Xi}^+)$ is given by the following expression*

$$\tilde{U}_{(r, \theta)}^+(dz, d\phi) = \frac{(1 - e^{r-z})^{\alpha/2}}{((1 - e^{r-z})^2 + |\phi - e^{r-z}\theta|^2)^{d/2}} dz d\phi \quad z > r, \theta, \phi \in \mathbb{R}^{d-1}. \quad (3.41)$$

Proof. We are going to identify this potential measure by relating the ascending ladder of the censored process with the ascending ladder process of the uncensored stable given in equation (3.11). Start by noticing that the ascending ladder process in the direction of the $x^{(1)}$ -axis of the censored stable $\tilde{X}$ is that of the uncensored stable X ,

as long as the starting point of these processes is above the hyperplane $\mathbb{H}_0$. Then, by the cylindrical Lamperti-Kiu transform we can express functionals of the ladder MAP $(\tilde{\xi}^+, \tilde{\Xi}^+)$, as functionals of the ladder stable $(H, X^{(2:d)+})$ in the following way. For $r \in \mathbb{R}, \theta \in \mathbb{R}^{d-1}$ and measurable bounded $f : \mathbb{R}^d \rightarrow \mathbb{R}$,

$$\mathbf{E}_{r,\theta} \left[\int_0^\infty f \left(e^{\tilde{\xi}_u^+}, e^{\tilde{\xi}_u^+} \tilde{\Xi}_u^+ \right) e^{\alpha/2 \tilde{\xi}_u^+} du \right] = \mathbb{E}_{e^r, e^r \theta} \left[\int_0^\infty f(H_u^+, X_u^{(2:d)+}) du \right],$$

recalling that $\alpha/2$ is the index of self-similarity of $(H, X^{(2:d)+})$, and the factor $e^{\alpha/2 \tilde{\xi}_u^+}$ is obtained from the inverse time-change in its cylindrical Lamperti Kiu transform.

Let us assume preemptively that the ascending ladder MAP potential possesses a density $\tilde{u}$ with respect to the Lebesgue measure. We now manipulate the right-hand side of the last expression for matching the arguments in the test functions to obtain

$$\begin{aligned} & \int_r^\infty \int_{\mathbb{R}^{d-1}} \tilde{u}_{r,\theta}^+(z, \phi) f(e^z, e^z \phi) e^{\alpha/2 z} dz d\phi \\ &= \int_{e^r}^\infty \int_{\mathbb{R}^{d-1}} U_{e^r, e^r \theta}^+(dy) f(y) \\ (y = e^z(1, \phi)) &= \int_r^\infty \int_{\mathbb{R}^{d-1}} u_{e^r, e^r \theta}^+(e^z, e^z \phi) f(e^z, e^z \phi) e^{dz} dz d\phi \\ &= \int_r^\infty \int_{\mathbb{R}^{d-1}} \left[\frac{(e^z - e^r)^{\alpha/2}}{((e^z - e^r)^2 + |e^z \phi - e^r \theta|^2)^{d/2}} \right] f(e^z, e^z \phi) e^{dz} dz d\phi, \end{aligned}$$

where in the last equality we used the ascending ladder potential density for the uncensored isotropic α -stable Lévy process given in Proposition 2.8. This implies that Lebesgue a.e.,

$$\begin{aligned} \tilde{u}_{r,\theta}^+(z, \phi) &= e^{-\alpha/2 z} \left[\frac{e^{dz} (e^z - e^r)^{\alpha/2}}{((e^z - e^r)^2 + |e^z \phi - e^r \theta|^2)^{d/2}} \right] \\ &= \frac{(1 - e^{r-z})^{\alpha/2}}{((1 - e^{r-z})^2 + |\phi - e^{r-z} \theta|^2)^{d/2}}. \end{aligned}$$

□

Ascending ladder MAP excursion occupation functional

Let us consider the structure of the occupation measure of $\tilde{X}$ under its excursion measure from the maximum of $\tilde{X}^{(1)}$. Let us work under the framework of excursions from the maximum of $\tilde{\xi}$. Consider $g_t = \sup\{s < t : \tilde{\xi}_s = \sup_{u < s} \tilde{\xi}_u\}$ and $d_t = \inf\{s > t : \tilde{\xi}_s = \sup_{u < s} \tilde{\xi}_u\}$ the starting and ending instants of the excursions from the

maximum of $(\tilde{\xi}, \tilde{\Xi})$. Thus, the excursions are the processes defined by

$$\left(\epsilon_t(s), \tilde{\Xi}_t^\epsilon(s)\right) = \left(\tilde{\xi}_{g_t} - \tilde{\xi}_{g_t+s}, \tilde{\Xi}_{g_t+s}\right), \quad \text{for } 0 \leq s \leq d_t - g_t =: \zeta_t, \quad (3.42)$$

and denote the excursion measures of the canonical excursion $\left((\epsilon(s), \tilde{\Xi}^\epsilon(s)) : s \in [0, \zeta]\right)$ by $(\tilde{\mathbf{n}}_\theta^+ : \theta \in \mathbb{R}^{d-1})$, see Section 0.4 for reference.

Theorem 3.27. *For $\theta \in \mathbb{R}^{d-1}$ and bounded measurable $f : [0, \infty) \times \mathbb{R}^{d-1} \rightarrow \mathbb{R}$ we have that, up to a multiplicative constant,*

$$\tilde{\mathbf{n}}_\theta^+ \left(\int_0^\zeta f(-\epsilon(s), \tilde{\Xi}^\epsilon(s)) ds \right) = \int_0^\infty \int_{\mathbb{R}^{d-1}} \frac{(e^r - 1)^{\alpha/2} e^{\alpha/2 r}}{((e^r - 1)^2 + |\phi - e^r \theta|^2)^{d/2}} f(-r, \phi) dr d\phi.$$

Proof. Standard excursion theory tells us that if $\mathbf{G}^+ := \cup_{t>0} g_t$ and $T_z := \inf\{t > 0 : \xi_t > z\}$, then

$$\begin{aligned} & \mathbf{E}_{0,\theta} \left[\int_0^{T_z} f(\tilde{\xi}_t, \tilde{\Xi}_t) dt \right] \\ &= \mathbf{E}_{0,\theta} \left[\sum_{g \in \mathbf{G}^+} \mathbf{1}_{\{\tilde{\xi}_g \leq z\}} \int_0^{\zeta_g} f(\tilde{\xi}_g + (\tilde{\xi}_{g+s} - \tilde{\xi}_g), \tilde{\Xi}_{g+s}) ds \right] \\ &= \mathbf{E}_{0,\theta} \left[\int_0^\infty \mathbf{1}_{\{\tilde{\xi}_t^+ \leq z\}} \tilde{\mathbf{n}}_{\tilde{\Xi}_t^+}^+ \left(\int_0^\zeta f(u - \epsilon(s), \tilde{\Xi}^\epsilon(s)) ds \right) \Big|_{u=\tilde{\xi}_t^+} dt \right] \\ &= \int_0^z \int_{\mathbb{R}^{d-1}} \tilde{U}_{(0,\theta)}^+(dr, d\phi) \tilde{\mathbf{n}}_\phi^+ \left(\int_0^\zeta f(r - \epsilon(s), \tilde{\Xi}^\epsilon(s)) ds \right). \end{aligned} \quad (3.43)$$

Substituting the explicit expression for the ascending potential in Theorem 3.26 we have

$$\begin{aligned} & \mathbf{E}_{0,\theta} \left[\int_0^{T_z} f(\tilde{\xi}_t, \tilde{\Xi}_t) dt \right] \\ &= \int_0^z \int_{\mathbb{R}^{d-1}} \frac{(1 - e^{-r})^{\alpha/2} dr d\phi}{((1 - e^{-r})^2 + |\phi - e^{-r} \theta|^2)^{d/2}} \tilde{\mathbf{n}}_\phi^+ \left(\int_0^\zeta f(r - \epsilon(s), \tilde{\Xi}^\epsilon(s)) ds \right). \end{aligned} \quad (3.44)$$

On the other hand, we can express this in terms of completely explicit formulas for the

(uncensored) stable process. We can write

$$\begin{aligned}
& \mathbf{E}_{0,\theta} \left[\int_0^{T_z} f(\tilde{\xi}_t, \tilde{\Xi}_t) dt \right] \\
&= \mathbb{E}_{(1,\theta)} \left[\int_0^{\tau_{e^z}^\wedge} f(\log |\tilde{X}_{\tilde{\varphi}_t^{-1}}^{(1)}|, \tilde{X}_{\tilde{\varphi}_t^{-1}}^{(2:d)} / |\tilde{X}_{\tilde{\varphi}_t^{-1}}^{(1)}|) dt \right] \\
&= \mathbb{E}_{(1,\theta)} \left[\int_0^{\tau_{e^z}^\wedge} f(\log |\tilde{X}_u^{(1)}|, \tilde{X}_u^{(2:d)} / |\tilde{X}_u^{(1)}|) |\tilde{X}_u^{(1)}|^{-\alpha} du \right] \\
&= \mathbb{E}_{(1,\theta)} \left[\int_0^{\tau_{e^z}^\wedge} f(\log X_u^{(1)}, X_u^{(2:d)} / X_u^{(1)}) (X_u^{(1)})^{-\alpha} \mathbf{1}_{\{\text{sign}(X_u^{(1)})=1\}} du \right],
\end{aligned}$$

where we have used (3.40) and the cylindrical Lamperti-Kiu decomposition. Now, using excursion theory from the maximum in the direction of the $x^{(1)}$ -axis for the isotropic α -stable Lévy process X , see equation (2.13) and the discussion below, as in (3.43), and making

$$f^*(x) = f(\log x^{(1)}, x^{(2:d)} / x^{(1)}) (x^{(1)})^{-\alpha} \mathbf{1}_{\{\text{sign}(x^{(1)})=1\}}$$

we have

$$\begin{aligned}
& \mathbb{E}_{(1,\theta)} \left[\int_0^{\tau_{e^z}^\wedge} f^*(X_u) du \right] \\
&= \int_1^{e^z} \int_{\mathbb{R}^{d-1}} U_{(1,\theta)}^+(dy) \mathbb{N}_{y^{(2:d)}}^+ \left(\int_0^{\zeta_\alpha} f^*(y^{(1)} - \epsilon^{(1)}(s), \epsilon^{(2:d)}(s)) ds \right) \\
&\propto \int_1^{e^z} \int_{\mathbb{R}^{d-1}} \frac{(y^{(1)} - 1)^{\alpha/2} dy^{(2:d)} dy^{(1)}}{|(y^{(1)} - 1)^2 + |y^{(2:d)} - \theta|^2|^{d/2}} \mathbb{N}_{y^{(2:d)}}^+ \left(\int_0^{\zeta_\alpha} f^*(y^{(1)} - \epsilon^{(1)}(s), \epsilon^{(2:d)}(s)) ds \right),
\end{aligned}$$

where in the last equality we used Proposition 2.8. Notice that, the excursion process (from its maximum) for the uncensored stable is denoted by $(\epsilon^{(1)}, \epsilon^{(2:d)})$ and must not be confused with $(\epsilon, \tilde{\Xi}^\epsilon)$ the excursion process (from the maximum) for the censored MAP. Making $y^{(1)} = e^r$ and $y^{(2:d)} = e^r \phi$ in the last integral we have

$$\begin{aligned}
& \mathbf{E}_{0,\theta} \left[\int_0^{T_z} f(\tilde{\xi}_t, \tilde{\Xi}_t) dt \right] \\
&= \int_0^z \int_{\mathbb{R}^{d-1}} \frac{(e^r - 1)^{\alpha/2} e^{dr} dr d\phi}{|(e^r - 1)^2 + |e^r \phi - \theta|^2|^{d/2}} \mathbb{N}_{e^r \phi}^+ \left(\int_0^{\zeta_\alpha} f^*(e^r - \epsilon^{(1)}(s), \epsilon^{(2:d)}(s)) ds \right) \\
&= \int_0^z \int_{\mathbb{R}^{d-1}} \frac{(1 - e^{-r})^{\alpha/2} dr d\phi}{|(1 - e^{-r})^2 + |\phi - e^{-r} \theta|^2|^{d/2}} e^{r\alpha/2} \mathbb{N}_{e^r \phi}^+ \left(\int_0^{\zeta_\alpha} f^*(e^r - \epsilon^{(1)}(s), \epsilon^{(2:d)}(s)) ds \right).
\end{aligned}$$

Comparing this expression with (3.44), by the positivity of $\frac{(1-e^{-r})^{\alpha/2} dr d\phi}{|(1-e^{-r})^2 + |\phi - e^{-r} \theta|^2|^{d/2}}$, for

any $r \in [0, z)$ and $\phi \in \mathbb{R}^{d-1}$ we can conclude that Lebesgue-a.e.

$$\tilde{\mathbf{n}}_\phi^+ \left(\int_0^\zeta f(r - \epsilon(s), \tilde{\Xi}^\epsilon(s)) ds \right) = e^{r\alpha/2} \mathbb{N}_{e^r\phi}^+ \left(\int_0^{\zeta_\alpha} f^*(e^r - \epsilon^{(1)}(s), \epsilon^{(2:d)}(s)) ds \right).$$

By taking $r = 0$ we can develop the right-hand side as follows. First, by Silverstein's identity (Lemma 2.12) in the case of excursions from the maximum in the direction of $x^{(1)}$

$$\begin{aligned} \tilde{\mathbf{n}}_\phi^+ \left(\int_0^\zeta f(-\epsilon(s), \tilde{\Xi}^\epsilon(s)) ds \right) &= \mathbb{N}_\phi^+ \left(\int_0^{\zeta_\alpha} f^*(1 - \epsilon^{(1)}(s), \epsilon^{(2:d)}(s)) ds \right) \\ &= \int_{-\infty}^1 \int_{\mathbb{R}^{d-1}} U_{(1,\phi)}^-(dz) f^*(z). \end{aligned}$$

Next, by the explicit expression for U^- given in Proposition 2.8 and substituting the definition of f^* , the integral is given by

$$\begin{aligned} &\int_{-\infty}^1 \int_{\mathbb{R}^{d-1}} \frac{(1 - z^{(1)})^{\alpha/2}}{((1 - z^{(1)})^2 + |\phi - z^{(2:d)}|^2)^{d/2}} f(\log z^{(1)}, \frac{z^{(2:d)}}{z^{(1)}}) (z^{(1)})^{-\alpha} \mathbf{1}_{\{z^{(1)} > 0\}} dz \\ &= \int_0^1 \int_{\mathbb{R}^{d-1}} \frac{(1 - z^{(1)})^{\alpha/2} (z^{(1)})^{-\alpha}}{((1 - z^{(1)})^2 + |\phi - z^{(2:d)}|^2)^{d/2}} f(\log z^{(1)}, \frac{z^{(2:d)}}{z^{(1)}}) dz \\ &= \int_0^\infty \int_{\mathbb{R}^{d-1}} \frac{(e^r - 1)^{\alpha/2} e^{\alpha/2r}}{((e^r - 1)^2 + |\phi - e^r \phi|^2)^{d/2}} f(-r, \phi) dr d\phi. \end{aligned}$$

where the last expression derives from the changes of variable $-r = \log z^{(1)}$ and $z^{(2:d)} = e^{-r}\phi$. This finishes the proof of the theorem. $\square$

We aim to use recent advancements in fluctuation theory and excursions of MAPs, as established in [29], particularly Theorem 4.1. This theorem connects a ladder potential (specifically U^- in their context, and $\tilde{u}^+$ in ours) with the occupation functional under the excursion measure of the dual process. To apply this result, we must first ensure that the necessary assumptions hold.

(A 2.2) $\tilde{\Xi}$ is a Hunt process and is quasi-left continuous.

(A 2.3) Upwards regularity of $\tilde{\xi}$.

(A 2.4) There exists a Markov process which is linked to $(\tilde{\xi}, \tilde{\Xi})$ via a weak reversibility property.

(A 2.5) The MAP ordinate satisfies $\lim_{t \rightarrow \infty} \xi_t = -\infty$ and the modulator is positive re-

current with invariant distribution $\tilde{\pi}$.

(A 2.6) The invariant measure $\hat{\tilde{\pi}}^+$ exists and $\hat{\mathbf{E}}_{0,\tilde{\pi}}[\bar{L}_1^{-1}] = \hat{\mathbf{n}}_{\tilde{\pi}}^+(\zeta) < \infty$

We were unable to prove assumptions (A 2.4)–(A 2.6).

Theorem 3.28 (Theorem 4.1 in [29] adapted to excursions from the maximum). *Suppose that the assumptions (A 2.2)–(A 2.6) are satisfied. We have for bounded measurable functions $f, h : \mathbb{R}^{d-1} \rightarrow \mathbb{R}$, and $g : [0, \infty) \rightarrow \mathbb{R}$, up to multiplicative constants,*

$$\begin{aligned} & \int_{\mathbb{R}^{d-1}} \hat{\tilde{\pi}}^+(d\theta) h(\theta) \hat{\mathbf{n}}^+ \left(\int_0^\zeta f(\tilde{\Xi}^\epsilon(s)) g(\epsilon(s)) ds \right) \\ &= \int_{\mathbb{R}^{d-1}} \tilde{\pi}(d\nu) f(\nu) \int_{[0,\infty) \times \mathbb{R}^{d-1}} \tilde{u}_\nu^+(y, \theta) dy d\theta g(y) h(\theta). \end{aligned} \quad (3.45)$$

In equation (3.45), the terms on the right-hand side are specified by Theorems 3.25 and 3.26. However, the relationship between the terms on the left-hand side and the occupation functional under the excursion measure from the minimum of $\tilde{\xi}$ remains unclear. Said functional is one of the two ingredients necessary to find an explicit expression for the potential in Corollary 3.23.

Similarly, Theorem 3.28 could be reformulated for excursions from the minimum, allowing us to attempt using Theorems 3.24 and 3.27 to obtain the descending ladder potential $\tilde{u}^-$. The approach for achieving this, however, is also uncertain.

Chapter 4

Numerics of the first entry/exit of an infinite strip

In this chapter we will consider two problems related to the isotropic α -stable process in $\mathbb{R}^d$ and the infinite strip $S = (-1, 1) \times \mathbb{R}^{d-1}$. The problems consist in finding the distribution of the position of first entry to S when started in $\mathbb{R} \setminus S$ and the distribution of the first exit of S when started in S . More specifically, if $\tau_S = \inf\{t > 0 : X_t \in S\}$ and $\tau_{S^c} = \inf\{t > 0 : X_t \in \mathbb{R}^d \setminus S\}$, we are interested in the laws

$$\mu_x^S(dy) := \mathbb{P}_x(X_{\tau_S} \in dy, \tau_S < \infty), \quad y \in S, x \notin S, \quad (4.1)$$

and

$$\mu_x^{S^c}(dy) := \mathbb{P}_x(X_{\tau_{S^c}} \in dy), \quad y \notin S, x \in S. \quad (4.2)$$

Notice that in the second case, the event $\{\tau_S < \infty\}$ is omitted due to the oscillatory behaviour of the first-coordinate process $X^{(1)}$, that is, almost surely

$$\limsup_{t \rightarrow \infty} X_t^{(1)} = -\liminf_{t \rightarrow \infty} X_t^{(1)} = +\infty,$$

which guarantees $\mathbb{P}_x(\tau_{S^c} < \infty) = 1$.

Our probabilistic motivation to study the problems (4.1) and (4.2) lies in their connection with a theory of excursions from the minimal distance to the hyperplane $\mathbb{H}_0$, and the promise of obtaining a generalisation of the results of [38] by following an analogous train of thought to that in said article. In addition to this, there is also an analytic motivation to solve the same problems. In the analytic jargon, the distributions involved

are standardised measures known as the α -harmonic measures, due to their relation with the Poisson equation for the fractional Laplacian in the domains S and $\mathbb{R}^d \setminus S$.

There exists a vast literature in regards the two previous problems in the one-dimensional case. For example, in [6] both problems are solved in the $d = 1$ isotropic (symmetric) case. In [55] and [26], problem (4.2) is solved in the $d = 1$, symmetric case. The motivation to solve these problems and the approaches they follow vary, but they all coincide in the fact that the solution satisfies an integral equation, derived by the Markov property. A decade later, and inspired in [26], the problem (4.2) in dimension one was solved in [52] for the asymmetric case. More recently, [36] and [48] have found solutions for the problem (4.1) in the one-dimensional asymmetric case. In the former, a probabilistic approach using the Lamperti transform of the so-called censored process is used, whilst in the latter, an integral equation is verified, in a similar spirit to how it is done in [6].

Extending the results on hitting and exiting S from $d = 1$ to $d \geq 2$ is highly non-trivial. For instance, if we attempt to follow the approach in [6], which involves solving an integral equation, the resulting integral equations—solved through an educated guess of the solutions—require the verification of rather difficult integrals, which are difficult to solve explicitly. An alternative, more accessible route is the probabilistic method introduced in [36], which begins by defining a censored process in higher dimensions. While setting up such a process is relatively straightforward, expressing it in a form suitable for practical computations again leads to complicated integrals, similar to those encountered in the analytic approach.

The aim of this chapter is to use numerical methods in order to understand the distributions in problems (4.1) and (4.2). In particular we will develop sampling algorithms which theoretically converge to a sampling algorithm of the distribution of the first hit into/out S .

4.1 Simulating the first hit into the strip

The aim of this section is to give a partial distributional decomposition of the density of the first hit into the infinite slab $S = (-1, 1) \times \mathbb{R}^{d-1}$ of $(X, \mathbb{P})$, together with a Monte Carlo method for numerically determining it. Pre-emptively assuming it exists, the density we are interested in is the bivariate function $H(x, y)$ such that for $x \in \mathbb{R}^d \setminus S$,

$$\mathbb{P}_x(X_{\tau_S} \in dy, \tau_S < \infty) = H(x, y)dy, \quad y \in (-1, 1) \times \mathbb{R}^{d-1}.$$

It is known that in the case $\alpha \in (1, 2)$ the process X will hit the slab with probability one because $X^{(1)}$, its first coordinate, will hit any interval with probability one; cf. [36]. However, in the case $\alpha \in (0, 1)$, we must necessarily work on the event $\{\tau_S < \infty\}$.

Because stable processes always make first passage into a half space by a jump (i.e. no creeping) we know that the coordinate $X^{(1)}$ must make a random number of crossings of $(-1, 1)$ before entering it. To this end, without loss of generality, we can start with $x^{(1)} > 1$ and define $\sigma_0 = 0$. Then for $k = 1, 2, \dots$, iteratively define,

$$\sigma_k = \begin{cases} \inf\{t > \sigma_{k-1} : X_t^{(1)} > -1\} & \text{if } k \text{ is even} \\ \inf\{t > \sigma_{k-1} : X_t^{(1)} < 1\} & \text{if } k \text{ is odd,} \end{cases}$$

and the random index N is such that $\{N = k\}$ agrees with $\tau_S = \sigma_k$.

Our first result concerns the conditional distribution of $X_{\tau_S}^{(2:d)}$ given $(X_t^{(1)}, t \leq \tau_S)$. More precisely, we are interested in the conditional distribution of $X_{\tau_S}^{(2:d)}$ given $(X_{\sigma_k}^{(1)}, k = 0, 1, \dots, N)$. To this end, let us define $\text{Cauchy}_{d-1}(0, \gamma)$, a $(d-1)$ -dimensional Cauchy distribution centred at 0 with scaling $\gamma > 0$. That is to say, a random variable on $\mathbb{R}^{d-1}$ with density

$$\frac{\Gamma(d/2)}{\pi^{d/2}} \frac{\gamma}{(\gamma^2 + x^2)^{d/2}}, \quad x \in \mathbb{R}^{d-1}. \quad (4.3)$$

Theorem 4.1 (Conditional law of first entry into a slab). *We have*

$$\text{Law} \left(X_{\tau_S}^{(2:d)} \middle| (X_t^{(1)}, t \leq \tau_S) \right) \sim x^{(2:d)} + \text{Cauchy}_{d-1} \left(0, \sum_{k=1}^N |X_{\sigma_k}^{(1)} - X_{\sigma_{k-1}}^{(1)}| \right).$$

Proof. By Lemma 2.5 in the case $d \geq 2$,

$$\mathbb{P}_x(X_{\tau_r^\vee} \in dy) = C_{\alpha,d} |x - y|^{-d} \frac{|r - x^{(1)}|^{\alpha/2}}{|r - y^{(1)}|^{\alpha/2}} dy, \quad \text{for } y^{(1)} < r < x^{(1)}$$

where $C_{\alpha,d} = \pi^{-(d/2+1)} \Gamma(d/2) \sin(\alpha\pi/2)$. Recalling that the same formula is true for the case $d = 1$; cf. Corollary 3.5 in Chapter 3 of [35] we have,

$$\mathbb{P}_x(X_{\tau_r^\vee}^{(1)} \in dy^{(1)}) = C_{\alpha,1} |x^{(1)} - y^{(1)}|^{-1} \frac{|r - x^{(1)}|^{\alpha/2}}{|r - y^{(1)}|^{\alpha/2}} dy^{(1)}, \quad \text{for } y^{(1)} < r < x^{(1)}.$$

Thus for $y^{(2:d)} \in \mathbb{R}^{d-1}$,

$$\mathbb{P}_x(X_{\tau_r^\vee}^{(2:d)} \in dy^{(2:d)} | X_{\tau_r^\vee}^{(1)} = y^{(1)}) = \frac{\Gamma(d/2)}{\pi^{d/2}} \frac{|x^{(1)} - y^{(1)}|}{((x^{(1)} - y^{(1)})^2 + |x^{(2:d)} - y^{(2:d)}|^2)^{d/2}} dy^{(2:d)}. \quad (4.4)$$

That is to say, under $\mathbb{P}_x$,

$$\text{Law} \left(X_{\tau_r^\vee}^{(2:d)} - x^{(2:d)} \middle| X_{\tau_r^\vee}^{(1)} = y^{(1)} \right) \sim \text{Cauchy}_{d-1} \left(0, |x^{(1)} - y^{(1)}| \right). \quad (4.5)$$

By symmetry we can show that for $x^{(1)} < r < y^{(1)}$, $\text{Law}(X_{\tau_r^\wedge}^{(2)} | X_{\tau_r^\wedge}^{(1)} = y^{(1)})$ under $\mathbb{P}_x$, follows a centered Cauchy distribution with scale parameter $|y^{(1)} - x^{(1)}|$.

Since we know that

$$\text{Cauchy}_{d-1}(0, \gamma) \sim \gamma \times \text{Cauchy}_{d-1}(0, 1),$$

we can thus identify, for $k = 1, \dots, N$, with $\sigma_0 = 0$, and hence $x^{(1)} = X_{\sigma_0}$,

$$\text{Law} \left(X_{\sigma_k}^{(2:d)} - x^{(2:d)} | X_{\sigma_j}^{(1)}, j = 0, 1, \dots, k \right) \sim \sum_{i=1}^k |X_{\sigma_i}^{(1)} - X_{\sigma_{i-1}}^{(1)}| \times \text{Cauchy}_{d-1}^{(i)}(0, 1),$$

where $\text{Cauchy}_{d-1}^{(i)}(0, 1)$ are iid copies of $\text{Cauchy}_{d-1}(0, 1)$ and the result follows by infinite divisibility and scaling of the Cauchy distribution. $\square$

The proof of Theorem 4.1 inspires the following numerical approach to simulating the law of X_{τ_S} using an approach which is reminiscent of the so-called ‘walk-on-spheres’ Monte Carlo approach described for stable processes in [33]. As such we refer to it as the ‘walk-on-half-spaces’ Monte Carlo approach.

Algorithm 1 (simulation of X_{τ_S} under $\mathbb{P}_x$ for $x \in \mathbb{R}^d$ with $x^{(1)} > 1$, $\alpha \in [1, 2)$).

sample x_1 from the distribution of $X_{\tau_1^\vee}$ under $\mathbb{P}_x$;

set $k = 1$;

while $x_k \notin S$

```

if  $x_k^{(1)} < -1$  then
    sample  $x_{k+1}$  from the distribution of  $X_{\tau_{-1}^\wedge}$  under
     $\mathbb{P}_{x_k}$ ;
    set  $k = k + 1$ ;
end if

if  $x_k^{(1)} > 1$  then
    sample  $x_{k+1}$  from the distribution of  $X_{\tau_1^\vee}$  under
     $\mathbb{P}_{x_k}$ ;
    set  $k = k + 1$ ;

end if

end while

return  $N = k$ ;

return  $X_{\tau_S} = x_k$ ;

```

More generally, the algorithm above stores the information $(x_1, \dots, x_N)$, which corresponds precisely to a simulation of $(X_{\sigma_1}, \dots, X_{\sigma_N})$. Theorem 4.1 implies that we can simulate the value of the $(2 : d)$ coordinates at τ_S by simply storing the jumps $\{x_{\sigma_k}^{(1)}\}_{k=1}^N$, and finally generate $X^{(2:d)}$ as a Cauchy instance centered at $x^{(2:d)}$ and with scale parameter given by $\sum_{k=1}^N |x_k^{(1)} - x_{k-1}^{(1)}|$.

4.1.1 Results of the simulations of first entry to the strip

We consider the implementation of the walk-on-half-spaces algorithm for a stable process on $\mathbb{R}^2$. To sample from the distribution of $X_{\tau_1^\vee} = (X_{\tau_1^\vee}^{(1)}, X_{\tau_1^\vee}^{(2)})$ under $\mathbb{P}_x$, $x^{(1)} > 1$, we split the procedure into two steps.

First, we simulate an instance of $X_{\tau_1^\vee}^{(1)}$. This can be done by treating this as a first passage problem of the one-dimensional symmetric stable process $X^{(1)}$ and noting from Lemma 2.5 that

$$\mathbb{E}_{x^{(1)}} \left[f \left(\frac{1 - X_{\tau_1^\vee}^{(1)}}{x^{(1)} - X_{\tau_1^\vee}^{(1)}} \right) \right] = \frac{\sin(\alpha\pi/2)}{\pi} \int_0^1 f(u) u^{-\alpha/2} (1-u)^{\alpha/2-1} du,$$

which said in another way means $(1 - X_{\tau_1^\vee}^{(1)})(X_{\tau_1^\vee}^{(1)} - x^{(1)})^{-1}$, when $X_0^{(1)} = x^{(1)}$, follows a $\text{Beta}(1 - \alpha/2, \alpha/2)$ distribution. Observe that this transform is a bijection, hence we can simulate from the Beta kernel and map it back to obtain $X_{\tau_1^\vee}^{(1)}$.

Second, we use equation (4.4), which under the event $\{X_{\tau_1^\vee}^{(1)} = y_1^{(1)}\}$, allows us to sample $X_{\tau_1^\vee}^{(2)} - x^{(2:d)}$ by sampling from a $\text{Cauchy}_1(0, |X_{\tau_1^\vee}^{(1)} - x^{(1)}|)$ distribution. By symmetry, the same can be done for sampling from $X_{\tau_{-1}^\wedge}$ under $\mathbb{P}_{x^{(1)}}, x^{(1)} < -1$. Except for these details on how to simulate $X_{\tau_1^\vee}$ and $X_{\tau_{-1}^\wedge}$, the rest of the algorithm implementation is straightforward, see Figure 4-1.

In the numerical examples, cf. Figure 4-2, we consider, without loss of generality, that $x^{(2)} = 0$. To illustrate the effect of choosing different parameters on the empirical distribution of X_{τ_S} we vary the point of issue $x^{(1)}$ and the scale parameter α . In Figure 4-2 (left) we observe the empirical distribution of 5 million first-hitting points in the strip, for 4 combinations of parameters $x^{(1)}$ and α . We observe that an initial point closer to the boundary of the strip leads to a sample more concentrated in the horizontal axis neighbouring the starting point. Similarly, a scale index closer to $\alpha = 2$ leads to a first-hitting point more concentrated around the starting position, but also increases the probability of hitting the strip near its boundaries. These effects can also be seen in the marginal histograms of these samples, Figure 4-2 (centre). It is worth noting that the marginal distribution of the first coordinate is known explicitly (cf. Theorem 1.1 in [36]). The code for these simulations can be found in the following github repository <https://github.com/sdaphnis/WoHS/>.

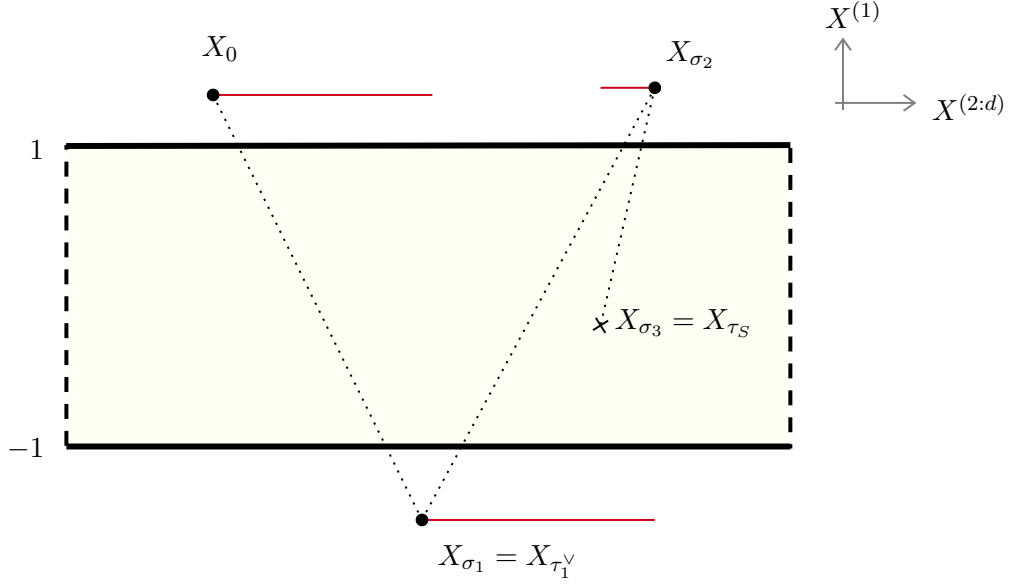


Figure 4-1: Representation of the Walk on half-spaces algorithm. In this illustration the process starts above the strip, the first jump out of $\mathbb{H}_{1\uparrow}$ misses the strip, we start the algorithm again taking as initial position the first hit into $\mathbb{H}_{-1\downarrow}$. The process repeats and after $N = 3$ crossings the process finally enters the strip. The red lines correspond to the Cauchy_1 horizontal displacements which add up to $X_{\tau_S}^{(2:d)}$.

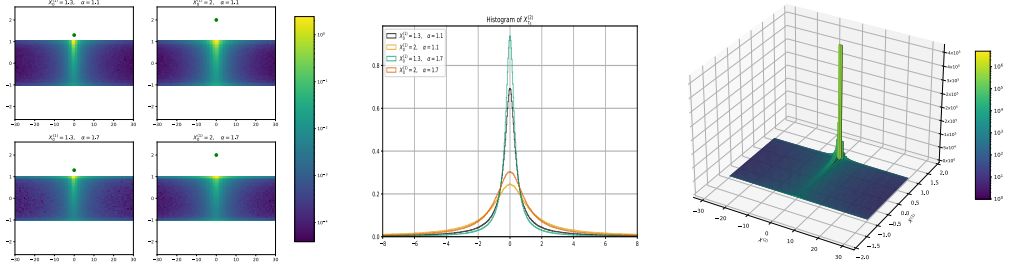


Figure 4-2: (left): Histograms of the simulations of first-hits into the strip for different values of $X_0 = (x^{(1)}, 0)$ and α . The green point represents the starting point X_0 . (centre): Histograms of the second coordinate at the instant of first-hit into the strip. Different values for the point of issue and α are considered. Instead of drawing solid bars, the histograms are drawn as step-like curves that outline the shape of the histogram, i.e. we see the edges of each bin without any filled area underneath. (right): A 3d histogram of the first hits into the strip.

4.1.2 Solving the theoretical non-convergence of the case $\alpha \in (0, 1)$

One of the problems with Algorithm 1 is that, in the setting $\alpha \in (0, 1)$, the while loop may never end on account of the fact that, by Corollary 1.2 of [36], $\mathbb{P}_x(\tau_S = \infty) > 0$. Nonetheless, it is possible to ‘force’ a stable process to hit S in the case $\alpha \in (0, 1)$ improving the efficiency of the walk-on-half-spaces algorithm without compromising the exactness of the algorithm. Such technique consists in conditioning the process to hit the hyperplane $\mathbb{H}_0$. This is done by considering the h -transform $h(x) = |x^{(1)}|^{\alpha-1}$ for $x \in \mathbb{R}^d$ (cf. Section 2.3.2). Let us call $\mathbb{P}^\circ$ to such conditional law, for which now is clear that $\mathbb{P}_x^\circ(\tau_S < \infty) = 1$ or in other words

$$\mathbb{P}_x^\circ(X_S \in dy, \tau_S < \infty) = \mathbb{P}_x^\circ(X_S \in dy).$$

We can calculate the distribution of the first hit into a half-space under $\mathbb{P}^\circ$ just multiplying by the quotient of the h -transforms as follows

$$\mathbb{P}_x^\circ(X_{\tau_1^\vee} \in dy) = \frac{|y^{(1)}|^{\alpha-1}}{(x^{(1)})^{\alpha-1}} C_{\alpha,d} |x - y|^{-d} \frac{(x^{(1)} - 1)^{\alpha/2}}{(1 - y^{(1)})^{\alpha/2}} dy, \quad \text{for } y^{(1)} < 1 < x^{(1)}. \quad (4.6)$$

Symmetrically, it is possible to obtain the distribution of $X_{\tau_{-1}^\wedge}$:

$$\mathbb{P}_x^\circ(X_{\tau_{-1}^\wedge} \in dy) = \frac{|y^{(1)}|^{\alpha-1}}{(-x^{(1)})^{\alpha-1}} C_{\alpha,d} |x - y|^{-d} \frac{(-1 - x^{(1)})^{\alpha/2}}{(y^{(1)} + 1)^{\alpha/2}} dy, \quad \text{for } x^{(1)} < -1 < y^{(1)}.$$

Naturally, these two distributions allow us to use the same steps of the walk-on-halfspaces algorithm to estimate $\mathbb{P}_x^\circ(X_{\tau_S} \in dy) = \frac{|y^{(1)}|^{\alpha-1}}{|x^{(1)}|^{\alpha-1}} \mathbb{P}_x(X_{\tau_S} \in dy, \tau_S < \infty)$ and therefore, to obtain the desired density $h(x, y)$ we need to ‘correct’ by the quotient $\frac{|y^{(1)}|^{\alpha-1}}{|x^{(1)}|^{\alpha-1}}$.

Simulating from $X_{\tau_1^\vee}$ and $X_{\tau_{-1}^\wedge}$ under $\mathbb{P}^\circ$

To simulate from $X_{\tau_1^\vee}$ under $\mathbb{P}_x^\circ$ means sampling an instance of the distribution (4.6). This new kernel cannot be put in correspondence with a Beta density through an elemental transformation as we did in the unconditional case. However, it is possible to use a simulation from a Gauss hypergeometric distribution (cf.[2]). We say that a random variable Z has a Gauss hypergeometric distribution with parameters a, b, c and w ($a > 0, b > 0$), if it has a continuous distribution whose probability density function is

$$GH(z|a, b, c, w) = C \frac{z^{a-1}(1-z)^{b-1}}{(1+wz)^c}, \quad \text{for } 0 \leq z \leq 1, \quad (4.7)$$

and $GH(z|a, b, c, w) = 0$ otherwise, where the proportionality constant C is given by

$$C^{-1} = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)} {}_2F_1(c, a; a+b; -w),$$

and we write $Z \sim GH(a, b, c, w)$.

Lemma 4.2. *Let $Y_1 = [X_{\tau_1^\vee}^{(1)} | X_{\tau_1^\vee}^{(1)} \in (0, 1)]$ and $Y_2 = [X_{\tau_1^\vee}^{(1)} | X_{\tau_1^\vee}^{(1)} \in (-\infty, 0)]$, and $x \in \mathbb{R}^d$ s.t. $x^{(1)} > 1$ then under $\mathbb{P}_x^\circ$*

$$Y_1 \sim GH(\alpha, 1 - \alpha/2, 1, -1/x^{(1)}),$$

$$\frac{1}{1 - Y_2} \sim GH(1 - \alpha/2, \alpha, 1, x^{(1)} - 1).$$

Moreover,

$$\mathbb{P}_x^\circ(X_{\tau_1^\vee}^{(1)} \in (0, 1)) = \frac{(x^{(1)})^{-\alpha}(x^{(1)} - 1)^{\alpha/2}\Gamma(\alpha)}{\Gamma(1 + \alpha/2)\Gamma(\alpha/2)} {}_2F_1\left(1, \alpha; 1 + \alpha/2; 1/x^{(1)}\right).$$

This lemma helps us simulate an instance of $X_{\tau_1^\vee}$ under $\mathbb{P}^\circ$ by simulating first the occurrence of the event $\{X_{\tau_1^\vee}^{(1)} \in (0, 1)\}$ and subsequently simulate from Y_1 or Y_2 respectively of if the event occurred or if it did not.

Lemma 4.3. *Similarly, if $Z_1 = [X_{\tau_{-1}^\wedge}^{(1)} | X_{\tau_{-1}^\wedge}^{(1)} \in (-1, 0)]$ and $Z_2 = [X_{\tau_{-1}^\wedge}^{(1)} | X_{\tau_{-1}^\wedge}^{(1)} \in (1, \infty)]$, and $x \in \mathbb{R}^d$ s.t. $x^{(1)} < -1$ then under $\mathbb{P}_x^\circ$*

$$-Z_1 \sim GH(\alpha, 1 - \alpha/2, 1, 1/x^{(1)}),$$

$$\frac{1}{1 + Z_2} \sim GH(1 - \alpha/2, \alpha, 1, -(x^{(1)} + 1)).$$

and,

$$\mathbb{P}_x^\circ\left(X_{\tau_{-1}^\wedge}^{(1)} \in (-1, 0)\right) = \frac{(-x^{(1)})^{\alpha/2}(-1 - x^{(1)})^{\alpha/2}\Gamma(\alpha)}{\Gamma(1 + \alpha/2)\Gamma(\alpha/2)} {}_2F_1\left(1, \alpha; 1 + \alpha/2; -1/x^{(1)}\right).$$

Proof of Lemmas 4.2 and 4.3. We will prove only the assertions in Lemma 4.3, those of Lemma 4.2 follow by symmetry. Consider f continuous and bounded and $x^{(1)} < -1$,

then

$$\begin{aligned}
& \mathbb{E}_{x^{(1)}}^{\circ} [f(Z_1)] \\
&= \frac{C_{\alpha,1}}{\mathbb{P}_{x^{(1)}}^{\circ} \left(X_{\tau_{-1}^{\wedge}}^{(1)} \in (-1, 0) \right)} \int_{-1}^0 f(z) \left(\frac{-z}{-x^{(1)}} \right)^{\alpha-1} \left(\frac{-1-x^{(1)}}{z+1} \right)^{\alpha/2} (z-x^{(1)})^{-1} dz \\
&= \frac{C_{\alpha,1}(-x^{(1)})^{-\alpha}(1-x^{(1)})^{\alpha/2}}{\mathbb{P}_{x^{(1)}}^{\circ} \left(X_{\tau_{-1}^{\wedge}}^{(1)} \in (-1, 0) \right)} \int_0^1 f(-u) u^{\alpha-1} (1-u)^{-\alpha/2} \left(1 + \frac{u}{x^{(1)}} \right)^{-1} du.
\end{aligned}$$

Comparing the kernel in the second equality against (4.7), we conclude that $-Z_1 \sim GH(\alpha, 1-\alpha/2, 1, \frac{1}{x^{(1)}})$, and equating constants we obtain

$$\mathbb{P}_{x^{(1)}}^{\circ} \left(X_{\tau_{-1}^{\wedge}}^{(1)} \in (-1, 0) \right) = \frac{(-x^{(1)})^{\alpha/2}(-1-x^{(1)})^{\alpha/2}\Gamma(\alpha)}{\Gamma(1+\alpha/2)\Gamma(\alpha/2)} {}_2F_1\left(1, \alpha; 1+\alpha/2; -1/x^{(1)}\right).$$

Finally, let us consider the expectation of $f(1/(1+Z_2))$ under $\mathbb{P}_{x^{(1)}}^{\circ}$

$$\begin{aligned}
& \mathbb{E}_x^{\circ} \left[f\left(\frac{1}{1+Z_2} \right) \right] \\
&= \int_0^{\infty} f\left(\frac{1}{1+z} \right) \frac{C_{\alpha,1}}{\mathbb{P}_{x^{(1)}}^{\circ} \left(X_{\tau_{-1}^{\wedge}}^{(1)} \in (0, \infty) \right)} \left(\frac{z}{-x^{(1)}} \right)^{\alpha-1} \left(\frac{-1-x^{(1)}}{z+1} \right)^{\alpha/2} (z-x)^{-1} dz \\
&\propto \int_0^1 f(u) u^{-\alpha/2} (1-u)^{\alpha-1} \left(1 - u(x^{(1)} + 1) \right)^{-1} du.
\end{aligned}$$

The second equality is given by an elemental change of variable, and the kernel allows us to conclude that $1/(1+Z_2) \sim GH(1-\alpha/2, \alpha, 1, -(x^{(1)}+1))$.

The symmetry property helps us to state that for $x^{(1)} > 1$, Y_1 under $\mathbb{P}_{x^{(1)}}^{\circ}$ is equal in distribution to $-Z_1$ under $\mathbb{P}_{-x^{(1)}}^{\circ}$. Analogously, Y_2 under $\mathbb{P}_{x^{(1)}}^{\circ}$ is equal in distribution to $-Z_2$ under $\mathbb{P}_{-x^{(1)}}^{\circ}$. $\square$

4.2 Simulating the first exit from the strip

To simulate a realisation from the kernel $\mu_x^{S^c}(\mathrm{d}y)$, $x \in S, y \notin S$, it is possible to use the Walk-on-spheres algorithm [33].

The algorithm (see Figure 4-3) consists in sampling from the first exit of the biggest sphere centered in x and inscribed in S , if the sample is still in the strip, then we sample the first exit of the biggest sphere centered in the previous realisation. Once the sample exits the strip, this is equivalent to a realisation from $\mu_x^{S^c}$. It is easy to see that the biggest sphere inscribed in the strip, and centered in x will always have a radius equal to the minimum between $1 - x^{(1)}$, the distance from x to the upper boundary of the strip, and $x^{(1)} - (-1)$, the distance from x to the bottom lid of the strip. Consider $\tau_{B_r(x)}^\wedge := \{t > 0 : |X_t - x| > r\}$ the first exit of the ball centered in x and with radius r .

Algorithm 2 (simulation of $X_{\tau_{S^c}}$ under $\mathbb{P}_x$ for $x \in \mathbb{R}^d$ with $x^{(1)} \in (-1, 1)$).

set $r_0 = \min\{1 - x^{(1)}, x^{(1)} + 1\}$

sample x_1 from the distribution of $X_{\tau_{B_{r_0}}^\wedge(x)}$ under $\mathbb{P}_x$;

set $k = 1$;

while $x_k \in S$:

set $r_k = \min\{1 - x_k^{(1)}, x_k^{(1)} + 1\}$;

sample x_{k+1} from the distribution of $X_{\tau_{B_{r_k}}^\wedge(x_k)}$ under $\mathbb{P}_{x_k}$;

set: $k = k + 1$;

end while

return $N = k$;

return $X_{\tau_{S^c}} = x_k$.

4.2.1 Simulating the exit from the center of any circle

In order to implement Algorithm 2 we use the following sequential way of simulating instances of $X_{\tau_{B_r}^\wedge}$ under $\mathbb{P}_x$, $r > 0, x \in \mathbb{R}^d$. We follow the next steps:

1. Simulate a realisation from $T \sim \text{Beta}(\alpha/2, 1 - \alpha/2)$.

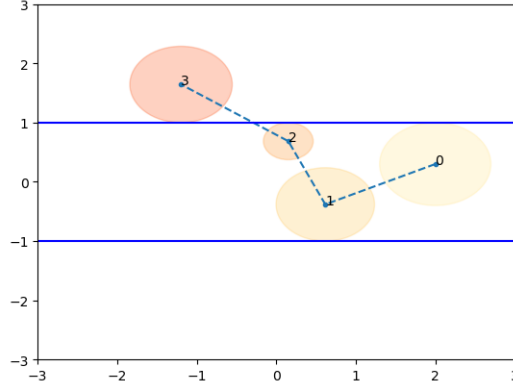


Figure 4-3: Walk on spheres algorithm to simulate the first exit from the infinite strip.

2. Take the transformation $Y = \frac{1}{\sqrt{T}}$. This is a simulation from $|X_{\tau_{B_1(0)}}^\wedge|$ under $\mathbb{P}$.
3. Simulate from $\Theta \sim \text{Unif}(\mathbb{S}_1)$. This is a simulation from $\left[\arg(X_{\tau_{B_1(0)}}^\wedge) || X_{\tau_{B_1(0)}}^\wedge \right]$.
4. By considering $Z = Y\Theta$, we obtain a simulation from $X_{\tau_{B_1(0)}}^\wedge$ under $\mathbb{P}$.
5. Finally, we shift and scale, $X = rZ + x$. This is a simulation of $X_{\tau_{B_r(x)}}^\wedge$ under $\mathbb{P}_x$.

The validity of these steps is ensured by the next lemma, whose proof is immediate from an elemental change of variables in Theorem A of [6].

Lemma 4.4. *For continuous bounded $g : (0, \infty) \rightarrow \mathbb{R}$ and $h : \mathbb{S}^{d-1} \rightarrow \mathbb{R}$,*

$$\begin{aligned}
& \mathbb{E}[g(1/|X_{\tau_{B_1(0)}}^\wedge|^{1/2})h(\arg(X_{\tau_{B_1(0)}}^\wedge))] \\
&= \int_{(0,\infty)} \int_{\mathbb{S}^{d-1}} g(1/\sqrt{r})h(\theta) \frac{\sin(\alpha\pi/2)}{\pi} (r^2 - 1)^{-\alpha/2} r^{-1} \sigma(d\theta) dr \\
&= \int_{(0,1)} \int_{\mathbb{S}^{d-1}} g(s)h(\theta) \frac{\sin(\alpha\pi/2)}{\pi} s^{\alpha/2-1} (1-s)^{-\alpha/2} \sigma(d\theta) ds.
\end{aligned}$$

where $\sigma(d\theta)$ is the normalized Lebesgue measure on $\mathbb{S}^{d-1}$.

The previous lemma ensures steps 1 to 4 of the simulation process. Using the shifting invariance of the α -stable Lévy process we have for continuous bounded $f : \mathbb{R}^d \rightarrow \mathbb{R}$, $r > 0$ and $x \in \mathbb{R}^d$

$$\mathbb{E}_x \left[f(X_{\tau_{B_r(x)}}^\wedge) \right] = \mathbb{E}[f(X_{\tau_{B_r(0)}}^\wedge + x)].$$

Therefore, by the scaling property we obtain that step 5 is also valid, i.e.

$$\mathbb{E}[f(X_{\tau_{B_r(0)}^\wedge} + x)] = \mathbb{E}\left[f(rX_{\tau_{B_1(0)}^\wedge} + x)\right].$$

4.2.2 Results of the simulations of first exit from the strip

The simulations mentioned in Algorithm 2 were done in dimension $d = 2$ using the library Scipy from Python. For the figures in this section, a sample of five million simulations was used.

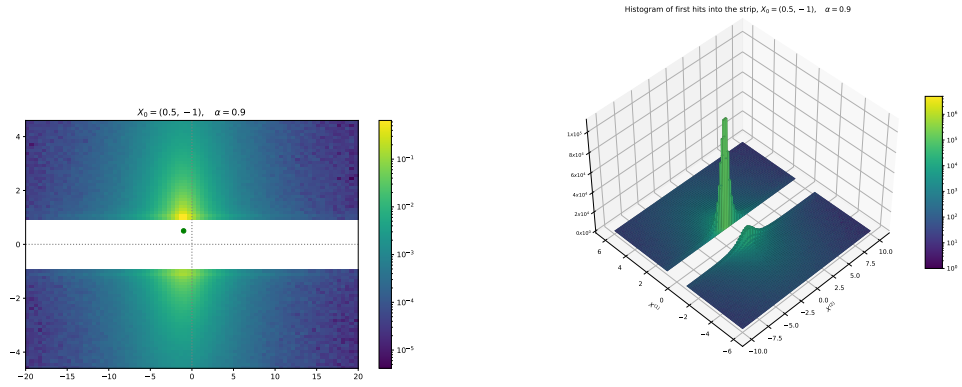


Figure 4-4: **Left:** Two-dimensional histogram of the first exit from the strip. The point of issue is $(0.5, -1)$. The colors correspond to the relative frequency in logarithmic scale as indicated in the bar to the right. **Right:** Three-dimensional plot of the histogram of $X_{\tau_{Sc}}$. The height of the bar corresponds to absolute frequency. The total number of simulations is $n = 5,000,000$.

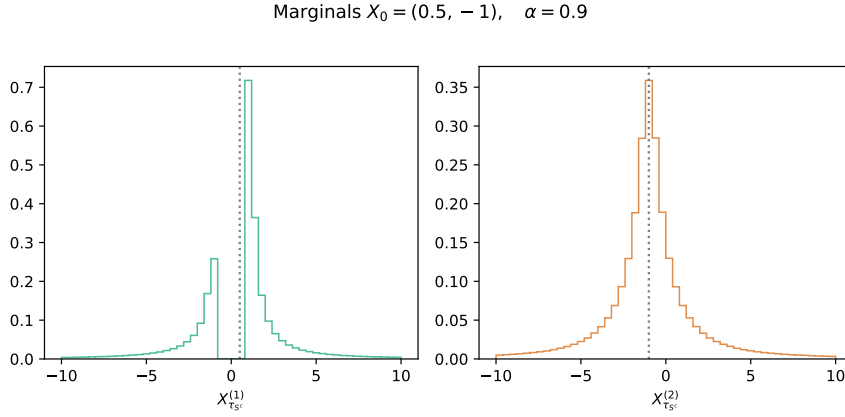


Figure 4-5: Marginal distribution of the first and second coordinates. The one in the left-hand side corresponds to Theorem A in [6], whilst the one in the right is unknown in closed formula.

4.2.3 Cauchy representation of the first exit from the strip

Let us assume preemptively that a density for $\mu_x^{S^c}(dy)$ with respect to Lebesgue measure exists, i.e.

$$\mathbb{P}_x(X_{\tau_{S^c}} \in dy) = h_{S^c}(x, y)dy, \quad x \in S, y \in S^c. \quad (4.8)$$

In this section we will prove that, $h_{S^c}(x, y)$ can be expressed as an infinite series of kernels corresponding to convolutions of the distribution of first hit into a half-space $(\mathbb{P}_x(X_{\tau_1^\wedge} \in dy))$. Moreover, we can give a representation of the law of the $(2 : d)$ coordinates at τ_{S^c} , as the combination of $(d - 1)$ -dimensional Cauchy distributions, whose scale parameters depend on the values of $X^{(1)}$ at the instants of crossing between $\mathbb{H}_{1\uparrow}$ and $\mathbb{H}_{-1\downarrow}$. Define, for $k \geq 1$ and $0 \leq j \leq k$

$$\sigma_{j,k} = \begin{cases} \inf\{t > \sigma_{j-1,k} : X_t^{(1)} > 1\} & \text{if } j \text{ is odd} \\ \inf\{t > \sigma_{j-1,k} : X_t^{(1)} < -1\} & \text{if } j \text{ is even,} \end{cases} \quad \text{if } k \text{ is odd,}$$

and

$$\sigma_{j,k} = \begin{cases} \inf\{t > \sigma_{j-1,k} : X_t^{(1)} < -1\} & \text{if } j \text{ is odd} \\ \inf\{t > \sigma_{j-1,k} : X_t^{(1)} > 1\} & \text{if } j \text{ is even,} \end{cases} \quad \text{if } k \text{ is even,}$$

where $\sigma_{0,k} = 0$ for all k .

Theorem 4.5. Let $x = (x^{(1)}, x^{(2:d)}) \in S$ and $y = (y^{(1)}, y^{(2:d)}) \in \mathbb{H}_{1\uparrow}$,

$$\begin{aligned} & \mathbb{P}_x(X_{\tau_{Sc}} \in dy) \\ &= \sum_{k=1}^{\infty} \left\{ (-1)^{k-1} \int_{z_1^{(1)}, \dots, z_{k-1}^{(1)}} \mathbb{P}_{x^{(1)}} \left(X_{\sigma_{1,k}}^{(1)} \in dz_1^{(1)}, \dots, X_{\sigma_{k-1,k}}^{(1)} \in dz_{k-1}^{(1)}, X_{\sigma_{k,k}}^{(1)} \in dy^{(1)} \right) \right. \\ & \quad \left. \times \mathbf{P} \left(x^{(2:d)} + \mathbf{Cauchy}_{d-1}(0, \gamma_k) \in dy^{(2:d)} \right) \right\}, \end{aligned} \quad (4.9)$$

where $\mathbf{Cauchy}_{d-1}(0, \gamma_k)$ are independent Cauchy random vectors with probability $\mathbf{P}$ and density (4.3), $z_0 = 0$, $z_k^{(1)}$ are integrated on $(1, \infty)$ or $(-\infty, -1)$ according to the domain of $X_{\sigma_{j,k}}^{(1)}$, which is $(1, \infty)$ if $j+k$ is even or $(-\infty, -1)$ if $j+k$ is odd, and

$$\gamma_k = \sum_{j=1}^{k-1} |z_j^{(1)} - z_{j-1}^{(1)}| + (y^{(1)} - z_{k-1}^{(1)}).$$

The result extends to the case $y \in \mathbb{H}_{-1\downarrow}$ just by swapping k even with k odd, in the definition of $\sigma_{j,k}$.

Remark 4.6. The formula in the previous theorem extends, to $\alpha \in (0, 2]$, the explicit distribution of first exit from the infinite strip for Brownian motion, viz.

$$\begin{aligned} & \mathbb{P}_x \left(B_{\tau_{(-1,1)}} \in dy \right) = \left(\delta_1(y^{(1)})(1 - x^{(1)})/2 + \delta_{-1}(y^{(1)})(x^{(1)} + 1)/2 \right) \\ & \quad \times \frac{\Gamma(d/2)}{\pi^{d/2}} \sum_{k=-\infty}^{\infty} \left[\frac{(2 - (1 - x^{(1)}) + 4k)}{(|y^{(2:d)}|^2 + (2 - (1 - x^{(1)}) + 4k)^2)^{d/2}} \right. \\ & \quad \left. + \frac{(2 - (1 + x^{(1)}) + 4k)}{(|y^{(2:d)}|^2 + (2 - (1 + x^{(1)}) + 4k)^2)^{d/2}} \right] dy^{(2:d)}, \end{aligned} \quad (4.10)$$

where δ_z denotes the Dirac measure at z . This density is easily obtained from the independence of $B^{(1)}$ and $B^{(2:d)}$ and a similar calculation to that in (3.24).

Formula (4.9) then clarifies what the dependency between $X_{\tau_{Sc}}^{(1)}$ and $X_{\tau_{Sc}}^{(2:d)}$ is in the isotropic α -stable Lévy case. Moreover, formula (4.9) is a generalisation of a well-known application of the so-called ‘reflection principle’ of Brownian Motion commonly used to characterise Brownian Motion killed in an interval (cf. Section 7.4 in [42]). Equation (4.9) is also a generalisation of Equation (8) in [52].

Proof. Noticing that $\tau_{Sc} = \min\{\tau_1^\wedge, \tau_{-1}^\vee\}$, taking $y \in \mathbb{H}_{1\uparrow}$ and partitioning we get

$$\mathbb{P}_x(X_{\tau_1^\wedge} \in dy) = \mathbb{P}_x(X_{\tau_1^\wedge} \in dy, \tau_1^\wedge < \tau_{-1}^\vee) + \mathbb{P}_x(X_{\tau_1^\wedge} \in dy, \tau_1^\wedge > \tau_{-1}^\vee).$$

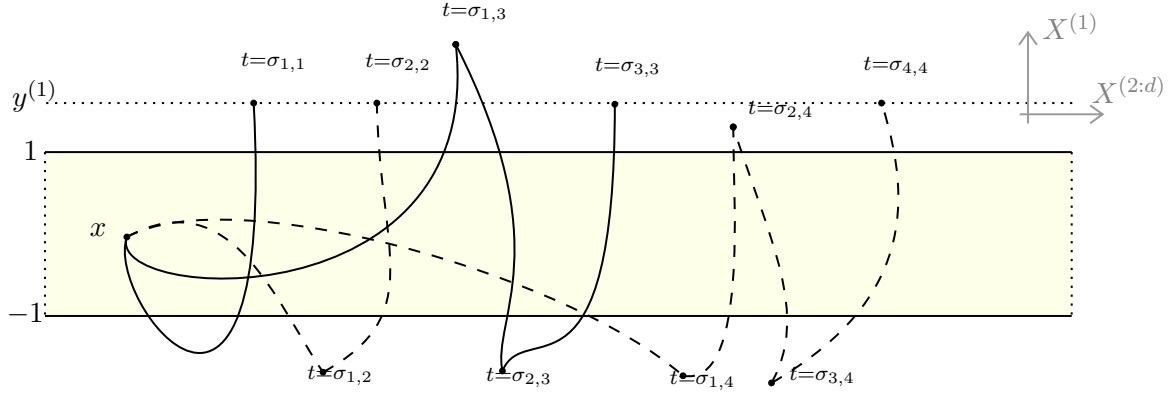


Figure 4-6: Examples of the first 4 events considered in expression (4.9). The probability of the paths with dashed line are to be subtracted from the probability of the paths with solid lines.

Noting that if $y^{(1)} > 1$, then $\mathbb{P}_x(X_{\tau_S^c} \in dy) = \mathbb{P}_x(X_{\tau_1^\wedge} \in dy, \tau_1^\wedge < \tau_{-1}^\vee)$, and using the strong Markov property at $\tau_{-1}^\vee$ on $\mathbb{P}_x(X_{\tau_1^\wedge} \in dy, \tau_1^\wedge > \tau_{-1}^\vee)$, we obtain

$$\mathbb{P}_x(X_{\tau_{Sc}} \in dy) = \mathbb{P}_x(X_{\tau_1^\wedge} \in dy) - \int_{\mathbb{H}_{-1\downarrow}} \mathbb{P}_x(X_{\tau_{-1}^\vee} \in du, \tau_{-1}^\vee < \tau_1^\wedge) \mathbb{P}_u(X_{\tau_1^\wedge} \in dy). \quad (4.11)$$

Similarly, for $u \in \mathbb{H}_{-1\downarrow}$ one can deduce

$$\mathbb{P}_x(X_{\tau_{Sc}} \in du) = \mathbb{P}_x(X_{\tau_{-1}^\vee} \in du) - \int_{\mathbb{H}_{1\uparrow}} \mathbb{P}_x(X_{\tau_1^\wedge} \in dz, \tau_1^\wedge < \tau_{-1}^\vee) \mathbb{P}_z(X_{\tau_{-1}^\vee} \in du). \quad (4.12)$$

Substituting (4.12) into (4.11), for $y^{(1)} > 1$

$$\begin{aligned} \mathbb{P}_x(X_{\tau_{Sc}} \in dy) &= \mathbb{P}_x(X_{\tau_1^\wedge} \in dy) - \int_{\mathbb{H}_{-1\downarrow}} \mathbb{P}_x(X_{\tau_{-1}^\vee} \in du) \mathbb{P}_u(X_{\tau_1^\wedge} \in dy) \\ &\quad + \int_{\mathbb{H}_{-1\downarrow}} \int_{\mathbb{H}_{1\uparrow}} \mathbb{P}_x(X_{\tau_1^\wedge} \in dz, \tau_1^\wedge < \tau_{-1}^\vee) \mathbb{P}_z(X_{\tau_{-1}^\vee} \in du) \mathbb{P}_u(X_{\tau_1^\wedge} \in dy). \end{aligned} \quad (4.13)$$

Notice that the first probability after the two integral signs in (4.13) corresponds to the same probability in the left-hand side of the same equation albeit evaluated in z instead of y . Hence, iterating $n - 1$ times and using the strong Markov property, we

obtain for $n \geq 1$

$$\begin{aligned}
& \mathbb{P}_x(X_{\tau_{Sc}} \in dy) \\
&= \sum_{k=1}^{2n} (-1)^{k-1} \int_{z_1, \dots, z_{k-1}} \mathbb{P}_x(X_{\sigma_{1,k}} \in dz_1, \dots, X_{\sigma_{k-1,k}} \in dz_{k-1}, X_{\sigma_{k,k}} \in dy) \\
&\quad + \mathcal{J}^{(2n+1)}(x, y),
\end{aligned} \tag{4.14}$$

where $X_{\sigma_{j,k}} \in \mathbb{H}_{1\uparrow}$ if $k+j$ is even, and in $\mathbb{H}_{-1\downarrow}$ otherwise. Hence, z_k is integrated in $\mathbb{H}_{1\uparrow}$ or in $\mathbb{H}_{-1\downarrow}$ accordingly, and for $n \geq 1$

$$\begin{aligned}
\mathcal{J}^{(2n+1)}(x, y) &= \int_{u, z_1, \dots, z_{2n-1}} \mathbb{P}_x(X_{\tau_{Sc}} \in du, \tau_1^\wedge < \tau_{-1}^\vee) \\
&\quad \times \mathbb{P}_u(X_{\sigma_{1,2n}} \in dz_1, \dots, X_{\sigma_{2n-1,2n}} \in dz_{n-1}, X_{\sigma_{2n,2n}} \in dy).
\end{aligned} \tag{4.15}$$

Let us prove that the factor $\mathcal{J}^{(2n+1)}$ tends to zero as $n \rightarrow \infty$. Fix $u, y \in \mathbb{H}_{1\uparrow}$, and consider the factor $\mathbb{P}_u(X_{\sigma_{1,2n}} \in dz_1, \dots, X_{\sigma_{2n-1,2n}} \in dz_{2n-1}, X_{\sigma_{2n,2n}} \in dy)$. By applying the strong Markov property we can rewrite it as

$$\mathbb{P}_u(X_{\tau_{-1}^\vee} \in dz_1) \mathbb{P}_{z_1}(X_{\tau_1^\wedge} \in dz_2) \times \dots \times \mathbb{P}_{z_{2n-2}}(X_{\tau_{-1}^\vee} \in dz_{2n-1}) \mathbb{P}_{z_{2n-1}}(X_{\tau_1^\wedge} \in dy)$$

which by the multiplication rule can be written as

$$\begin{aligned}
& \mathbb{P}_{u^{(1)}}(X_{\tau_{-1}^\vee}^{(1)} \in dz_1^{(1)}) \mathbb{P}_{u^{(2:d)}}(X_{\tau_{-1}^\vee}^{(2:d)} \in dz_1^{(2:d)} | X_{\tau_{-1}^\vee}^{(1)} \in dz_1^{(1)}) \\
& \quad \times \mathbb{P}_{z_1^{(1)}}(X_{\tau_1^\wedge}^{(1)} \in dz_2^{(1)}) \mathbb{P}_{z_1^{(2:d)}}(X_{\tau_1^\wedge}^{(2:d)} \in dz_2^{(2:d)} | X_{\tau_1^\wedge}^{(1)} \in dz_2^{(1)}) \times \\
& \quad \cdot \cdot \cdot \\
& \quad \times \mathbb{P}_{z_{2n-1}^{(1)}}(X_{\tau_1^\wedge}^{(1)} \in dy^{(1)}) \mathbb{P}_{z_{2n-1}^{(2:d)}}(X_{\tau_1^\wedge}^{(2:d)} \in dy^{(2:d)} | X_{\tau_1^\wedge}^{(1)} \in dy^{(1)}).
\end{aligned}$$

All the conditional probabilities in the previous expression are, by equation (4.5), probabilities of a $(d-1)$ -dimensional Cauchy random variable. Indeed, we can rewrite the

previous as

$$\begin{aligned}
& \mathbb{P}_{u^{(1)}}(X_{\tau_1^\vee}^{(1)} \in dz_1^{(1)}) \mathbb{P}(\text{Cauchy}_{d-1}(u^{(2:d)}, u^{(1)} - z^{(1)}) \in dz_1^{(2:d)}) \\
& \quad \times \mathbb{P}_{z_1^{(1)}}(X_{\tau_1^\wedge}^{(1)} \in dz_2^{(1)}) \mathbb{P}(\text{Cauchy}_{d-1}(z_1^{(2:d)}, z_2^{(1)} - z_1^{(1)}) \in dz_2^{(2:d)}) \times \\
& \quad \cdot \cdot \cdot \\
& \quad \times \mathbb{P}_{z_{2n-1}^{(1)}}(X_{\tau_1^\wedge}^{(1)} \in dy^{(1)}) \mathbb{P}(\text{Cauchy}_{d-1}(z_{2n-1}^{(2:d)}, y^{(1)} - z_{2n-1}^{(1)}) \in dy^{(2:d)}).
\end{aligned}$$

Ignoring the factors that do not depend on the $(2:d)$ coordinates and integrating with respect to $z_1^{(2:d)}, z_2^{(2:d)}, \dots, z_{2n-1}^{(2:d)}$ we observe, by the infinite divisibility of the Cauchy distribution, that the integral with respect to $z_1^{(2:d)}, z_2^{(2:d)} \dots z_{2n-1}^{(2:d)}$ in the second line of Equation (4.15) is equal to

$$\begin{aligned}
& \mathbb{P}\left(\text{Cauchy}_{d-1}\left(u^{(2:d)}, b_n\right) \in dy^{(2:d)}\right) \\
& = \frac{\Gamma(d/2)}{\pi^{d/2}} \frac{b_n}{|b_n^2 + |y^{(2:d)} - u^{(2:d)}|^2|^{d/2}} dy^{(2:d)}, \tag{4.16}
\end{aligned}$$

where $b_n = \sum_{k=1}^{2n-1} |z_k^{(1)} - z_{k-1}^{(1)}| + (y^{(1)} - z_{2n-1}^{(1)})$. Finally, notice that by the jumps across $\mathbb{H}_{1\uparrow}$ and $\mathbb{H}_{-1\downarrow}$ in the sequence $(z_k)_{k=1}^{2n-1}$ imply that $b_n > 2(2n-1)$, therefore the dominated convergence theorem guarantees that $\mathcal{J}^{2n+1}(x, y)$ tends to zero as $n \rightarrow \infty$.

Theorem 4.5 is proved after observing that we can use, in the same way we did above, the product rule, equation (4.5) and the infinite divisibility of the Cauchy distribution in

$$\sum_{k=1}^{2n} (-1)^{k-1} \int_{z_1, \dots, z_{k-1}} \mathbb{P}_x(X_{\sigma_{1,k}} \in dz_1, \dots, X_{\sigma_{k-1,k}} \in dz_{k-1}, X_{\sigma_{k,k}} \in dy)$$

and take the limit as $n \rightarrow \infty$ in (4.14).

□

4.2.4 More about the WoS and Monte Carlo simulation

In principle, although it was not originally formulated for unbounded sets, one could use the walk-on-spheres method for sampling the first hit into the slab $S = (-1, 1) \times \mathbb{R}$. However, even realising this, one of the reasons to prefer the walk-on-half-spaces algorithm over the WoS is its improved efficiency. For example, consider $x = (x^{(1)}, 0, \dots, 0)$ above the slab, i.e. $x^{(1)} > 1$, the probability that the WoS requires only one simulation

to get inside the slab is

$$\begin{aligned}\mathbb{P}_x(X_{\tau_{B_{x^{(1)}-1}(x)}} \in S) &= \mathbb{P}\left(X_{\tau_{B_1(0)}} \in \left(\frac{-1-x^{(1)}}{x^{(1)}-1}, -1\right) \times \mathbb{R}^{d-1}\right) \\ &= \int_{\{y \in \left(\frac{-1-x^{(1)}}{x^{(1)}-1}, -1\right) \times \mathbb{R}^{d-1}\}} C_{\alpha,d} |1 - |y|^2|^{-\alpha/2} |x - y|^{-d} dy, \quad (4.17)\end{aligned}$$

where $C_{\alpha,d} = \pi^{-(d/2+1)} \Gamma(d/2) \sin(\alpha\pi/2)$ and in the first equality we used shifting and scaling as in Lemma 4.4. This integral is bounded by 1/2, due to the 1/2 probability of exiting the ball $B_{x^{(1)}-1}(x)$ by its upper hemisphere.

On the other hand, in the WoHS algorithm, the probability of entering the slab in the first jump is given by

$$\mathbb{P}_{x^{(1)}}(X_{\tau_1^\vee} \in (-1, 1)) = I_{\frac{2}{x^{(1)}+1}}(\alpha/2, 1 - \alpha/2),$$

where $I_z(a, b)$ is the CDF of a Beta(a, b) distribution evaluated in $z \in (0, 1)$.

It is not obvious that the previous probability is always greater than the integral (4.17), however, we can compare these expressions numerically as functions of $x^{(1)}$ and for different values of α (see Fig.4-7), and conclude that for this domain the WoHS produces a first simulation faster than WoS.

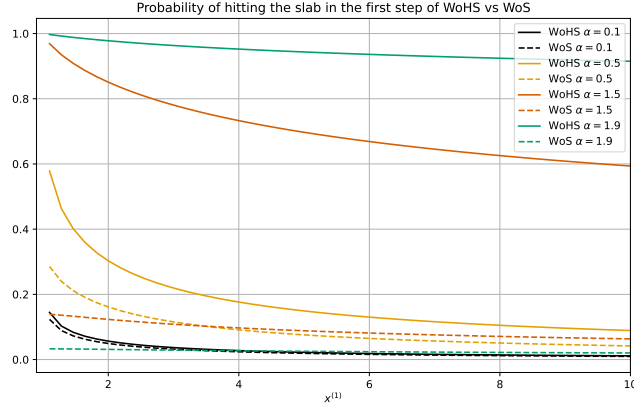


Figure 4-7: Comparison of the rate of acceptance of the first jump in WoHS vs WoS for simulating the first hit into the slab.

Notice in Fig.4-7 that there is a monotonic behaviour in terms of α for the walk-on-half-spaces algorithm, whilst for the WoS when α gets closer to 2 the probability of hitting

the slab decreases significantly. This occurs because for $\alpha \approx 2$ the law of $X_{\tau_{B_{x^{(1)}}-1}(x)}}$ is highly concentrated near the boundary of the ball, making that the only likely point to be hit near the slab is the tangent point between slab and sphere. In closing this chapter on numerical methods for determining the distribution

$$\mathbb{P}_x(X_{\tau_D} \in dy \mid \tau_D < \infty), \quad x \notin D, y \in D,$$

where $\tau_D = \inf\{t > 0 : X_t \in D\}$ for a suitably well-behaved domain D , we emphasize the potential of hybrid approaches, such as combining the Walk-on-Spheres (WoS) and Walk-on-Halfspaces (WoHS) methods, to improve computational efficiency.

Several avenues for further exploration arise naturally from this work. First, can we rigorously establish that the WoHS method provides an unbiased estimate of the solution to a fractional Dirichlet problem? While similar results have been shown in the context of non-fractional cases (e.g., $\alpha = 2$), extending these proofs to fractional settings presents challenges. Notably, the existing results by Kyprianou et al. rely heavily on Bucur's work [11], which is constrained to scenarios involving exiting balls. Broadening these proofs to more general domains remains an open question of significant theoretical interest.

Additionally, the notion of computational efficiency warrants further clarification, particularly through practical examples. For instance, consider a domain shaped like a puzzle piece, characterized by a combination of flat and concave edges. Accurately and efficiently estimating the distribution of first hitting points on such a complex domain underscores the practical utility of the combined WoS and WoHS methods, not only in the fractional case, but also in the classical Dirichlet problem (case $\alpha = 2$).

These considerations highlight the intersection of theory and computational innovation, setting the stage for future work to refine and expand the applicability of these methods to broader classes of domains and stochastic processes.

References

- [1] Larbi Alili et al. “Inversion, duality and Doob h -transforms for self-similar Markov processes”. In: *Electronic Journal of Probability* 22 (2017), pp. 1–18.
- [2] C Armero and MJ Bayarri. “Prior assessments for prediction in queues”. In: *Journal of the Royal Statistical Society Series D: The Statistician* 43.1 (1994), pp. 139–153.
- [3] Søren Asmussen. *Applied probability and queues*. Vol. 2. Springer, 2003.
- [4] Jean Bertoin. *Lévy processes*. Vol. 121. Cambridge university press Cambridge, 1996.
- [5] LE Blumenson. “A derivation of n -dimensional spherical coordinates”. In: *The American Mathematical Monthly* 67.1 (1960), pp. 63–66.
- [6] Robert M Blumenthal, Ronald K Gettoor, and DB Ray. “On the distribution of first hits for the symmetric stable processes”. In: *Transactions of the American Mathematical Society* 99.3 (1961), pp. 540–554.
- [7] Robert McCallum Blumenthal and Ronald Kay Gettoor. *Markov processes and potential theory*. Courier Corporation, 2007.
- [8] Krzysztof Bogdan and T Żak. “On Kelvin transformation”. In: *Journal of Theoretical Probability* 19.1 (2006), pp. 89–120.
- [9] Krzysztof Bogdan et al. *Potential analysis of stable processes and its extensions*. Springer Science & Business Media, 2009.
- [10] Andrei N Borodin and Paavo Salminen. *Handbook of Brownian motion-facts and formulae*. Springer Science & Business Media, 2015.
- [11] Claudia Bucur. “Some observations on the Green function for the ball in the fractional Laplace framework”. In: *arXiv preprint arXiv:1502.06468* (2015).

- [12] Tomasz Byczkowski, J Małecki, and Michal Ryznar. “Bessel potentials, hitting distributions and Green functions”. In: *Transactions of the American Mathematical Society* 361.9 (2009), pp. 4871–4900.
- [13] Maria Emilia Caballero and Loïc Chaumont. “Conditioned stable Lévy processes and the Lamperti representation”. In: *Journal of Applied Probability* 43.4 (2006), pp. 967–983.
- [14] ME Caballero, Juan-Carlos Pardo, and JL Pérez. “Explicit identities for Lévy processes associated to symmetric stable processes”. In: *Bernoulli* 17.1 (2011), pp. 34–59.
- [15] Djalil Chafaï, Edward B Saff, and Robert S Womersley. “On the solution of a Riesz equilibrium problem and integral identities for special functions”. In: *Journal of Mathematical Analysis and Applications* 515.1 (2022), p. 126367.
- [16] Kai Lai Chung. “Probabilistic approach in potential theory to the equilibrium problem”. In: *Annales de l’institut Fourier*. Vol. 23. 3. 1973, pp. 313–322.
- [17] Erhan Çinlar. “Entrance-exit distributions for Markov additive processes”. In: *Stochastic Systems: Modeling, Identification and Optimization, I* (1976), pp. 22–38.
- [18] Erhan Çinlar. *Lévy systems of Markov additive processes*. Tech. rep. Discussion Paper, 1973.
- [19] Erhan Çinlar. “Markov additive processes. II”. In: *Zeitschrift für Wahrscheinlichkeitstheorie und verwandte Gebiete* 24.2 (1972), pp. 95–121.
- [20] Lawrence C Evans. *Partial differential equations*. Vol. 19. American Mathematical Society, 2022.
- [21] Nobuyuki Ikeda and Shinzo Watanabe. “On some relations between the harmonic measure and the Lévy measure for a certain class of Markov processes”. In: *Journal of Mathematics of Kyoto University* 2.1 (1962), pp. 79–95.
- [22] Kiyosi Itô, P Henry Jr, et al. *Diffusion processes and their sample paths: Reprint of the 1974 edition*. Springer Science & Business Media, 1996.
- [23] Jevgenijs Ivanovs and Jakob D Thøstesen. “Lévy processes conditioned to stay in a half-space with applications to directional extremes”. In: *Modern Stochastics: Theory and Applications* 10.1 (2022), pp. 59–75.
- [24] H Kaspi. “Excursion laws of Markov processes in classical duality”. In: *The Annals of Probability* 13.2 (1985), pp. 492–518.

- [25] H Kaspi. “On the symmetric Wiener-Hopf factorization for Markov additive processes”. In: *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* 59.2 (1982), pp. 179–196.
- [26] Harry Kesten. “Random walks with absorbing barriers and Toeplitz forms”. In: *Illinois Journal of Mathematics* 5.2 (1961), pp. 267–290.
- [27] Alexey Kuznetsov, Andreas E Kyprianou, and Juan C Pardo. “Meromorphic Lévy processes and their fluctuation identities”. In: (2012).
- [28] Mateusz Kwaśnicki et al. “Oscillatory attraction and repulsion from a subset of the unit sphere or hyperplane for isotropic stable Lévy processes”. In: *A Lifetime of Excursions Through Random Walks and Lévy Processes: A Volume in Honour of Ron Doney’s 80th Birthday* (2021), pp. 283–313.
- [29] AE Kyprianou, M Motala, and V Rivero. “Williams’ path decomposition for self-similar Markov processes in $\mathbb{R}^d$ ”. In: *Electronic Journal of Probability* 29 (2024), pp. 1–31.
- [30] Andreas Kyprianou et al. “Entrance laws at the origin of self-similar Markov processes in high dimensions”. In: *Transactions of the American Mathematical Society* 373.9 (2020), pp. 6227–6299.
- [31] Andreas E Kyprianou. “Deep factorisation of the stable process”. In: *Electronic Journal of Probability* 21 (2016), pp. 1–28.
- [32] Andreas E Kyprianou. *Introductory lectures on fluctuations of Lévy processes with applications*. Springer Science & Business Media, 2006.
- [33] Andreas E Kyprianou, Ana Osojnik, and Tony Shardlow. “Unbiased ‘walk-on-spheres’ Monte Carlo methods for the fractional Laplacian”. In: *IMA Journal of Numerical Analysis* 38.3 (2018), pp. 1550–1578.
- [34] Andreas E Kyprianou, Sandra Palau, and Tsogzolmaa Saizmaa. “Attraction to and repulsion from a subset of the unit sphere for isotropic stable Lévy processes”. In: *Stochastic Processes and their Applications* 137 (2021), pp. 272–293.
- [35] Andreas E Kyprianou and Juan Carlos Pardo. *Stable Lévy processes via Lamperti-type representations*. Vol. 7. Cambridge University Press, 2022.
- [36] Andreas E Kyprianou, Juan Carlos Pardo, and Alexander R Watson. “Hitting distributions of α -stable processes via path censoring and self-similarity”. In: *The Annals of Probability* 42.1 (2014), pp. 398–430.
- [37] Andreas E Kyprianou, Victor Rivero, and Weerapat Satitkanitkul. “Deep factorisation of the stable process III: the view from radial excursion theory and the point of closest reach”. In: *Potential Analysis* (2019), pp. 1–29.

- [38] Andreas E Kyprianou, Victor Rivero, and Batı Şengül. “Deep factorisation of the stable process II: Potentials and applications”. In: *Annales de l’Institut Henri Poincaré, Probabilités et Statistiques*. Vol. 54. 1. Institut Henri Poincaré. 2018, pp. 343–362.
- [39] Andreas E Kyprianou and Alexander R Watson. “Potentials of stable processes”. In: *Séminaire de Probabilités XLVI*. Springer, 2014, pp. 333–343.
- [40] John Lamperti. “Semi-stable Markov processes. I”. In: *Zeitschrift für Wahrscheinlichkeitstheorie und verwandte Gebiete* 22.3 (1972), pp. 205–225.
- [41] Bernard Maisonneuve. “Exit systems”. In: *The Annals of Probability* (1975), pp. 399–411.
- [42] Peter Mörters and Yuval Peres. *Brownian motion*. Vol. 30. Cambridge University Press, 2010.
- [43] Masao Nagasawa. “Time reversions of Markov processes”. In: *Nagoya Mathematical Journal* 24 (1964), pp. 177–204.
- [44] MM Osypchuk and MI Portenko. “On the distribution of a rotationally invariant α -stable process at the hitting time of a given hyperplane”. In: *Reports of the National Academy* 12 (2018), pp. 14–20.
- [45] Sidney Port. *Brownian motion and classical potential theory*. Elsevier, 2012.
- [46] Sidney C Port. “A remark on hitting places for transient stable process”. In: *The Annals of Mathematical Statistics* 39.2 (1968), pp. 365–371.
- [47] Sidney C Port. “The first hitting distribution of a sphere for symmetric stable processes”. In: *Transactions of the American Mathematical Society* 135 (1969), pp. 115–125.
- [48] Christophe Profeta and Thomas Simon. “On the harmonic measure of stable processes”. In: *Séminaire de probabilités XLVIII*. Springer, 2016, pp. 325–345.
- [49] Daniel Revuz and Marc Yor. *Continuous martingales and Brownian motion*. Vol. 293. Springer Science & Business Media, 2013.
- [50] LCG Rogers. “Williams’ characterisation of the Brownian excursion law: proof and applications”. In: *Séminaire de Probabilités XV 1979/80*. Springer, 1981, pp. 227–250.
- [51] Leonard CG Rogers and David Williams. *Diffusions, Markov processes, and martingales: Volume 1, foundations*. Cambridge University Press, 2000.

- [52] Boris Alekseevich Rogozin. “The distribution of the first hit for stable and asymptotically stable walks on an interval”. In: *Teoriya Veroyatnostei i ee Primeneniya* 17.2 (1972), pp. 342–349.
- [53] Yozo Tamura and Hiroshi Tanaka. “On a formula on the potential operators of absorbing Lévy processes in the half space”. In: *Stochastic processes and their applications* 118.2 (2008), pp. 199–212.
- [54] John B Walsh. “Markov processes and their functional in duality”. In: *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* 24.3 (1972), pp. 229–246.
- [55] Harold Widom. “Stable processes and integral equations”. In: *Transactions of the American Mathematical Society* 98.3 (1961), pp. 430–449.
- [56] M Yor. “À propos de l’inverse du mouvement brownien dans $R^n, n \geq 3$ ”. In: *Annales de l’IHP Probabilités et statistiques*. Vol. 21. 1. 1985, pp. 27–38.
- [57] Marc Yor. “On some exponential functionals of Brownian motion”. In: *Advances in applied probability* 24.3 (1992), pp. 509–531.